Phase 1 PDAC IND:
AI Generation
Draft 1.0

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Cover Letter

ChemicalQDevice
San Diego, California
July 1, 2026

Food and Drug Administration
Center for Drug Evaluation and Research
Division of Oncology 1
5901 to 5913 Ammendale Road
Beltsville, Maryland 20705 to 1266

Re: Initial Investigational New Drug Application, Serial 0000 - Daraxonrasib (RMC-6236), Phase 1, LLM-Directed Robotic Whipple in KRAS-Mutated PDAC

Dear Division of Oncology 1 Reviewer:

On behalf of ChemicalQDevice, and in my capacity as sponsor-investigator, I am submitting this initial Investigational New Drug Application (IND v1.0) under 21 CFR §312.23 for a Phase 1, first-in-human, prospective open-label single-arm study of daraxonrasib (RMC-6236), an oral once-daily RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor, administered in the perioperative setting of a large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) for patients with resectable or borderline-resectable, KRAS G12 mutated pancreatic ductal adenocarcinoma (PDAC) [1, 2]. This application is assigned serial number 0000 as the initial submission and is transmitted with FDA Form 1571 as the cover sheet for the submission and the sponsor agreement, together with FDA Form 3674 (clinicaltrials.gov certification) and FDA Form 1572 [3]. The mechanistic premise is well established: daraxonrasib binds GTP-bound active mutant RAS in complex with cyclophilin A and thereby suppresses downstream effector signaling, and the agent received FDA Breakthrough Therapy designation in June 2025 for previously treated metastatic PDAC bearing KRAS G12 alterations [1, 4].

This submission is combined in nature. It is concurrently an IND under 21 CFR Part 312 for the investigational drug daraxonrasib and an Investigational Device Exemption (IDE) under 21 CFR Part 812 for the significant-risk, Class II collaborative on-premises robotic surgical device that directs the operative procedure as an early-feasibility study [2, 5]. The LLM functions strictly as a hash-pinned, second-opinion advisory oracle isolated outside the vendor kinematic stack; full device autonomy is prohibited in accordance with 21 CFR §312.21(e), and a verification-before-generation ten-gate suite governs every advisory output (ACCEPT, BLOCK, or ESCALATE) consistent with H.R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026 [6, 7]. The dual regulatory pathways are presented in an integrated dossier so that the drug and device safety considerations are reviewed together rather than in isolation, and so that any drug-device interaction is visible to a single reviewing team.

We respectfully request the standard 30-day safety review period under 21 CFR §312.40, after which, absent a clinical hold under 21 CFR §312.42, the sponsor-investigator intends to initiate the study at a single academic site [8]. The dose-escalation design follows a 3+3 schema across three oral once-daily dose levels (DL1 160 mg, DL2 220 mg, DL3 300 mg, the RASolute 302 recommended Phase 2 dose) with a 28-day dose-limiting toxicity window, enrolling up to 18 treated subjects from approximately 36 screened, with staggered sentinel enrollment and an accrual target of roughly one to two subjects per month [1].

Study at a glance. Table C.1 summarizes the principal design parameters so that the reviewer has the operative facts of the investigation on the first page of the submission.

Parameter	Specification
Investigational drug	Daraxonrasib (RMC-6236), oral RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor, once daily
Investigational device	On-premises LLM-directed eight-arm collaborative robotic Whipple (8 arms $\times$ 7 degrees of freedom = 56 degrees of freedom; 640 sensor channels), Class II, significant-risk, early-feasibility
Indication	Resectable or borderline-resectable PDAC with documented KRAS G12 (G12D, G12V, or G12R); ECOG 0 to 1; Whipple candidate
Phase and design	Phase 1, first-in-human, prospective open-label single-arm; combined IND (21 CFR Part 312) and IDE (21 CFR Part 812)
Dose levels	DL1 160 mg, DL2 220 mg, DL3 300 mg once daily; 3+3 escalation; 28-day dose-limiting toxicity window
Sample size	Up to 18 treated subjects (approximately 36 screened); 1 academic site; 1 to 2 enrollments per month
Co-primary endpoints	Device and procedure SAE incidence through 30 days; MTD and RP2D of daraxonrasib; technical feasibility (task completion without unplanned conversion attributable to device malfunction or unsafe behavior)
LLM role	Hash-pinned second-opinion advisory oracle only; full autonomy prohibited (21 CFR §312.21(e)); ten-gate verification-before-generation suite

Table C.1. Principal design parameters of the Phase 1 LLM-directed robotic Whipple study of daraxonrasib in KRAS G12 mutated PDAC.

Contents enclosed. The following components are submitted in the ReGARDD IND order, with the Cover Letter and FDA Forms 1571 and 3674 preceding the generated Table of Contents:

Item	Component
1	FDA Forms 1571 and 3674 (3674 Box B selected; PHS Act 402(j) registration requirements addressed)
2	Table of Contents (generated)
3	Introduction, including drug name, pharmacological class, structure, formulation, route, and duration
4	General Investigational Plan, including the 3+3 escalation rationale and first-year plan
5	Investigator Brochure
6	Proposed Clinical Research (study protocol v1.0.0, informed consent, Form 1572, financial disclosures)
7	Chemistry, Manufacturing, and Control information (drug substance, drug product, placebo, labeling, environmental assessment)
8	Pharmacology and Toxicology information, including the integrated computational evidence base
9	Previous Human Experience (RASolute 302, RASolve 301, Breakthrough Therapy designation)
10	Additional Information (drug dependence, radioactive drugs, pediatric studies)
11	Relevant Information and References with Back Matter

Table C.2. Enclosed components of the initial IND, in ReGARDD order, with the Cover Letter and FDA Forms preceding the generated Table of Contents.

Regulatory basis for each enclosure. Each component answers a specific content requirement of 21 CFR §312.23(a), and Table C.3 maps the enclosure to its codified citation so that the reviewer can confirm completeness against the regulation directly. The mapping also makes explicit where the combined drug and device structure of this submission adds content beyond the conventional drug-only IND, in particular the integrated Physical AI safety, governance, and verification material that supports the IDE arm [3, 5].

IND content element	21 CFR citation	Where addressed
Cover sheet and sponsor agreement	§312.23(a)(1)	FDA Form 1571, serial 0000
Table of contents	§312.23(a)(2)	Generated Table of Contents
Introductory statement and general investigational plan	§312.23(a)(3)	Introduction and General Investigational Plan
Investigator brochure	§312.23(a)(5)	Investigator Brochure
Protocols	§312.23(a)(6)	Proposed Clinical Research, protocol v1.0.0
Chemistry, manufacturing, and control	§312.23(a)(7)	CMC information
Pharmacology and toxicology	§312.23(a)(8)	Pharmacology and Toxicology, with computational evidence base
Previous human experience	§312.23(a)(9)	Previous Human Experience
Additional information	§312.23(a)(10)	Additional Information
Significant-risk device and safety reporting	21 CFR Part 812; §312.32; §312.404	IDE arm; integrated Physical AI safety reporting

Table C.3. Cross-reference of each enclosed IND component to its governing provision of 21 CFR §312.23(a) and to the supporting device and safety regulations engaged by the combined submission.

Sponsor-investigator and contact. The sponsor-investigator is Kevin Kawchak, Chief Executive Officer, ChemicalQDevice, San Diego, California, who may be reached at kevink@chemicalqdevice.com. The sponsor-investigator's financial interest is disclosed under 21 CFR Part 54 and mitigated by a three-tier independent oversight structure comprising a Data Safety Monitoring Board with halt authority at cohort boundaries and on triggered review, an Independent Safety Monitor with real-time per-procedure stop authority, and a Physical AI Safety Review Committee operating on a cadence of no more than 90 days [9]. Binding halt rules require Data Safety Monitoring Board review and possible halt upon any device or procedure serious adverse event in at least one of three subjects within a cohort, or upon two device-related deaths at any time, providing an explicit and conservative stopping boundary for this first-in-human Physical AI investigation.

Safety reporting commitments. The sponsor-investigator will comply with the expedited reporting timelines of 21 CFR §312.32 and with the Physical AI reporting framework engaged by the device arm. Table C.4 states the controlling clocks that will govern the conduct of the study from first dose onward [3].

Reportable event	Clock	Basis
Fatal or life-threatening suspected adverse reaction	7 calendar days	21 CFR §312.32(c)(1)(i)(A)
Other serious and unexpected suspected adverse reaction	15 calendar days	21 CFR §312.32(c)(1)(i)(B)
Serious Physical AI adverse event	7 or 15 calendar days	21 CFR §312.32(f), (g)
Cybersecurity incident with potential for patient harm	7 calendar days	21 CFR §312.32(g)
Cybersecurity incident without patient-harm potential	15 calendar days	21 CFR §312.32(g)
Sustained AI model degradation, sim-to-real divergence beyond twice tolerance, or digital-twin discrepancy	15 calendar days	21 CFR §312.32(g)

Table C.4. Expedited safety reporting clocks the sponsor-investigator commits to under 21 CFR §312.32, including the six Physical AI reporting triggers engaged by the investigational device.

AI-assisted assembly and real-time monitorability. This IND was assembled under a single master prompt that executed a four-stage mermaid to draft to full to final build, with each generated file committed and pushed to one continuously updated GitHub pull request in real time, so that the complete authoring history of the submission is human-monitorable as it is produced and is reproducible from a hash-pinned deterministic commit (repository v4.3.0) [10, 11]. The same verification-before-generation discipline applied to clinical advisory outputs governs document generation: each figure is reproduced exactly from a grayscale Mermaid source, each numeric value is grounded in the protocol and the computational evidence base, and each codified citation is checked against the regulation [7, 12]. Figure 1 shows the single-prompt build pipeline and Figure 2 shows the ReGARDD Table-of-Contents architecture that maps each numbered IND section to exactly one source file. The repository is available at `https://github.com/kevinkawchak/physical-ai-oncology-trials/tree/main/trial-ind`.

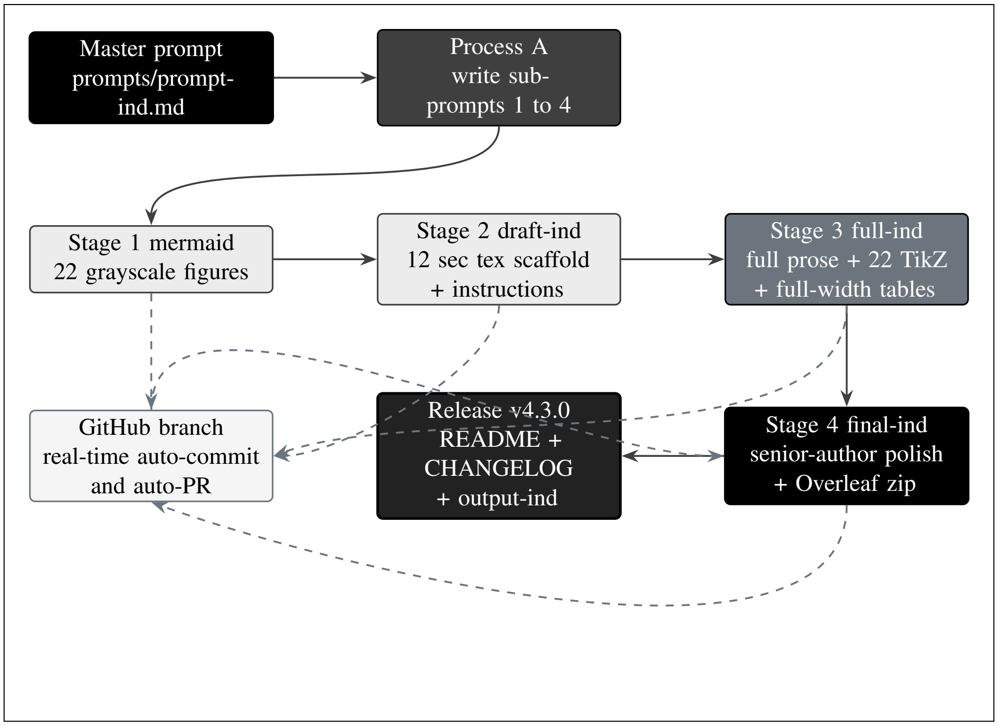

Figure 1. The single-prompt mermaid to draft to full to final build pipeline for the Phase 1 PDAC IND. One master prompt drives Process A (write the four sub-prompts) and then Process B (execute them as Stages 1 to 4). Dashed edges show that every generated file is committed and pushed to one continuously updated pull request in real time, and the feedback edge shows author review returning into the final stage, all without manual intervention.

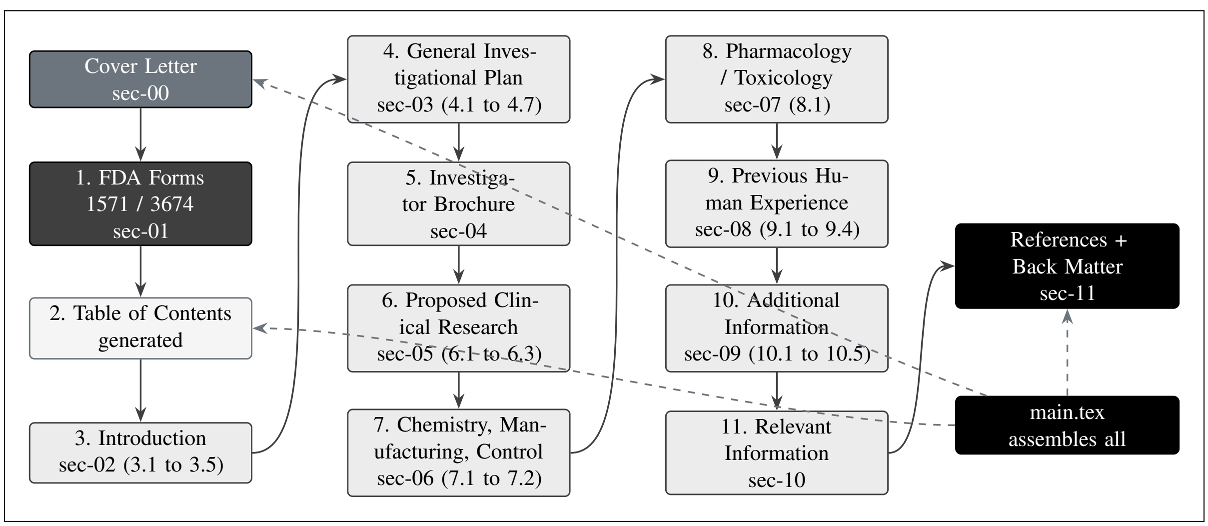

Figure 2. The ReGARDD IND Table of Contents reproduced as the file architecture of the build. The Cover Letter and the FDA Forms 1571 and 3674 precede the generated Table of Contents; the eleven numbered IND sections and the References and Back Matter each become exactly one sections/sec tex file assembled by main.tex, so the document structure and the repository structure match one-to-one.

Scientific and computational support. The pharmacology and toxicology section is supported by an integrated computational evidence base developed by the sponsor-investigator across 2025 and 2026, including a 60-second PDAC Whipple and daraxonrasib simulation, an AI digital-twin PDAC simulation, a quantitative systems pharmacology metastatic-PDAC model with verification, validation, and uncertainty quantification, and a 100,000-patient in silico Phase 3 five-arm PDAC study performed in triplicate [13, 14, 15, 16]. These analyses do not substitute for the empirical safety review but provide quantitative priors that inform the dose levels, the perioperative pause-and-restart advisory, and the sim-to-real tolerances that the device must satisfy before any patient-facing robotic activity. The clinical rationale rests on the severity of the indication: PDAC carries a five-year overall survival below 13 percent and ranks third in United States and fourth in European cancer mortality, so even a first-in-human feasibility study is justified by the magnitude of unmet need and by the precision of the proposed estimation framework [4, 17].

Statistical framing. The study is descriptive and non-confirmatory in design, consistent with ICH E9 and TRIPOD+AI, with no formal power calculation; precision is reported through exact Clopper-Pearson 95 percent confidence intervals [18, 12]. As a worked example grounded in the maximum sample size, an observed rate of 0 events in 18 treated subjects corresponds to a point estimate of 0 percent with an exact 95 percent interval of 0 to 18 percent, while a single event in 18 subjects corresponds to 5.6 percent with an interval of 0.1 to 27 percent; at the cohort level, 0 of 6 corresponds to an interval of 0 to 46 percent. These intervals make explicit the limits of inference at first-in-human scale and frame all supportive two-sided testing at alpha 0.05 without multiplicity adjustment [18]. Informed consent will be obtained under 21 CFR Part 50 and ICH E6(R3), with a Physical AI opt-out under 21 CFR §312.60(f), explicit treatment of therapeutic misconception, and a plain-language explanation of staged autonomy, which for this Phase 1 is limited to clinician-controlled teleoperation and supervised operation only [19].

We appreciate the Division's review and welcome any questions during the 30-day period. The sponsor-investigator is available for a teleconference at the Division's convenience and will respond promptly to any request for additional information.

Respectfully submitted,

Kevin Kawchak
Chief Executive Officer and Sponsor-Investigator
ChemicalQDevice, San Diego, California
kevink@chemicalqdevice.com
IND v1.0; repository v4.3.0

1 FDA Forms 1571 and 3674

The administrative gateway to this initial Investigational New Drug (IND) application is the pair of executed Food and Drug Administration (FDA) forms that open every Center for Drug Evaluation and Research (CDER) submission: FDA Form 1571 (Investigational New Drug Application) and FDA Form 3674 (Certification of Compliance, under 42 U.S.C. §282(j)). These two forms are positioned, together with the Cover Letter, ahead of the Table of Contents so that a reviewer encounters the sponsor agreement, the serial number, the contents map, and the clinicaltrials.gov certification before any scientific module. They are the regulatory envelope around the LLM-Directed PDAC Robotic Daraxonrasib Phase 1 protocol (protocol v1.0.0), a first-in-human, prospective, open-label, single-arm, early-feasibility study that combines an IND under 21 CFR Part 312 with an Investigational Device Exemption (IDE) framework under 21 CFR Part 812 for an on-premises, LLM-directed, eight-arm robotic pancreaticoduodenectomy (Whipple) device used together with oral daraxonrasib (RMC-6236), an oral RAS(ON) multi-selective (pan-RAS) inhibitor [1, 2]. This section summarizes the field-by-field content and the governing logic of each form; the executed, signed PDF versions of FDA Form 1571 and FDA Form 3674 are assembled with the physical submission and are not reproduced inside this LaTeX source, in keeping with FDA's expectation that the original signed forms accompany the package [8, 3].

The two forms discharge two distinct legal functions that together constitute the administrative predicate for FDA's acceptance of the file. Form 1571 is the instrument through which the sponsor-investigator, ChemicalQDevice (Kevin Kawchak, Chief Executive Officer, as the named sponsor-investigator), formally undertakes the obligations of a sponsor under 21 CFR Part 312, subpart D, and simultaneously supplies the cover-sheet metadata (serial number, contents, monitoring and safety responsibilities) that FDA's document-control system uses to file each piece of the record. Form 3674 is the instrument through which the sponsor discharges the separate statutory duty, created by section 402(j) of the Public Health Service (PHS) Act and enforced through 21 U.S.C. §355 and 42 U.S.C. §282(j), to certify the clinicaltrials.gov status of every clinical investigation referenced in the application. A defect in either form, an unsigned box 13 on Form 1571 or an unselected certification box on Form 3674, is a clerical deficiency that can stall acceptance of an otherwise complete scientific package, which is why this section treats both forms with the same rigor applied to the protocol and the Investigator's Brochure [8, 9].

Figure 3 traces the cover-sheet flow from the Cover Letter through Form 1571 (its serial logic and dual purpose) into Form 3674 (its three mutually exclusive certification boxes) and onward to the assembled IND that starts the 30-day FDA review clock under 21 CFR §312.40.

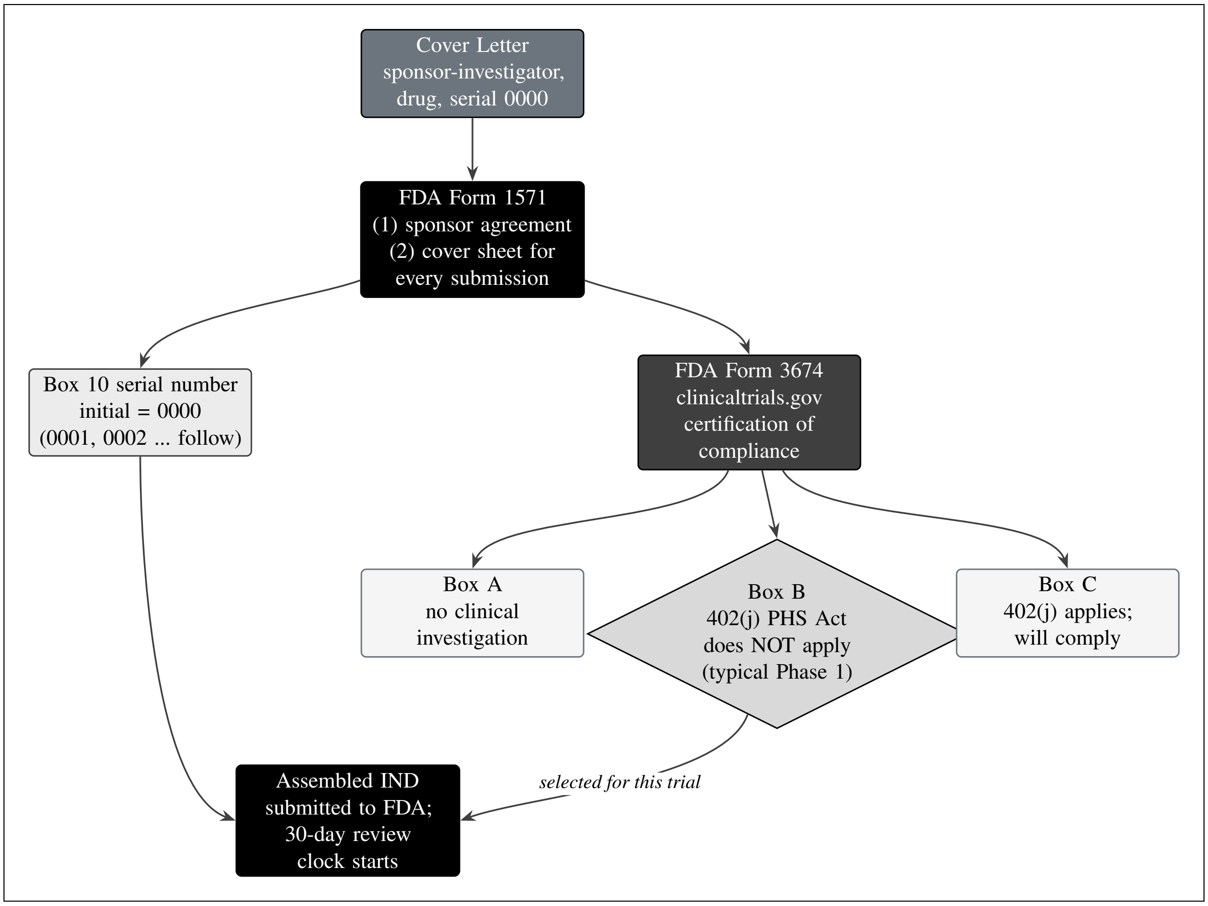

Figure 3. The Cover Letter, FDA Form 1571, and FDA Form 3674 cover-sheet flow. Form 1571 serves the dual purpose of a sponsor-investigator agreement and a cover sheet for every submission; the initial serial number in box 10 is 0000. Form 3674 certifies clinicaltrials.gov compliance through one of Box A (no clinical investigation), Box B (an investigation to which 42 U.S.C. §282(j) / section 402(j) of the PHS Act does not apply, the typical Phase 1 selection), or Box C (the requirements apply). The Cover Letter and the FDA forms precede the Table of Contents.

1.1 FDA Form 1571 - Investigational New Drug Application

FDA Form 1571 serves a dual purpose that defines its central place in the submission. First, it is the binding sponsor-investigator agreement: by signing box 13, the sponsor-investigator, ChemicalQDevice (Kevin Kawchak, Chief Executive Officer, as the named sponsor-investigator), commits that the investigation will not begin until 30 days after FDA receives the IND (unless notified sooner that the studies may proceed), that a reviewing Institutional Review Board (IRB) compliant with 21 CFR Part 56 will be responsible for the initial and continuing review of each study, that the investigation will be conducted in accordance with all applicable regulatory requirements, and that an investigator will obtain informed consent under 21 CFR Part 50 [19, 18]. Second, and on a recurring basis, Form 1571 is the cover sheet that accompanies every subsequent submission to the IND file, including protocol amendments, information amendments, IND safety reports, annual reports, and general correspondence; each such submission carries its own Form 1571 bearing the next sequential serial number so that the agency can reconstruct the complete history of the file [8].

The form opens with sponsor identification and contact fields. Box 1 (Name of Sponsor) and boxes 2 through 5 (Address, including street, city, state, and ZIP) carry the sponsor-investigator entity ChemicalQDevice and its San Diego, California place of business; box 6 (Telephone Number) and the associated electronic-contact information complete the routing so that the assigned project manager and the safety contact can be reached. Box 7 captures the name(s) of any contract research organization, which for this single-academic-site study is limited to the named clinical site and the entities providing investigational product and device support. The IND Number field is left blank on the initial submission because the number is assigned by FDA upon receipt; once assigned, that pre-assigned or receipt-assigned IND number is carried on the Form 1571 of every later serial submission. Box 9 (Name of Drug) records daraxonrasib (RMC-6236), the oral RAS(ON) multi-selective small-molecule inhibitor that binds the GTP-bound active state of mutant RAS in complex with cyclophilin A and suppresses downstream effector signaling, administered once daily [1, 4].

The general-information block describes the proposed research and the phase of investigation. The form indicates that this is a Phase 1 investigation (the first-in-human checkbox), names the indication as histologically or cytologically confirmed resectable or borderline-resectable pancreatic ductal adenocarcinoma (PDAC) with a documented KRAS G12 mutation (G12D, G12V, or G12R) in patients who are candidates for pancreaticoduodenectomy, and lists the protocol title and the combined IND and IDE nature of the application. The contents-of-the- application checkboxes (box 12) are marked to indicate which items of 21 CFR §312.23 are included in this serial submission: the Table of Contents, the Introductory Statement and General Investigational Plan, the Investigator's Brochure, the study protocol(s), Chemistry, Manufacturing, and Controls (CMC) information, pharmacology and toxicology information, previous human experience, and additional information. Box 11 designates the individuals responsible for monitoring the conduct and progress of the clinical investigation and the person(s) responsible for review and evaluation of information relevant to the safety of the drug; for this study those responsibilities map onto the three-tier oversight structure (the Data and Safety Monitoring Board, the Independent Safety Monitor, and the Physical AI Safety Review Committee) described in the protocol [5, 7].

Because the present application is a combined drug-device study, the box 12 contents block carries an additional editorial burden relative to a drug-only Phase 1 IND. The CMC item must cross-reference not only the drug substance and drug product modules for daraxonrasib but also the device-description and engineering-control material that establishes the on-premises eight-arm robotic Whipple platform as a significant-risk investigational device, including its 56 degrees of freedom (eight arms at seven degrees of freedom each), its 640 sensor channels (80 per arm), its positional accuracy of 0.05 mm RMS at 1200 mm/s, and its layered force limits (no more than 3 N per arm and no more than 18 N cumulative). The pharmacology and toxicology item must reference the computational evidence base (the 60-second PDAC Whipple-plus-daraxonrasib simulation, the artificial-intelligence digital-twin PDAC simulation, the quantitative systems pharmacology metastatic-PDAC simulation with verification, validation, and uncertainty quantification, and the 100,000-patient in silico Phase 3 five-arm PDAC study) that supports the integrated risk assessment [14, 13, 15, 16]. Marking box 12 accurately therefore tells the reviewer, in a single glance, that the file is internally complete across both the drug and the device dimensions of the study.

Box 10 is the serial number. FDA requires every submission to an IND to be numbered consecutively, and the initial submission is serial number 0000. Each subsequent submission to this IND file is numbered sequentially as 0001, 0002, 0003, and so forth, in the chronological order in which the agency receives it, regardless of the submission type. For this initial IND the serial number is therefore 0000, and that value appears both in box 10 of Form 1571 and in the Cover Letter so that a reviewer can immediately confirm that this is the originating submission that starts the 30-day review period under 21 CFR

§312.40 [8, 3]. Table 1.1 summarizes the principal Form 1571 fields and the value or status of each for this initial submission.

Form 1571 field	Description and content	Value / status (serial 0000)
Boxes 1 to 6: Sponsor and contact	Sponsor-investigator name, mailing address, and telephone for the responsible entity.	ChemicalQDevice; San Diego, California; sponsor-investigator Kevin Kawchak, CEO.
IND Number	Assigned by FDA on receipt; carried on every later serial submission.	Blank on initial submission (to be assigned).
Box 7: CRO / site	Contract research organizations and clinical site(s) supporting the study.	Single academic clinical site; no separate CRO.
Box 9: Name of Drug	Investigational drug and active ingredient.	Daraxonrasib (RMC-6236), oral RAS(ON) pan-RAS inhibitor.
Phase / indication	Phase of investigation and the disease studied.	Phase 1, first-in-human; resectable or borderline-resectable KRAS G12 PDAC.
Box 10: Serial Number	Consecutive numbering of every submission to the file.	0000 (initial); 0001, 0002 ... follow.
Box 11: Monitoring / safety	Persons responsible for monitoring conduct and for safety review.	DSMB, Independent Safety Monitor, Physical AI Safety Review Committee.
Box 12: Contents	Checkboxes for the 21 CFR §312.23 items included.	Introductory statement, general plan, IB, protocol, CMC, pharm/tox, prior human experience, additional information.
Box 13: Sponsor agreement	Signed commitments (30-day wait, IRB review, GCP, informed consent).	Signed by sponsor-investigator; executed PDF assembled with submission.

Table 1.1. Field-by-field summary of FDA Form 1571 for the initial IND (serial number 0000). The executed and signed PDF of the form is assembled with the physical submission and is not reproduced in this source.

Box 13 deserves separate treatment because it is the legally operative clause of the entire form. The four commitments enumerated there map directly onto specific provisions of 21 CFR Part 312 and Part 50, and the sponsor-investigator's signature converts them from regulatory text into a personal undertaking enforceable by the agency. The 30-day wait corresponds to 21 CFR §312.40(b), which prohibits initiation of the clinical investigation until 30 days after FDA receives the IND unless the agency notifies the sponsor that studies may begin earlier; for this first-in-human Physical AI study the sponsor does not seek early initiation and will hold all patient-facing activity until the full 30-day period has elapsed and the Phase 0 simulation-validation gate (a unified safety level of at least 7.0, at least 1000 simulations across at least two frameworks, sim-to-real trajectory under 2 mm and tip-force under 0.5 N) has been satisfied [3, 17]. The IRB commitment corresponds to 21 CFR Part 56 and is reinforced by the protocol's reliance on a single reviewing IRB. The general-compliance commitment incorporates the good clinical practice expectations of ICH E6(R3), and the informed-consent commitment corresponds to 21 CFR Part 50 as supplemented by the Physical AI opt-out provision the protocol grounds in 21 CFR §312.60(f) [19, 18]. Table 1.2 sets out the four box-13 commitments alongside their codified anchors and the study-specific manner of discharge.

Box 13 commitment	**Codified anchor**	**How discharged in this study**
Investigation will not begin before 30 days after FDA receipt.	21 CFR §312.40(b)	No early-initiation request; all patient-facing activity held until the 30-day clock elapses and the Phase 0 simulation gate passes.
A Part 56 IRB will perform initial and continuing review.	21 CFR Part 56	Single reviewing IRB engaged for the academic site; continuing review at the protocol-specified cadence with cohort-boundary checkpoints.
The investigation will follow all applicable requirements.	21 CFR Part 312; ICH E6(R3)	Three-tier oversight (DSMB, Independent Safety Monitor, Physical AI Safety Review Committee) plus binding halt rules enforce ongoing compliance.
An investigator will obtain informed consent.	21 CFR Part 50; §312.60(f)	Consent under Part 50 and ICH E6(R3) with a Physical AI opt-out, therapeutic-misconception language, and staged-autonomy disclosure.

Table 1.2. The four box 13 sponsor-investigator commitments, their codified anchors in 21 CFR, and the manner in which each is discharged in the LLM-directed PDAC robotic daraxonrasib Phase 1 study.

The dual purpose of Form 1571 has a practical consequence for the lifecycle of this Physical AI study. Because each protocol amendment, information amendment, and IND safety report under 21 CFR §312.32 will itself be filed under a new serial number on its own Form 1571, the form becomes the running index of the entire investigational record, including the Physical AI specific reporting events. The protocol defines six Physical AI reporting triggers under 21 CFR §312.32(g) (a serious Physical AI adverse event reported on the 7-day or 15-day clock; a system-drug interaction within 15 days; a cybersecurity incident within 15 days, or within 7 days where there is potential for patient harm; sustained

AI model degradation over at least three consecutive procedures or 24 cumulative hours; sim-to-real divergence exceeding twice tolerance, that is a trajectory deviation greater than 4 mm or a force deviation greater than 1.0 N; and a digital-twin discrepancy), and each such report will enter the file as a serially numbered submission cover-sheeted by Form 1571 [5, 6]. In this way the form binds the administrative numbering scheme to the safety-surveillance scheme of the trial.

A worked example makes the serial-numbering discipline concrete. Suppose the initial IND (serial 0000) clears the 30-day review and the study opens; the first protocol amendment that adds a clarifying eligibility criterion would be filed as serial 0001, an information amendment updating the Investigator's Brochure as serial 0002, and the first IND safety report (for instance, a fatal or life-threatening suspected adverse reaction filed on the 7 calendar day clock under 21 CFR §312.32(c)(1)(i)) as serial 0003, each cover-sheeted by its own Form 1571 carrying the FDA-assigned IND number and the next consecutive serial. If a sustained AI model degradation trigger fires (model performance below threshold over at least three consecutive procedures or 24 cumulative hours), the resulting Physical AI safety report under §312.32(g) would take the next serial in turn. The annual report required by 21 CFR §312.33 would likewise be a serially numbered Form 1571 submission. Because the serial sequence is strictly chronological and type-agnostic, a reviewer can reconstruct the full procedural history of the file, drug and device events interleaved, simply by reading the serial numbers in order [8, 9].

The recurring cover-sheet role of Form 1571 should not be confused with FDA Form 1572 (Statement of Investigator), which is a separate document addressed elsewhere in this IND under proposed clinical research. Form 1571 binds the sponsor and indexes the file; Form 1572 binds each clinical investigator to the protocol and to the obligations of 21 CFR §312.60. For this single-site, sponsor-investigator study the two roles converge in practice on the same organization, but the forms remain distinct instruments with distinct signatories and distinct regulatory functions, and conflating them is a common cause of refuse-to-file findings that this submission is structured to avoid [8, 5].

1.2 FDA Form 3674 - Certification of Compliance

FDA Form 3674 is the certification, required by section 402(j) of the Public Health Service (PHS) Act (codified at 42 U.S.C. §282(j)), that the sponsor has complied with the registration and results-reporting requirements of clinicaltrials.gov to the extent those requirements apply to the submission. The form presents three mutually exclusive certification boxes, exactly one of which must be selected, and the selection determines what the sponsor is affirming about the clinicaltrials.gov status of the studies referenced in the application [8]. The logic of the three boxes is summarized in Table 1.3.

Box	What the box certifies	Applicability to this Phase 1

Box	What the box certifies	Applicability to this Phase 1
Box A	The submission does not reference any clinical trial; that is, there is no clinical investigation at all.	Not selected. This submission references a Phase 1 clinical investigation.
Box B	The submission references a clinical trial, but the requirements of 42 U.S.C. §282(j) (section 402(j) of the PHS Act) do not apply to that trial.	Selected. This is a Phase 1 investigation; a Phase 1 trial of a drug is excluded from the "applicable clinical trial" definition that triggers mandatory 402(j) registration and results reporting.
Box C	The submission references a clinical trial that is an applicable clinical trial to which 42 U.S.C. §282(j) does apply, and the sponsor certifies it has complied (or will comply) with registration and results reporting.	Not selected for this initial Phase 1. Would apply to a later applicable (for example, Phase 2 or later controlled) trial.

Table 1.3. The Box A / B / C certification logic of FDA Form 3674 and the selection for this initial Phase 1 IND. Exactly one box is checked; Box B is selected with the rationale stated in the text.

For this initial Phase 1 IND, Box B is selected. The rationale is that the study referenced in this application is a Phase 1, first-in-human investigation of daraxonrasib together with the LLM-directed robotic Whipple device, and a Phase 1 trial of a drug falls outside the statutory definition of an "applicable clinical trial" that triggers the mandatory clinicaltrials.gov registration and results-reporting obligations of 42 U.S.C. §282(j). Under that definition, the controlled clinical investigations of a drug or biologic that are subject to mandatory registration are generally those other than Phase 1 trials; because the present study is Phase 1 (up to n = 18 treated subjects, with approximately 36 screened, at a single academic site, using a 3+3 dose escalation across dose levels of 160 mg, 220 mg, and 300 mg once daily with a 28-day dose-limiting-toxicity window), the section 402(j) requirements do not apply to it, and Box B is therefore the correct and typical Phase 1 selection [8, 10].

It is worth stating precisely why the "applicable clinical trial" test resolves to Box B here, because the device dimension of this study could otherwise invite confusion. The section 402(j) definition reaches two families of investigations: applicable drug clinical trials, which are controlled clinical investigations (other than Phase 1 investigations) of a drug subject to section 505 of the Federal Food, Drug, and Cosmetic Act; and applicable device clinical trials, which are prospective clinical studies of health outcomes comparing an intervention against a control in a device subject to section 510(k), 515, or 522, together with certain pediatric postmarket surveillance. The present study is a single-arm, open-label, non-comparative early-feasibility study with no control arm and no comparison of health outcomes between groups; it does not meet the "controlled clinical investigation comparing an intervention with a control" element of either family, and its drug component is expressly a Phase 1 investigation excluded from the drug-trial definition. On both the drug axis and the device axis, therefore, the trial falls outside the mandatory-registration perimeter, and Box B is the legally accurate selection rather than a mere convention [8, 7]. Table 1.4 records this analysis as a compact decision trail.

Definitional element of 402(j)	**Met by this study?**	**Reason**
Controlled investigation of a drug, other than Phase 1.	No	The drug component is expressly a Phase 1, first-in-human investigation, an enumerated exclusion.
Controlled device study comparing an intervention with a control on health outcomes.	No	Single-arm, open-label, non-comparative early-feasibility design with no control group and no between-group outcome comparison.
Applicable pediatric postmarket device surveillance.	No	Adult population (age greater than or equal to 18); no postmarket surveillance mandate is involved.
Net result: applicable clinical trial under 402(j)?	No	Box B (requirements do not apply) is the legally accurate selection.

Table 1.4. Decision trail showing why the study is not an "applicable clinical trial" under 42 U.S.C. §282(j) on either the drug axis or the device axis, supporting the Box B selection on Form 3674.

Selecting Box B does not signify that the trial will go unregistered. The sponsor records that, notwithstanding the inapplicability of the mandatory 402(j) requirements, voluntary registration on clinicaltrials.gov is intended and is encouraged as a matter of transparency, and that any human clinical trial supported in whole or in part by National Institutes of Health (NIH) funding is subject to the NIH policy on clinical-trial registration and results information submission, which independently requires registration on clinicaltrials.gov regardless of the Form 3674 box selected. The first-in-human Physical AI nature of this study, and the public-trust considerations attached to an LLM-directed surgical device operating under verification-before- generation safeguards aligned with H.R. 9510 (the Verification Before Generation in Physical AI Oncology Trials Act of 2026), further support voluntary, prospective registration so that the design, the staged autonomy limits, and the safety endpoints are publicly visible before the first dose [6, 7, 17]. Registering the design prospectively also exposes, to public scrutiny, the binding halt rules (DSMB review and potential halt on a device or procedure serious adverse event in at least one of three subjects within a cohort, or on two device-related deaths at any time) and the three co-primary endpoints (device and procedure serious-adverse-event incidence through 30 days, the maximum tolerated dose and recommended Phase 2 dose of daraxonrasib via the 3+3 design, and technical feasibility), so that the trial's pre-specified stopping logic cannot be revised silently after results emerge [5, 10].

The voluntary-registration posture interacts with the program's planned trajectory in a way that the sponsor anticipates on the face of this submission. Should the program advance to a later applicable clinical trial (for example, a randomized or otherwise controlled Phase 2 study of daraxonrasib with the robotic platform), that later study would meet the "controlled clinical investigation other than Phase 1" element of the 402(j) drug-trial definition, Box C would then be selected on that submission's Form 3674, and the corresponding clinicaltrials.gov registration (within 21 days of enrolling the first subject) and results reporting (generally within one year of the primary completion date) would be certified. Maintaining a single, continuously updated clinicaltrials.gov record from the Phase 1 stage forward therefore eases the eventual transition from voluntary Box B registration to mandatory Box C certification without a gap in the public record [8, 9]. Table 1.5 contrasts the present Phase 1 posture with the anticipated later-phase posture across the dimensions a reviewer is most likely to check.

Dimension	**This Phase 1 IND (serial 0000)**	**Anticipated later applicable trial**
Form 3674 box	Box B (requirements do not apply).	Box C (requirements apply; compliance certified).
Trigger for registration	Voluntary, plus NIH policy if NIH-funded.	Mandatory under 42 U.S.C. §282(j).
Registration timing	Prospective, before first dose, as a transparency measure.	Within 21 days of first-subject enrollment.
Results reporting	Voluntary; supports public trust in the Physical AI design.	Generally within one year of primary completion date.
Design feature driving the result	Phase 1, single-arm, non-comparative early feasibility.	Controlled or comparative design (for example, randomized Phase 2).

Table 1.5. Form 3674 posture for the present Phase 1 IND contrasted with the anticipated posture for a later applicable clinical trial, showing the planned transition from a voluntary Box B registration to a mandatory Box C certification without a gap in the public record.

As with Form 1571, the certification on Form 3674 is provided on the executed, signed PDF that is assembled with the physical submission; the signature attests that the information on the certification is true and complete and acknowledges that a knowing and willful false certification is a criminal offense under 18 U.S.C. §1001, and that a false certification may also subject the certifier to civil monetary penalties and to the withholding or recovery of grant funds under the enforcement provisions of 42 U.S.C. §282(j)(5). Because Form 3674 must accompany an IND submission that references one or more clinical trials, a current Form 3674 is filed with the initial IND (serial 0000) and is re-filed, with the box selection re-evaluated, whenever a later submission adds or materially changes a referenced clinical investigation, keeping the clinicaltrials.gov certification synchronized with the evolving scope of the program [8, 9]. In practice the sponsor treats any amendment that introduces a new protocol, converts the single-arm design to a controlled design, or adds a comparator as an event that requires a fresh Form 3674 and a fresh evaluation of the Box A / B / C selection, mirroring the serial-numbering discipline that Form 1571 imposes on the file.

Taken together, the executed Form 1571 and Form 3674, placed ahead of the Table of Contents with the Cover Letter, complete the administrative envelope that initiates FDA's review of this LLM-directed

PDAC robotic daraxonrasib Phase 1 application. Form 1571 supplies the binding sponsor agreement and the serial-numbered cover sheet that will index every future drug and device event in the file; Form 3674 supplies the clinicaltrials.gov certification, here Box B, with a documented decision trail and a stated voluntary-registration posture that anticipates the eventual Box C transition. The fidelity of these two forms, verified field by field against 21 CFR §312.23 and the section 402(j) definition, is the clerical foundation on which the scientific modules that follow are allowed to rest [8, 3, 6].

2 Table of Contents

3 Introduction

3.1 Introductory Statement

This Investigational New Drug (IND) application (IND v1.0, repository v4.3.0) supports a Phase 1, first-in-human, combined IND and Investigational Device Exemption (IDE) clinical investigation of on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) with perioperative daraxonrasib (RMC-6236) in patients with KRAS G12-mutated pancreatic ductal adenocarcinoma (PDAC). The investigation is filed under 21 CFR §312 for the investigational drug and is coupled to a significant-risk early-feasibility device study under 21 CFR §812 for the LLM-directed robotic platform; the two pathways are coordinated within a single integrated protocol because the device conducts the resection that generates the operative and pathology inputs to the daraxonrasib restart advisory [1, 2]. The sponsor-investigator is ChemicalQDevice (Kevin Kawchak, Chief Executive Officer), and the conduct, oversight, and reporting commitments described throughout this dossier are made in accordance with the applicable provisions of 21 CFR §312.23 and the FDA Phase 1 content and format guidance [8, 3].

This Introduction is structured to satisfy the introductory-statement requirement of 21 CFR §312.23(a)(3)(i), which directs the sponsor to provide a brief introductory statement and general investigational plan that names the drug and its active ingredients, identifies the pharmacological class, states the structural formula, describes the dosage form and route of administration, gives the broad objectives and planned duration of the investigation, and summarizes the previous human experience and the status of the drug in other countries. Each of these codified content elements is addressed in a dedicated subsection below, in the ReGARDD content order, so that a reviewer can map every regulatory expectation onto a labeled location in this dossier. Where a content element is developed at greater depth elsewhere (for example the full structural characterization in the Chemistry, Manufacturing, and Control information, or the full safety architecture in the General Investigational Plan and the Investigator's Brochure), this Introduction states the substance concisely and gives the explicit cross-reference rather than duplicating the downstream text, consistent with the FDA expectation that the IND read as one internally consistent document [8].

The clinical problem this investigation addresses is severe and time-sensitive. PDAC is the third leading cause of cancer death in the United States and the fourth in the European Union, with total five-year overall survival below 13 percent [4]. The pancreaticoduodenectomy is the most technically demanding of all curative-intent abdominal oncology operations: it requires resection and reconstruction of four luminal structures (the pancreas, the duodenum, the common bile duct, and the stomach or pylorus) and dissection around five named vessels (the superior mesenteric vein, the portal vein, the hepatic artery, the celiac axis, and the superior mesenteric artery). The leading lethal postoperative sequence is the fistula-sepsis cascade: a clinically relevant pancreatic fistula that progresses to intra-abdominal infection, hemorrhage, sepsis, and failure to rescue. The 2025 Dutch nationwide cohort of 1000 robotic pancreaticoduodenectomies, which anchors the contemporary human baseline, reported a conversion rate of 10.1 percent, Clavien-Dindo grade III or higher complications of 41.3 percent, International Study Group on Pancreatic Surgery (ISGPS) grade B or C fistula of 24.4 percent, and 30-day mortality of 3.9 percent. Even expert, high-volume robotic surgery therefore leaves a substantial residual of conversions, high-grade complications, and clinically relevant fistulae. The intervention studied here pairs a precise, auditable, on-premises LLM-directed robotic Whipple with the RAS(ON) multi-selective inhibitor daraxonrasib, targeting an in-window R0 resection while layering hard cyber-physical safety limits and optimizing the perioperative drug-restart window.

To make the contemporary human baseline explicit for the reviewer, Table 3.1 collects the principal Dutch nationwide robotic-cohort outcomes against which the secondary feasibility and safety endpoints

Table 6: Contemporary human baseline for robotic pancreaticoduodenectomy. The 2025 Dutch nationwide cohort of 1000 robotic pancreaticoduodenectomies anchors the benchmark against which the secondary feasibility and safety endpoints of this Phase 1 investigation are interpreted descriptively, not as a powered comparator.

Outcome	Dutch 2025 cohort	Relevance to this investigation
Unplanned conversion to open	10.1%	Anchors the technical-feasibility co-primary endpoint (completion without unplanned device-attributable conversion)
Clavien-Dindo grade III or higher	41.3%	Anchors the 30-day device-related and procedure-related serious-adverse-event co-primary endpoint
ISGPS grade B or C pancreatic fistula	24.4%	Anchors the secondary fistula endpoint and informs the restart-advisory fistula input
30-day mortality	3.9%	Anchors the 30-day and 90-day mortality secondary endpoints

of this investigation are interpreted. These figures are not used as a formal statistical comparator; the Phase 1 design is descriptive and non-confirmatory and enrolls only up to 18 treated participants, which is far too small to power a comparison against a national registry. The baseline is presented because it frames the unmet need quantitatively and because it defines the order of magnitude of the conversion, high-grade complication, and fistula rates that the integrated drug-device intervention is designed to reduce over its staged-autonomy lifecycle. A first-in-human study cannot and does not promise a reduction in these endpoints; it establishes whether the combined intervention can be delivered safely and feasibly at all, which is the necessary precondition for any later efficacy claim.

Table 3.1. Contemporary human baseline for robotic pancreaticoduodenectomy from the 2025 Dutch nationwide cohort. These outcomes frame the unmet need quantitatively and contextualize the secondary endpoints; they are not used as a powered statistical comparator in this first-in-human study.

The device that conducts the resection is an eight-arm parallel coelomic surgical platform carrying 56 mechanical degrees of freedom (eight arms with seven degrees of freedom each) and a 640-channel mixed-rate sensor stack (80 channels per arm), with force channels sampling at 100 kHz and all remaining channels at the 10 kHz command rate, positioning accuracy of 0.05 mm root mean square at a peak tip velocity of 1200 mm/s. An on-premises repository-pinned LLM acts as a second-opinion advisory oracle held strictly outside the robot vendor kinematic stack and hash-pinned to a deterministic seed and a specific commit, never as an autonomous controller; full autonomy is prohibited consistent with 21 CFR §312.21(e). The hard cyber-physical safety envelope comprises a per-arm tip-force cap of 3 N or less, a cumulative cross-arm tip-force cap of 18 N or less enforced at the heartbeat boundary, a 10 kHz heartbeat coordination bus broadcasting a 64-byte frame on a 100 microsecond per-arm watchdog deadline, an emergency arm park within 50 microseconds of a missed or out-of-parameter frame, a cross-arm emergency stop that halts the full platform in 3 ms or less, and a software-independent hardware emergency stop in 500 ms or less consistent with 21 CFR §312.404. A vascular no-fly gate samples at 10 kHz around five named vessels (the superior mesenteric vein, portal vein, hepatic artery, celiac axis, and superior mesenteric artery), with a soft-warning advisory zone at 3.0 mm, a no-fly blocking zone at 1.0 mm for the superior mesenteric and portal veins and 1.5 mm for the hepatic artery,

Table 7: Hard cyber-physical safety envelope enforced below the advisory layer. Every limit is enforced in hardware or deterministic firmware; the on-premises advisory LLM cannot relax or override any limit. Response times are upper bounds.

Safety mechanism	Quantitative limit	Regulatory anchor and function
Per-arm tip-force cap	3 N or less per arm	Bounds local tissue and vascular load on each of the eight arms
Cumulative cross-arm tip-force cap	18 N or less, all arms	Enforced at the heartbeat boundary across the parallel platform
Heartbeat coordination bus	10 kHz, 64-byte frame, 100 microsecond watchdog	Deterministic per-arm coordination; missed frame triggers protection
Emergency arm park	50 microseconds or less	Parks an arm on a missed or out-of-parameter frame
Cross-arm emergency stop	3 ms or less to 0.0 N	Halts the full eight-arm platform from any arm event
Hardware emergency stop	500 ms or less, software-independent	21 CFR §312.404; floor below all firmware and the advisory layer
Vascular no-fly gate	soft 3.0 mm, block 1.0 to 1.5 mm, hard-stop 5.0 mm, 10 kHz	Distance-keyed protection around five named vessels

celiac axis, and superior mesenteric artery, and a hard-stop emergency-stop zone at 5.0 mm. In this Phase 1 the device operates only at the clinician-controlled and supervised-semiautonomous levels under continuous human oversight with immediate manual override, consistent with the staged-autonomy posture in which higher autonomy is reserved for future stages [1, 2, 7].

The hard cyber-physical safety envelope is presented quantitatively in Table 3.2 so that the numeric limits the device enforces, the response times it guarantees, and the regulatory anchor for each are visible in one place at the outset of the dossier. The envelope is deliberately constructed so that every limit is enforced in hardware or in deterministic firmware below the LLM advisory layer; the advisory LLM cannot relax, override, or widen any of these limits, and a missed or out-of-parameter heartbeat frame triggers the protective response regardless of any advisory output. This separation of a deterministic, software-independent safety floor from a non-autonomous advisory layer is the central design principle that makes the integrated drug-device intervention suitable for a first-in-human study, and it is developed in full in the General Investigational Plan and the Investigator's Brochure.

Table 3.2. Hard cyber-physical safety envelope. The per-arm and cumulative tip-force caps, the 10 kHz heartbeat with 100 microsecond watchdog, the graded emergency-stop response times, and the distance-keyed vascular no-fly gate are enforced deterministically below the non-autonomous advisory layer, with the software-independent hardware emergency stop providing the floor under 21 CFR §312.404.

The eligible population is adults (18 years or older) with histologically or cytologically confirmed, resectable or borderline-resectable PDAC; documented KRAS G12 mutation (G12D, G12V, or G12R); ECOG performance status 0 or 1; and candidacy for pancreaticoduodenectomy, eligible under the broad pan-KRAS criteria of the RASolute 302 program. An exemplar screening subject, PAT-PDAC-0001, is a 68-year-old with a 3.4 cm head-of-pancreas tumor abutting the superior mesenteric vein at 75 degrees, KRAS G12D, microsatellite-stable disease, CA 19-9 of 412 U/mL, ECOG 1, and four cycles of neoadjuvant modified FOLFIRINOX, illustrating the borderline-resectable phenotype the investigation targets. This exemplar is a worked illustration of the eligibility logic and is not an enrolled participant; it is used throughout the dossier to make the screening, simulation sign-off, operative, and restart-advisory pathways concrete for the reviewer. For PAT-PDAC-0001 the 75-degree venous abutment falls within the borderline-resectable definition (tumor-vessel contact of 180 degrees or less without occlusion), the KRAS G12D genotype satisfies the molecular criterion, the ECOG 1 status satisfies the performance criterion, and the completion of four cycles of neoadjuvant modified FOLFIRINOX satisfies the candidacy criterion, so the subject would proceed to the Phase 0 simulation-validation sign-off and the baseline Unified Safety Level verification before any patient-facing robotic activity.

The investigational plan rests on a three-part acceleration thesis that is the animating premise of the present repository. A patient already enrolled in a Phase 1 PDAC study has little survival margin to spare on administrative delay; the clinical and operational time of a trial (recruitment, treatment, and waiting for progression-free and overall survival to mature) cannot be compressed by writing faster, and neither can the fixed regulatory review clocks, but the administrative and preparation time spent authoring the large documents between steps can be compressed substantially [10, 17]. Faster, internally consistent paperwork therefore means faster treatment: a complete and consistent package reaches the Institutional Review Board (IRB) and the FDA sooner, the 30-day IND safety clock starts sooner, sites activate sooner, and the next dose-escalation cohort opens sooner. This IND, like the protocol and the supporting publication from which it is assembled, was generated through a single-prompt mermaid to draft to full to final pipeline whose every artifact is committed and reviewable in real time, and the same mechanism that produced this dossier maintains the safety reports, amendments, and study reports that the trial will require throughout its lifecycle.

The acceleration thesis distinguishes three categories of trial time with sharp care, because conflating them is the most common way that claims about AI-assisted trial conduct overreach. The first category is clinical and biological time: the months a participant is treated, the interval to disease progression, and the time for overall survival to mature. This time is fixed by human physiology and the natural history of PDAC and cannot be shortened by any authoring tool; a faster document does not make a tumor respond faster. The second category is regulatory review time: the statutory 30-day IND safety review under 21 CFR §312.40(b), the IRB review interval, and the cohort-boundary safety reviews of the Data Safety Monitoring Board. These intervals are bounded by regulation and by the deliberate pace of safety review and are not compressed by faster authoring. The third category is administrative and preparation time: the weeks spent drafting, cross-checking, formatting, and revising the large regulated documents that sit between every clinical and regulatory step. Only the third category is addressed by the single-prompt workflow, and the acceleration claim is bounded strictly to it. Because the third category nonetheless sits on the critical path before the 30-day clock can start and between every cohort, compressing it advances the start of the fixed clocks rather than shortening them, which is the precise and defensible form of the acceleration thesis [10, 17].

Figure 4 places this IND within the full Phase 1 oncology document landscape it inhabits. Before submission the repository LLM generates the IND dossier, the Investigator's Brochure, the clinical protocol, and the informed consent form; during the trial it produces the 7-day and 15-day safety reports, the annual Development Safety Update Report (DSUR), protocol amendments, and dose-escalation minutes; and after the trial it assembles the tables, listings, and figures, the ICH E3 clinical study report, and the downstream manuscripts and lay summaries. The IND package is the first and rate-limiting artifact in this relay, which is why the acceleration thesis is most consequential here [10].

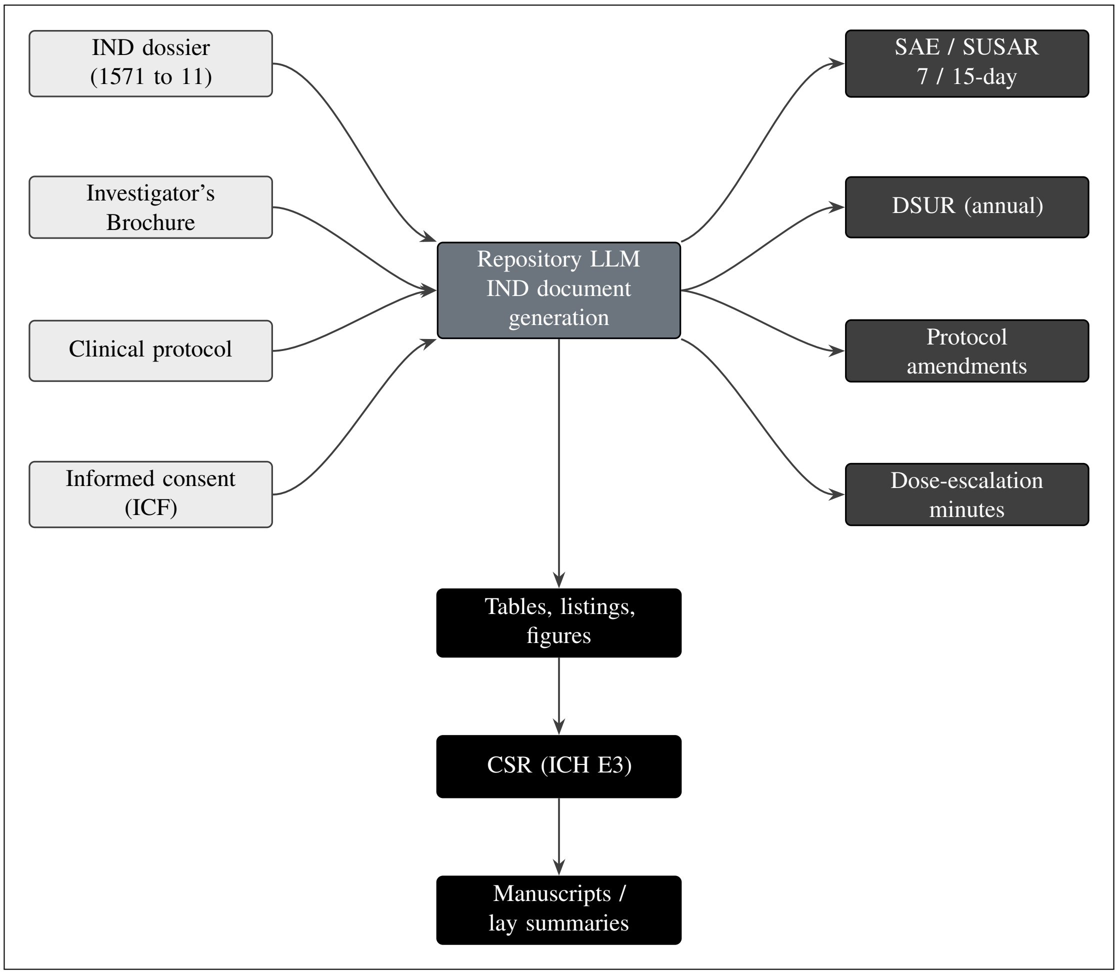

Figure 4. The large Phase 1 oncology document landscape around the IND, grouped into before-submission inputs (the IND dossier, Investigator's Brochure, protocol, and consent), during-trial process documents (7 and 15-day safety reports, the annual DSUR, amendments, and dose-escalation minutes), and after-trial outputs (tables, listings, figures, the ICH E3 clinical study report, and manuscripts and lay summaries), all generated by the repository LLM around the central hub. This figure renders in the Introduction to place the IND in the full document lifecycle the same method maintains.

The relay structure of Figure 4 has a direct regulatory consequence that motivates the entire dossier. The before-submission inputs (the IND dossier, the Investigator's Brochure, the protocol, and the consent form) must be complete, mutually consistent, and cross-referenced before the 30-day IND safety clock under 21 CFR §312.40(b) can begin; any internal inconsistency between, for example, the dose levels stated in the Introduction, the escalation rule stated in the General Investigational Plan, and the dose-modification table in the protocol is a source of review questions that delays the clock. The single-prompt workflow addresses precisely this failure mode by generating every artifact from one pinned source of trial facts, so that the dose levels, the safety-reporting clocks, the eligibility criteria, and the endpoint definitions are identical wherever they appear. The during-trial process documents (the 7-day and 15-day safety reports, the annual DSUR, the amendments, and the dose-escalation minutes) inherit the same fact base, so a protocol amendment automatically propagates a consistent change to the consent form and the safety-reporting plan rather than requiring manual reconciliation across documents. The after-trial outputs (the tables, listings, and figures, the ICH E3 clinical study report, and the manuscripts and lay summaries) close the loop by drawing from the same committed record. The reviewer therefore encounters a dossier in which every number that appears more than once appears identically, which is the practical meaning of the internal-consistency expectation in the FDA Phase 1 guidance [8, 10].

The remainder of this Introduction follows the ReGARDD IND content sequence. It names the drug and its active ingredients (§3.1.1), states the pharmacological class (§3.1.2) and structural description (§3.1.3), summarizes the dosage form (§3.1.4), the route of administration (§3.1.5), and the duration of the proposed clinical investigation (§3.1.6); it then summarizes previous human experience (§3.2), the status of the drug in other countries (§3.3), and the preclinical and computational evidence (§3.4), with the supporting references collected in §3.5. Quantitative and design detail introduced here is developed in full in the General Investigational Plan (§4), the Investigator's Brochure (§5), the proposed clinical research (§6), the Chemistry, Manufacturing, and Control information (§7), and the Pharmacology and Toxicology information (§8).

3.1.1 Name of the Drug and All Active Ingredients

The investigational drug is daraxonrasib, also known by its development code RMC-6236. Daraxonrasib is the single active ingredient of the drug product; there are no additional active pharmaceutical ingredients. It is an orally bioavailable, RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor that binds the active, guanosine-triphosphate-bound (ON) state of mutant RAS in a tri-complex with the intracellular chaperone cyclophilin A, suppressing downstream effector signaling across the common KRAS oncoprotein variants found in PDAC [1, 2]. The molecular target population for this investigation is defined by the KRAS G12 alleles (G12D, G12V, and G12R) that predominate in pancreatic ductal adenocarcinoma; daraxonrasib provides broad pan-KRAS coverage that spans these alleles, which is the property that makes it suitable for the protocol's molecularly defined eligibility [1].

The nomenclature used throughout this dossier is fixed here to avoid ambiguity. The nonproprietary (generic) name is daraxonrasib; the development code is RMC-6236; the two terms refer to the same single chemical entity and are used interchangeably, with daraxonrasib preferred in clinical text and RMC-6236 retained where the source material uses the code. Because the drug product contains daraxonrasib as its only active ingredient and no second active pharmaceutical ingredient, the IND requires no separate active-ingredient characterization for a fixed-dose combination, and the combination character of the intervention is created not by a second drug but by the pairing of the single drug with the investigational LLM-directed device. This distinction is material to the regulatory framing: the integrated intervention is a drug-device combination in the operational sense that the device generates the operative and pathology inputs to the drug-restart advisory, but the drug product itself remains a single-active-ingredient oral tablet, and the device carries no pharmacologic dose of its own.

Figure 5 reproduces the mechanism and pharmacological-class schematic for daraxonrasib that grounds §3.1.1 through §3.1.3. Daraxonrasib forms the tri-complex with cyclophilin A, binds the active GTP-bound state of mutant RAS, intercepts the KRAS G12 variants that define the trial population, suppresses downstream MAPK and PI3K effector signaling, and thereby reduces tumor-cell proliferation and survival; it is dosed orally once daily at the three escalation levels (DL1 160 mg, DL2 220 mg, DL3 300 mg) and carries FDA Breakthrough Therapy designation granted in June 2025 for previously treated metastatic KRAS G12 PDAC [1, 2].

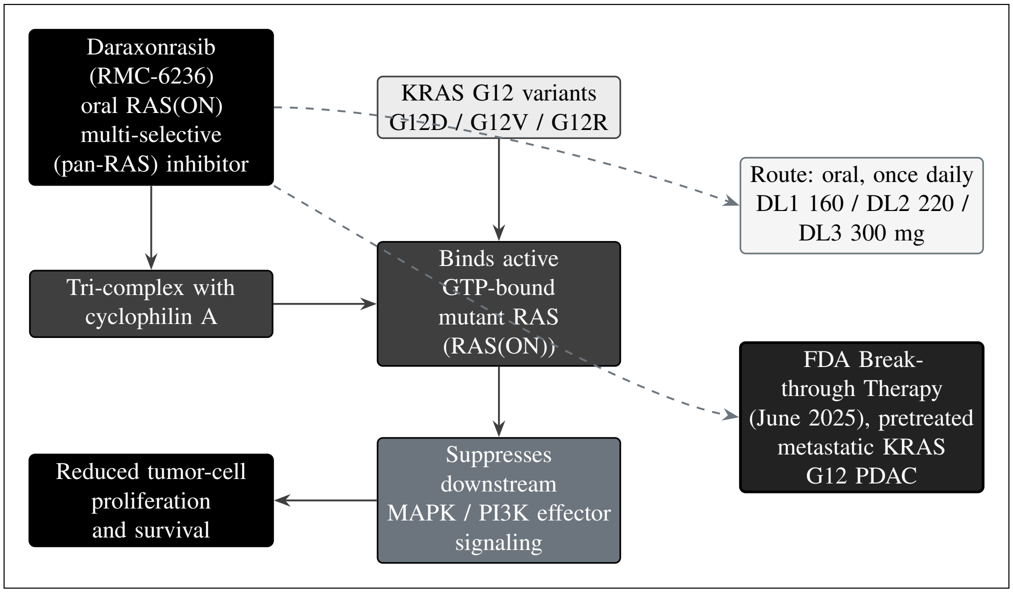

Figure 5. Daraxonrasib (RMC-6236) is an oral RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor. It forms a tri-complex with cyclophilin A and binds the active, guanosine-triphosphate-bound state of mutant RAS, including the KRAS G12 variants (G12D, G12V, G12R) that define the trial population, suppressing downstream MAPK and PI3K effector signaling and reducing tumor-cell proliferation and survival. It is dosed orally once daily at DL1 160 mg, DL2 220 mg, and DL3 300 mg, and holds FDA Breakthrough Therapy designation (June 2025) for previously treated metastatic KRAS G12 PDAC. This figure renders in §3.1.1 (Name of Drug and Active Ingredients), §3.1.2 (Pharmacological Class), and §3.1.3 (Structural Formula).

In the present investigation daraxonrasib is studied in a perioperative, curative-intent surgical context that is distinct from its late-phase metastatic program: it is administered orally once daily in a defined preoperative course, paused across the operative window, and restarted postoperatively on a day chosen by the LLM-bound, human-confirmed advisory described in §3.1.5 and developed in the proposed clinical research. Because the drug is paired with an investigational device in the drug-delivery pathway, the combined IND and IDE submission carries a full Physical AI System Description meeting the eleven fields of 21 CFR §312.23(g) [3], the substance of which is presented in the CMC and device sections of this dossier.

The eleven-field Physical AI System Description required under 21 CFR §312.23(g) is the structural anchor that ties the drug and device components together for the reviewer, and its fields are introduced here so the named subsections that develop them can be located. The description records the intended use and indication; the system architecture, including the eight-arm platform, the 640-channel sensor stack, and the on-premises advisory LLM held outside the vendor kinematic stack; the autonomy level, fixed at the clinician-controlled and supervised-semiautonomous tiers with full autonomy prohibited; the hardware and software safety architecture, including the heartbeat, the watchdog, the graded emergency stops, and the vascular no-fly gate; the verification-before-generation gate suite that screens every advisory output; the human-factors and oversight model; the cybersecurity and audit-trail provisions under 21 CFR part 11; the data-governance and model-pinning provisions; the change-control and revalidation provisions; the reporting triggers under 21 CFR §312.32(g); and the record-retention provisions. Each field is developed in the General Investigational Plan, the Investigator's Brochure, and the CMC information, and the cross-references are given at each point of first mention so the reviewer can trace any field to its full treatment.

3.1.2 Pharmacological Class of the Drug

Daraxonrasib belongs to the RAS(ON) multi-selective small-molecule inhibitor class. Pharmacologically, the agent is a tri-complex inhibitor: it does not act on the inactive, guanosine-diphosphate-bound (OFF) conformation that earlier-generation allele-specific inhibitors target, but instead engages the active, GTP-bound (ON) conformation of mutant RAS by recruiting the abundant intracellular chaperone cyclophilin A to form a high-affinity ternary complex on the RAS surface [1, 2]. This ON-state, chaperone-mediated mechanism confers multi-selective (pan-RAS) coverage across KRAS, NRAS, and HRAS variants, and in particular across the KRAS G12 alleles (G12D, G12V, G12R) that account for the large majority of RAS-driven PDAC. By occupying the active oncoprotein, the inhibitor blocks engagement of downstream effectors and suppresses the mitogen activated protein kinase (MAPK) and phosphoinositide 3-kinase (PI3K) signaling axes, with the consequence of reduced tumor-cell proliferation and survival, as summarized in Figure 5.

The class distinction between ON-state multi-selective inhibition and the allele-specific OFF-state inhibition of earlier agents is not a matter of taxonomy alone; it determines both the breadth of the eligible population and the anticipated safety profile, and it is therefore developed here in the depth the reviewer needs to interpret the dose-finding design. Allele-specific OFF-state inhibitors bind a single mutant conformation (for example a specific G12C pocket) and consequently cover only one allele, which excludes the majority of PDAC patients whose tumors carry G12D, G12V, or G12R. An ON-state pan-RAS inhibitor that engages the active oncoprotein through a cyclophilin A tri-complex covers the spectrum of KRAS G12 alleles in one agent, which is exactly the property that lets this protocol adopt the broad pan-KRAS eligibility of the RASolute 302 program rather than restricting to a single allele. The breadth of coverage is the molecular reason the eligible population is defined by the presence of any KRAS G12 mutation rather than by a single variant.

The class assignment is material to the safety profile anticipated in this Phase 1 study and to the dose-finding rationale. As a multi-selective RAS pathway inhibitor, daraxonrasib carries the on-target, mechanism-based toxicities characteristic of broad RAS and MAPK pathway suppression, including gastrointestinal effects (nausea, diarrhea, stomatitis), dermatologic and mucocutaneous events (rash), and hepatic enzyme elevations; these are the toxicity categories that the 3+3 escalation is specifically designed to detect and bound and that inform the dose-limiting toxicity (DLT) definitions in §4. The class also informs the perioperative pause-and-restart logic, because broad pathway inhibition during the immediate postoperative window carries a theoretical risk to wound and anastomotic healing if the restart is mistimed, which is why the restart day is keyed to fistula grade and drug trough rather than fixed.

Table 3.3 organizes the anticipated mechanism-based toxicity categories that follow from the pharmacological class, the physiological basis for each, and the way each category is detected and bounded in this Phase 1 design. The table is presented here because the class assignment is the single most important determinant of what the 3+3 escalation is built to find: the dose-limiting toxicity definitions in the General Investigational Plan are constructed around exactly these categories, and the perioperative pause-and-restart logic exists to manage the wound-healing concern that the same broad pathway suppression creates. Presenting the categories at the level of the Introduction lets the reviewer see the through-line from pharmacological class to DLT definition to restart advisory before encountering each in its full subsection.

Table 8: Anticipated mechanism-based toxicity categories of RAS(ON) multi-selective inhibition and how each is detected and bounded in this Phase 1 design. The categories follow from the pharmacological class and define what the 3+3 escalation and the dose-limiting-toxicity definitions are constructed to find.

Toxicity category	Mechanistic basis	Detection and bounding in this design
Gastrointestinal (nausea, diarrhea, stomatitis)	Broad MAPK pathway suppression in rapidly dividing gut epithelium	Graded continuously; contributes to DLT definitions across the 28-day window
Dermatologic and mucocutaneous (rash)	On-target effect of RAS and MAPK suppression in skin and mucosa	Graded continuously; severe events meet DLT thresholds in the escalation
Hepatic enzyme elevation	Pathway-related hepatocellular effect of broad RAS inhibition	Laboratory monitoring across the DLT window; thresholds in the DLT definition
Wound and anastomotic healing	Theoretical impairment of healing under broad pathway suppression	Managed by the perioperative pause and the fistula-keyed restart advisory

Table 3.3. Mechanism-based toxicity categories of RAS(ON) multi-selective inhibition. The gastrointestinal, dermatologic and mucocutaneous, and hepatic categories define the dose-limiting-toxicity targets of the 3+3 escalation, and the wound-healing concern motivates the perioperative pause-and-restart advisory that keys the restart day to fistula grade and drug trough.

3.1.3 Structural Formula of the Drug

Daraxonrasib (RMC-6236) is a synthetic, non-peptidic small-molecule organic compound of defined molecular structure designed to bind the RAS-cyclophilin A interface and engage the active GTP-bound state of mutant RAS. It is a single, well-characterized chemical entity rather than a mixture, a biologic, or a nucleic-acid therapeutic, and it is the sole active ingredient of the drug product. The molecule is the macrocyclic tri-complex inhibitor scaffold characteristic of the RAS(ON) class, engineered for oral bioavailability and once-daily exposure, and its structure supports the broad pan-KRAS coverage that defines the molecularly selected trial population.

The structural class carries two functional implications that the reviewer should hold while reading the CMC information. First, the macrocyclic scaffold is the structural feature that allows the molecule to present a binding surface complementary to both the cyclophilin A chaperone and the active RAS oncoprotein simultaneously, which is the structural basis for the tri-complex mechanism and therefore for the pan-RAS coverage; the structure and the mechanism are not separable claims. Second, the engineering for oral bioavailability and once-daily exposure is the structural basis for the oral tablet dosage form and the once-daily route developed in §3.1.4 and §3.1.5, so the structural description, the dosage form, and the route of administration form one coherent account of how the single active ingredient is delivered. The Introduction states these implications qualitatively; the quantitative structural characterization that substantiates them is consolidated in the CMC information.

The complete structural formula of daraxonrasib, including the molecular formula, molecular weight, stereochemistry, the structural diagram of the macrocyclic scaffold, the chemical name, and the characterization data establishing identity and purity, is provided in the Chemistry, Manufacturing, and Control information of this submission at §7.1.1 (Drug Substance), consistent with 21 CFR §312.23(a)(7)

Table 9: Daraxonrasib dose-level structure for the Phase 1 3+3 escalation. DL3 (300 mg once daily) corresponds to the pan-KRAS RASolute 302 recommended Phase 2 dose; DL1 and DL2 sit below it to preserve a conservative first-in-human margin. The DLT window is 28 days, and the recommended Phase 2 dose (RP2D) is selected at or below the maximum tolerated dose (MTD).

Dose level	Daily dose	DLT window	3+3 cohort	Role and rationale
DL1	160 mg QD	28 days	3, expand to 6 if 1 DLT	Conservative starting dose, two steps below the RASolute 302 reference
DL2	220 mg QD	28 days	3, expand to 6 if 1 DLT	Intermediate level, one step below the reference dose
DL3	300 mg QD	28 days	3, expand to 6 if 1 DLT	RASolute 302 recommended Phase 2 dose; highest planned level

[8, 3]. Presenting the full structural characterization in the CMC section avoids duplication and keeps the structural, physicochemical, and manufacturing detail consolidated where the reviewer expects it; this Introduction provides only the qualitative structural description and the cross-reference. The mechanism schematic in Figure 5 summarizes the functional relationship between the structure and its pharmacological target.

3.1.4 Formulation of the Dosage Form(s) to be Used

Daraxonrasib is supplied as oral tablets in the strengths required to compose the three once-daily dose levels evaluated in the 3+3 escalation: DL1 at 160 mg, DL2 at 220 mg, and DL3 at 300 mg. The 300 mg once-daily level (DL3) corresponds to the recommended Phase 2 dose established in the pan-KRAS RASolute 302 program, and the escalation begins one and two steps below it (160 mg and 220 mg) to preserve a conservative first-in-human margin for the novel perioperative surgical context [1, 2]. Tablets are swallowed whole with water and are not crushed, split, or chewed unless a future protocol amendment provides validated instructions. No compounding or reconstitution is required; preparation consists of the site pharmacist verifying participant identity, the assigned dose level, and the expiry, then dispensing the once-daily oral dose with administration instructions.

Tablets are packaged in child-resistant, light-protective containers labeled in accordance with 21 CFR §312.6, bearing the investigational caution statement “Caution: New Drug. Limited by Federal (or United States) law to investigational use,” together with the protocol number, lot number, storage conditions, and a unique container identifier; because the study is open-label, the participant-facing label identifies the product and the assigned daily dose. The product is stored at controlled room temperature in a secured, access-controlled, temperature-monitored location within the investigational-product pharmacy, in its original light-protective packaging, with continuous temperature monitoring and documented excursion handling, and is dispensed only within its labeled expiry or retest date. The dose-level structure that the dosage form must support is summarized in Table 3.4; the full formulation composition, appearance, packaging, stability, and labeling detail is provided in §7.1.2 (Drug Product) and §7.1.4 (Labeling).

Table 3.4. Daraxonrasib oral dose levels and the 3+3 escalation structure. At each level three participants are observed across the 28-day DLT window; the cohort escalates on 0 of 3, expands to 6 on 1 of 3, escalates on 1 of 6, and fixes the MTD at the preceding level on 2 or more of 6.

The dose-level spacing is chosen so that the escalation explores a clinically meaningful exposure range while preserving a conservative first-in-human margin in the novel perioperative surgical context. The step from DL1 to DL2 is 60 mg (a 37.5 percent relative increment over the 160 mg starting dose), and the step from DL2 to DL3 is 80 mg (a 36.4 percent relative increment over the 220 mg intermediate dose), so the two steps are of comparable relative magnitude and neither introduces a large exposure jump between adjacent cohorts. Because DL3 is the established RASolute 302 recommended Phase 2 dose rather than a dose above prior human experience, the escalation does not seek a new maximum tolerated dose above the known tolerable exposure; instead it confirms tolerability of the established exposure within the new perioperative surgical setting and identifies the recommended Phase 2 dose at or below it. This is the conservative posture appropriate to a first-in-human study that combines a drug with established late-phase exposure and a device with no prior human experience: the drug exposure stays within the prior human range while the novel combination is characterized for safety and feasibility.

3.1.5 Route of Administration

Daraxonrasib is administered by the oral route, once daily, on a perioperative schedule comprising a defined preoperative course, a protocol-mandated pause spanning the operative window, and a postoperative restart whose timing is governed by an advisory. The preoperative course establishes systemic exposure before surgery; the pause spans the immediate operative and early-recovery window during which broad RAS pathway suppression could impair wound and anastomotic healing; and the restart returns systemic therapy at the earliest safe point to suppress micrometastatic outgrowth. The drug is not delivered by the device, which has no pharmacologic dose of its own; the device conducts the resection and reconstruction and supplies the operative and pathology inputs that the restart advisory consumes [1].

The perioperative pause-and-restart advisory is LLM-bound and human-confirmed, never autonomous. It takes two inputs and produces one of three restart recommendations. The inputs are the pancreaticojejunostomy ISGPS postoperative pancreatic-fistula grade (A, B, or C) and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold. The outputs are a restart on postoperative day T+7d, T+14d, or T+21d, or a continued hold. In the prespecified 32-iteration deterministic simulation sweep, baseline serum was 0.45 ng/mL at the operative timepoint across all iterations (below the 0.5 ng/mL trough), and the advisory recommended T+7d in 29 of 32 iterations (90.6 percent, Grade A fistula with subthreshold trough), T+14d in 3 of 32 iterations (9.4 percent, Grade B fistula or a delayed serum rise), and T+21d in 0 of 32 iterations; Grade C fistula maps to T+21d or a continued hold and to the drug stop rule. Because the pancreaticojejunostomy is the dominant determinant of postoperative pancreatic-fistula risk, the fistula-grade input to the drug advisory is based on that anastomosis. The advisory is one of only two coupling points between the drug and device components of the integrated intervention, and it keeps the restart decision human-confirmed and fully traceable in the audit trail. The detailed advisory logic and the simulation sweep are presented in the proposed clinical research and Investigator's Brochure sections of this dossier.

The mapping from the two inputs to the four possible outputs is fully prespecified and deterministic, which is what allows the 32-iteration sweep to characterize the advisory before any human exposure. Table 3.5 sets out the decision logic so that the reviewer can see exactly how each combination of fistula grade and serum trough maps to a restart day or a hold, and can verify that the simulation distribution (29 of 32 at T+7d, 3 of 32 at T+14d, 0 of 32 at T+21d) follows from the logic given the subthreshold baseline serum of 0.45 ng/mL observed across all iterations. Working the exemplar through

Table 10: Perioperative daraxonrasib pause-and-restart advisory logic. The advisory takes the pancreaticojejunostomy ISGPS fistula grade and the serum trough relative to 0.5 ng/mL and recommends a restart day or a hold; every recommendation is human-confirmed and preserved in the audit trail. The 32-iteration deterministic sweep distribution follows from this logic at the sub-threshold 0.45 ng/mL baseline.

Fistula grade	Serum trough	Recommendation	Sweep frequency and notes
Grade A	Below 0.5 ng/mL	Restart T+7d	29 of 32 (90.6%); modal outcome at 0.45 ng/mL baseline
Grade B	At or above 0.5 ng/mL, or delayed rise	Restart T+14d	3 of 32 (9.4%); Grade B or delayed serum rise
Grade B or A	Delayed threshold crossing	Restart T+21d	0 of 32 in the sweep; reserved for late rise
Grade C	Any	Restart T+21d or hold	Engages the drug stop rule; continued hold permitted

the logic makes this concrete: for a participant who achieves a Grade A pancreaticojejunostomy result with a serum trough of 0.45 ng/mL (below the 0.5 ng/mL threshold), the advisory recommends restart at T+7d, the modal outcome in the sweep, subject to human confirmation and the audit trail. A Grade B result, or a delayed serum rise crossing the threshold, shifts the recommendation to T+14d, and a Grade C result maps to T+21d or a continued hold and engages the drug stop rule. In every case the recommendation is advisory and human-confirmed; the LLM does not start, hold, or change the dose, and the recorded decision and its inputs are preserved in the hash-chained audit trail.

Table 3.5. Perioperative pause-and-restart advisory decision logic. The pancreaticojejunostomy fistula grade and the serum trough relative to 0.5 ng/mL map to a restart day (T+7d, T+14d, or T+21d) or a continued hold; the 32-iteration deterministic sweep yields 29 of 32 at T+7d, 3 of 32 at T+14d, and 0 of 32 at T+21d at the sub-threshold 0.45 ng/mL baseline, with every recommendation human-confirmed.

3.1.6 Duration of the Proposed Clinical Investigation(s)

The duration of the proposed clinical investigation is defined at two levels: the per-participant timeline and the overall study timeline. For each participant, the investigation comprises up to 28 days of screening (Day -28 to Day -1) during which histologic confirmation, KRAS G12 confirmation, staging, and eligibility are established; a baseline day (Day -1) that carries the Phase 0 simulation sign-off and the Unified Safety Level (USL) and safety-matrix verification; surgery on Day 0; an acute window (Day 1 to Day 7) carrying the ISGPS fistula grading and the daraxonrasib restart advisory; the Day 30 primary safety window during which device-related and procedure-related serious adverse events, including Clavien-Dindo grade III or higher complications, are ascertained; Day 90 pathology and mortality assessment, which carries the R0 resection determination and the 90-day mortality endpoint; and long-term follow-up every 12 weeks to the 24-month overall-survival endpoint, during which progression-free survival (PFS) and overall survival (OS) are assessed by RECIST v1.1 [1].

Table 3.6 lays out the per-participant timeline as a single schedule so the reviewer can see how the screening, baseline, operative, acute, primary-safety, pathology, and long-term-follow-up windows align with the assessments and the regulatory anchors that govern each. The timeline is constructed so that no patient-facing robotic activity can occur before the Phase 0 simulation-validation gate is satisfied at the baseline day, so that the acute window carries both the fistula grading and the drug-restart

Table 11: Per-participant timeline of the proposed investigation. Each window carries the assessments and gates shown; no patient-facing robotic activity occurs before the Phase 0 simulation-validation gate is satisfied at the baseline day.

Window	Timing	Principal assessments and gates
Screening	Day -28 to -1	Histologic and KRAS G12 confirmation; staging; eligibility determination
Baseline	Day -1	Phase 0 simulation sign-off; Unified Safety Level and safety-matrix verification
Surgery	Day 0	LLM-directed eight-arm robotic Whipple; operative parameter capture
Acute window	Day 1 to 7	ISGPS fistula grading; daraxonrasib restart advisory; serum trough
Primary safety	Day 30	Device-related and procedure-related SAE, including Clavien-Dindo III or higher
Pathology and mortality	Day 90	R0 resection determination; 90-day mortality endpoint
Long-term follow-up	Every 12 weeks to 24 months	PFS and OS by RECIST v1.1

advisory, and so that the Day 30 window ascertains the primary safety endpoint while the Day 90 window carries the R0 determination and the 90-day mortality endpoint. The long-term follow-up then runs on a 12-week cadence to the 24-month overall-survival endpoint.

Table 3.6. Per-participant timeline from screening through the 24-month overall-survival endpoint. The Phase 0 simulation-validation gate at the baseline day precedes any patient-facing robotic activity; the acute window carries the fistula grading and the restart advisory; and the Day 30 and Day 90 windows carry the primary safety endpoint and the R0 and mortality endpoints respectively.

The daraxonrasib DLT observation window is 28 days from the start of dosing at each dose level, which governs the cadence of the 3+3 escalation. At the study level, the investigation enrolls up to 18 treated participants (approximately 36 screened) at a single academic site, with an anticipated accrual of approximately one to two participants per month under sentinel staggered enrollment; with three dose levels of up to six DLT-evaluable participants each, the maximum of 18 follows directly (three levels times six). Adding the per-participant follow-up to 24-month OS to the accrual period, the total study duration is governed primarily by the time required to mature the overall-survival endpoint in the last-enrolled participant. A mandatory Phase 0 simulation-validation gate (at least 1000 simulated procedures across at least two independent frameworks, with a USL rating of at least 7.0 and a sim-to-real trajectory gap below 2 mm and tip-force gap below 0.5 N) precedes any patient-facing robotic activity and is re-validated after any qualifying change, so the patient-facing portion of the investigation does not begin until that gate is satisfied. The first-year operational plan and the projected enrollment cadence are developed in the General Investigational Plan (§4).

A worked accrual illustration grounds the study-level duration. At an anticipated accrual of one to two participants per month under sentinel staggered enrollment, and with the maximum of 18 treated participants distributed across three dose levels of up to six DLT-evaluable participants each, the accrual period spans roughly nine to eighteen months in the absence of escalation pauses; sentinel staggering and the 28-day DLT window between cohort decisions add deliberate intervals that lengthen this span, because a cohort must complete its DLT window and pass the cohort-boundary safety review before the next opens. The total study duration is then dominated not by accrual but by the per-participant follow-up to the 24-month overall-survival endpoint in the last-enrolled participant: even if accrual completes near month eighteen, the last participant's overall-survival endpoint matures roughly twenty-four months later, so the overall-survival readout governs the study end date. This is the expected structure for a first-in-human study whose primary endpoints are safety, dose, and feasibility but whose exploratory survival endpoints require long maturation.

The investigation is statistically descriptive and non-confirmatory, consistent with ICH E9 principles for early-phase trials and with TRIPOD+AI for the AI-enabled components; there is no power calculation and no comparator arm. Estimation uses exact Clopper-Pearson 95 percent confidence intervals, and the maximum sample of 18 participants yields the precision appropriate to a first-in-human dose-finding and early-feasibility study: an observed proportion of 0 of 18 corresponds to 0 percent (95 percent CI 0 to 18 percent), 1 of 18 corresponds to 5.6 percent (95 percent CI 0.1 to 27 percent), 0 of 3 to a CI of 0 to 71 percent, and 0 of 6 to a CI of 0 to 46 percent. The three co-primary endpoints are the incidence of device-related and procedure-related serious adverse events through the Day 30 window (including Clavien-Dindo grade III or higher), the maximum tolerated dose and recommended Phase 2 dose of daraxonrasib determined by the 3+3 escalation across the 28-day DLT window, and technical feasibility defined as the proportion of participants completing the assigned operative tasks without unplanned conversion attributable to device malfunction or unsafe behavior. Binding halt rules govern progression: a device-related or procedure-related serious adverse event in at least one of three participants within a cohort, or two device-related deaths at any time, triggers Data Safety Monitoring Board review and possible halt. Safety reporting follows the 7-calendar-day clock for fatal or life-threatening suspected adverse reactions and the 15-calendar-day clock for other serious and unexpected reactions under 21 CFR §312.32, augmented by six Physical AI reporting triggers under 21 CFR §312.32(g) and a -24 hour to +72 hour audit-trail preservation window around each Physical AI adverse event [8, 3].

The precision of the maximum-sample estimates deserves explicit interpretation because it is the quantitative reason a first-in-human study of this size is descriptive rather than confirmatory. With 18 treated participants and exact Clopper-Pearson intervals, an observed event count of zero across the whole study still admits an upper 95 percent confidence bound near 18 percent, and a single observed event yields a point estimate of 5.6 percent with an interval that extends to roughly 27 percent. At the cohort level the precision is necessarily wider still: zero of three events admits an upper bound near 71 percent, and zero of six admits an upper bound near 46 percent. These wide intervals are not a deficiency of the design; they are the honest reflection of what a sample of this size can establish, and they are the reason the study claims only safety, dose, and feasibility and reserves any efficacy inference for the later, larger Phase 2. Table 3.7 collects these worked precision figures so the reviewer can read the attainable certainty directly.

Table 12: Worked precision of the maximum-sample and cohort-level estimates under exact Clopper-Pearson 95 percent confidence intervals. The wide intervals reflect the descriptive, non-confirmatory character of a first-in-human study and are the quantitative basis for reserving efficacy inference to the later Phase 2.

Observed	Point estimate	95% Clopper-Pearson CI	Interpretation
0 of 18	0%	0 to 18%	Whole-study zero-event upper bound
1 of 18	5.6%	0.1 to 27%	Whole-study single-event estimate
0 of 3	0%	0 to 71%	Cohort-level zero-event upper bound
0 of 6	0%	0 to 46%	Expanded-cohort zero-event upper bound

Table 3.7. Worked precision of the maximum-sample and cohort-level estimates. Even a whole-study zero-event observation admits an upper 95 percent confidence bound near 18 percent, and a single event yields a point estimate of 5.6 percent; these bounds are the quantitative basis for the descriptive, non-confirmatory framing of the first-in-human design.

3.2 Summary of Previous Human Experience

Daraxonrasib has an active and advancing clinical development program in RAS-mutated solid tumors, with the most mature human experience in pancreatic cancer. The FDA granted daraxonrasib Breakthrough Therapy designation in June 2025 for previously treated metastatic PDAC with KRAS G12 mutations, reflecting preliminary clinical evidence of a meaningful advantage over available therapy in this population [1, 2]. The late-phase program comprises RASolute 302, a Phase 3 study in previously treated metastatic PDAC, and RASolve 301, a first-line program; the broad pan-KRAS eligibility of RASolute 302 defines the molecular eligibility adopted in this protocol, and 300 mg once daily is the established recommended Phase 2 dose carried forward as the highest dose level (DL3) in the present escalation [1]. The previous human experience with daraxonrasib therefore establishes a pan-KRAS antitumor rationale, a tolerability and exposure basis for the dose levels studied here, and a regulatory recognition (Breakthrough Therapy) of unmet need in the same molecular population. The perioperative pause-and-restart logic additionally respects any RASolve 301 checkpoint-inhibitor combination context.

The present investigation differs from the prior daraxonrasib human experience in two material respects that define the scope of the first-in-human determination for this IND. First, it studies the drug in a perioperative, curative-intent surgical setting (resectable and borderline-resectable disease) rather than in the metastatic, palliative setting of the late-phase program, so the wound-healing and anastomotic-healing considerations are new and motivate the pause-and-restart advisory. Second, it couples the drug to an investigational LLM-directed robotic device, which has no prior human experience and which is evaluated here as a significant-risk early-feasibility Class II collaborative platform under the IDE pathway; the device-related and procedure-related safety endpoints in §4 are specific to this novel combination. The full marketed-experience, prior-clinical, and clinical-care experience analysis is presented in the Previous Human Experience section of this dossier (§9).

It is worth stating precisely what carries over from the prior human experience and what does not, because the first-in-human determination for this IND turns on that boundary. What carries over is the drug-level exposure and tolerability basis: the 300 mg once-daily recommended Phase 2 dose has been characterized in the late-phase metastatic program, so DL3 in this escalation is an established exposure rather than a dose beyond prior human experience, and the lower levels DL1 and DL2 sit below it. The pan-KRAS antitumor rationale and the molecular eligibility likewise carry over, because the RASolute 302 program established the breadth of the pan-KRAS coverage and the molecular criterion this protocol adopts. What does not carry over is everything specific to the new combination: the perioperative surgical setting with its wound-healing and anastomotic-healing considerations is new, the LLM-directed robotic device has no prior human experience, and the integrated drug-device coupling through the restart advisory is novel. The first-in-human determination is therefore made with respect to the combination and the surgical setting, while the drug exposure itself remains within the prior human range.

Figure 6 threads this IND into the real-world daraxonrasib PDAC program. The KRAS-mutated PDAC patient who receives daraxonrasib with the robotic Whipple is the origin of a document thread that runs from this initial Phase 1 IND, through the 3+3 cohort-review packages that establish the recommended Phase 2 dose, the amendments with synchronized consent, and the Phase 1-to-2 clinical study report and end-of-Phase-2 briefing, to the Phase 2 randomized protocol at the 300 mg recommended Phase 2 dose, with each document built by the same single-prompt workflow that produced this dossier.

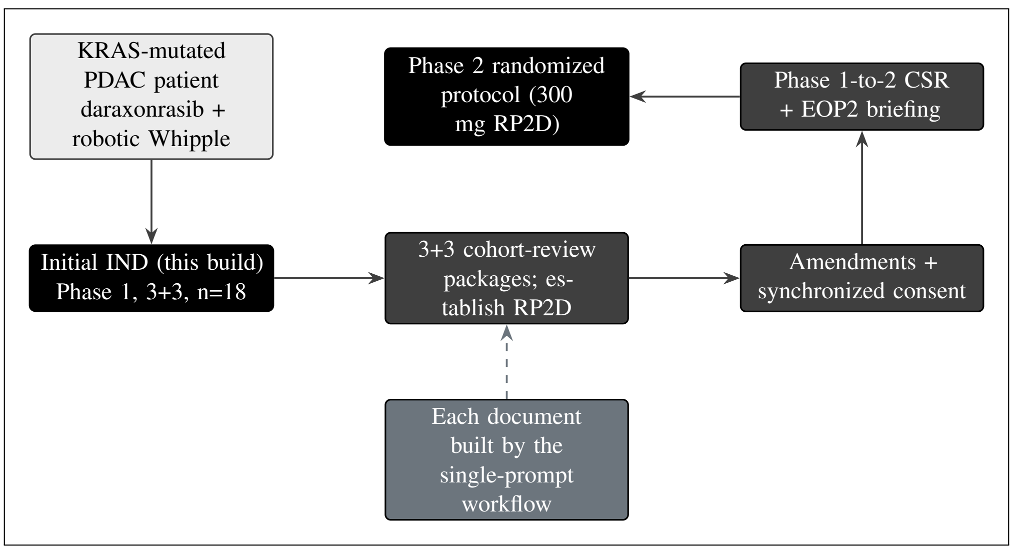

Figure 6. The real-world daraxonrasib PDAC document thread, from the KRAS-mutated patient and the intervention through this initial Phase 1 IND, the 3+3 cohort-review packages that establish the recommended Phase 2 dose, the amendments with synchronized consent, and the Phase 1-to-2 clinical study report and end-of-Phase-2 briefing, to the Phase 2 randomized protocol at the 300 mg recommended Phase 2 dose, each document built by the single-prompt workflow. This figure renders in the Introduction (§3.2) to thread the IND into the real program.

The document thread of Figure 6 is the practical embodiment of the previous-human-experience account: the prior late-phase program establishes the 300 mg recommended Phase 2 dose that this Phase 1 carries forward, and this Phase 1 in turn establishes the recommended Phase 2 dose for the perioperative combination that a subsequent Phase 2 randomized protocol would adopt. Each link in the thread is a regulated document that the same single-prompt workflow generates and maintains, so the dose, the eligibility criteria, and the safety architecture remain consistent from the initial IND through the cohort-review packages and amendments to the Phase 1-to-2 clinical study report and the end-of-Phase-2 briefing. The reviewer can therefore read the IND not as an isolated artifact but as the first, rate-limiting document of a coherent program in which every downstream document inherits the same pinned trial facts.

The investigational design is further grounded by a body of computational and in silico evidence developed by the sponsor-investigator throughout 2025 and 2026, which supplies the simulation scaffolding adapted in this protocol and independently complements the daraxonrasib drug-development program. This evidence comprises a 60-second PDAC robotic Whipple and daraxonrasib simulation; an AI digital-twin pancreatic-cancer simulation for in silico FDA compliance and cost efficiency [14]; a quantitative systems pharmacology (QSP) metastatic-PDAC trial simulation with code, verification, validation, and uncertainty quantification, and a reproducible playbook [13]; a 100,000-patient in silico Phase 3 five-arm PDAC trial triplicate [15]; and end-to-end PDAC digital-twin clinical-trial proposals [16]. The sponsor-investigator's record of LLM evaluation and trust over August 2024 to June 2026 further grounds the device governance, including a comparative code-generation evaluation of 16 proprietary versus open-source LLMs against the FDA Adverse Event Reporting System [12], the on-premises advisory-LLM and verification-before-generation design [7], and the legislative framing aligned with H.R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026 [6]. The previous human experience and the supporting computational evidence together motivate the favorable benefit-risk determination for this low-survival, limited-alternative population.

Table 3.8 collects the previous-human-experience and computational-evidence base in one place, distinguishing the drug-level clinical experience, the regulatory recognition, and the in silico evidence, and recording the role each plays in grounding the present design. The table makes explicit that the human experience and the regulatory recognition pertain to the drug and its molecular target, while the computational evidence pertains to the device, the integrated combination, and the simulation scaffolding that substitutes for and supplements conventional preclinical testing for the Physical AI system. Reading these together is the basis for the benefit-risk determination in a population with five-year overall survival below 13 percent and limited alternatives.

Table 3.8. Previous human experience and computational evidence base. The drug-level clinical experience and the Breakthrough Therapy recognition ground the pharmacologic rationale, the eligibility, and the dose levels; the computational and LLM-evaluation evidence grounds the device, the integrated combination, and the simulation scaffolding that precedes any human exposure.

3.3 Status of Drug in Other Countries

Daraxonrasib is an investigational agent that has not received marketing approval in the United States or in any other jurisdiction for any indication; it is studied worldwide under investigational authorizations within its sponsor's RAS(ON) development program, and its U.S. Breakthrough Therapy designation (June 2025) is a development-stage designation rather than a marketing authorization [1, 2]. The drug has therefore not been withdrawn from marketing in any country for reasons of safety or effectiveness, and there is no adverse foreign regulatory action to report under 21 CFR §312.23(a)(5). Within the present investigation, the LLM-directed robotic device component is similarly investigational and is studied under the IDE pathway as a significant-risk Class II collaborative device; it carries no marketing clearance or approval in any jurisdiction.

The negative statement required by 21 CFR §312.23(a)(5) is made affirmatively here: there is no country in which daraxonrasib has been withdrawn from marketing for any reason, because it has never been marketed in any country; there is no adverse foreign regulatory action of any kind to report; and there is no marketing authorization, clearance, or approval for either the drug or the device in any jurisdiction. The U.S. Breakthrough Therapy designation is a development-stage designation that expedites review and does not constitute or imply a marketing authorization. This affirmative negative statement is the

Table 13: Previous human experience and computational evidence grounding the investigation. Drug-level clinical experience and regulatory recognition pertain to daraxonrasib and its molecular target; the in silico evidence pertains to the device, the integrated combination, and the simulation scaffolding.

Evidence element	Type	Role in grounding this design
RASolute 302 (Phase 3, pretreated metastatic PDAC)	Drug clinical experience	Defines pan-KRAS eligibility; establishes 300 mg RP2D carried forward as DL3
RASolve 301 (first-line program)	Drug clinical experience	Informs combination context respected by the restart logic
Breakthrough Therapy designation (June 2025)	Regulatory recognition	Recognizes unmet need in the same KRAS G12 molecular population
60-second PDAC Whipple and daraxonrasib simulation	Computational	Simulation scaffolding adapted in the protocol
AI digital-twin PDAC simulation [14]	Computational	In silico FDA-compliance and cost-efficiency basis
QSP metastatic-PDAC simulation with VVUQ [13]	Computational	Code, verification, validation, uncertainty quantification; reproducible playbook
100,000-patient in silico Phase 3 triplicate [15]	Computational	Large reproducible in silico trial experience
End-to-end PDAC digital-twin proposals [16]	Computational	Digital-twin clinical-trial design basis
16-LLM code-generation evaluation vs FDA AERS [12]	LLM evaluation	Grounds device and authoring governance

precise content the regulation requires, and it is recorded here so the reviewer can confirm the status-in-other-countries element is satisfied without ambiguity.

Although this Phase 1 investigation is conducted at a single United States academic site under the combined IND and IDE, the dossier is authored to align with the principal international device and trial frameworks so that the same documentation can support future multi-jurisdiction extension without re-derivation. The cross-jurisdiction alignment that the device safety architecture and the Physical AI System Description are designed to satisfy is summarized in Table 3.9. The hard cyber-physical safety envelope (the 10 kHz heartbeat with a 100 microsecond watchdog, the cross-arm emergency stop of 3 ms or less, the per-arm tip-force cap of 3 N or less and cumulative cap of 18 N or less, and the vascular no-fly gating) and the 21 CFR part 11 hash-chained audit trail are constructed to map onto the corresponding requirements of the European Union Medical Device Regulation, the China National Medical Products Administration, the Japan Pharmaceuticals and Medical Devices Agency, the United Kingdom Medicines and Healthcare products Regulatory Agency, Health Canada, and the Australian Therapeutic Goods Administration [11, 3].

Table 14: Cross-jurisdiction regulatory status and alignment. Daraxonrasib is investigational with no marketing authorization in any country; the device is investigational under the IDE. The dossier is authored to align with the principal international frameworks for future multi-jurisdiction extension.

Jurisdiction	Framework	Status and alignment for this investigation
United States	FDA, 21 CFR §312 / §812	Combined IND / IDE; daraxonrasib investigational with Breakthrough Therapy designation (June 2025); device significant-risk Class II under IDE
European Union	EU MDR 2017/745	No marketing authorization; safety envelope and audit trail authored to map onto MDR essential requirements
China	NMPA	No marketing authorization; device safety architecture aligned for future NMPA submission
Japan	PMDA	No marketing authorization; Physical AI System Description aligned to PMDA expectations
United Kingdom	MHRA	No marketing authorization; trial and device documentation aligned to MHRA framework
Canada	Health Canada	No marketing authorization; dossier structured for future Health Canada extension
Australia	TGA	No marketing authorization; safety and audit-trail design aligned to TGA requirements

Table 3.9. Cross-jurisdiction status of daraxonrasib and the LLM-directed device. The agent and device are investigational in every jurisdiction; the documentation is authored once to align with the principal international device and trial frameworks so it can support future multi-jurisdiction extension.

The cross-jurisdiction alignment in Table 3.9 is a documentation-authoring posture, not a regulatory submission to any of the named authorities; no application has been filed with the European Union, China, Japan, the United Kingdom, Canada, or Australia, and none is implied by the alignment. The purpose of authoring the safety architecture and the Physical AI System Description against the corresponding international requirements is to avoid re-derivation if the program later extends to those jurisdictions: because the hard cyber-physical safety envelope and the 21 CFR part 11 hash-chained audit trail are constructed to map onto the essential requirements of the European Union Medical Device Regulation and the expectations of the other named authorities, a future submission would reuse the same architecture rather than re-engineer it. The reviewer should read the table as evidence that the single-site United States investigation has been designed with international portability in mind, while the regulatory status today remains investigational in every jurisdiction with no marketing authorization anywhere.

3.4 Overview of Preclinical Data

The preclinical and computational basis for this investigation supports both the pharmacologic rationale for daraxonrasib in KRAS G12-mutated PDAC and the safety and feasibility rationale for the LLM-directed robotic device, and is presented in full in the Pharmacology and Toxicology information (§8) and the Investigator's Brochure (§5). On the pharmacology side, daraxonrasib acts through the mechanism-based, on-target suppression of RAS pathway signaling described in §3.1.2: by engaging the active GTP-bound state of mutant RAS in a tri-complex with cyclophilin A, it suppresses MAPK and PI3K effector signaling and reduces tumor-cell proliferation and survival across the KRAS G12 alleles that define the trial population [1]. The drug-distribution, exposure, and tolerability assumptions that anchor the three dose levels (DL1 160 mg, DL2 220 mg, DL3 300 mg) are carried forward from the pan-KRAS RASolute 302 program, in which 300 mg once daily is the recommended Phase 2 dose, and the anticipated mechanism-based toxicity profile (gastrointestinal, dermatologic and mucocutaneous, and hepatic) informs the DLT definitions and the perioperative pause-and-restart logic [2].

On the device and combined-intervention side, the preclinical evidence is dominated by a structured program of high-fidelity simulation and digital-twin work that substitutes for and supplements conventional preclinical testing for the Physical AI system, consistent with the verification, validation, and uncertainty quantification (VVUQ) discipline that the protocol applies. The mandatory Phase 0 simulation-validation gate requires at least 1000 simulated procedures across at least two independent frameworks (for example Isaac, MuJoCo, Gazebo, and PyBullet), a Unified Safety Level rating of at least 7.0, and a sim-to-real gap below 2 mm in trajectory and below 0.5 N in tip force, with re-validation after any qualifying change; this gate is the principal preclinical safety bar that the device must clear before any patient-facing activity. The supporting computational evidence base includes the 60-second PDAC robotic Whipple and daraxonrasib simulation, the AI digital-twin pancreatic-cancer simulation for in silico FDA compliance and cost efficiency [14], the QSP metastatic-PDAC trial simulation with code, VVUQ, and a reproducible playbook [13], the 100,000-patient in silico Phase 3 five-arm PDAC trial triplicate [15], and the end-to-end PDAC digital-twin clinical-trial proposals [16]. Together these establish that the integrated drug-device intervention has been characterized in silico across a large and reproducible body of simulated experience before any human exposure, and that the on-premises advisory LLM operates strictly as a bounded, hash-pinned, second-opinion oracle with full autonomy prohibited consistent with 21 CFR §312.21(e) [7, 6]. The integrated pharmacology and toxicology summary, including the structure of the pharmacology summary, the integrated toxicology summary, and the full data tabulation, is presented in §8.

The Phase 0 simulation-validation gate deserves explicit treatment because it is the preclinical safety bar that takes the place, for the Physical AI system, of the conventional animal-toxicology package that would precede a drug-only first-in-human study. The gate is quantitative and prespecified: it requires at least 1000 simulated procedures across at least two independent simulation frameworks, so that the result does not depend on the idiosyncrasies of any single simulator; a Unified Safety Level rating of at least 7.0, which is the composite safety threshold the device must reach; and a sim-to-real gap below 2 mm in trajectory and below 0.5 N in tip force, which bounds how far the simulated behavior may diverge from the physical platform before the validation is considered to hold. The gate is re-validated after any qualifying change, so that a modification to the platform, the firmware, or the advisory model cannot reach a patient without re-clearing the bar. Figure 7 shows the pre-submission authoring workflow that assembles the dossier in which this gate and the rest of the device safety architecture are documented.

Figure 7 documents how the present submission was authored. The principal investigator's clinical rationale for KRAS-mutated PDAC, the biostatistician's 3+3 and exact-confidence-interval rules, and the ReGARDD IND template under 21 CFR part 312 are compiled by the repository LLM under medical-writer review in an electronic document-management system into the eleven IND modules (the FDA 1571 through the relevant-information section), the protocol and Investigator's Brochure, and the informed consent form at a sixth to eighth grade reading level with the Physical AI opt-out, all of which start the 30-day FDA clock on submission.

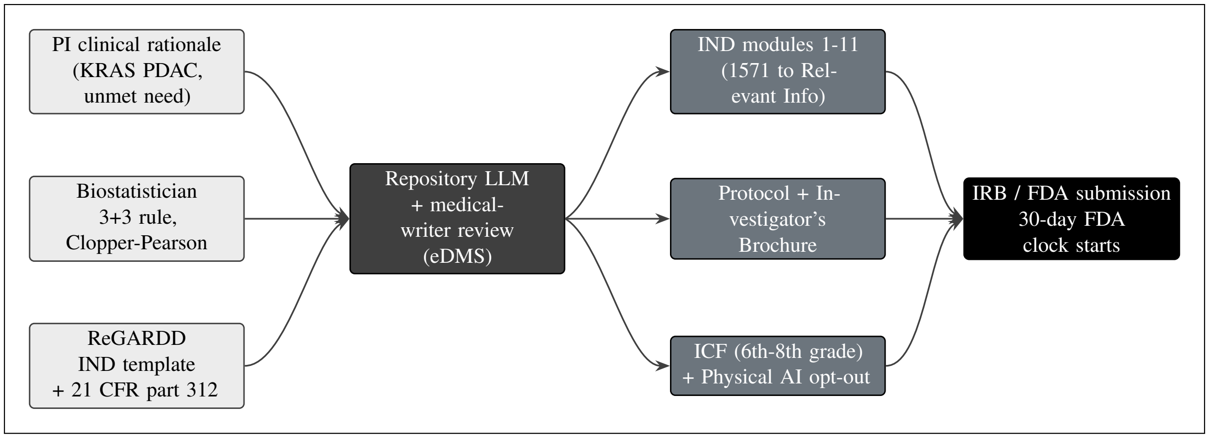

Figure 7. Pre-submission authoring. The principal investigator's clinical rationale, the biostatistician's 3+3 and exact-confidence-interval rules, and the ReGARDD IND template under 21 CFR part 312 are compiled by the repository LLM under medical-writer review into the eleven IND modules, the protocol and Investigator's Brochure, and the informed consent form with the Physical AI opt-out, which start the 30-day FDA clock on submission. This figure renders in the Introduction and §6.1 (Study Protocol) to document how the submission was authored.

The preclinical and computational basis, the pre-submission authoring workflow, and the verification-before-generation discipline together support the central claim that the integrated drug-device intervention has been characterized in silico across a large and reproducible body of simulated experience before any human exposure, and that every advisory output is screened by a verification-before-generation gate suite that issues an accept, block, or escalate verdict, with the on-premises LLM bound strictly to a second-opinion advisory role and full autonomy prohibited under 21 CFR §312.21(e) [7, 6]. The integrated pharmacology and toxicology summary, including the structure of the pharmacology summary, the integrated toxicology summary, and the full data tabulation, is presented in §8, and the device safety architecture and the Physical AI System Description are developed in §4 and §5.

3.5 References

The references cited throughout this Introduction are drawn from the IND reference library and are collected, formatted, and rendered in the References and Back Matter section of this dossier. The principal sources grounding the clinical subject, the daraxonrasib mechanism and class, the device safety architecture, the indication, the duration timeline, and the acceleration thesis include the FDA Phase 1 content and format guidance [8], the oncology-trial LLM authoring basis [10], the daraxonrasib robotic Whipple and on-premises device sources [1, 2], the 21 CFR Part 312, Part 50, and ICH Physical AI unification adaptations [3, 19, 18], the PDAC epidemiology and acceleration sources [4, 17, 11], the verification-before-generation and clinical-trust Physical AI sources [6, 7, 5, 9], the comparative LLM evaluation against the FDA Adverse Event Reporting System [12], and the computational digital-twin and QSP evidence base [14, 13, 15, 16]. Each citation key resolves against the IND reference library, and the assembled list renders in the References and Back Matter section.

4 General Investigational Plan

This General Investigational Plan describes, at the level of detail required by 21 CFR §312.23(a)(3)(iv), the rationale for the investigation, the indication to be studied, the general approach to evaluating the investigational articles, the trials to be conducted during the first year, the estimated number of subjects to be exposed, and the foreseeable drug-related and device-related risks together with their mitigations. The investigation is a single, integrated, first-in-human study of two regulated articles administered in combination: perioperative oral daraxonrasib (RMC-6236), an investigational pan-RAS(ON) multi-selective small-molecule inhibitor evaluated under the investigational new drug pathway of 21 CFR part 312, and an on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) system, a significant-risk Class II collaborative device evaluated under the investigational device exemption pathway of 21 CFR part 812. The study carries a single protocol, a single informed consent bearing the Physical AI opt-out, and a single three-tier safety governance structure. The plan that follows is intentionally quantitative so that the reviewer can assess the design, the dose schedule, the operating characteristics, and the benefit-risk balance from the numbers themselves [8, 1].

This plan is written to be self-contained at the level of the General Investigational Plan while cross-referencing the modules that elaborate each element: the Investigator Brochure (§5) for the integrated nonclinical and clinical summary of daraxonrasib and the device, the proposed clinical research module (§6) for the full protocol and informed consent, the Chemistry, Manufacturing and Control information (§7), and the Pharmacology and Toxicology information (§8). Where this plan states a numeric specification, that specification is the binding value carried forward consistently into those modules and into the protocol, so the reviewer encounters one set of numbers throughout the submission rather than several. The General Investigational Plan is, by the design of 21 CFR §312.23(a)(3)(iv), a brief description; here it is expanded to the quantitative depth that a combined drug-device first-in-human investigation in a lethal indication warrants, because the benefit-risk judgment the reviewer must make turns on the exact dose schedule, the exact safety envelope, and the exact operating characteristics rather than on a qualitative summary [8, 2].

4.1 Rationale

The scientific rationale rests on a population of profound unmet need and a mechanistically matched intervention. Pancreatic ductal adenocarcinoma (PDAC) remains among the most lethal of solid tumors: the five-year overall survival is below 13 percent, and the disease is the third leading cause of cancer death in the United States and the fourth in the European Union [4]. The great majority of PDAC is driven by an activating KRAS mutation at codon 12, most commonly KRAS G12D, G12V, or G12R, and until recently this oncogenic dependency was considered pharmacologically intractable. Daraxonrasib (RMC-6236) is an oral once-daily RAS(ON) multi-selective (pan-RAS) inhibitor that binds the GTP-bound, active conformation of mutant RAS in a tri-complex with cyclophilin A and thereby suppresses downstream effector signaling through the MAPK and PI3K pathways. The agent received FDA Breakthrough Therapy designation in June 2025 for previously treated metastatic PDAC harboring a KRAS G12 mutation, and the 300 mg once-daily dose is the established recommended Phase 2 dose of the RASolute 302 pan-KRAS program [1, 10]. Pairing a genotype-matched systemic agent with a margin-negative curative-intent resection in the resectable and borderline-resectable setting is a scientifically coherent strategy for a disease in which the resection margin is the strongest modifiable determinant of survival.

The biological logic of the pairing deserves to be stated precisely because it shapes the entire dose schedule. In the resectable and borderline-resectable setting the principal lever on survival is a microscopically margin-negative (R0) resection, and the principal threat to that lever is residual disease at the retroperitoneal and superior-mesenteric-vessel margins. A pan-RAS inhibitor that suppresses the dominant oncogenic driver in the perioperative window has the potential to reduce the viable tumor burden at those margins and to address micrometastatic disease that an operation alone cannot reach, while the operation removes the bulk disease that systemic therapy alone cannot durably control. The two modalities are therefore complementary rather than redundant, and neither is a substitute for the other. The investigational article on the device side is not a second drug but the instrument that delivers the operation to a higher and more reproducible standard of margin control and vascular safety, which is why the IND and the IDE are evaluated as one integrated investigation rather than as two parallel studies [1, 2].

The second component of the rationale is the surgical platform itself. The Whipple operation is the most technically demanding curative-intent abdominal procedure, and its early morbidity is dominated by pancreatic fistula, intra-abdominal hemorrhage, sepsis, and failure to rescue. The investigational eight-arm robotic system is engineered to perform this operation within hard cyber-physical limits under continuous human oversight: eight arms each with seven degrees of freedom (56 degrees of freedom total), 640 sensor channels (80 per arm) sampled at 100 kHz for force and 10 kHz for other modalities, a positional accuracy of 0.05 mm RMS at 1200 mm/s, per-arm tip-force caps at or below 3 N and a cumulative cross-arm cap at or below 18 N, a 10 kHz safety heartbeat (a 64-byte frame) with a 100 microsecond watchdog, arm-park within 50 microseconds, a cross-arm emergency stop within 3 ms to 0.0 N, and a hardware emergency stop within 500 ms consistent with 21 CFR §312.404 [2, 1]. A vascular no-fly gate continuously protects five named vessels (the superior mesenteric vein, portal vein, hepatic artery, celiac axis, and superior mesenteric artery) with a 3.0 mm advisory soft-warning zone, a 1.0 mm no-fly zone for the superior mesenteric and portal veins and a 1.5 mm no-fly zone for the arterial structures, and a 5.0 mm hard-stop zone that triggers an emergency stop. The on-premises LLM operates strictly as a second-opinion advisory oracle, hash-pinned to a deterministic seed and a repository commit, isolated outside the vendor kinematic stack, and prohibited from full autonomy consistent with 21 CFR §312.21(e); every advisory passes a ten-gate verification-before-generation suite returning ACCEPT, BLOCK, or ESCALATE, aligned with H.R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026 [6, 7].

The operation is decomposed into eight prespecified operative phases so that the device-feasibility endpoint can be assessed task by task rather than as an all-or-nothing whole. Phases P1 through P4 cover the dissection and specimen removal; phase P5 is the pancreaticojejunostomy reconstructed as a duct-to-mucosa anastomosis with a ring tension held between 0.30 N and 0.60 N; phase P6 is the hepaticojejunostomy reconstructed end-to-side with a working tension between 0.20 N and 0.50 N; phase P7 is the gastrojejunostomy reconstructed antecolic with a tension between 0.40 N and 0.80 N; and phase P8 closes the case with imaging, hemostasis, and drain placement. These per-phase tension envelopes are not decorative: each is a real-time hard constraint enforced by the same force-capping logic that bounds the per-arm and cross-arm tip force, so a reconstruction that would exceed its phase envelope is prevented before it can injure tissue. The phase decomposition is also the unit at which the LLM advisory operates, because each anastomotic phase presents a discrete decision point at which a verified second opinion is most useful [2, 1].

The schedule rationale is distinct from, and complementary to, the clinical rationale. This IND is itself the highest-value product of a verified single-prompt authoring method whose purpose is to compress the administrative and preparation time that separates clinical milestones, without ever altering tumor biology or any fixed regulatory clock. Figure 8 ranks the six document targets where faster, validated authoring yields the greatest schedule value. The initial IND and IRB package, the subject of this build, is the highest-value target because it sits on the critical path; one input, faster validated authoring, feeds all six targets, and each contributes to a common compression of administrative and preparation time measured in months to a year. Crucially, the gains are capped by clinical events and by fixed review clocks: faster authoring cannot shorten the 30-day FDA review or manufacture missing toxicology or stability data [17, 11].

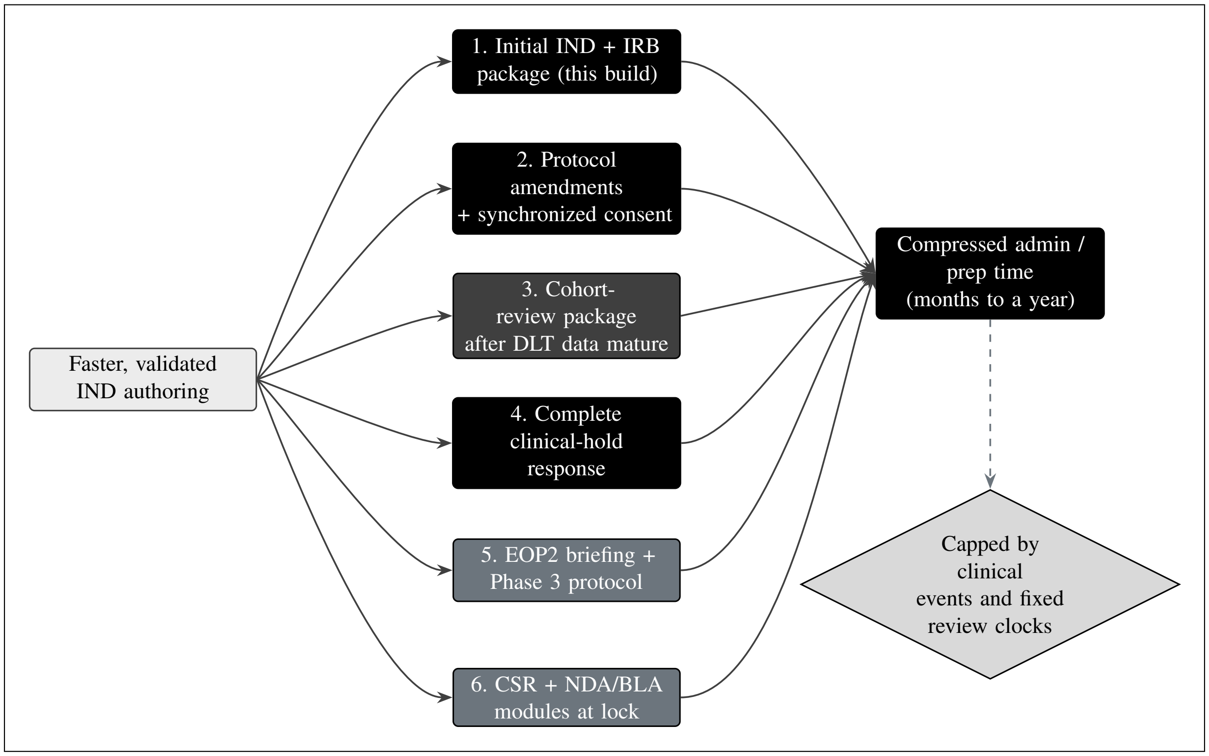

Figure 8. The six greatest-acceleration IND document targets, ranked, with the initial IND and IRB package the highest-value target and the subject of this build. One input, faster validated authoring, feeds all six; each contributes to a common compression of administrative and preparation time of months to a year. The schedule value is highest on the critical path, and the gains are capped by clinical events and fixed review clocks (they cannot shorten the 30-day FDA review or manufacture missing data).

The mechanism of the schedule benefit is specific and is shown in Figure 9. A clinical-trial timeline divides into three buckets: a clinical and operational bucket (recruitment, surgery, the dose-limiting-toxicity window, and progression-free and overall-survival maturation), which is not compressible; an administrative and preparation bucket (data cleaning, analysis, and document authoring), which is compressible; and a fixed regulatory-review bucket (the 30-day IND review and the 30-day clinical-hold review), which is fixed. The repository LLM acts on the middle bucket only. The exact per-document savings are: the initial IND and IRB package falls from 8 to 12 weeks to 1 to 4 days; a protocol amendment with synchronized consent falls from 2 to 4 weeks to 1 to 2 days; a cohort-review package falls from 1 to 2 weeks to about 1 day; a complete clinical-hold response falls from 3 to 6 weeks to 2 to 4 days; an annual development safety update report (DSUR) falls from 4 to 8 weeks to 2 to 3 days; and a clinical study report under ICH E3 falls from 3 to 6 months to 1 to 2 weeks, summing to months to a year saved cumulatively [17, 9].

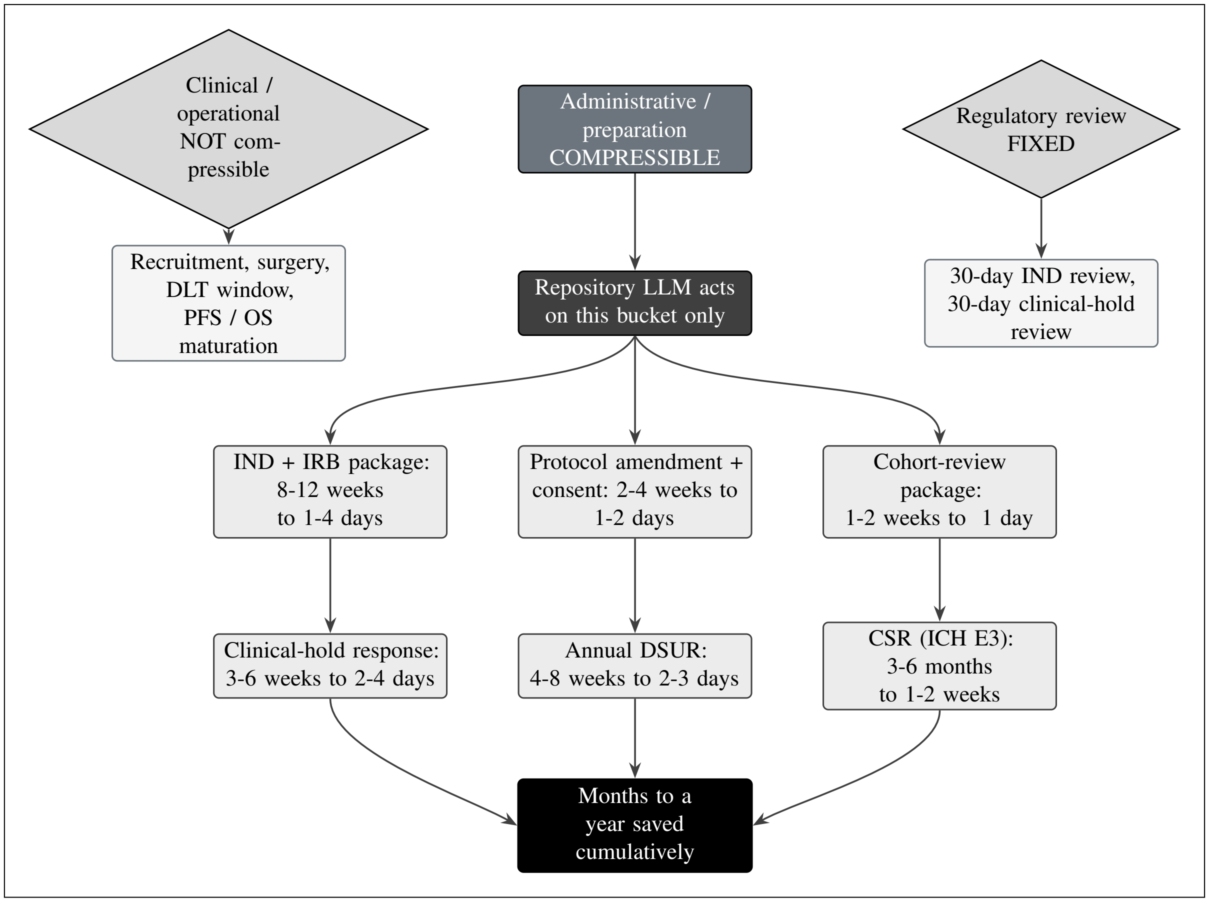

Figure 9. The three timeline buckets. Only the administrative and preparation bucket compresses; the clinical and operational bucket and the fixed regulatory review clocks do not. The repository LLM acts on the middle bucket only, with exact per-document savings: initial IND and IRB package 8 to 12 weeks to 1 to 4 days; protocol amendment with synchronized consent 2 to 4 weeks to 1 to 2 days; cohort-review package 1 to 2 weeks to about 1 day; complete clinical-hold response 3 to 6 weeks to 2 to 4 days; annual DSUR 4 to 8 weeks to 2 to 3 days; clinical study report 3 to 6 months to 1 to 2 weeks, summing to months to a year saved cumulatively.

To make the bucket argument auditable rather than rhetorical, Table 4.1 tabulates the same six documents with their traditional and method-assisted durations, the implied compression factor, and the bucket each occupies. The compression factor is computed at the midpoint of each stated range so that the reader can reproduce it: for the initial IND and IRB package the traditional midpoint is 10 weeks (70 days) and the method-assisted midpoint is about 2.5 days, a factor near 28; for the clinical study report the traditional midpoint is about 4.5 months (135 days) and the method-assisted midpoint is about 10.5 days, a factor near 13. Every entry in the table sits in the administrative and preparation bucket, which is the only bucket the method touches; the clinical and operational bucket and the fixed regulatory-review bucket are listed in the final rows precisely to show that they carry a compression factor of one by construction [17, 9].

Document or interval	Traditional	Method-assisted	Factor	Bucket
Initial IND + IRB package (this build)	8 to 12 weeks	1 to 4 days	≈28x	Administrative / preparation (compressible)
Protocol amendment + synchronized consent	2 to 4 weeks	1 to 2 days	≈14x	Administrative / preparation (compressible)
Cohort-review package (post-DLT)	1 to 2 weeks	about 1 day	≈11x	Administrative / preparation (compressible)
Complete clinical-hold response	3 to 6 weeks	2 to 4 days	≈11x	Administrative / preparation (compressible)
Annual DSUR	4 to 8 weeks	2 to 3 days	≈17x	Administrative / preparation (compressible)
Clinical study report (ICH E3)	3 to 6 months	1 to 2 weeks	≈13x	Administrative / preparation (compressible)
30-day IND review	30 days	30 days	1x	Regulatory review (fixed)
Recruitment, surgery, DLT window, PFS / OS maturation	not compressible	not compressible	1x	Clinical / operational (not compressible)

Table 4.1. Per-document acceleration with the compression factor computed at the midpoint of each stated range. Every accelerated item sits in the administrative and preparation bucket; the fixed regulatory-review clock and the clinical and operational interval carry a compression factor of one by construction and are shown for contrast.

These two rationales meet at the patient. Figure 10 traces the patient time-saved cascade: faster validated authoring advances each step from IND and IRB submission and site activation, through the elapse of the 30-day clock and the first dose, to faster 3+3 cohort turnover toward the recommended Phase 2 dose, shortening the white space between phases and, for this low-survival population, extending progression-free and overall survival, without changing tumor biology or any fixed review clock. The cascade is the patient-level expression of the schedule benefit and is the bridge between the rationale here and the favorable benefit-risk balance developed in §4.6 [17, 4].

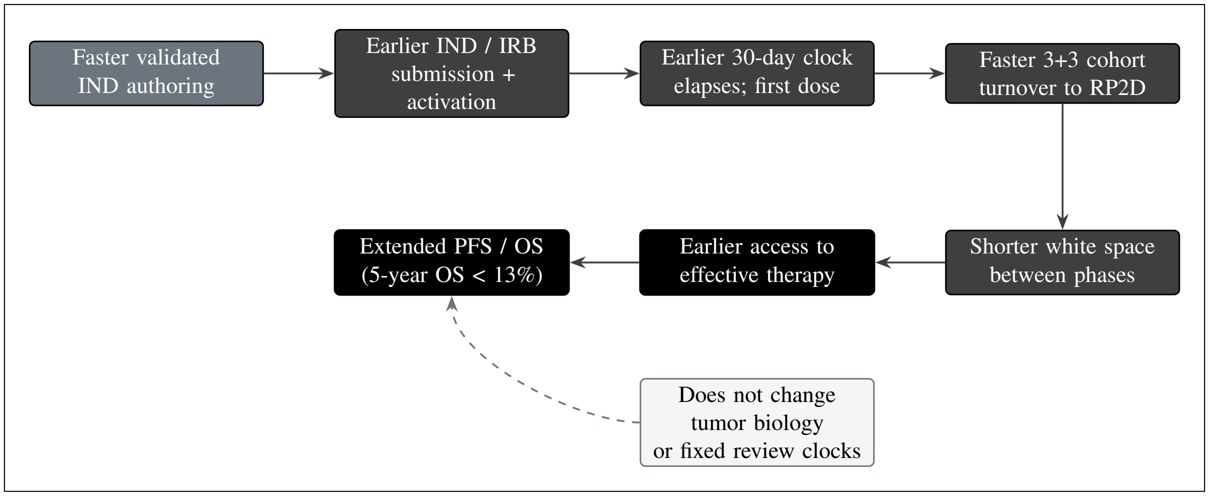

Figure 10. The patient time-saved cascade. Faster validated authoring advances each step from IND submission to first dose, then through faster 3+3 cohort turnover to the recommended Phase 2 dose, shortening the white space between phases and, for this low-survival population (five-year overall survival under 13 percent), extending progression-free and overall survival, without changing tumor biology or the fixed regulatory review clocks.

A worked schedule example fixes the magnitude of the cascade in days. Suppose the administrative authoring of the initial IND and IRB package is compressed from a traditional 70 days (the 10-week midpoint) to about 3 days. The fixed 30-day FDA review clock is unchanged, the operation and the 28-day DLT window are unchanged, and the survival follow-up clocks are unchanged. The net effect is that the first dose and the first operation are reached approximately 67 days earlier than they would otherwise have been, and that 67-day advance propagates through every downstream cohort because each cohort's start depends on the prior cohort's completion. Across three cohorts and the inter-phase authoring that connects this study to a later-phase program, the accumulated advance reaches the months-to-a-year figure stated in Figure 9. None of this advance is borrowed from biology or from patient safety: the DLT window, the sentinel staggering, and the review clocks remain exactly as specified [17, 7].

4.2 Indication to be Studied

The indication to be studied is histologically or cytologically confirmed pancreatic ductal adenocarcinoma that is resectable or borderline-resectable on multidisciplinary radiologic review, harbors a documented KRAS G12 mutation (for example G12D, G12V, or G12R) on a validated tumor genotyping assay, occurs in an adult of at least 18 years with an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1, and is anatomically compatible with the eight-arm robotic system and with a planned pancreaticoduodenectomy. Eligibility for perioperative daraxonrasib follows the broad pan-KRAS criteria of the RASolute 302 program [1]. The indication is deliberately narrow at this first-in-human stage so that the safety and feasibility readouts are interpretable: distant metastatic disease, vascular involvement precluding a margin-negative resection, prior pancreatic resection precluding the planned reconstruction, uncontrolled intercurrent illness, pregnancy or lactation, and known contraindication to daraxonrasib are exclusionary. Table 4.2 summarizes the eligibility frame that defines the indication; the full inclusion and exclusion criteria are set out in the protocol and reproduced in the proposed clinical research module of this IND.

Domain	Inclusion (must meet all)	Exclusion (any one disqualifies)
Disease and stage	Histologically or cytologically confirmed PDAC; resectable or borderline-resectable on multidisciplinary review.	Distant metastatic disease on staging; vascular involvement precluding R0 resection or anatomy incompatible with the eight-arm system.
Genotype	Documented KRAS G12 mutation (G12D, G12V, or G12R) on a validated assay; eligible under pan-KRAS RASolute 302 criteria.	Absence of a qualifying KRAS G12 mutation.
Performance and organ function	ECOG performance status 0 or 1; adequate hematologic, hepatic, and renal function for major pancreaticobiliary surgery and daraxonrasib.	Uncontrolled cardiac, pulmonary, hepatic, renal, or active infectious disease making surgery or dosing unsafe.
Surgical candidacy	Candidate for pancreaticoduodenectomy; tumor and vascular anatomy compatible with port placement and the working volume.	Prior pancreatic resection or prior surgery precluding the planned reconstruction.
Age and consent	Age at least 18 years; signed informed consent including explicit election or declination of the Physical AI opt-out.	Inability to consent or to comply with the Schedule of Activities.
Other	Measured baseline CA 19-9 recorded for serial assessment (no threshold imposed).	Pregnancy or lactation; known contraindication or hypersensitivity to daraxonrasib or a required excipient; unmanageable drug interaction.

Table 4.2. Eligibility summary defining the indication to be studied. Selection is equitable and is determined solely by the scientific and safety criteria; limited English proficiency is not an exclusion, and qualified interpreters and translated consent materials are provided.

The indication is positioned within the larger PDAC treatment landscape so that the reviewer can see what is and is not claimed. The resectable and borderline-resectable population is a minority of all PDAC at presentation, because most patients present with locally advanced or metastatic disease; it is, however, the population in which a curative-intent resection is anatomically possible and therefore the population in which margin control and perioperative systemic therapy can change the survival trajectory. The genotype restriction to KRAS G12 is mechanistic rather than arbitrary: daraxonrasib acts on the GTP-bound active conformation of mutant RAS, so a documented qualifying mutation is a prerequisite for the drug to have its intended target. The anatomic restriction to tumors compatible with the eight-arm working volume and with the five-vessel no-fly geometry is a device-safety prerequisite, not a comment on resectability in general. The indication is thus the intersection of an oncologic window, a genotype, and an anatomic envelope, and each restriction maps to a specific element of the intervention [1, 2].

Equity in selection is an explicit design commitment. The eligibility frame imposes no criterion that is not scientifically or safety-justified; limited English proficiency is not an exclusion, and qualified interpreters and translated consent materials are provided so that language is never the reason a patient cannot enroll. No threshold is imposed on the baseline CA 19-9; it is measured and recorded only so that the serial trend can be followed, because a substantial fraction of PDAC is CA 19-9 non-secretory and a threshold would exclude those patients without scientific justification. The screen-to-treat ratio, approximately 36 screened to reach up to 18 treated, reflects the genotype and anatomic restrictions rather than any socioeconomic filter, and the screening log is retained so that the equity of the selection can be audited [1, 7].

A synthetic reference exemplar, designated PAT-PDAC-0001, illustrates a representative eligible participant and is used for design illustration only: a 68-year-old with a 3.4 cm head-of-pancreas tumor abutting the superior mesenteric vein at 75 degrees, KRAS G12D, microsatellite stable, a baseline CA 19-9 of 412 U/mL, ECOG performance status 1, after a completed neoadjuvant modified FOLFIRINOX course of four cycles. This exemplar does not represent an enrolled individual but fixes the anatomic and oncologic context in which the eligibility frame and the device no-fly gating are exercised. The 75-degree venous abutment, for instance, is exactly the geometry in which the 1.0 mm no-fly zone for the superior mesenteric and portal veins and the 3.0 mm soft-warning zone are most relevant, because the dissection plane runs along the vessel for a substantial arc; the exemplar therefore serves as a concrete test case for the device-safety argument rather than as a generic patient sketch [1, 2].

4.3 General Approach for Evaluation of Treatment

The general approach is a prospective, open-label, single-arm, first-in-human study that combines a standard Phase 1 3+3 dose-finding evaluation of perioperative daraxonrasib with an early-feasibility evaluation of the LLM-directed eight-arm robotic Whipple system. A single-arm, open-label design is scientifically and ethically appropriate, and a randomized comparator arm is not, because the first-in-human objectives, characterizing early safety, identifying a recommended Phase 2 dose, and demonstrating that a novel surgical system can perform its intended function within hard cyber-physical limits, are not comparative-efficacy questions and are answered most safely in a small, intensively monitored cohort with sentinel staggered enrollment and case-by-case data and safety monitoring board (DSMB) review. The early-feasibility pathway recognized by FDA for significant-risk devices is explicitly structured around small first-in-human cohorts that generate the device-performance and safety evidence needed before any controlled comparison [8]. The 2025 Dutch nationwide cohort of 1000 robotic pancreaticoduodenectomies provides a robust external human benchmark for the secondary surgical-quality endpoints, so the absence of an internal control does not leave the surgical results uninterpretable [1].

The choice of an external benchmark over an internal control is a deliberate design decision with explicit limits. A nationwide cohort of 1000 consecutive robotic pancreaticoduodenectomies captures the contemporary standard of robotic Whipple practice across many surgeons and the learning-curve gradient that accompanies any new surgical platform, and it supplies stable comparator rates for the four secondary surgical-quality endpoints that the present study cannot internally control: an ISGPS grade B/C fistula rate of 24.4 percent, a Clavien-Dindo grade III or higher complication rate of 41.3 percent, an unplanned conversion rate of 10.1 percent, and a 30-day mortality of 3.9 percent. These comparators are used for descriptive contextualization only. The single-arm design supports estimation of the study rates with exact confidence intervals; it does not support inferential between-group testing against the Dutch cohort, because the populations, the case mix, and the perioperative systemic therapy differ, and because a first-in-human cohort of up to 18 is far too small to power such a comparison. The benchmark therefore answers the question "is the observed rate in the expected neighborhood of contemporary practice" and not the question "is this system superior" [1, 8].

4.3.1 Objectives and Endpoints

The study carries three co-primary objectives, one for each axis of the combined IND/IDE investigation, beneath which sit secondary oncologic-surgical quality endpoints benchmarked against the Dutch 2025 cohort and exploratory non-confirmatory endpoints. The first co-primary objective characterizes the early safety of the combined intervention; its endpoint is the incidence of device-related or procedure-related serious adverse events through 30 days after the procedure, including Clavien-Dindo grade III or higher complications, with device or procedure attribution adjudicated by an independent reviewer blinded to the daraxonrasib dose level. The second co-primary objective determines the maximum tolerated dose (MTD) and recommended Phase 2 dose (RP2D) of perioperative daraxonrasib, defined operationally by the rate of dose-limiting toxicities (DLTs) within the 28-day DLT window of the 3+3 escalation. The third co-primary objective establishes the technical feasibility of the eight-arm robotic Whipple under continuous human oversight; its endpoint is the proportion of procedures that complete all prespecified assigned operative tasks without unplanned conversion caused by device malfunction or unsafe device behavior. Figure 11 renders the objective-to-endpoint hierarchy, and Table 4.3 sets out the full mapping with justifications and the Dutch 2025 comparators [1, 8].

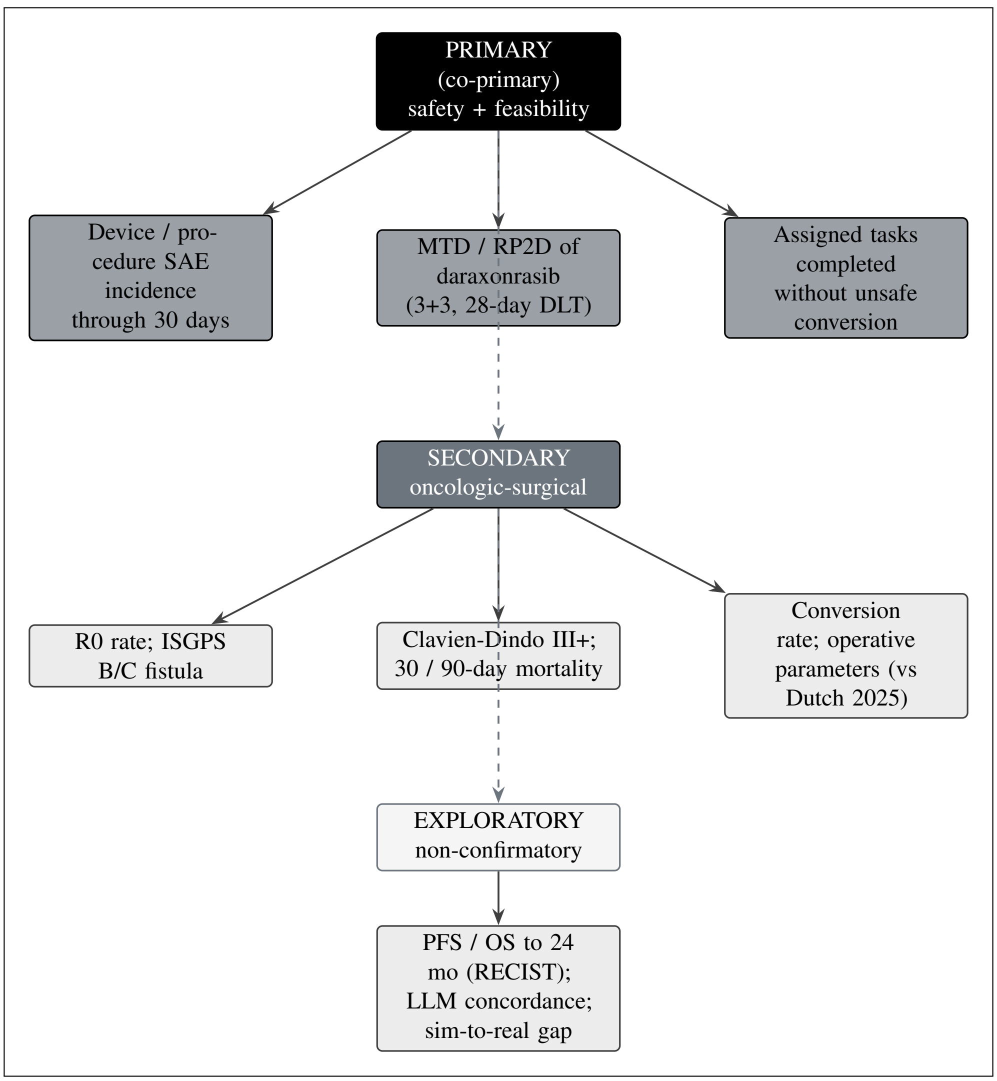

Figure 11. The objective-to-endpoint hierarchy of the general investigational plan. Three co-primary objectives (device or procedure serious-adverse-event incidence through 30 days, the maximum tolerated dose and recommended Phase 2 dose of daraxonrasib by 3+3 over the 28-day window, and completion of assigned operative tasks without unsafe conversion) sit above secondary oncologic-surgical quality endpoints (R0 rate, ISGPS grade B/C fistula, Clavien-Dindo grade III or higher, 30 and 90-day mortality, conversion rate, and operative parameters benchmarked against the Dutch 2025 robotic cohort) and exploratory, non-confirmatory endpoints (progression-free and overall survival to 24 months by RECIST, LLM advisory concordance, and the sim-to-real gap).

Tier	**Objective and endpoint**	**Justification / comparator**
Co-primary		
Safety	Device-related or procedure-related SAE incidence through 30 days, including Clavien-Dindo grade III+, dose-blinded attribution.	Standard first-in-human safety anchor; captures the early lethal cascade of fistula, hemorrhage, sepsis, and failure to rescue.
Dose-finding	MTD and RP2D of perioperative daraxonrasib by the DLT rate within the 28-day window of the 3+3 across DL1 to DL3.	Established dose-finding design for targeted oncology agents; anchored to the RASolute 302 pan-KRAS program.
Feasibility	Proportion of procedures completing all assigned operative tasks without unplanned conversion from device malfunction or unsafe behavior.	FDA-recognized early-feasibility readout for a significant-risk device.
Secondary (descriptive, Dutch 2025 comparators)		
Oncologic adequacy	R0 (microscopically margin-negative) resection rate at day-90 pathology.	Strongest modifiable surgical determinant of survival; descriptive comparison with Dutch 2025.
Fistula	ISGPS grade B/C postoperative pancreatic fistula rate.	Leading driver of post-Whipple morbidity; Dutch 2025 comparator 24.4 percent.
Major complications	Clavien-Dindo grade III+ complication rate through 90 days.	Validated severity threshold; Dutch 2025 comparator 41.3 percent.
Mortality	30-day and 90-day all-cause mortality.	Standard short-term survival safety measures; Dutch 2025 30-day comparator 3.9 percent.
Conversion	Unplanned conversion rate from the robotic system to open or conventional approach.	Contextualizes feasibility; Dutch 2025 comparator 10.1 percent.
Operative parameters	Estimated blood loss, operative time, and length of stay.	Mean and median with range; Dutch 2025 learning-curve context.
Exploratory (non-confirmatory)		
Survival	PFS and OS assessed to 24 months by RECIST v1.1, Kaplan-Meier.	Preliminary, hypothesis-generating signal; the small single-arm sample precludes confirmatory inference.
Advisory reliability	Concordance between the LLM advisory output and the independent hard-coded	Characterizes the second-opinion oracle and informs autonomy-graduation decisions reserved for later phases.

Table 4.3. The objective-to-endpoint mapping with justifications and the Dutch 2025 nationwide robotic-pancreatoduodenectomy comparators. Comparisons are descriptive only; the single-arm design supports estimation, not inferential between-group testing.

The three co-primary objectives are weighted equally rather than ranked, because the investigation cannot be declared a success on any single axis alone. A favorable drug-safety and dose-finding result is uninformative if the device cannot complete the operation within its safety envelope, and an excellent device-feasibility result is unacceptable if the perioperative drug schedule is intolerable. The co-primary structure therefore requires that the early-safety, dose-finding, and feasibility readouts each be satisfactory before the combined intervention is judged ready to advance, and it is the reason the safety-attribution adjudication is performed blinded to the dose level, so that a device or procedure event is not silently attributed to the drug or the reverse. The secondary endpoints are descriptive and support the interpretation of the co-primary results; the exploratory endpoints are explicitly hypothesis-generating and are reported with the caveat that the sample size precludes confirmatory inference [8, 18].

4.3.2 Dose-Escalation Approach

Daraxonrasib is administered orally once daily. The escalation uses exact doses anchored to the RASolute 302 pan-KRAS program, in which the 300 mg once-daily dose is the established RP2D. To preserve a conservative first-in-human margin in the perioperative setting, the escalation begins below that reference dose and steps up to it: DL1 is 160 mg once daily (the starting dose), DL2 is 220 mg once daily, and DL3 is the 300 mg once-daily reference dose. Three participants enroll at each level and are observed over a 28-day DLT window. If no DLT occurs among the first three, the cohort escalates; if one of three experiences a DLT, the cohort expands to six; a single DLT among six permits escalation, whereas two or more DLTs at a level define that level as exceeding the MTD and fix the MTD at the preceding level. The MTD is the highest level at which fewer than two of six DLT-evaluable participants experience a DLT, and the RP2D is selected at or below the MTD with integration of all safety, tolerability, and available pharmacokinetic data. Figure 12 renders the decision automaton and Table 4.4 states the rule explicitly for each cohort outcome [1, 8].

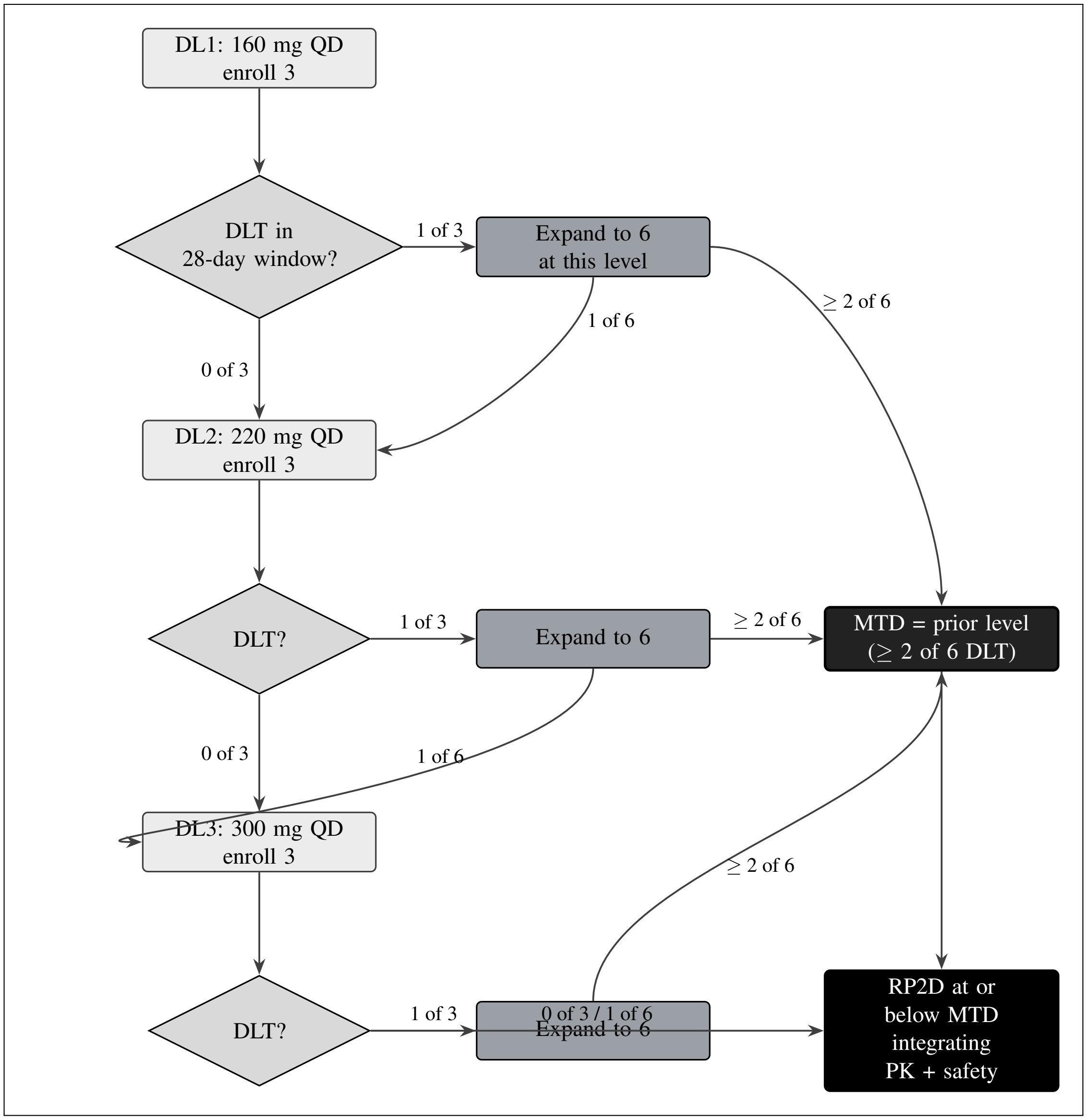

Figure 12. The standard 3+3 dose-escalation automaton across the three daraxonrasib dose levels (DL1 160 mg, DL2 220 mg, DL3 300 mg once daily). Three participants enroll at each level; with no dose-limiting toxicity in the 28-day window the cohort escalates, with one of three it expands to six, and two or more of six fixes the maximum tolerated dose at the prior level. The recommended Phase 2 dose is selected at or below the maximum tolerated dose, integrating pharmacokinetic and safety data. The design maximum is three levels times six, up to 18 treated.

Cohort observation	DLT count	Decision	Consequence
First 3 at a level	0 of 3	Escalate	Open the next higher dose level; current level deemed tolerated at the cohort of 3.
First 3 at a level	1 of 3	Expand to 6	Enroll 3 additional participants at the same level before any escalation decision.
First 3 at a level	≥ 2 of 3	MTD exceeded	Do not enroll further at this level; the MTD is the preceding (lower) level.
Expanded 6 at a level	1 of 6	Escalate	Single DLT in six permits escalation to the next higher level.
Expanded 6 at a level	≥ 2 of 6	MTD exceeded	The level exceeds the MTD; the MTD is the preceding level.
Highest level (DL3)	0 of 3 or 1 of 6	RP2D candidate	DL3 (300 mg) tolerated; RP2D selected at or below DL3 integrating PK and safety.
Any level	≥ 2 of 6	Declare MTD / RP2D	MTD fixed at prior level; RP2D at or below MTD.

Table 4.4. The 3+3 dose-escalation decision rule. The maximum tolerated dose is the highest level at which fewer than two of six DLT-evaluable participants experience a dose-limiting toxicity within the 28-day window; the recommended Phase 2 dose is selected at or below the MTD. The design maximum is three levels times six, up to 18 treated.

The dose spacing is itself a safety choice. The step from DL1 to DL2 is 60 mg (a 1.375-fold increment) and the step from DL2 to DL3 is 80 mg (a 1.364-fold increment), so neither escalation exceeds roughly a 1.4-fold increase in exposure, which is conservative for a targeted agent whose RP2D is already established. Because DL3 is the established RASolute 302 RP2D rather than an extrapolated supratherapeutic dose, the escalation never asks a participant to receive a dose above the dose at which the agent's tolerability is already characterized in the metastatic setting; the perioperative novelty is the timing and the surgical context, not an unprecedented exposure. This is the reason the starting dose is set below the reference RP2D rather than at it: in the perioperative setting the additional hazard of impaired wound healing and anastomotic integrity justifies beginning with a conservative margin and confirming tolerability before stepping up to the reference dose [1, 10].

A dose-limiting toxicity is defined for the 28-day window as a clinically significant toxicity attributable to daraxonrasib that meets the protocol's prespecified grade and persistence thresholds, including a perioperative event in which the drug is the probable contributor to impaired anastomotic healing. The 28-day window is chosen to span the early postoperative healing interval during which a drug-related effect on the pancreaticojejunal and hepaticojejunal anastomoses would manifest, and it aligns the drug-toxicity assessment with the same 30-day horizon over which the device-and-procedure serious-adverse-event endpoint is captured. The DLT-evaluable population comprises participants who either complete the full 28-day window or experience a DLT within it; a participant who is not evaluable for a non-toxicity reason is replaced so that each dose level reaches its required number of DLT-evaluable participants, which preserves the operating characteristics of the 3+3 rule [8, 18].

The perioperative pause-and-restart of daraxonrasib is an integral part of the dose schedule rather than a separate procedure. The drug is held before surgery, and the restart day is set by an LLM-bound, human-confirmed advisory whose inputs are the postoperative pancreatic fistula grade by the International Study Group of Pancreatic Surgery (ISGPS) classification (grade A, B, or C) and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold, with the four permitted outputs T+7 days, T+14 days, T+21 days, or hold. A 32-iteration deterministic sweep of this advisory returned a baseline serum of 0.45 ng/mL (sub-threshold) in all iterations, a T+7-day restart in 29 of 32 iterations (90.6 percent, corresponding to grade A and sub-threshold), a T+14-day restart in 3 of 32 iterations (9.4 percent, corresponding to grade B or a delayed trough), and no T+21-day restart. The sweep is reproduced in Table 4.5 so that the reviewer can see that the advisory's restart-day distribution is dominated by the earliest safe restart and that no iteration produced a prolonged hold, while the human confirmation step preserves clinician authority over every restart decision [1, 7].

Advisory output	**Iterations**	**Share**	**Driving condition**
Restart at T+7 days	29 of 32	90.6%	ISGPS grade A fistula and serum trough sub-threshold (0.45 ng/mL $<$ 0.5 ng/mL).
Restart at T+14 days	3 of 32	9.4%	ISGPS grade B fistula or a delayed serum trough; restart deferred one week.
Restart at T+21 days	0 of 32	0.0%	No iteration met the grade or trough condition for a three-week deferral.
Hold (no restart)	0 of 32	0.0%	No iteration met the hold condition; every iteration restarted within three weeks.

Table 4.5. The 32-iteration deterministic sweep of the perioperative daraxonrasib pause-and-restart advisory. The baseline serum trough was 0.45 ng/mL (sub-threshold) in every iteration. The restart-day distribution is dominated by the earliest safe restart (T+7 days in 90.6 percent), and every advisory is human-confirmed before it is acted upon.

4.3.3 Device Readiness and Statistical Framework

The device has no pharmacologic dose; its dose analogue is the readiness evidence required before any patient-facing robotic activity, escalated only by passing a mandatory Phase 0 simulation-validation gate. Three conditions must be met and documented before the first robotic case: a Unified Safety Level (USL) rating of at least 7.0 on the surgical-readiness scale; completion of at least 1000 simulated procedures across at least two independent frameworks (for example Isaac, MuJoCo, Gazebo, or PyBullet), so the finding is not an artifact of any single simulator; and a sim-to-real fidelity within prespecified limits, a trajectory gap less than 2 mm and a tip-force gap less than 0.5 N between the simulation prediction and the operative record. The gate is re-validated after any change to the system. These limits operate together with the intra-operative hard caps (a cross-arm emergency-stop budget at or below 3 ms, per-arm tip-force caps at or below 3 N, a cumulative cross-arm cap at or below 18 N, and vascular no-fly gating) so the device dose analogue is met only when both the simulation evidence and the real-time safety envelope are demonstrably in place [2, 1].

The cyber-physical safety envelope is specified numerically so that each limit is verifiable rather than aspirational, and Table 4.6 collects the binding values in one place. The temporal hierarchy of the safety responses is itself meaningful: the 10 kHz heartbeat samples the system every 100 microseconds and a 100 microsecond watchdog detects a missed frame within one cycle; an arm parks within 50 microseconds; a cross-arm emergency stop drives all arms to 0.0 N within 3 ms; and a hardware emergency stop completes within 500 ms. Each successive response operates at a coarser time scale and a higher level of authority, so a fault is caught at the finest scale at which it can be caught and escalates only if it is not resolved. The vascular no-fly gate adds a spatial hierarchy on top of this temporal one: a 3.0 mm soft-warning zone issues an advisory, a 1.0 mm (veins) or 1.5 mm (arteries) no-fly zone blocks motion toward the vessel, and a 5.0 mm hard-stop zone triggers an emergency stop. In a representative 100-tick sampling window the gate returned 83 clear verdicts, 6 soft-warning verdicts, 6 no-fly verdicts, and 5 hard-stop verdicts, which illustrates that the gate is active and discriminating rather than dormant [2, 1].

Parameter	**Binding value**	**Function**
Degrees of freedom	8 arms x 7 = 56	Working envelope for the eight-arm collaborative system.
Sensor channels	640 (80 per arm)	100 kHz force sampling; 10 kHz for other modalities.
Positional accuracy	0.05 mm RMS at 1200 mm/s	Trajectory precision under motion.
Per-arm tip-force cap	$\leq$ 3 N	Bounds the force any single arm applies to tissue.
Cumulative cross-arm cap	$\leq$ 18 N	Bounds the total force applied across all arms.
Safety heartbeat / watchdog	10 kHz (64-byte frame) / 100 microsecond watchdog	Detects a missed frame within one cycle.
Arm-park latency	$\leq$ 50 microseconds	Parks an arm on a local fault.
Cross-arm emergency stop	$\leq$ 3 ms to 0.0 N	Drives all arms to zero force.
Hardware emergency stop	$\leq$ 500 ms (21 CFR §312.404)	Final hardware-level stop authority.
Vascular no-fly zones	3.0 mm warn; 1.0 mm vein / 1.5 mm artery no-fly; 5.0 mm hard-stop	Protects SMV, PV, HA, CA, SMA.

Table 4.6. The binding cyber-physical safety envelope of the eight-arm robotic Whipple system. Each value is a hard, verifiable limit; the temporal responses form a hierarchy from a 100 microsecond watchdog to a 500 ms hardware emergency stop, and the vascular no-fly gate adds a spatial hierarchy of warn, no-fly, and hard-stop zones.

The statistical framework is that of a small, intensively monitored, single-arm investigation whose primary purpose is to characterize early safety and to identify a recommended dose, not to test a comparative-efficacy hypothesis. The analyses are predominantly descriptive and estimation-based, follow ICH E9, and are reported in accordance with TRIPOD+AI for the model-assisted advisory component [18]. No power calculation is performed because the study tests no confirmatory efficacy hypothesis. Every proportion is accompanied by an exact (Clopper-Pearson) two-sided 95 percent confidence interval rather than a normal-approximation interval, which is unreliable near the bounds and at small sample sizes; time-to-event variables are summarized by Kaplan-Meier with the number at risk; missing data are not imputed for the primary or safety analyses; and a nominal alpha of 0.05 two-sided applies to supportive estimation only, with no multiplicity adjustment because the analyses are descriptive and non-confirmatory. Four analysis populations are applied consistently: the Safety population (all who received any intervention), the DLT-evaluable population (all dosed participants who completed the 28-day window or had a DLT within it), the Per-Protocol population, and the modified intention-to-treat population [18, 8].

The choice of the exact (Clopper-Pearson) interval over the normal-approximation (Wald) interval is consequential at these sample sizes and is therefore stated explicitly. The Wald interval is constructed as a symmetric band around the point estimate and, when the observed proportion is zero or one, collapses to a degenerate interval of zero width that falsely implies certainty; at a denominator of three or six this failure is not a rare edge case but the expected situation, because a clean safety record is the hoped-for result. The Clopper-Pearson interval, by inverting the exact binomial tail probabilities, always returns a non-degenerate interval that honestly reflects how little a small denominator can exclude. The TRIPOD+AI reporting discipline is applied to the LLM advisory component so that the advisory's inputs, outputs, deterministic seed, pinned commit, and concordance with the hard-coded gate are all reported in a form that an external reviewer can audit, which is the model-facing analogue of the participant-level data tabulation applied to the clinical results [18, 7].

4.4 Description of First Year Trial(s)

During the first year, a single trial is conducted at a single academic medical center: the combined IND/IDE first-in-human study of perioperative daraxonrasib and the LLM-directed eight-arm robotic Whipple. Enrollment is planned for up to 18 treated participants, allocated across the three daraxonrasib dose levels (DL1 through DL3) by the 3+3 rule with staggered sentinel enrollment, from approximately 36 screened. Recruitment proceeds at approximately one to two participants per month through the institutional multidisciplinary pancreatic-cancer program, with eligible individuals identified at tumor-board review and described to them by a member of the research team who is not the treating surgeon, to reduce undue influence. The planned accrual rate, the sentinel staggered enrollment of the 3+3 design, and the Phase 0 readiness gate together give an enrollment period sufficient to treat 18 participants from approximately 36 screened [1, 8].

Sentinel staggering is a binding feature of the first-year conduct: no two participants in a cohort undergo surgery within the same DLT-evaluation interval until the supervising surgeon and the DSMB have reviewed the preceding case. After each participant completes the 28-day DLT-evaluation window and before the next participant in the cohort undergoes surgery, the accumulating safety data are reviewed; after each dose-level cohort is complete, the DSMB convenes for a formal interim safety review and an escalation, expansion, hold, or halt decision. The prespecified halt rules are quantitative and binding: enrollment and escalation halt for a mandatory DSMB review if the device-related or procedure-related serious-AE rate reaches or exceeds one third within a cohort, or if two device-related deaths occur at any point. Additional pause triggers include any unexpected device-related serious AE, any breach of a hard cyber-physical safety limit (emergency-stop latency, positional, or force envelope), and any failure of the pre-procedure safety-matrix verification. The DSMB may halt, pause, modify, or permit continuation, and a halt for an immediate hazard is implemented at once [1].

Oversight is delivered through a three-tier structure with distinct cadences and authorities. The DSMB operates at cohort boundaries and on triggered review and holds the authority to halt the study; the Independent Safety Monitor operates per procedure in real time and holds the authority to stop an individual case; and the Physical AI Safety Review Committee operates on a cadence of at most 90 days and reviews the device-and-advisory performance, including the advisory-to-gate concordance and any model-degradation signal. The reviewing IRB provides ethical oversight, and the sponsor-investigator is ChemicalQDevice (Kevin Kawchak, CEO), whose conflict of interest is disclosed under 21 CFR part 54 and mitigated precisely by the independence of these three tiers, so that no halt, stop, or escalation decision rests with the financially interested party. The structure is designed so that the finest-grained safety authority (the per-case stop) sits closest to the patient and the broadest authority (the study halt) sits with the body that sees the accumulating cohort data [1, 7, 5].

The device operates within a phase-graduated staged-autonomy model. In this Phase 1 study the system runs only at the clinician-controlled (teleoperation or supervised) level and the supervised semi-autonomous level, both under continuous human oversight with an immediate manual override; full autonomy is prohibited at this phase consistent with 21 CFR §312.21(e). Any future move to higher autonomy (Stage 2 supervised semiautonomous expansion or Stage 3 higher autonomy) is reserved for a later phase and gated by independent safety review, a USL reassessment, and DSMB concurrence. The perioperative daraxonrasib pause-and-restart advisory is exercised in every case: daraxonrasib is paused before surgery and the restart day is set by an LLM-bound, human-confirmed advisory whose inputs are the postoperative pancreatic fistula grade by ISGPS (A, B, or C) and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold, with outputs of T+7 days, T+14 days, T+21 days, or hold. A 32-iteration deterministic sweep returned a baseline serum of 0.45 ng/mL (sub-threshold) in all iterations, a T+7-day restart in 29 of 32 (90.6 percent, Grade A and sub-threshold), a T+14-day restart in 3 of 32 (9.4 percent, Grade B or delayed), and no T+21-day restart [1, 7].

Safety reporting during the first year follows the 21 CFR §312.32 expedited clocks, augmented by the Physical AI triggers. A fatal or life-threatening suspected adverse reaction is reported within 7 calendar days and other serious and unexpected suspected adverse reactions within 15 calendar days. Six additional Physical AI reporting triggers under §312.32(g) apply: a serious Physical AI adverse event on the 7 or 15-day clock; a system-drug interaction within 15 days; a cybersecurity incident within 15 days, or within 7 days where patient harm is possible; a sustained AI model degradation across at least three consecutive procedures or a cumulative 24-hour interval; a sim-to-real divergence exceeding twice tolerance (a trajectory gap above 4 mm or a force gap above 1.0 N); and a digital-twin discrepancy. For every Physical AI adverse event an audit trail is preserved from 24 hours before to 72 hours after the event under 21 CFR part 11. Seven Physical AI record types are retained for a minimum of two years after approval or discontinuation, with HIPAA Safe Harbor de-identification applied to any disclosed record [3, 6].

The first-year objectives map onto the endpoint hierarchy of Figure 11: the co-primary safety, dose-finding, and feasibility readouts are accrued as cohorts complete, the secondary surgical-quality endpoints are estimated against the Dutch 2025 benchmark as the per-protocol set matures, and the exploratory survival, advisory-concordance, and sim-to-real measures begin to accumulate through the day-90 pathology review and the 24-month survival follow-up. The realistic first-year deliverable is completion of the DL1 and DL2 cohorts and initiation of the DL3 cohort, with an interim determination of the MTD and a provisional RP2D if DL3 is reached and tolerated, together with the device-feasibility and telemetry summaries that support progression to a later-phase efficacy study [1].

A representative first-year accrual scenario makes the timeline concrete without overstating precision. At an accrual of approximately one to two participants per month, and with the sentinel constraint that each participant's 28-day DLT window must close (and the supervising surgeon and DSMB must review the case) before the next participant in the cohort proceeds to surgery, a cohort of three DLT-evaluable participants occupies on the order of three to four months including the staggering intervals, and an expanded cohort of six occupies correspondingly longer. Under the most favorable realization, in which no level requires expansion, the three cohorts of three traverse DL1, DL2, and DL3 within the first year and yield a provisional RP2D; under a realization in which one level expands to six, the first year completes DL1 and DL2 and initiates DL3, which is the realistic deliverable stated above. In either case the Phase 0 readiness gate (USL at least 7.0, at least 1000 simulations across at least two frameworks, sim-to-real trajectory below 2 mm and force below 0.5 N) is cleared before the first robotic case, so the first-year clock starts only after device readiness is documented [8, 2].

4.5 Number of Subjects to be Evaluated

The planned sample size is up to 18 treated participants, allocated across three daraxonrasib dose levels (DL1 through DL3) by the 3+3 rule, from approximately 36 screened. This number is not derived from a power calculation, because the study tests no confirmatory efficacy hypothesis; it is determined jointly by the dose-finding design and by the early-feasibility precedent for significant-risk devices, and it is justified by its operating characteristics. The 3+3 rule enrolls cohorts of three: a level with no DLT in three escalates, a level with one DLT in three expands to six, and a level with two or more DLTs in up to six is exceeded, defining the dose below it as the MTD. The maximum exposure at any single dose level is therefore six participants, and three escalation levels give a design maximum of three times six, equal to 18 treated participants; many realizations terminate with fewer. This ceiling is consistent with the FDA early-feasibility convention of generally fewer than ten participants per device-feasibility cohort while providing the per-dose exposure the drug-arm 3+3 escalation requires [8].

Rather than power, the design is characterized by its operating characteristics and by the precision of estimation at the planned sample sizes. The probability that the 3+3 rule escalates past a given dose level is a function of the true DLT probability at that level: a dose with a true DLT rate near 10 percent is very likely to be tolerated and escalated, whereas a dose with a true DLT rate of 40 percent or higher is very likely to be identified as excessively toxic and to define the MTD at the level below. The exact-interval precision is summarized in Table 4.7. Zero events in three yields an exact two-sided 95 percent confidence interval of approximately 0 to 71 percent; zero in six, approximately 0 to 46 percent; zero in 18, approximately 0 to 18 percent; and one event in 18, a point estimate of 5.6 percent with an interval of approximately 0.1 to 27 percent. These intervals make explicit both the value and the limits of a first-in-human cohort: a clean safety record across the full cohort of 18 bounds the true event rate below roughly one in five, while a single event still excludes rates above roughly one quarter [8, 18].

Quantity	Planned value	Operating characteristic / precision
Cohort size	3, expand to 6	3+3 rule: escalate on 0/3; expand on 1/3; exceed dose on $\geq$ 2/6.
Per-dose maximum	6	Maximum exposure at any one dose level before the MTD is declared.
Dose levels	3 (DL1 to DL3)	Three-level escalation; design maximum 3 x 6 = 18 treated.
Total treated	up to 18 (approx. 36 screened)	At or below the early-feasibility convention of generally fewer than 10 per device cohort, summed across dose levels.
0 events / 3	point estimate 0%	Exact two-sided 95% CI approximately 0% to 71%.
0 events / 6	point estimate 0%	Exact two-sided 95% CI approximately 0% to 46%.
0 events / 18	point estimate 0%	Exact two-sided 95% CI approximately 0% to 18%.
1 event / 18	point estimate 5.6%	Exact two-sided 95% CI approximately 0.1% to 27%.
Significance level	$\alpha = 0.05$ (two-sided)	Applies to supportive estimation only; no α is spent on dose selection.

Table 4.7. Operating characteristics and exact-interval (Clopper-Pearson) precision at the planned sample sizes. The dose-finding behavior is that of the standard 3+3 rule; no statistical power is computed because the study tests no confirmatory efficacy hypothesis.

The relationship between the true DLT rate and the escalation behavior of the 3+3 rule can be made quantitative, and Table 4.8 illustrates it for the first cohort of three at a level. If the true DLT probability at a level is p, the probability that none of three participants experiences a DLT (and the level escalates outright) is the cube of one minus p; for p near 0.10 this is approximately 0.73, for p near 0.20 approximately 0.51, and for p near 0.40 approximately 0.22. The complementary probability that at least one of three experiences a DLT (and the level either expands to six or is exceeded) rises correspondingly. The table shows why the 3+3 rule is conservative at low true rates and decisive at high true rates: a level whose true DLT rate is genuinely near one in ten is escalated most of the time, whereas a level whose true DLT rate is near two in five triggers expansion or exceedance most of the time, which is the desired behavior for a first-in-human dose-finding design [8, 18].

True DLT rate p	P(0 of 3): escalate	P($\geq$1 of 3): expand or exceed	Interpretation
0.10	≈ 0.73	≈ 0.27	A genuinely low-toxicity level escalates most of the time.
0.20	≈ 0.51	≈ 0.49	An intermediate level expands or escalates about evenly.
0.30	≈ 0.34	≈ 0.66	A higher-toxicity level usually triggers expansion or exceedance.
0.40	≈ 0.22	≈ 0.78	A toxic level rarely escalates outright; the MTD is likely set below it.

Table 4.8. Escalation behavior of the 3+3 rule for the first cohort of three as a function of the true dose-limiting-toxicity rate p, computed as the binomial probability of zero events in three. The rule escalates a genuinely low-toxicity level most of the time and triggers expansion or exceedance at a toxic level most of the time, the intended first-in-human behavior.

The four analysis populations apportion the 18 participants across the analyses without altering the total exposure: the Safety population comprises all who received any intervention and is the primary population for safety and primary-endpoint analyses; the DLT-evaluable population comprises those who completed the 28-day window or had a DLT within it and drives the escalation and MTD determination; the Per-Protocol population supports the secondary surgical-quality analyses; and the modified intention-to-treat population supports the supportive and exploratory efficacy summaries. Because the cohort is small and intensively monitored, individual participant data are tabulated in full, so that every aggregate statistic can be traced to its contributing records [18].

4.6 Drug Related Risks

The foreseeable risks of the investigation arise from three sources, the investigational drug, the investigational device, and the surgical procedure, and each is bounded by a specific control rather than left to clinical judgment alone. The anticipated risks of daraxonrasib are those characteristic of RAS-pathway inhibition and of the RASolute 302 experience, including gastrointestinal effects (nausea, diarrhea, and stomatitis), dermatologic effects (rash), hepatic enzyme elevation, and the class effects of MAPK-pathway suppression; in the perioperative setting the additional concern is impaired wound healing and anastomotic integrity, which is the explicit reason for the pause-and-restart advisory that holds daraxonrasib before surgery and resumes it only on a human-confirmed, fistula-grade-and-trough-informed schedule (T+7, T+14, T+21 days, or hold). Drug toxicity is captured through the 28-day DLT window, the dose-blinded SAE attribution, the exact-interval safety reporting, and the binding cohort-level halt rules [1, 10]. Table 4.9 summarizes the principal risks, their sources, and their mitigations.

Risk source	Foreseeable risk	Mitigation
Daraxonrasib (drug)	GI effects (nausea, diarrhea, stomatitis), rash, hepatic enzyme elevation, MAPK-pathway class effects; impaired perioperative wound healing.	28-day DLT window; conservative 160 mg starting dose below the 300 mg RP2D; pause-and-restart advisory keyed to ISGPS grade and 0.5 ng/mL trough; cohort-level halt rules.
Robotic system (device)	Device malfunction, unsafe trajectory, excessive tip force, vascular injury near the SMV, PV, HA, CA, or SMA.	Force caps ≤ 3 N per arm and ≤ 18 N cumulative; cross-arm E-stop ≤ 3 ms; hardware E-stop ≤ 500 ms; 10 kHz heartbeat with 100 microsecond watchdog; vascular no-fly gating; Phase 0 gate (USL ≥ 7.0, ≥ 1000 sims, sim-to-real < 2 mm and < 0.5 N).
LLM advisory layer	Erroneous or unverified advisory output; over-reliance (therapeutic misconception); model degradation.	Second-opinion advisory only; full autonomy prohibited (§312.21(e)); ten-gate verification-before-generation (ACCEPT/BLOCK/ESCALATE); hash-pinned deterministic seed and commit; Physical AI opt-out; advisory-to-gate concordance monitored.
Whipple procedure	Pancreatic fistula, hemorrhage, intra-abdominal infection, sepsis, failure to rescue, unplanned conversion.	Sentinel staggered enrollment with case-by-case DSMB review; three-tier oversight (DSMB, Independent Safety Monitor, Physical AI Safety Review Committee); conversion pathway with continuous human override.
Schedule and population	Delay to effective therapy in a low-survival disease.	Verified single-prompt authoring compresses only the administrative bucket, advancing first dose without altering tumor biology or fixed review clocks (Figure 10).

Table 4.9. Principal drug-related, device-related, procedural, and schedule-related risks with their controls. Each risk is bounded by a specific, prespecified mitigation; the residual risk after these controls is judged acceptable for the population and the stage of investigation.

The device-related and procedural risks are bounded by the cyber-physical safety envelope and the three-tier oversight structure. The hard caps (per-arm tip force at or below 3 N, cumulative cross-arm force at or below 18 N, cross-arm emergency stop within 3 ms to 0.0 N, and hardware emergency stop within 500 ms), the 10 kHz heartbeat with the 100 microsecond watchdog, and the five-vessel no-fly gating limit the magnitude of any single excursion, while the Phase 0 gate (USL at least 7.0, at least 1000 simulations across at least two frameworks, and a sim-to-real gap below 2 mm and 0.5 N) ensures the system is demonstrably ready before any patient-facing activity. The LLM advisory risks (an erroneous or unverified advisory and the therapeutic misconception of over-reliance on the system) are bounded by the advisory-only role, the prohibition on full autonomy under 21 CFR §312.21(e), the ten-gate verification-before-generation suite, the hash-pinned deterministic seed and commit, the Physical AI opt-out in the consent, and the monitoring of advisory-to-gate concordance. Safety reporting follows the 21 CFR §312.32 clocks (fatal or life-threatening suspected adverse reactions within 7 calendar days and other serious and unexpected reactions within 15 calendar days), augmented by the six Physical AI reporting triggers under §312.32(g) and an audit trail preserved from 24 hours before to 72 hours after each Physical AI adverse event under 21 CFR part 11 [3, 6].

The therapeutic misconception merits a specific control because it is the characteristic ethical hazard of a study that pairs a novel device with an advisory model. A participant might over-attribute capability to the robotic system or to the LLM advisory and consent on the mistaken belief that the technology guarantees a better outcome than standard care. This is addressed at three points: in the informed consent under 21 CFR part 50 and ICH E6(R3), which explains that the device operates only at clinician-controlled and supervised semiautonomous levels with continuous human oversight and that the LLM is a second opinion rather than a decision-maker; in the Physical AI opt-out under 21 CFR §312.60(f), which lets a participant decline the Physical AI advisory layer without losing access to the study's surgical and drug components; and in the staged-autonomy explanation, which makes clear that full autonomy is prohibited at this phase. The consent is described to the participant by a research-team member who is not the treating surgeon, reinforcing that the decision is voluntary and free of the treating surgeon's influence [19, 7].

The benefit-risk balance is favorable for this population and at this stage. The population is vulnerable, with a lethal malignancy whose five-year survival is below 13 percent and for whom the resection margin is the strongest modifiable determinant of survival; the intervention pairs a genotype-matched systemic agent with a margin-negative curative-intent resection delivered within hard cyber-physical limits under continuous human oversight. The schedule benefit reaches the patient through the cascade of Figure 10: faster validated authoring advances IND and IRB submission and site activation, the elapse of the 30-day clock and the first dose, and the 3+3 cohort turnover toward the RP2D, shortening the white space between phases and, for this low-survival population, extending progression-free and overall survival, without changing tumor biology or any fixed review clock. For patients whose disease will not pause, the alternative of slower sequential authoring carries its own, often larger, cost in delayed treatment. With each foreseeable risk bounded by a prespecified control, with the conservative 160 mg starting dose, the sentinel staggered enrollment, the binding halt rules, and the three-tier oversight, the probable benefit of the investigation exceeds its probable risk, and the General Investigational Plan supports the proposed first-year conduct [4, 1, 7].

4.7 References

The clinical, statistical, and acceleration content of this plan is grounded in the trial protocol and the supporting evidence base. The Phase 1 first-in-human guidance and the early-feasibility framework inform the single-arm dose-finding design and the operating characteristics [8]. The daraxonrasib robotic Whipple program and the on-premises eight-arm system specifications supply the drug, device, and advisory facts [1, 2, 10]. The Physical AI adaptations of 21 CFR Part 312 and Part 50 and the ICH guidance govern the safety reporting, consent, and statistical conventions [3, 19, 18]. H.R. 9510 and the clinical-trust framework establish the verification-before-generation discipline [6, 7]. The PDAC epidemiology, the acceleration mechanism, and the supporting computational evidence are drawn from the population and document-acceleration sources [4, 17, 11, 9, 5, 12, 14, 13, 15, 16]. The full bibliography is consolidated in the references and back-matter section of this IND.

5 Investigator Brochure

This section is the investigational new drug (IND) level summary of the Investigator's Brochure (IB) for the large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy with perioperative daraxonrasib Phase 1 study. It is provided here as a clearly labeled supporting document under 21 CFR §312.23(a)(5) and §312.55, drawing together, in one place and at IND-summary depth, the physical, chemical, and pharmaceutical properties of the investigational drug daraxonrasib (RMC-6236); the available nonclinical pharmacology, pharmacokinetics, and toxicology that underpin the conservative 160 mg first-in-human starting dose; a full description of the investigational eight-arm collaborative surgical device (56 degrees of freedom, 640 sensor channels, the per-arm and cumulative force caps, the layered emergency-stop chain, and the five-vessel no-fly gate); and a structured summary of the data with explicit guidance for the investigator. The full, standalone Investigator's Brochure is a separate bound document submitted with this IND and incorporated by reference; the present section is the IND-level synopsis and the pointer to that document, consistent with the ReGARDD convention of using a distinct section to indicate and include a supporting document. The clinical-conduct detail that the IB summarizes is carried in the study protocol (§6 of this IND), the chemistry, manufacturing, and control information (§7), and the pharmacology and toxicology information (§8), and the investigator is directed to those sections and to the bound IB for the controlling text [8, 1].

The Investigator's Brochure occupies a defined place in the architecture of an IND. Under 21 CFR §312.23(a)(5) the sponsor must furnish, for a study to be conducted by an investigator other than the sponsor, a copy of the IB; and under 21 CFR §312.55 the sponsor must give each participating investigator a current brochure and must keep each investigator informed of new safety information as it emerges. The brochure is, in the language of the International Council for Harmonisation (ICH) E6(R3) Good Clinical Practice guideline, a compilation of the clinical and nonclinical data on the investigational article that are relevant to its study in human subjects, assembled to provide the investigators and others involved in the trial with the information needed to understand the rationale for, and to comply with, the key features of the protocol. For a combined drug-device intervention of the kind studied here the brochure must do double duty, providing the reference safety information for both the drug evaluated under 21 CFR part 312 and the device evaluated under 21 CFR part 812, and the present section is written to reflect that dual structure throughout [18, 3].

Supporting Document Status and Scope

The Investigator's Brochure assembles, for the clinical investigator and the reviewing institutional review board (IRB), all reference safety and efficacy information needed to conduct the study and to assess the expectedness of adverse events. Because the intervention has two inseparable components, an investigational drug evaluated under 21 CFR part 312 and an investigational device evaluated under 21 CFR part 812 in a single combined IND and investigational device exemption (IDE) submission, the IB has two coordinated parts: the daraxonrasib drug brochure and a Physical AI device supplement that functions as the reference safety information for the device and the on-premises advisory large language model. An adverse event is judged unexpected, and therefore potentially reportable on an expedited timeline, when it is not listed at the observed specificity or severity in the daraxonrasib IB (for drug events) or in the Physical AI supplement and System Description (for device events), as specified in §8 of this IND and in the protocol assessment section. The IB is reviewed and reissued at least annually and whenever new safety information materially changes the risk profile, and currently enrolled participants are re-consented when such a change occurs. The version of the IB in effect at the time of each procedure is recorded in the case report form so that expectedness determinations are traceable to a fixed reference text [1, 3].

The scope of this section is deliberately bounded. It is the IND-level synopsis of the brochure and the formal pointer to the bound document; it is not the controlling clinical-conduct text, which is the protocol (§6), and it is not the controlling chemistry or nonclinical text, which is carried in the chemistry, manufacturing, and control information (§7) and the pharmacology and toxicology information (§8). Where this synopsis and any of those modules might appear to differ, the bound IB and the cited module govern, and the values restated here are restated precisely so that no discrepancy arises. The brochure and this synopsis share one internally consistent set of quantitative facts: the three protocol dose levels of 160 mg, 220 mg, and 300 mg once daily; the 28-day dose-limiting toxicity (DLT) observation window; the 0.5 ng/mL serum trough threshold that governs the perioperative pause-and-restart advisory; and the device safety envelope with its 3 N per-arm and 18 N cumulative tip-force caps, its 10 kHz heartbeat coordination bus, its layered emergency-stop chain, and its five-vessel no-fly gate. These values appear once in the brochure and are carried forward unchanged into every module of the IND [1, 2].

The composition of the initial IND and IRB package into which this brochure is bound is shown in Figure 14. The clinical protocol, the Investigator's Brochure, the nonclinical, chemistry, and stability data, the informed consent and recruitment materials, the investigator forms (FDA Forms 1571, 1572, and 3674), and the safety information are assembled into one internally consistent package whose submission starts the fixed 30-day review clock of 21 CFR §312.40(b). Faster, internally consistent assembly starts that clock and all participating sites sooner, but it cannot shorten the statutory 30-day period and it cannot generate toxicology or stability data that do not exist; the brochure is one indispensable component of that package and is the reference safety text against which every adverse event is later judged [8, 9].

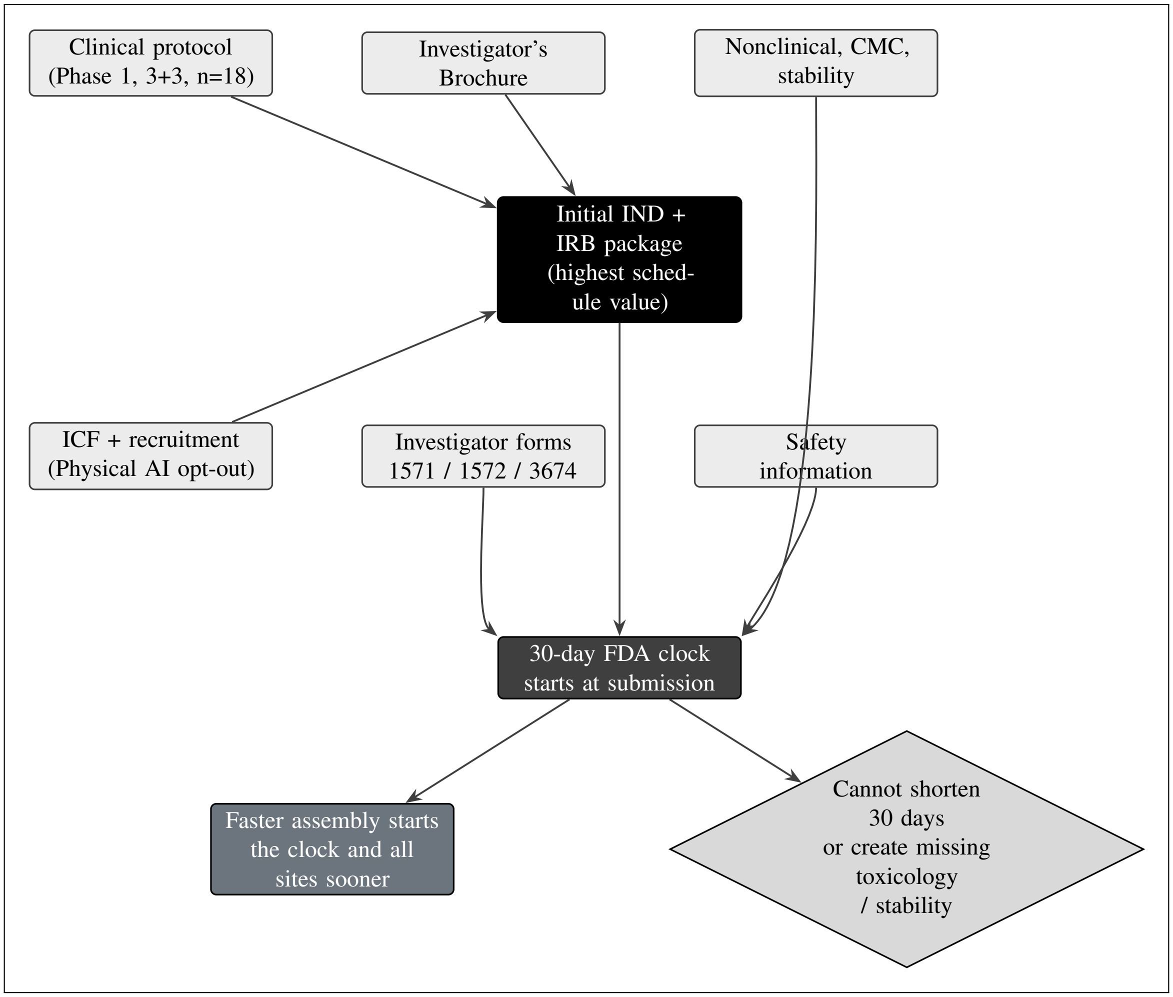

Figure 14. Composition of the initial IND and IRB package. The clinical protocol, the Investigator's Brochure, the nonclinical, chemistry, and stability data, the informed consent and recruitment materials, the investigator forms (1571, 1572, 3674), and the safety information are assembled into one internally consistent package. Faster, internally consistent assembly starts the 30-day FDA clock of 21 CFR §312.40(b) and all participating sites sooner but cannot shorten the fixed period or generate missing toxicology or stability data.

Physical, Chemical, and Pharmaceutical Properties of Daraxonrasib

Daraxonrasib (development code RMC-6236) is an orally bioavailable, small-molecule, RAS(ON) multi-selective (pan-RAS) inhibitor. Its pharmacological mechanism is a tri-complex mode of action: the molecule binds the active, guanosine triphosphate (GTP) bound conformation of mutant RAS in a ternary complex with the abundant intracellular chaperone cyclophilin A, and in doing so it sterically and allosterically blocks the engagement of downstream effectors, suppressing the mitogen-activated protein kinase (MAPK) and phosphoinositide 3-kinase (PI3K) signaling that drives pancreatic ductal adenocarcinoma (PDAC). Because it targets the shared GTP-bound state rather than a single codon substitution, daraxonrasib is active across the spectrum of common oncogenic KRAS variants relevant to PDAC, including the G12D, G12V, and G12R substitutions that define the eligible population of this study. The agent is administered as an oral tablet, once daily, and is supplied in the strengths required to compose the three protocol dose levels of 160 mg, 220 mg, and 300 mg once daily. Tablets are swallowed whole with water and are not crushed, split, or chewed; they are packaged in child-resistant,

Table 15: Daraxonrasib (RMC-6236) physical, chemical, and pharmaceutical properties material to the investigator.

Property	Description
Development code	RMC-6236
Pharmacological class	Oral RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor
Molecular mechanism	Tri-complex inhibitor: binds the GTP-bound active conformation of mutant RAS in a ternary complex with cyclophilin A, blocking effector engagement and suppressing MAPK and PI3K signaling
KRAS variant coverage	Active across common oncogenic KRAS codon-12 variants, including G12D, G12V, and G12R (the eligible PDAC genotypes)
Dosage form	Oral tablet, swallowed whole with water; not crushed, split, or chewed
Route and schedule	Oral, once daily (continuous, except across the protocol perioperative pause)
Protocol dose levels	160 mg, 220 mg, and 300 mg once daily (DL1, DL2, DL3)
Storage and packaging	Child-resistant, light-protective containers; controlled room temperature; secured, temperature-monitored, access-controlled
Investigational-use labeling	21 CFR §312.6 caution statement, protocol number, lot number, storage conditions
Regulatory standing	FDA Breakthrough Therapy (June 2025) for previously treated metastatic PDAC with KRAS G12; investigational in the perioperative curative-intent surgical context here

Table 5.1. Physical, chemical, and pharmaceutical properties of daraxonrasib restated at IND-summary depth; the controlling chemistry, manufacturing, and control text is in §7 of this IND and in the bound IB.

light-protective containers labeled under 21 CFR §312.6 with the investigational-use caution statement, the protocol number, the lot number, and the storage conditions, and are stored at controlled room temperature in a secured, temperature-monitored, access-controlled location until dispensed. The detailed chemistry, manufacturing, and control characterization of the drug substance and drug product, including appearance, composition, container-closure system, and the stability program that governs the labeled expiry, is provided in §7 of this IND; the present summary restates only the properties most relevant to the investigator at the point of care [1, 4].

The properties most material to the investigator are collected in Table 5.1. The distinguishing pharmacological feature of daraxonrasib relative to mutation-selective KRAS inhibitors is its breadth: a single agent addresses the three KRAS G12 variants that together account for the great majority of PDAC, removing the need to genotype to a single allele before treatment and matching the agent to the broad pan-KRAS eligibility of the RASolute 302 program from which the protocol dose assumptions are drawn. The mechanistic class, the route, the schedule, the dose levels, and the clinical-context investigational status are stated together so that the investigator can read the drug's identity and its place in this study from a single tabulation [1, 11].

The clinical and regulatory standing of daraxonrasib supports its use as the drug component of this first-in-human perioperative study. The agent received FDA Breakthrough Therapy designation in June 2025 for previously treated metastatic PDAC harboring a KRAS G12 mutation, and the dose and exposure assumptions of this protocol are anchored to the pan-KRAS RASolute 302 program, in which 300 mg once daily is the established recommended Phase 2 dose (RP2D). In the present study the drug is investigational specifically in the perioperative, curative-intent surgical context and for its use in combination with the investigational device; the escalation deliberately begins one and two steps below the established RP2D to preserve a conservative first-in-human margin in this new clinical setting [1, 11].

The dose levels and their relationship to the established RP2D are worth making explicit because the entire first-in-human safety argument rests on them. The three dose levels are DL1 at 160 mg once daily, DL2 at 220 mg once daily, and DL3 at 300 mg once daily, with the highest level set equal to the established RASolute 302 RP2D. The intermediate level represents an increment of 60 mg, or about 37.5 percent, above the starting dose, and the top level a further increment of 80 mg, or about 36.4 percent, above the intermediate level. The starting dose of 160 mg is therefore about 53.3 percent of the established RP2D, leaving roughly a factor of 1.9 between first exposure in this surgical setting and the dose already characterized for safety in the metastatic program. That margin is intentional: in the perioperative window the drug is paused before surgery and restarted only on a fistula-grade-keyed and trough-keyed schedule, so the conservative starting dose protects the participant during the period of greatest physiologic stress while the 3+3 escalation establishes the maximum tolerated dose (MTD) and the RP2D in this new context [1, 10].

Nonclinical Pharmacology, Pharmacokinetics, and Toxicology

The pharmacology of daraxonrasib is summarized in §8.1.1 of this IND and rests on the RAS(ON) tri-complex mechanism described above: selective engagement of the GTP-bound mutant RAS in complex with cyclophilin A, with downstream suppression of effector signaling and consequent antiproliferative and pro-apoptotic activity in KRAS-mutant PDAC models. Drug distribution and pharmacodynamics support a once-daily oral schedule with sustained target coverage. The pharmacokinetic behavior most relevant to this protocol is the relationship between the daily dose, the serum trough concentration, and the 0.5 ng/mL trough threshold that governs the perioperative pause-and-restart advisory. A 32-iteration deterministic pharmacokinetic sweep, performed to characterize the perioperative window, returned a baseline serum daraxonrasib concentration of 0.45 ng/mL at the operative timepoint across all 32 iterations, that is, below the 0.5 ng/mL trough threshold in every iteration, consistent with the protocol-mandated pre-operative pause; the advisory then recommended a postoperative restart on day T+7 in 29 of 32 iterations (90.6 percent, corresponding to a Grade A fistula with a sub-threshold trough), on day T+14 in 3 of 32 iterations (9.4 percent, corresponding to a Grade B or delayed serum rise), and on day T+21 in 0 of 32 iterations. This sweep is described in the study protocol (§6 of this IND) and is summarized in §8 and §3.4 [13, 14, 16].

The perioperative sweep deserves a worked reading because it is the quantitative backbone of the restart advisory. The baseline serum concentration of 0.45 ng/mL at the operative timepoint sits 0.05 ng/mL, or 10 percent, below the 0.5 ng/mL threshold, and it does so in all 32 of 32 iterations rather than on average; the sub-threshold finding is therefore robust to the parameter variation the sweep explores, which is exactly the property a perioperative pause rule needs. The restart-day distribution of 29 iterations at T+7, 3 iterations at T+14, and 0 iterations at T+21 maps directly onto the fistula-grade logic: the common case is a Grade A fistula with a sub-threshold trough, which the advisory resolves to the earliest safe restart at day T+7, while the minority case of a Grade B or delayed-rise trajectory defers the restart to day T+14, and no iteration in this deterministic sweep reached the T+21 tier, which is reserved for the slower trajectories and, with a Grade C fistula, maps instead to a continued hold and

Table 16: 32-iteration deterministic perioperative pharmacokinetic sweep and the resulting restart-day distribution.

Restart tier	Iterations	Share	Clinical correlate
Baseline at operative timepoint	32 of 32	100%	Serum 0.45 ng/mL, below the 0.5 ng/mL threshold in every iteration; supports the pre-operative pause
Restart day T+7	29 of 32	90.6%	Grade A fistula with a sub-threshold trough; earliest safe restart
Restart day T+14	3 of 32	9.4%	Grade B fistula or a delayed serum rise; restart deferred one week
Restart day T+21	0 of 32	0%	Reserved for slower trajectories; with a Grade C fistula maps to a continued hold and the stopping rule

Table 5.2. The deterministic perioperative pharmacokinetic sweep underpinning the pause-and-restart advisory. The baseline serum concentration of 0.45 ng/mL falls below the 0.5 ng/mL trough threshold in all 32 iterations, and the restart day is keyed to the pancreaticojejunostomy ISGPS fistula grade and the serum trough; the controlling text is the protocol (§6) and §8 of this IND.

to the drug stopping rule. Table 5.2 lays the sweep out so that the advisory's behavior can be read at a glance [13, 14, 16].

Table 17: Daraxonrasib class toxicity profile, expectedness reference, and protocol management pathway.

Class effect	Expected behavior	Management and reporting reference
Diarrhea	Common, dose-related, reversible; listed in the daraxonrasib IB	Early antidiarrheals; dose interruption if severe; a DLT-defining event in the 28-day window engages the 3+3 dose-reduction rules
Nausea	Common, dose-related; listed in the IB	Antiemetic prophylaxis and treatment; dose interruption if severe; graded by CTCAE v5.0
Rash (dermatologic)	Common, dose-related; listed in the IB	Aggressive dermatologic care begun early; dose interruption if severe; reduction per protocol
Unlisted event	Not listed at the observed specificity or severity	Unexpected; assess against 21 CFR §312.32 (7-day fatal or life-threatening; 15-day other serious and unexpected)

Table 5.3. The drug class toxicity profile that serves as the expectedness reference for daraxonrasib. Gastrointestinal and dermatologic events at the listed severities are expected and managed per protocol; an event not listed at the observed specificity or severity is unexpected and is assessed against the regulatory reporting clocks.

The toxicology integrated summary (§8.1.2) states the safety margins that underpin the conservative DL1 160 mg first-in-human starting dose. The class toxicity profile of RAS(ON) multi-selective inhibition is dominated by gastrointestinal effects (diarrhea, nausea) and dermatologic effects (rash), which are managed symptomatically and through the protocol dose-interruption and dose-reduction rules; rescue management of these class effects, including antidiarrheals and aggressive dermatologic care, is specified in the protocol. The 160 mg starting dose sits a full step below the 220 mg intermediate level and two steps below the 300 mg established RP2D, providing a deliberate margin for the perioperative surgical setting in which the drug is paused across the operative window and restarted only on a fistula-grade-keyed and trough-keyed schedule. The full data tabulation (§8.1.3) tabulates the supporting studies; for this AI-generated submission the supporting nonclinical and in silico evidence base includes the author's quantitative-systems-pharmacology metastatic-PDAC simulation with verification, validation, and uncertainty quantification; the artificial intelligence digital-twin PDAC simulations; the 100,000-patient in silico Phase 3 five-arm PDAC evaluation; and the end-to-end PDAC digital-twin trial work, each grounded in the references identified in the figure below. The pharmacology and toxicology integrated-summary structure that organizes §8 is reproduced in Figure 13 [13, 14, 16, 15].

The class toxicity profile is the reference against which drug-related expectedness is judged, so the investigator should hold it clearly in mind. The dominant events are gastrointestinal (diarrhea and nausea) and dermatologic (rash), and these are anticipated, dose-related, and reversible with the protocol management rules; their appearance at the listed severities is expected and does not by itself trigger expedited reporting, whereas an event that is not listed at the observed specificity or severity in the daraxonrasib IB is unexpected and is assessed against the 7-day and 15-day clocks of 21 CFR §312.32. Management proceeds through symptomatic care, dose interruption, and, if a DLT-defining event occurs within the 28-day window, the dose-reduction and 3+3 cohort rules; antidiarrheals and aggressive dermatologic care are deployed early to keep these class effects below the severity at which they would interrupt dosing. Table 5.3 sets out the class effects, their expected behavior, and the management pathway so that the investigator can distinguish an expected, manageable class effect from a reportable unexpected event [1, 18].

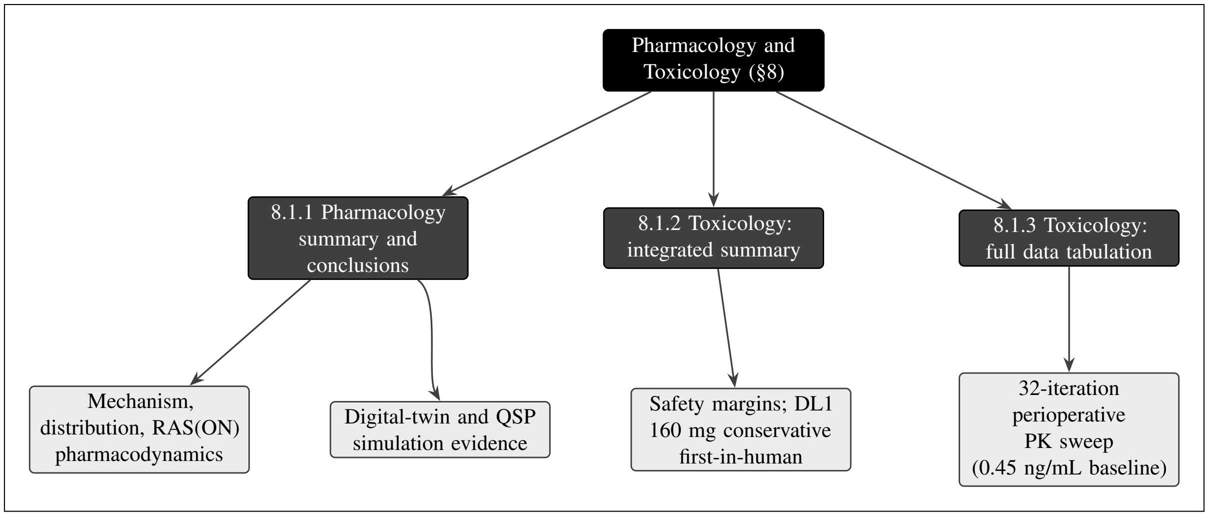

Figure 13. The pharmacology and toxicology integrated-summary structure for §8. The pharmacology summary covers the RAS(ON) mechanism, drug distribution, and pharmacodynamics; the integrated toxicology summary states the safety margins underpinning the conservative DL1 160 mg first-in-human starting dose; and the full data tabulation tabulates the supporting studies, here grounded in the author's digital-twin and quantitative-systems-pharmacology simulation evidence and the 32-iteration perioperative pharmacokinetic sweep (0.45 ng/mL baseline serum trough at the operative timepoint across all iterations).

The supporting in silico evidence base named in the full data tabulation deserves to be enumerated because it is the principal nonclinical foundation of this AI-generated submission. The quantitative-systems-pharmacology metastatic-PDAC simulation supplies a mechanistic model of drug exposure and effect with explicit verification, validation, and uncertainty quantification; the artificial-intelligence digital-twin PDAC simulations supply patient-level trajectory modeling that informs the perioperative window; the 100,000-patient in silico Phase 3 five-arm PDAC evaluation supplies a large-scale comparative-effectiveness context; and the end-to-end PDAC digital-twin trial work supplies the integrated trial-simulation framework from which the perioperative sweep is drawn. Each is a computational rather than a wet-laboratory study, and the brochure presents it as such; the reviewer is directed to §8.1.3 for the full tabulation and to the cited records for the underlying methods and results [13, 14, 15, 16].

Investigational Device Description: the Eight-Arm Robotic System

The investigational device is an on-premises, LLM-directed, eight-arm parallel coelomic surgical platform that conducts the pancreaticoduodenectomy and generates the operative and pathology inputs to the perioperative drug-restart advisory. The on-premises, repository-pinned LLM acts strictly as a second-opinion advisory oracle held outside the robot vendor kinematic stack and is never an autonomous controller; full autonomy is prohibited consistent with 21 CFR §312.21(e), and in this Phase 1 study the device operates only at the clinician-controlled teleoperated and supervised semiautonomous autonomy levels under continuous human oversight with immediate manual override. The platform carries 56 mechanical degrees of freedom, composed of eight arms with seven degrees of freedom each (8 multiplied by 7), and a 640-channel mixed-rate sensor stack of 80 channels per arm. Force channels sample at 100 kHz and all remaining channels sample at the 10 kHz command rate. Positioning accuracy is 0.05 mm root mean square at a peak tip velocity of 1,200 mm/s. The cyber-physical safety envelope comprises a per-arm tip-force cap of no more than 3 N and a cumulative cross-arm tip-force cap of no more than 18 N enforced at the heartbeat boundary; a 10 kHz heartbeat coordination bus broadcasting a 64-byte frame on a 100 microsecond per-arm watchdog deadline; an emergency arm park within 50 microseconds of a missed or out-of-parameter frame; a cross-arm emergency stop (E-stop) that halts the full platform within 3 ms at a 0.0 N force clamp; and a software-independent hardware E-stop that meets the 21 CFR §312.404 requirement of no more than 500 ms. The device specifications are summarized in Table 5.4, and the layered safety envelope, including the five-vessel no-fly gate, is summarized in Table 5.5 [1, 2].

The numeric structure of the platform is internally consistent and worth stating as an arithmetic identity: the 56 mechanical degrees of freedom are exactly eight arms multiplied by seven degrees of freedom per arm, and the 640 sensor channels are exactly eight arms multiplied by 80 channels per arm. The mixed-rate sensing partitions cleanly: the force channels, which carry the per-arm and cumulative tip-force signals that the safety caps police, sample at 100 kHz, a factor of ten faster than the 10 kHz command rate at which the remaining channels and the heartbeat coordination bus operate. That ten-to-one ratio is deliberate, because force is the physical quantity whose excursion must be caught fastest. The positioning accuracy of 0.05 mm root mean square is maintained at a peak tip velocity of 1,200 mm/s, and both are verified in the pre-procedure safety matrix at the start of each case rather than assumed from a prior calibration [1, 2].

The eight cooperating arms are coordinated by the 10 kHz heartbeat bus and carry phase-keyed tool assignments across the eight operative phases. Phases P1 through P4 comprise the dissection and specimen removal: Arms 1 and 2 perform the active dissection under the vessel control of Arm 3 (bipolar plus retractor) and the near-infrared indocyanine-green imaging of Arm 4, with Arm 7 providing suction and irrigation throughout. The specimen is removed at the end of P4 and reconstruction begins. The three reconstructive anastomoses are then constructed in sequence under closed-loop ring-tension control by the Arm 5 anastomosis probe, which issues a soft warning whenever measured ring tension falls outside its controller band: the duct-to-mucosa pancreaticojejunostomy (PJ) in P5 with a band of 0.30 to 0.60 N, the end-to-side hepaticojejunostomy (HJ) in P6 with a band of 0.20 to 0.50 N, and the antecolic gastrojejunostomy (GJ) in P7 with a band of 0.40 to 0.80 N; P8 covers final imaging, hemostasis, and drain placement. The pancreaticojejunostomy is the dominant determinant of postoperative pancreatic-fistula risk and is therefore the anastomosis whose ISGPS fistula grade supplies the input to the perioperative drug-restart advisory [1, 9].

The phase-keyed tool assignment and the per-anastomosis ring-tension bands are collected in Table 5.6 so that the investigator can see the operative choreography and its closed-loop controls in one place. The three ring-tension bands are anastomosis-specific because the three reconstructions differ in tissue and in consequence: the pancreaticojejunostomy band of 0.30 to 0.60 N reflects the soft, friable pancreatic remnant and the fistula risk that makes this anastomosis the keyed input to the restart advisory; the hepaticojejunostomy band of 0.20 to 0.50 N reflects the thinner-walled biliary tissue; and the gastrojejunostomy band of 0.40 to 0.80 N reflects the more robust gastric and jejunal walls. A measured tension outside the controller band raises a soft warning from the Arm 5 anastomosis probe rather than halting the platform, because a tension excursion is an advisory signal to the operating surgeon, not a vascular-proximity hazard; the hard halts are reserved for the no-fly gate and the heartbeat watchdog described below [1, 2].

Table 18: Eight-arm LLM-directed robotic Whipple device specifications (Investigator's Brochure device supplement; 21 CFR §312.23(g) hardware fields).

Specification	Value	Notes
Architecture	Eight-arm parallel coelomic platform	On-premises LLM advisory oracle outside vendor kinematic stack
Degrees of freedom	56 total (8 arms x 7 DOF)	Cooperating arms, phase-keyed tool assignment P1 to P8
Sensor channels	640 total (80 per arm)	Force at 100 kHz; all other channels at 10 kHz command rate
Positioning accuracy	0.05 mm RMS at 1,200 mm/s	Verified in the pre-procedure safety matrix each case
Per-arm tip-force cap	3 N maximum	Enforced continuously per arm
Cumulative force cap	18 N maximum	Enforced at the heartbeat boundary across all arms
Heartbeat bus	10 kHz, 64-byte frame	100 microsecond per-arm watchdog deadline
Emergency arm park	50 microseconds maximum	On a missed or out-of-parameter frame
Cross-arm E-stop	3 ms maximum, 0.0 N clamp	Software escalation halting the full platform
Hardware E-stop	500 ms maximum	Software-independent; meets 21 CFR §312.404
Autonomy level (Phase 1)	Teleoperated / supervised	Continuous human oversight; full autonomy prohibited (§312.21(e))

Table 5.4. Device specifications carried in the Investigator's Brochure device supplement and the Physical AI System Description; the controlling hardware, software, and safety-system text is in §6 and §7 of this IND and in the bound IB.

Safety Envelope: Force Caps, E-Stop Chain, and the No-Fly Gate

The device safety envelope is enforced before and during every procedure rather than reconstructed afterward. The vascular safety-zone gate ticks at 10 kHz and evaluates the instrument-tip proximity to each of five named vessels, the superior mesenteric vein (SMV), the portal vein (PV), the hepatic artery (HA), the celiac axis (CA), and the superior mesenteric artery (SMA), against three concentric thresholds: a soft-warning zone at 3.0 mm that raises an advisory but does not block the command, a no-fly zone that blocks the command (1.0 mm at the SMV and PV; 1.5 mm at the HA, CA, and SMA), and a hard-stop zone at 5.0 mm that forces a cross-arm E-stop within 3 ms. Across a representative 100-tick verdict sample the gate returned clear on 83 ticks, soft warning on 6, no-fly on 6, and hard-stop on 5, exercising all five vessels across all eight operative phases. The halt chain that backs the gate is layered: an on-time heartbeat frame continues the command state, whereas a missed or out-of-parameter frame parks the affected arm within 50 microseconds and escalates to the cross-arm E-stop, and the software-independent hardware E-stop backs the software chain at the 500 ms outer bound. The safety envelope is summarized in Table 5.5 [1, 7].

Table 19: Phase-keyed tool assignment and closed-loop ring-tension control across the eight operative phases.

Phase	Activity and arm assignment	Closed-loop control
P1 to P4	Dissection and specimen removal: Arms 1 and 2 dissect under Arm 3 vessel control (bipolar plus retractor) and Arm 4 near-infrared ICG imaging; Arm 7 suction and irrigation throughout	Five-vessel no-fly gate active at 10 kHz; force caps enforced; specimen removed at the end of P4
P5 (PJ)	Duct-to-mucosa pancreaticojejunostomy by the Arm 5 anastomosis probe	Ring-tension band 0.30 to 0.60 N; soft warning outside band; ISGPS grade keys the restart advisory
P6 (HJ)	End-to-side hepaticojejunostomy	Ring-tension band 0.20 to 0.50 N; soft warning outside band
P7 (GJ)	Antecolic gastrojejunostomy	Ring-tension band 0.40 to 0.80 N; soft warning outside band
P8	Final imaging, hemostasis, and drain placement	Imaging confirmation; no-fly gate and force caps remain active

Table 5.6. The eight operative phases, the phase-keyed arm and tool assignment, and the per-anastomosis closed-loop ring-tension bands. The pancreaticojejunostomy (P5) is the dominant fistula-risk determinant and supplies the ISGPS grade to the perioperative drug-restart advisory.

The geometry of the no-fly gate is worth stating precisely because it is asymmetric by design. The soft-warning radius of 3.0 mm and the hard-stop radius of 5.0 mm are uniform across all five vessels, but the blocking no-fly radius is vessel-specific: 1.0 mm at the two veins (SMV and PV) and the larger 1.5 mm at the three arteries (HA, CA, and SMA). The arterial margin is set wider because an arterial injury is the more catastrophic event, so the gate blocks the command at a greater standoff for the hepatic artery, the celiac axis, and the superior mesenteric artery than for the two veins. The ordering of the three radii is fixed and monotone, with the hard-stop radius (5.0 mm) outside the soft-warning radius (3.0 mm), which in turn lies outside the blocking radius (1.0 or 1.5 mm); a tip approaching a vessel therefore crosses the hard-stop boundary first and triggers an E-stop before it could reach the blocking or warning radii, which is the conservative ordering a vascular-safety gate requires. The representative 100-tick verdict sample partitions exactly into 83 clear, 6 soft warning, 6 no-fly, and 5 hard-stop ticks, summing to 100, and exercises all five vessels across all eight operative phases [1, 7].

Table 20: Device safety envelope: force caps, layered emergency-stop chain, and the five-vessel no-fly gate.

Safety element	Threshold / action	Basis and verdict sample
Per-arm force cap	3 N maximum per arm	Continuous; force channels at 100 kHz
Cumulative force cap	18 N maximum across arms	Enforced at the heartbeat boundary
Heartbeat and watchdog	10 kHz bus, 100 microsecond deadline	On-time frame continues command state
Arm park	50 microseconds maximum	Triggered by a missed or out-of-parameter frame
Cross-arm E-stop	3 ms maximum, 0.0 N clamp	Software escalation; halts full platform
Hardware E-stop	500 ms maximum	Independent of software; 21 CFR §312.404
Soft-warning zone	3.0 mm, all five vessels	Advisory only; 6 of 100 ticks
No-fly zone (block)	1.0 mm SMV/PV; 1.5 mm HA/CA/SMA	Command blocked; 6 of 100 ticks
Hard-stop zone	5.0 mm, all five vessels	Forces E-stop; 5 of 100 ticks
Clear verdict	Outside all zones	Command proceeds; 83 of 100 ticks

Table 5.5. The cyber-physical safety envelope and the five-vessel no-fly gate (SMV, PV, hepatic artery, celiac axis, SMA), sampled at 10 kHz; the representative 100-tick verdict sample partitions into 83 clear, 6 soft warning, 6 no-fly, and 5 hard-stop ticks. The controlling text is in the protocol assessment section (§6 and §8 of this IND).

Every advisory command, every safety-gate verdict, every force and trajectory sample at the recorded rates, and every E-stop event is written to a 21 CFR part 11 hash-chained, append-only audit trail, so that compliance with the cyber-physical envelope is verifiable for each case and any deviation is time-stamped and detectable. The audit trail also satisfies the preservation requirement from 24 hours before to 72 hours after any reportable Physical AI adverse event. Before any patient-facing robotic activity, the platform must clear the Phase 0 simulation validation gate (a Unified Safety Level of at least 7.0; at least 1,000 simulations across at least two simulation frameworks; a sim-to-real trajectory gap below 2 mm; and a tip-force gap below 0.5 N), and the gate is re-cleared after any change. The on-premises LLM is hash-pinned to a deterministic seed and repository commit and is governed by a verification-before-generation gate suite, aligned with H.R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026 [6, 7, 5].

The Phase 0 simulation validation gate is a hard precondition rather than a target, and its four criteria are conjunctive: the Unified Safety Level must reach at least 7.0; at least 1,000 simulations must be run across at least two independent simulation frameworks; the sim-to-real trajectory gap must fall below 2 mm; and the sim-to-real tip-force gap must fall below 0.5 N. All four must hold simultaneously before any patient-facing robotic activity, and the gate is re-cleared in full after any change to the platform, the model, or the safety configuration, so that no modification is allowed to reach a patient without a fresh validation. The sim-to-real tolerances connect directly to the Physical AI reporting trigger for sim-to-real divergence: a trajectory gap beyond twice the 2 mm tolerance (that is, above 4 mm) or a tip-force gap beyond twice the 0.5 N tolerance (that is, above 1.0 N) is a reportable divergence under the six Physical AI triggers of 21 CFR §312.32(g). The investigator should read the Phase 0 gate, the run-time safety envelope, and the reporting triggers as one continuous chain of defenses, each checked against a fixed numeric criterion [6, 7].

Summary of Data and Guidance for the Investigator

The investigator should treat the daraxonrasib drug brochure and the Physical AI device supplement as the two controlling reference texts for expectedness and risk. For the drug, the established 300 mg once-daily RP2D and the class toxicity profile (gastrointestinal and dermatologic effects) define the expected events; the study deliberately starts at 160 mg once daily to preserve a conservative first-in-human margin in the perioperative setting, and the 28-day dose-limiting toxicity window, the 3+3 escalation, and the dose-interruption and dose-reduction rules govern management. For the device, the force caps, the layered E-stop chain, and the five-vessel no-fly gate define the expected safety behavior; any robotic malfunction, AI prediction error, sensor failure, communication loss, unauthorized access, digital-twin divergence, or event requiring activation of the E-stop is a Physical AI adverse event under 21 CFR §312.32(f) and is assessed against this supplement for expectedness. The perioperative pause-and-restart advisory is LLM-bound and human-confirmed and never autonomous; the investigator confirms the restart day keyed to the pancreaticojejunostomy ISGPS fistula grade and the serum trough relative to 0.5 ng/mL, with a Grade C fistula mapping to T+21 or a continued hold and to the drug stopping rule. All adverse events are graded by CTCAE version 5.0 and reported on the regulatory timelines (no later than 7 calendar days for an unexpected fatal or life-threatening suspected adverse reaction and no later than 15 calendar days for any other serious and unexpected suspected adverse reaction), with the six Physical AI reporting triggers of §312.32(g) layered on top [1, 3, 18].

The six Physical AI reporting triggers of 21 CFR §312.32(g) layered on top of the ordinary drug-safety clocks are stated in Table 5.7 so that the investigator can match an observed device or model event to the correct reporting clock without ambiguity. The first trigger is a serious Physical AI adverse event, reported on the ordinary 7-day or 15-day clock by seriousness and expectedness; the second is a system-drug interaction, reported within 15 days; the third is a cybersecurity incident, reported within 15 days or within 7 days if there is potential for patient harm; the fourth is a sustained AI model degradation, defined as degradation across at least three consecutive procedures or at least 24 cumulative hours; the fifth is a sim-to-real divergence beyond twice the tolerance (trajectory above 4 mm or tip-force above 1.0 N); and the sixth is a digital-twin discrepancy. These triggers operate in addition to, not instead of, the ordinary serious-adverse-reaction reporting of §312.32(c), and the device supplement is the expectedness reference for each [3, 7, 5].

Table 21: The six Physical AI reporting triggers of 21 CFR §312.32(g) and their clocks, layered on the ordinary drug-safety reporting.

#	Trigger	Reporting clock and definition
1	Serious Physical AI adverse event	7 days (unexpected fatal or life-threatening) or 15 days (other serious and unexpected), as for a drug suspected adverse reaction
2	System-drug interaction	Within 15 days
3	Cybersecurity incident	Within 15 days, or within 7 days if there is potential for patient harm
4	Sustained AI model degradation	Reportable; degradation across at least 3 consecutive procedures or at least 24 cumulative hours
5	Sim-to-real divergence	Divergence beyond twice the tolerance: trajectory above 4 mm or tip-force above 1.0 N
6	Digital-twin discrepancy	Reportable discrepancy between the digital twin and the observed system

Table 5.7. The six Physical AI reporting triggers of 21 CFR §312.32(g), layered on top of the ordinary 7-day and 15-day drug-safety clocks. The Physical AI device supplement of the Investigator's Brochure is the expectedness reference against which each trigger is judged; the audit trail is preserved from 24 hours before to 72 hours after any reportable Physical AI adverse event.

The categories of event that constitute a Physical AI adverse event under 21 CFR §312.32(f) are enumerated so that the investigator does not have to infer them. A robotic malfunction, an AI prediction error, a sensor failure, a communication loss, an instance of unauthorized access, a digital-twin divergence, and any event that requires activation of the emergency stop each constitute a Physical AI adverse event and each is assessed against the device supplement for expectedness. The audit trail is the evidentiary record for that assessment: because every advisory command, every gate verdict, every force and trajectory sample, and every E-stop event is hash-chained and append-only, the sequence of events around any reportable occurrence can be reconstructed exactly, and the preservation window from 24 hours before to 72 hours after the event ensures the relevant record survives. The three-tier oversight structure (the data safety monitoring board with halt authority, the independent safety monitor with per-procedure real-time stop authority, and the Physical AI Safety Review Committee on a cadence of no more than 90 days) reviews these events, and the binding halt rules (one or more device or procedure serious adverse events within a cohort, or two device-related deaths at any time) trigger a board review and possible halt [3, 7, 5].

The investigator is reminded that the controlling clinical-conduct text is the study protocol and that the present section is an IND-level summary; the full standalone Investigator's Brochure is a separate bound document submitted with this IND and is the authoritative reference safety information for both the drug and the device. The investigator should hold a current copy of the bound IB, record its version in the case report form for each procedure, and consult §6, §7, and §8 of this IND for the drug-and-device intervention detail, the chemistry, manufacturing, and control information, and the pharmacology and toxicology information that the IB summarizes. The brochure is reviewed and reissued at least annually and whenever new safety information materially changes the risk profile, and currently enrolled participants are re-consented when such a change occurs, so that the reference text against which every expectedness determination is made remains current throughout the conduct of the study [8, 1, 10].

6 Proposed clinical research

The proposed clinical research is presented in this section of the investigational new drug (IND) application in the form required by 21 CFR §312.23(a)(6). It consists of the study protocol for the planned Phase 1 first-in-human investigation, the informed consent that governs each participant's enrollment, and the investigator and facilities information that establishes that the study will be conducted by qualified persons at adequate facilities. The investigation described here is a prospective, open-label, single-arm, first-in-human study that combines a standard 3+3 dose-finding evaluation of perioperative daraxonrasib (RMC-6236) with an early-feasibility evaluation of an on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) system, a significant-risk Class II collaborative device. Because the device participates in the delivery of the investigational drug, the study is conducted under a combined IND (21 CFR §312) and investigational device exemption (IDE, 21 CFR §812) submission with a single integrated protocol, a single informed consent carrying the Physical AI opt-out, and a single integrated safety governance structure. The full clinical protocol, version 1.0.0, is bound to this IND as a supporting document and is archived with the digital object identifier DOI 10.5281/zenodo.20780121 (https://doi.org/10.5281/zenodo.20780121); the synopsis and supporting tables in §6.1 summarize that bound document, and every quantitative value reported here is traceable to it [1, 2].

This section is written so that an FDA reviewer, an institutional review board (IRB) member, a data safety monitoring board (DSMB) member, or an independent monitor can read the proposed clinical research in isolation and reconstruct exactly how the study is conducted, how each participant is protected, and how every regulatory clock is met. The regulatory architecture is deliberately layered: the drug component satisfies the content requirements of 21 CFR §312.23(a)(6) for a Phase 1 study protocol, namely a statement of objectives and purpose, the names and addresses and qualifications of investigators, the criteria for participant selection and exclusion and an estimate of the number to be studied, a description of the design including the kind of control group, the method for determining doses to be administered, a description of observations and measurements to fulfill the objectives, and a description of the clinical procedures and laboratory tests undertaken to monitor the effects of the drug and to minimize risk. The device component layers on the IDE content of 21 CFR §812.25, namely the investigational plan, the report of prior investigations, and the monitoring procedures, and the Physical AI System Description meeting the eleven fields of 21 CFR §312.23(g) as adapted for autonomy-bearing investigations. Throughout, the convention is that the drug and the device are governed by one integrated protocol so that no safety signal on either side is reviewed in isolation from the other, and that the autonomy-bearing behavior of the system is held to a continuous human-oversight standard that prohibits full autonomy at this phase consistent with 21 CFR §312.21(e) [8, 6].

6.1 Study Protocol

Protocol identification and overview

The bound protocol is entitled the LLM-Directed PDAC Robotic Daraxonrasib Phase 1 study (protocol version 1.0.0, DOI 10.5281/zenodo.20780121). It is a prospective, open-label, single-arm, first-in-human dose-finding and early-feasibility study in adults with histologically or cytologically confirmed, resectable or borderline-resectable, KRAS G12-mutated pancreatic ductal adenocarcinoma (PDAC) who are candidates for pancreaticoduodenectomy. The central hypothesis is that the combined drug-device intervention can be delivered with an acceptable early safety profile and can complete the assigned operative tasks under continuous human oversight while a recommended Phase 2 dose (RP2D) of daraxonrasib is identified, supporting progression to a later-phase efficacy study. Enrollment is planned for up to $n = 18$ treated participants (approximately 36 screened) at a single academic medical center, assigned across three daraxonrasib dose levels by the 3+3 rule with staggered sentinel enrollment, at a planned accrual of approximately one to two participants per month. The investigation is structured around the early-feasibility pathway recognized for significant-risk devices, which generates the device-performance and safety evidence required before any controlled comparison is undertaken [8, 1].

The choice of a first-in-human dose-finding and early-feasibility design follows directly from the regulatory status of the two components. Daraxonrasib has not previously been studied in a perioperative schedule and has not previously been paired with an autonomy-bearing surgical device, so no prior human experience quantifies the safety of the combination; the device has never operated on a human participant; and the integrated drug-device combination has been characterized only in computational evidence. A small, intensively monitored, dose-escalating cohort with staggered sentinel enrollment and binding halt rules is therefore the design that minimizes the number of participants exposed before each successive safety question is answered, consistent with the FDA guidance on first-in-human and Phase 1 oncology studies and with the early-feasibility framework for significant-risk devices [8, 10]. The disease context reinforces the ethical case for a small, carefully escalated study: PDAC carries a total five-year survival below 13% and is the third leading cause of cancer death in the United States and the fourth in Europe, so the curative-intent operation under study is performed in a population for whom every avoidable surgical delay is consequential, and for whom the alternative to enrollment is the same demanding operation without the investigational advisory layer [4].

Objectives and endpoints

The protocol specifies three co-primary objectives, each evaluated with exact two-sided 95% Clopper-Pearson confidence intervals because the design is descriptive and non-confirmatory under ICH E9 with no formal power calculation and no multiplicity adjustment. The first co-primary endpoint is the incidence of device- or procedure-related serious adverse events through 30 days, including any Clavien-Dindo Grade III or higher complication. The second co-primary endpoint is the maximum tolerated dose (MTD) and the RP2D of perioperative daraxonrasib, determined by the 3+3 rule across a 28-day dose-limiting toxicity (DLT) window. The third co-primary endpoint is technical feasibility, defined as the proportion of participants completing the assigned operative tasks without unplanned conversion attributable to device malfunction or unsafe device behavior. The secondary endpoints are the margin-negative (R0) resection rate on day-90 pathology (per-protocol), the International Study Group of Pancreatic Surgery (ISGPS) Grade B or C postoperative pancreatic-fistula rate, the Clavien-Dindo Grade III or higher complication rate at 90 days, 30-day and 90-day mortality, the conversion rate, and operative parameters (estimated blood loss, operating-room time, and length of stay), each benchmarked descriptively against the 2025 Dutch nationwide robotic pancreaticoduodenectomy cohort (conversion 10.1%, Clavien-Dindo III+ 41.3%, ISGPS B/C 24.4%, 30-day mortality 3.9%). Exploratory endpoints are progression-free and overall survival to 24 months by RECIST version 1.1 with Kaplan-Meier estimation, the concordance of the LLM advisory with the hard-coded safety gate, and the sim-to-real gap in millimeters and newtons [1, 10].

Each endpoint is paired with an explicit definition, an analysis population, and a prespecified summary statistic so that the analysis is fully prespecified before the first participant is enrolled. Table 22 sets out the complete endpoint hierarchy with the definition, the population, and the summary measure for each endpoint. The three co-primary endpoints are deliberately heterogeneous: a device-and-procedure safety endpoint, a drug dose-finding endpoint, and a device feasibility endpoint. This structure reflects the combined IND/IDE nature of the investigation, in which the dose-finding question for the drug and the readiness and feasibility questions for the device must be answered together, because the device participates in delivering the drug and the perioperative restart of the drug depends on the operative and pathology results of the device procedure.

Table 23 presents the schedule of activities that governs the timing of every assessment, from screening through the operative episode and the 24-month overall-survival follow-up.

Study design and dose escalation

The drug component proceeds under the IND pathway and the device component under the IDE pathway, with a single integrated protocol and a single safety governance structure. The escalation uses three dose levels anchored to the pan-KRAS RASolute 302 program, in which 300 mg once daily is the established RP2D: DL1 is 160 mg once daily (starting dose, chosen below the reference dose to preserve a conservative first-in-human margin), DL2 is 220 mg once daily, and DL3 is the 300 mg once-daily reference dose. Three participants are enrolled at each level and observed across a 28-day DLT window. If no DLT occurs among the first three, the cohort escalates; if one of three experiences a DLT, the cohort expands to six; a single DLT among six permits escalation, whereas two or more DLTs at a level fix the MTD at the preceding level. The MTD is the highest level at which fewer than two of six DLT-evaluable participants experience a DLT, and the RP2D is selected at or below the MTD. With a maximum of three levels of six, the design treats up to 18 participants. The analysis populations are the Safety population (all dosed participants), the DLT-evaluable population, the Per-protocol population, and the modified intention-to-treat (mITT) population.

The 3+3 decision rule is applied uniformly at every level, and the precise transition logic is set out in Table 24. The rule is intentionally simple and fully deterministic so that the escalation decision can be reproduced by any reviewer from the observed DLT count alone, and so that no judgment beyond the DLT adjudication is required to choose the next cohort. A DLT is any drug-attributable toxicity meeting the protocol grading thresholds during the 28-day window after the first dose at a given level; events adjudicated as attributable to the surgery or the device rather than to the drug are captured under the device/procedure safety endpoint and the Physical AI safety stream rather than as DLTs, so that the dose-finding question for the drug is not contaminated by device-attributable events.

The estimation precision that this sample supports is illustrated by the exact Clopper-Pearson intervals, which are the appropriate interval for a small-sample binomial proportion because they guarantee at least nominal coverage without relying on a normal approximation. An observed rate of 0 of 18 yields 0% (95% CI 0 to 18%), 1 of 18 yields 5.6% (95% CI 0.1 to 27%), 0 of 3 yields 0% (95% CI 0 to 71%), and 0 of 6 yields 0% (95% CI 0 to 46%) [1]. These worked values are reproduced in Table 25 so that a reviewer can read off the inferential reach of each cohort size directly. The contrast between the 0-of-3 and 0-of-18 rows makes the rationale for staggered sentinel enrollment explicit: a single cleared cohort of three bounds the true serious-event rate only loosely (up to 71%), whereas the full 18-participant experience tightens the upper bound to 18%, so the study deliberately defers any claim of acceptable safety until the cumulative experience supports it.

The device has no pharmacologic dose; its dose analogue is the readiness evidence required before any patient-facing activity, escalated only by passing the mandatory Phase 0 simulation-validation gate. Three conditions must be met and documented before the first robotic case: a Unified Safety Level (USL) rating of at least 7.0 on the surgical-readiness scale; at least 1000 simulated procedures completed across at least two independent simulation frameworks (for example Isaac, MuJoCo, Gazebo, or PyBullet) so that the readiness finding is not an artifact of any single simulator; and a sim-to-real fidelity within prespecified limits, a trajectory gap of less than 2 mm and a tip-force gap of less than 0.5 N between the simulation prediction and the operative record. The gate is re-run after any software, model, or configuration change before any further patient-facing use, so that no model update, retraining, or reconfiguration reaches a participant without a fresh demonstration of readiness [2, 14]. Table 26 states the three Phase 0 gate conditions, their numeric thresholds, and the consequence of a failed condition.

Population and eligibility

The study population comprises adults with histologically or cytologically confirmed, resectable or borderline-resectable, KRAS G12-mutated PDAC who are eligible for perioperative daraxonrasib under the broad pan-KRAS RASolute 302 criteria and whose tumor and vascular anatomy are compatible with the eight-arm robotic system. Participant selection is equitable; eligibility is determined solely by the scientific and safety criteria, and limited English proficiency is not an exclusion criterion. The synthetic reference exemplar PAT-PDAC-0001 (a 68-year-old with a 3.4 cm head-of-pancreas tumor abutting the superior mesenteric vein at 75°, KRAS G12D, microsatellite stable, CA 19-9 of 412 U/mL, ECOG performance status 1, after a completed neoadjuvant modified FOLFIRINOX course of four cycles) illustrates a representative eligible participant and does not represent an enrolled individual. The full eligibility criteria are summarized in Table 27.

The eligibility criteria are constructed so that each criterion maps to an identified risk. The KRAS G12 genotype requirement establishes the molecular target for daraxonrasib and aligns the population with the pan-KRAS RASolute 302 reference program from which the dose levels are drawn; the resectable or borderline-resectable staging and the multidisciplinary resectability review ensure that the curative-intent operation is appropriate and that vascular involvement precluding an R0 resection is excluded before any device exposure; the ECOG 0 or 1 and adequate organ and marrow function requirements ensure that the participant can tolerate both the major pancreaticobiliary operation and the RAS(ON) inhibitor; and the anatomic-compatibility requirement ensures that the tumor and the five named vessels are within the operating envelope of the eight-arm platform. Limited English proficiency is explicitly not an exclusion criterion, and qualified interpreters and translated materials are provided, so that the equitable-selection principle of the governing human-subjects regulations is met and the population is not narrowed in a way that would compromise generalizability or fairness [19].

Intervention: drug and device

Daraxonrasib is an orally bioavailable RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor that binds the active, guanosine-triphosphate-bound state of mutant RAS in complex with cyclophilin A, suppressing downstream effector signaling across the common KRAS variants in PDAC; it received FDA Breakthrough Therapy designation in June 2025 for previously treated metastatic KRAS G12-mutant PDAC. In this study it is administered orally once daily in a perioperative schedule comprising a defined pre-operative course, a protocol-mandated pause across the operative window, and a postoperative restart governed by the advisory in §6.1 below. The device component is an eight-arm parallel coelomic surgical platform carrying 56 mechanical degrees of freedom (8 arms $\times$ 7 per-arm degrees of freedom) and a 640-channel mixed-rate sensor stack (80 channels per arm), with force channels sampling at 100 kHz and all remaining channels at the 10 kHz command rate, and a positioning accuracy of 0.05 mm root mean square at a peak tip velocity of 1,200 mm/s. An on-premises repository-pinned, open-weights LLM acts as a second-opinion advisory oracle held strictly outside the robot vendor kinematic stack and hash-pinned to a deterministic seed and a repository commit; it is never an autonomous controller, and full autonomy is prohibited at this phase consistent with 21 CFR §312.21(e). The advisory is gated by a verification-before-generation ten-gate suite returning ACCEPT, BLOCK, or ESCALATE, aligned with H.R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026 [6, 2].

The hard cyber-physical safety envelope comprises a per-arm tip-force cap of ≤ 3 N, a cumulative cross-arm tip-force cap of ≤ 18 N enforced at the heartbeat boundary, a 10 kHz heartbeat coordination bus broadcasting a 64-byte frame on a $100\,\mu$s per-arm watchdog deadline, an emergency arm park within $50\,\mu$s of a missed or out-of-parameter frame, a cross-arm emergency stop (E-stop) that halts the full platform in ≤ 3 ms at a 0.0 N force clamp, and a software-independent hardware E-stop meeting the 21 CFR §312.404 requirement of ≤ 500 ms. A vascular no-fly gate sampled at 10 kHz evaluates instrument-tip proximity to five named vessels (superior mesenteric vein, portal vein, hepatic artery, celiac axis, and superior mesenteric artery) against three concentric thresholds: a soft-warning zone at 3.0 mm raises an LLM advisory, a no-fly zone (1.0 mm at the superior mesenteric and portal veins; 1.5 mm at the hepatic artery, celiac axis, and superior mesenteric artery) blocks the command, and a hard-stop zone at 5.0 mm forces the ≤ 3 ms E-stop; across a representative 100-tick verdict sample the gate returned clear on 83 ticks, soft-warning on 6, no-fly on 6, and hard-stop on 5. The operation proceeds across eight operative phases: phases P1 through P4 comprise dissection and specimen removal, phase P5 the duct-to-mucosa pancreaticojejunostomy under a controller ring-tension band of 0.30 to 0.60 N, phase P6 the end-to-side hepaticojejunostomy at 0.20 to 0.50 N, phase P7 the antecolic gastrojejunostomy at 0.40 to 0.80 N, and phase P8 final imaging, hemostasis, and drain placement [2].

The numeric parameters of the safety envelope are consolidated in Table 28 so that a reviewer can confirm that every quoted limit, latency, and threshold is internally consistent and traceable to the bound protocol. The envelope is structured as a defense-in-depth chain in which a soft advisory layer (the 3.0 mm soft-warning zone and the LLM second opinion) is backed by a command-blocking layer (the no-fly zones and the force caps) which is in turn backed by an irreversible halt layer (the 5.0 mm hard-stop zone, the $\leq$ 3 ms cross-arm E-stop, and the software-independent hardware E-stop). No single layer is relied upon alone, and the hardware E-stop is independent of all software so that a software fault cannot disable the final halt path. The per-phase controller ring-tension bands constrain the anastomotic forces during the reconstruction phases P5 through P7, where the surgical risk of a leak is concentrated, and the resulting pancreaticojejunostomy fistula grade is the dominant input to the perioperative restart advisory and the drug stopping rule.

Perioperative daraxonrasib pause-and-restart advisory

The two components are coupled at one decision point only: the perioperative pause-and-restart advisory for the drug is informed by the operative and pathology results of the device procedure. Daraxonrasib is paused before surgery and restarted postoperatively on a day chosen by an advisory that is LLM-bound and human-confirmed, never autonomous. The advisory takes two inputs, the pancreaticojejunostomy ISGPS fistula grade (A, B, or C) and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold, and recommends a restart at postoperative day T+7d, T+14d, or T+21d (or a continued hold). In the 32-iteration deterministic simulation sweep, baseline serum was 0.45 ng/mL at the operative timepoint across all iterations (below the 0.5 ng/mL trough threshold), and the advisory recommended T+7d in 29 of 32 iterations (90.6%, Grade A fistula with sub-threshold trough), T+14d in 3 of 32 (9.4%, Grade B or delayed serum rise), and T+21d or continued hold in 0 of 32 (reserved for Grade C, which also maps to the drug stopping rule). Figure 15 reproduces the advisory logic and the sweep distribution.

The advisory illustrates the disciplined separation between the autonomy-bearing component and the human decision: the LLM proposes a restart day from the two inputs, but the restart is executed only after a qualified clinician confirms it, so that no drug-administration decision is ever made autonomously. The mapping is monotone in both inputs, with a lower fistula grade and a sub-threshold trough favoring the earliest restart (T+7d) and a higher grade or a rising trough favoring a later restart (T+14d) or a hold; a confirmed Grade C fistula precludes restart entirely and maps to the drug stopping rule. Table 29 sets out the worked input-to-output mapping together with the sweep frequencies, so that the advisory can be audited against the deterministic sweep distribution reproduced in the figure.

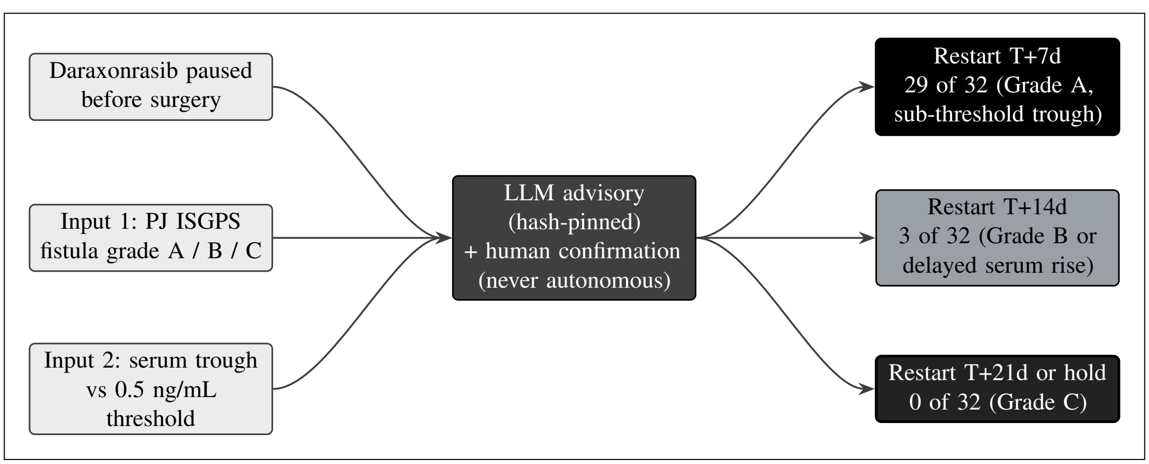

Figure 15. The perioperative daraxonrasib pause-and-restart advisory. The drug is paused before surgery and the postoperative restart day is set by a hash-pinned, human-confirmed LLM advisory (never autonomous) from two inputs: the pancreaticojejunostomy ISGPS fistula grade (A, B, or C) and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold. In the 32-iteration deterministic sweep the advisory recommended restart at T+7 days in 29 of 32 iterations (Grade A with sub-threshold trough), T+14 days in 3 of 32 (Grade B or delayed serum rise), and T+21 days or continued hold in 0 of 32 (reserved for Grade C).

Assessments and the vascular and halt safety chain

Clinical safety assessments comprise vital signs, a directed physical examination, hematology, serum chemistry including hepatic and pancreatic indices, coagulation studies, the serum daraxonrasib trough assay, and protocol-specified imaging, at the frequencies in the schedule of activities. In addition, the device generates a continuous safety telemetry stream that is a source of protocol observations under 21 CFR §312.23(g): per-arm and cumulative force profiles, trajectory accuracy against the 0.05 mm specification, every vascular safety-zone verdict, every heartbeat and watchdog event, and every E-stop activation, each captured at its recorded rate in the hash-chained audit trail. The halt chain is layered: the 10 kHz heartbeat bus broadcasts a 64-byte frame on a 100 μs per-arm watchdog deadline; a frame that is on time continues the command state, whereas a missed or out-of-parameter frame parks the affected arm within 50 μs and escalates to the cross-arm E-stop that halts the platform in ≤ 3 ms at a 0.0 N force clamp, backed by the software-independent hardware E-stop. Postoperative pancreatic fistula is graded by the ISGPS definition as a biochemical leak, Grade B (requiring a change in management), or Grade C (requiring reoperation or causing organ failure or death), and the pancreaticojejunostomy grade is the dominant determinant and the input to the restart advisory and the drug stopping rule [1, 10].

The vascular no-fly gate is the most safety-critical of the telemetry-driven controls, because the five named vessels border the operative field of a pancreaticoduodenectomy and an inadvertent vascular injury is among the most lethal intraoperative events. The gate is evaluated at 10 kHz against the three concentric thresholds, and the representative 100-tick verdict sample (clear 83, soft-warning 6, no-fly 6, hard-stop 5) demonstrates the expected distribution in which most ticks are clear, a minority raise a soft advisory, a smaller minority block a command, and the hard-stop verdicts force the E-stop. Table 30 sets out the three vascular zones, their per-vessel thresholds, the system action, and the corresponding verdict count in the representative sample, so that the proximity controls are fully specified and auditable against the telemetry record.

Safety reporting clocks and the six Physical AI triggers

Adverse events are captured through two parallel and equally binding streams, a conventional pharmacovigilance stream for clinical events graded by CTCAE version 5.0, and a Physical AI safety stream for device- and AI-related events under 21 CFR §312.32(f). Serious adverse events are reported by the investigator to the sponsor within 24 hours of awareness, and the sponsor submits IND/IDE safety reports on the regulatory clocks: any unexpected fatal or life-threatening suspected adverse reaction is reported as soon as possible but no later than 7 calendar days after initial receipt of the information, and any other serious and unexpected suspected adverse reaction is reported no later than 15 calendar days after the determination that the event is reportable (21 CFR §312.32). In addition, six Physical AI reporting triggers are reported under §312.32(g): a serious Physical AI adverse event on the 7- or 15-day timeline as applicable; a system-drug interaction causing incorrect drug administration, regardless of seriousness, within 15 days; a cybersecurity incident within 15 days (or 7 days if it has the potential to cause patient harm); sustained AI model degradation below the pre-specified acceptance criteria for three or more consecutive procedures or a cumulative 24 hours; a sim-to-real divergence beyond twice the validated gap tolerance (trajectory deviation greater than 4 mm or force discrepancy greater than 1.0 N); and a digital-twin discrepancy beyond the pre-specified accuracy thresholds. For every Physical

AI adverse event, the complete hash-chained audit trail from 24 hours before the event to 72 hours after the event is preserved under 21 CFR part 11 and made available to the FDA on request. Figure 16 reproduces the clocks and the six triggers [8, 3].

The six Physical AI triggers extend conventional pharmacovigilance to the autonomy-bearing failure modes that a drug-only IND does not contemplate, and each is mapped to a defined clock and a defined evidentiary preservation in Table 31. Two features distinguish the Physical AI stream from the conventional stream. First, several triggers fire regardless of clinical seriousness: a system-drug interaction causing incorrect drug administration is reportable even if no harm results, and a cybersecurity incident is reportable on a shortened 7-day clock when patient harm is merely possible, because the autonomy-bearing context makes a near-miss a leading indicator of a future serious event. Second, every Physical AI adverse event triggers preservation of the hash-chained audit trail from 24 hours before to 72 hours after the event under 21 CFR part 11, so that the complete command, telemetry, advisory, and override record surrounding the event is available for reconstruction. The two streams are equally binding, and a single event that is both a clinical serious adverse event and a Physical AI adverse event is reported on whichever clock is shorter.

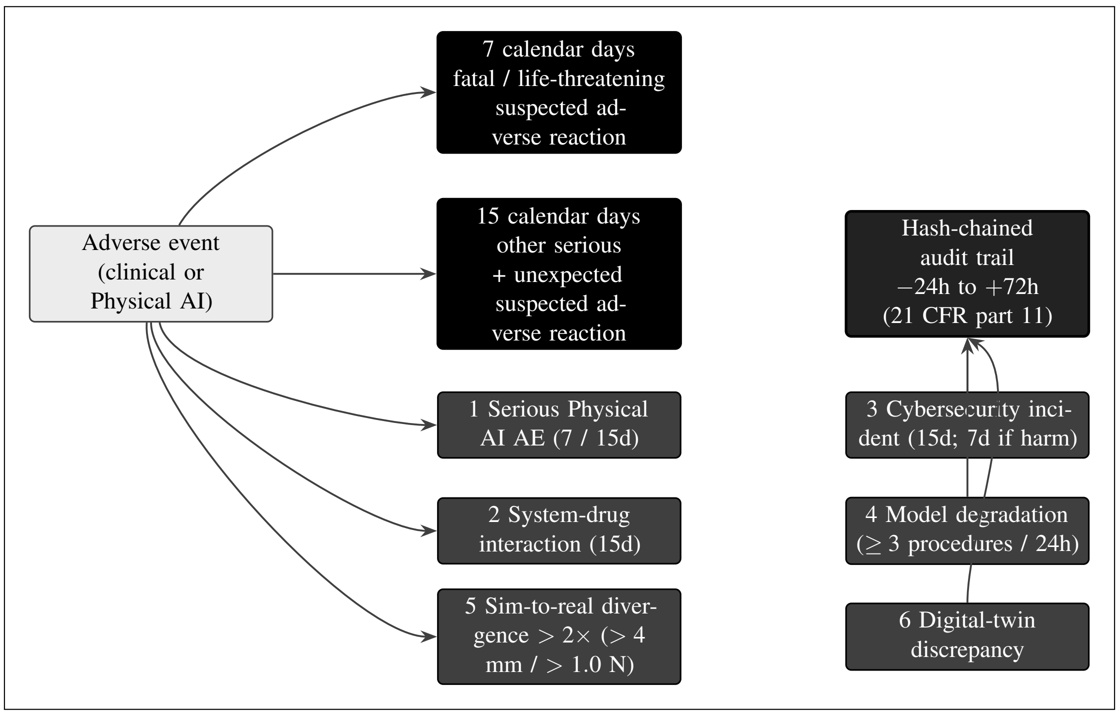

Figure 16. The safety-reporting clocks and the six Physical AI reporting triggers. A fatal or life-threatening suspected adverse reaction is reported within 7 calendar days and any other serious and unexpected suspected adverse reaction within 15 calendar days (21 CFR §312.32). The six Physical AI triggers of §312.32(g) are a serious Physical AI adverse event (7- or 15-day), a system-drug interaction (15-day), a cybersecurity incident (15-day, or 7-day if patient harm is possible), sustained model degradation (at least 3 consecutive procedures or 24 cumulative hours), a sim-to-real divergence exceeding twice the validated tolerance (trajectory over 4 mm or force over 1.0 N), and a digital-twin discrepancy. Each preserves the hash-chained audit trail from minus 24 to plus 72 hours under 21 CFR part 11.

Discontinuation and stopping rules

The protocol distinguishes discontinuation of an intervention from withdrawal from the study; a participant who stops one or both components for a protocol-defined reason remains in the study for safety and survival follow-up unless consent for that follow-up is withdrawn. The investigational device is discontinued for the index procedure, with conversion to the appropriate fallback technique, whenever any of five prespecified conditions occurs: a conversion to open or manual technique judged by the supervising surgeon to better serve patient safety or oncologic adequacy (conversion is always available and is never itself a protocol violation); repeated excursions against the $\leq$ 3 N per-arm or $\leq$ 18 N cumulative caps; any breach of a vascular no-fly zone (1.0 mm at the superior mesenteric and portal veins; 1.5 mm at the hepatic artery, celiac axis, and superior mesenteric artery) or a recurring pattern of no-fly approaches; any sim-to-real divergence beyond twice the validated tolerance (trajectory deviation greater than 4 mm or force discrepancy greater than 1.0 N), which also triggers a §312.32(g) report; and any hard-stop cross-arm E-stop event, after which device resumption requires that the cause be identified and the system re-pass the pre-procedure safety matrix. Daraxonrasib is discontinued in an individual participant for a drug-attributable DLT during the 28-day window, for any other unacceptable toxicity not resolving with protocol dose interruption and reduction, for a confirmed ISGPS Grade C fistula (which precludes restart and maps to the T+21d or continued-hold arm of the advisory), for pregnancy, for contraindicating intercurrent illness, or at the participant's request [2, 1].

Two binding halt rules govern the study as a whole and are applied by the data safety monitoring board (DSMB): a device- or procedure-related serious adverse event in at least one of three participants within a cohort, or two device-related deaths at any time, triggers DSMB review and a halt. On any hold or closure, enrollment and patient-facing robotic activity stop immediately, affected participants are transitioned to appropriate clinical care, and a controlled decommissioning and data-preservation procedure seals the complete hash-chained command and telemetry record for the regulatory retention period. Table 32 consolidates the device discontinuation conditions, the drug discontinuation conditions, and the two study-level halt rules, with the level at which each applies and the action that follows, so that the entire stopping logic can be read at one glance and audited against the source documents.

Statistical considerations and oversight

The statistical approach is descriptive and estimation-based, non-confirmatory, and follows ICH E9 and the TRIPOD+AI reporting standard for the model-assisted advisory component. No formal power calculation is performed; exact two-sided 95% Clopper-Pearson confidence intervals are reported for the binomial endpoints, and a two-sided alpha of 0.05 is used only in a supportive sense with no multiplicity adjustment. Independent safety oversight operates on three tiers: the DSMB reviews accumulating safety data at each cohort boundary and at any triggered review and holds halt authority; the Independent Safety Monitor (ISM) is designated for each procedure and provides real-time case-level oversight with the authority to call a stop; and the Physical AI Safety Review Committee meets on a cadence not exceeding 90 days (and ad hoc on any triggering event) to review the autonomy-bearing behavior of the system, the verification-before-generation gate-surface decisions, the USL rating, and any software change, and to recommend USL reassessment or oversight changes. Seven Physical AI record types (the LLM advisory record, the robot command record, the sensor and telemetry record, the safety-limit-event record, the verification-before-generation gate-surface record, the human-override and intervention record, and the binding hash-chained audit trail) are retained for the period required by 21 CFR §312.57, at least two years after a marketing application is approved or, if none is filed, two years after the investigation is discontinued and the FDA is notified [1, 7].

The benchmarking of the secondary endpoints against the 2025 Dutch nationwide robotic pancreaticoduodenectomy cohort is descriptive and is not a formal comparison; it situates the observed rates against a contemporary real-world reference so that an unexpectedly high conversion, complication, fistula, or mortality rate is recognized promptly even though the small sample cannot support a hypothesis test against the reference. The seven Physical AI record types are summarized in Table 33 with their content and retention basis, so that the records environment can be confirmed to satisfy 21 CFR §312.57 and 21 CFR part 11. Each record type is append-only and hash-chained where applicable, so that the provenance of every advisory verdict, every robot command, and every human override is reconstructable for the full retention period, and so that the FDA can audit the autonomy-bearing behavior of the system after the fact with the same fidelity as a contemporaneous observer [7, 9].

The full protocol, version 1.0.0, with its complete schedule of activities, statistical analysis plan, and Physical AI System Description meeting the eleven fields of 21 CFR §312.23(g), is the bound supporting document for this section and is archived at DOI 10.5281/zenodo.20780121 (https://doi.org/10.5281/zenodo.20780121).

6.2 Informed Consent

Informed consent is obtained by a qualified member of the research team who is not the treating surgeon, before any study-specific procedure, in language the participant can understand and with adequate time and opportunity to ask questions, consistent with 21 CFR part 50 and ICH E6(R3) [19, 18]. The consent process incorporates each of the basic and additional elements of informed consent required by 21 CFR §50.25, and a copy of the form and any participant information sheet are submitted with the protocol. The consent discussion covers the diagnosis, the alternatives (including standard open or conventional robotic pancreaticoduodenectomy without the investigational advisory layer), and the prognosis of a malignancy with a total five-year survival below 13%; the perioperative daraxonrasib regimen, its once-daily oral administration, the protocol-mandated operative pause, and the toxicities characteristic of the RAS(ON) inhibitor class; and, in addition to the standard elements, an explicit and distinct disclosure of the on-premises LLM-directed eight-arm robotic system, its role as a second-opinion advisory oracle held outside the vendor kinematic stack, its continuous human oversight with immediate manual override, and its system-specific risks. The voluntary nature of participation, the right to withdraw at any time without penalty and without loss of any benefit to which the participant is otherwise entitled, and the maintenance of confidentiality under the HIPAA Safe Harbor de-identification method are stated at consent and reaffirmed at each contact.

The consent process is mapped explicitly to the basic and additional elements of 21 CFR §50.25 in Table 34, so that an IRB reviewer can confirm that each required element is addressed and can see which elements carry Physical-AI-specific content beyond what a drug-only consent would contain. The additional elements specific to this trial class are the distinct disclosure of the autonomy-bearing device and its system-specific risks, the explanation of the staged-autonomy model, and the documented Physical AI opt-out, each of which is addressed in the dedicated paragraphs below. The consent form is written at an accessible reading level, and the consent conversation is conducted with adequate time, with qualified interpreters and translated materials where needed, so that the consent is meaningful and not merely a signature [19, 7].

The consent form addresses the therapeutic misconception directly. Because the population is vulnerable, facing the most technically demanding curative-intent abdominal operation for a lethal cancer, the form and the consent conversation make explicit that the study is a first-in-human, dose-finding, and early-feasibility investigation whose primary purpose is to characterize safety and feasibility rather than to confer individual therapeutic benefit, and that the LLM directs no autonomous action. The staged-autonomy model is explained: in this Phase 1 the system operates only at the clinician-controlled (teleoperation or supervised) and supervised semiautonomous levels under continuous human oversight, full autonomy is prohibited consistent with 21 CFR §312.21(e), and any future move to higher autonomy is reserved for a later phase and gated by independent review. An independent consent monitor is used when indicated, and qualified interpreters and translated materials are provided so that consent is meaningful for participants with limited English proficiency [7, 10].

A defining feature of this trial class is the documented Physical AI opt-out. Under 21 CFR §312.60(f), as adapted for Physical AI investigations, the participant is offered, and may exercise, a documented right to request administration of the operative intervention without the Physical AI advisory layer where clinically feasible, without penalty and without affecting eligibility for the remainder of the study or for clinical care [3]. The participant's election or declination of the opt-out is recorded in the source document and the case report form, and the schedule of activities (Table 23) captures that election at the consent visit. Where an unanticipated problem changes the risk profile of the combined intervention, including a relevant Physical AI safety event, the consent document is updated and currently enrolled participants are re-consented before any further study procedure, consistent with the original opt-out.

6.3 Investigator and Facilities Data

The investigation is conducted at a single academic medical center through its institutional multidisciplinary pancreatic-cancer program. Eligible individuals are identified at tumor-board review and at surgical and medical-oncology clinics, and the study is described to them by a member of the research team who is not the treating surgeon, to reduce the potential for undue influence. The site possesses an established high-volume pancreaticobiliary surgical service, a robotic surgical program, an investigational-product pharmacy with controlled, temperature-monitored storage and a drug-accountability log, an anatomic-pathology service applying a standardized synoptic margin protocol with masked independent adjudication, a central imaging capability for RECIST version 1.1 reads by a blinded reviewer, and the clinical laboratory capacity for the serum daraxonrasib trough assay and the hematologic, hepatic, renal, and pancreatic safety panels. The device facilities include the qualified clinical environment in which the eight-arm platform is maintained and calibrated between procedures, the operator control stations, the on-premises compute environment that hosts the repository-pinned advisory LLM in network isolation from the vendor kinematic stack under a deny-by-default authentication and authorization model, and the 21 CFR part 11 records environment that maintains the hash-chained, append-only audit trail. Day-to-day monitoring is delegated to a robotics- and artificial-intelligence-competent clinical research organization (CRO) over the site team, which comprises the operating surgeon, a primary operator, and a qualified backup operator ready to assume control or invoke the manual override at any time [2, 9].

The facilities are listed against their function and their governing regulatory basis in Table 35, so that the adequacy of the facilities required by 21 CFR §312.23(a)(6)(iii)(b) can be confirmed for both the drug and the device components. The site is selected for its combination of a high-volume pancreaticobiliary surgical service and an established robotic program, because the combined drug-device intervention demands surgeons and operators experienced in both the open fallback technique and the robotic approach, and because the early-feasibility nature of the device requires a site capable of maintaining, calibrating, and isolating the on-premises compute environment to the standard required by 21 CFR part 11. The network isolation of the advisory LLM from the vendor kinematic stack under a deny-by-default model is a facility-level control that enforces the architectural separation between the advisory oracle and the controller, so that the prohibition on autonomous control is realized in the infrastructure and not only in the protocol [2, 11].

The governance structure that supervises these investigators and facilities is the three-tier oversight architecture described in §6.1: the Sponsor-Investigator answers to a reviewing IRB and to the independent DSMB, convenes the Physical AI Safety Review Committee on a cadence not exceeding 90 days, designates the Independent Safety Monitor for each procedure, and delegates monitoring to the CRO. The governance and safety-oversight structure that binds the investigators, the facilities, and the independent reviewers together is presented in full in §10.4 (Figure 20), and is summarized for the present section in the oversight table below.

The three-tier oversight architecture is summarized in Table 36, with the membership, the review cadence, and the authority of each tier. The tiers are deliberately complementary in their temporal scope: the Independent Safety Monitor provides real-time, case-level oversight during each procedure with immediate stop authority; the DSMB provides periodic, cohort-level oversight at each cohort boundary and at any triggered review with binding halt authority; and the Physical AI Safety Review Committee provides autonomy-focused oversight on a cadence not exceeding 90 days and ad hoc on any triggering event, with authority to recommend USL reassessment and oversight changes. The combination ensures that an emerging safety signal is caught at whichever timescale it manifests, from a single intraoperative event to a slow degradation across procedures, and that the binding stop and halt authorities rest with bodies independent of the conduct of the study [1, 7].

6.3.1 FDA Form 1572

A Statement of Investigator (FDA Form 1572) is executed by each investigator responsible for the conduct of the study at the clinical site, in satisfaction of 21 CFR §312.53(c), and is assembled with the submission together with the investigator's current curriculum vitae and the documentation of relevant qualifications. The executed Form 1572 attests that the investigator has the training and experience to assume responsibility for the proper conduct of the investigation, identifies the name and address of the clinical site and of the clinical laboratory and other facilities to be used, names the reviewing IRB responsible for review and approval of the study, lists the sub-investigators (the operating surgeon, the primary and backup device operators, the medical oncologist, the pathologist, and the research coordinators) who will assist in the conduct of the investigation, and identifies the protocol by its title and version. By signing, the investigator commits to conduct the study in accordance with the protocol, to personally conduct or supervise the investigation, to inform participants that the drug and device are being used for investigational purposes and to ensure that consent and IRB requirements are met, to report adverse events to the sponsor in accordance with 21 CFR §312.64, to read and understand the investigator's brochure and the Physical AI System Description, to ensure that all associates and staff are informed of their obligations, to maintain adequate and accurate records available for inspection under 21 CFR §312.68, and to comply with all other requirements of 21 CFR part 312. The executed forms, with attachments, are bound into the submission and are updated whenever a sub-investigator or facility changes [8].

The commitments made by signing Form 1572 are summarized against their regulatory basis in Table 37, so that the completeness of the investigator's undertaking can be confirmed at a glance and so that the additional commitment specific to this trial class, to read and understand the Physical AI System Description as well as the investigator's brochure, is made visible alongside the conventional commitments. The Form 1572 commitments are the mechanism by which the sponsor's obligations are devolved to the investigator at the site, and the executed forms together with the curricula vitae establish that the investigation will be conducted by qualified persons as required by 21 CFR §312.23(a)(6)(iii)(b) [8, 9].

6.3.2 Disclosure of Financial Interests

Financial interests are disclosed and managed in accordance with 21 CFR part 54, Financial Disclosure by Clinical Investigators, and the certification and disclosure statements (FDA Forms 3454 and 3455 as applicable) are collected before participation and updated through one year after completion of the study. The disclosure makes explicit the principal conflict of interest in this investigation: the Sponsor-Investigator, Kevin Kawchak, is the chief executive officer of ChemicalQDevice, the entity that develops the on-premises LLM-directed eight-arm robotic Physical AI system under study and holds the combined IND and IDE. This relationship is disclosed in full to the FDA and to the reviewing IRB, and it is managed by the independent oversight structure described in §6.3 so that safety and escalation decisions are not made by a party with a financial interest in the outcome alone [3, 7].

The mitigation is structural and operates at every decision point that could be influenced by the conflict. The DSMB, composed of members independent of the conduct of the study with surgical, oncologic, biostatistical, and device-safety expertise, holds the binding halt authority and applies the prespecified halt rules (a device- or procedure-related serious adverse event in at least one of three participants within a cohort, or two device-related deaths at any time). The Independent Safety Monitor, designated for each procedure and independent of the operating team, holds real-time stop authority at the case level. The Physical AI Safety Review Committee reviews the autonomy-bearing behavior of the system, the verification-before-generation gate-surface decisions, the USL rating, and any software change on a cadence not exceeding 90 days, and recommends USL reassessment or oversight changes. Routine monitoring is delegated to a CRO independent of the device manufacturer, and audits may be conducted by a designee independent of the conduct of the study. Because the binding safety, escalation, and halt decisions rest with these independent bodies rather than with the Sponsor-Investigator, the financial conflict is disclosed and mitigated to a degree commensurate with the first-in-human, autonomy-bearing nature of the investigation, and any identified conflict is disclosed and managed before it can affect the conduct, analysis, or reporting of the study [1].

The conflict and its structural mitigation are summarized in Table 38, with the independent body to which each binding decision is assigned, so that an FDA or IRB reviewer can confirm that no binding safety, escalation, or halt decision rests with the conflicted Sponsor-Investigator. The guiding principle is that disclosure alone is insufficient for an autonomy-bearing, first-in-human investigation in which the Sponsor-Investigator develops and commercializes the device under study; the conflict is therefore managed by relocating every binding decision to an independent body, and by delegating routine monitoring to a CRO independent of the device manufacturer. The financial-disclosure certifications are updated through one year after completion of the study under 21 CFR part 54, so that any change in the financial relationships is captured for the full reporting period [3, 5].

Table 22: Table 6.1. Endpoint hierarchy for the bound Phase 1 protocol, with the definition, the analysis population, and the prespecified summary measure for each co-primary, secondary, and exploratory endpoint. All binomial endpoints are summarized with exact two-sided 95% Clopper-Pearson confidence intervals.

Endpoint	Definition	Population	Summary measure
Co-primary 1: device/procedure SAE	Incidence of device- or procedure-related serious adverse events through 30 days, including any Clavien-Dindo Grade III+ complication	Safety (all dosed)	Proportion with exact 95% CI
Co-primary 2: MTD and RP2D	Maximum tolerated dose and recommended Phase 2 dose by the 3+3 rule across the 28-day DLT window	DLT-evaluable	Dose level; DLT rate per level with exact 95% CI
Co-primary 3: technical feasibility	Proportion completing assigned operative tasks without unplanned conversion from device malfunction or unsafe behavior	Safety (all dosed)	Proportion with exact 95% CI
Secondary: R0 resection	Margin-negative resection on day-90 synoptic pathology	Per-protocol	Proportion with exact 95% CI
Secondary: ISGPS B/C fistula	Grade B or C postoperative pancreatic fistula by the ISGPS definition	Safety (all dosed)	Proportion vs Dutch 24.4%
Secondary: Clavien-Dindo III+	Grade III or higher complication at 90 days	Safety (all dosed)	Proportion vs Dutch 41.3%
Secondary: mortality	30-day and 90-day mortality	Safety (all dosed)	Proportion vs Dutch 3.9% (30-day)
Secondary: conversion	Conversion to open or manual technique	Safety (all dosed)	Proportion vs Dutch 10.1%
Secondary: operative parameters	Estimated blood loss, operating-room time, length of stay	Safety (all dosed)	Median with range
Exploratory: PFS and OS	Progression-free and overall survival to 24 months by RECIST v1.1	mITT	Kaplan-Meier estimate
Exploratory: advisory concordance	Concordance of the LLM advisory with the hard-coded safety gate	Safety (all dosed)	Proportion of concordant verdicts
Exploratory: sim-to-real gap	Trajectory gap (mm) and tip-force gap (N) between simulation and the operative record	Safety (all dosed)	Median (mm; N)

Table 23: Table 6.2. Schedule of activities (summary) for the bound Phase 1 protocol. The intensive safety-collection window spans consent through the 30-day postoperative visit and encompasses the 28-day daraxonrasib dose-limiting toxicity window.

Assessment	Screen	Pre-op / day 0	Post-op to day 30	Long-term follow-up
Informed consent and Physical AI opt-out election	X	-	-	re-consent on amendment
Eligibility, ECOG, KRAS G12 genotype, staging imaging	X	-	-	-
Multidisciplinary resectability review	X	X	-	-
Daraxonrasib pre-operative course (oral once daily)	-	X	-	-
Protocol-mandated operative pause of daraxonrasib	-	X	-	-
Robotic Whipple (8 operative phases P1 to P8)	-	X	-	-
Pre-procedure safety matrix and USL confirmation	-	X	-	re-check after any change
Perioperative pause-and-restart advisory (T+7/14/21d)	-	-	X	-
Vitals, hematology, chemistry, hepatic/pancreatic indices	X	X	X	per follow-up schedule
Serum daraxonrasib trough assay	-	X	X	as indicated
ISGPS fistula grading (PJ dominant)	-	-	X	-
Adverse-event and Physical AI AE capture	X	X	X	SAE/PAI AE to 24 months
RECIST v1.1 imaging (central read)	X	-	X	per follow-up schedule
Pathology, R0 margin (day-90 read, per-protocol)	-	-	X	day 90
Progression-free and overall survival	-	-	-	to 24 months (q12 weeks)

Table 24: Table 6.3. The 3+3 dose-escalation decision rule applied at each daraxonrasib dose level (DL1 160 mg, DL2 220 mg, DL3 300 mg once daily) across the 28-day dose-limiting toxicity window. The maximum tolerated dose is the highest level at which fewer than two of six dose-limiting-toxicity-evaluable participants experience a dose-limiting toxicity.

DLTs in first 3	Action	Resolution
0 of 3	Escalate to the next dose level	Current level cleared; move up one level (or declare RP2D at DL3)
1 of 3	Expand the current level to 6	Enroll 3 more at the same level and re-evaluate at 6
2 or 3 of 3	Stop escalation at this level	MTD exceeded; MTD is the preceding level
1 of 6 (after expansion)	Escalate to the next dose level	Level tolerated; move up one level (or declare RP2D at DL3)
2 or more of 6	Stop escalation at this level	MTD exceeded; MTD is the preceding level

Table 25: Table 6.4. Worked exact two-sided 95% Clopper-Pearson confidence intervals for representative observed event counts at the cohort and full-study sample sizes. The intervals quantify the descriptive precision of the non-confirmatory design and motivate the staggered sentinel enrollment.

Events	Denominator	Point estimate	Exact two-sided 95% Clopper-Pearson interval
0	3	0%	0 to 71%
0	6	0%	0 to 46%
0	18	0%	0 to 18%
1	18	5.6%	0.1 to 27%

Table 26: Table 6.5. The mandatory Phase 0 simulation-validation gate conditions that constitute the device readiness analogue of a drug dose. All three conditions must be met and documented before the first robotic case, and the gate is re-run after any software, model, or configuration change.

Condition	Threshold	Consequence if not met
Unified Safety Level (USL) rating	At least 7.0 on the surgical-readiness scale	No patient-facing robotic activity; remediation and re-rating required
Simulated procedure volume	At least 1000 simulated procedures across at least 2 independent frameworks (Isaac, MuJoCo, Gazebo, PyBullet)	Readiness finding not established; additional simulation required before any case
Sim-to-real fidelity	Trajectory gap less than 2 mm and tip-force gap less than 0.5 N versus the operative record	Gate fails; the system may not proceed and is re-validated after correction

Table 27: Table 6.6. Key inclusion and exclusion criteria for the bound Phase 1 protocol.

Inclusion criteria (all required)	Exclusion criteria (any one excludes)
Signed informed consent before any study-specific procedure, including explicit Physical AI opt-out election or declination	Distant metastatic disease on staging
Age 18 years or older at consent	Vascular involvement precluding a margin-negative (R0) resection, or anatomy incompatible with the eight-arm robotic approach
Histologically or cytologically confirmed resectable or borderline-resectable PDAC on multidisciplinary review	Prior pancreatic resection or prior surgery precluding the planned reconstruction
Documented KRAS G12 mutation (for example G12D, G12V, or G12R) on a validated assay	Uncontrolled intercurrent illness or comorbidity making major surgery or daraxonrasib unsafe
ECOG performance status 0 or 1	Pregnancy or lactation, or unwillingness to use adequate contraception where applicable
Adequate organ and marrow function for major pancreaticobiliary surgery and daraxonrasib	Known contraindication or hypersensitivity to daraxonrasib or a required excipient, or an unmanageable drug interaction
Measured baseline CA 19-9 recorded for serial assessment; eligible for daraxonrasib under RASolute 302 pan-KRAS criteria	Inability to provide informed consent or to comply with the Schedule of Activities
Tumor and vascular anatomy compatible with the eight-arm robotic system on preoperative imaging	Anatomy or comorbidity otherwise incompatible with safe conduct of the combined intervention

Table 28: Table 6.7. The hard cyber-physical safety envelope of the eight-arm robotic platform, with the numeric value and the regulatory or operative role of each parameter. Every value is traceable to the bound protocol and to the Physical AI System Description.

Parameter	Value	Role
Mechanical degrees of freedom	56 (8 arms $\times$ 7)	Operating envelope of the platform
Sensor channels	640 (80 per arm)	100 kHz force; 10 kHz all other
Positioning accuracy	0.05 mm RMS at 1,200 mm/s	Trajectory specification benchmarked by the sim-to-real gap
Per-arm tip-force cap	$\leq$ 3 N	Hard force limit per arm
Cumulative cross-arm cap	$\leq$ 18 N	Hard force limit across arms, enforced at the heartbeat boundary
Heartbeat coordination bus	10 kHz, 64-byte frame	100 μs per-arm watchdog deadline
Emergency arm park	within 50 μs	On a missed or out-of-parameter frame
Cross-arm E-stop	$\leq$ 3 ms at 0.0 N clamp	Halts the full platform
Hardware E-stop	$\leq$ 500 ms	Software-independent; 21 CFR §312.404
Soft-warning vascular zone	3.0 mm	Raises an LLM advisory
No-fly vascular zone	1.0 mm (SMV, PV); 1.5 mm (HA, CA, SMA)	Blocks the command
Hard-stop vascular zone	5.0 mm	Forces the $\leq$ 3 ms E-stop

Table 29: Table 6.8. The perioperative daraxonrasib pause-and-restart advisory mapping and the 32-iteration deterministic sweep distribution. The two inputs are the pancreaticojejunostomy ISGPS fistula grade and the serum daraxonrasib trough relative to the 0.5 ng/mL threshold; the output is a human-confirmed restart day. Baseline serum was 0.45 ng/mL (sub-threshold) across all iterations.

Restart output	Input condition	Sweep frequency	Disposition
T+7d	Grade A fistula with sub-threshold trough (below 0.5 ng/mL)	29 of 32 (90.6%)	Earliest restart; resume daraxonrasib
T+14d	Grade B fistula or a delayed serum rise	3 of 32 (9.4%)	Deferred restart; resume after re-assessment
T+21d or continued hold	Grade C fistula	0 of 32	Restart precluded; maps to the drug stopping rule

Table 30: Table 6.9. The vascular no-fly gate sampled at 10 kHz against five named vessels, with the per-vessel proximity threshold, the system action, and the verdict count in a representative 100-tick sample. SMV, superior mesenteric vein; PV, portal vein; HA, hepatic artery; CA, celiac axis; SMA, superior mesenteric artery.

Zone	Threshold	System action	Sample verdicts
Clear	Beyond all thresholds	Command proceeds normally	83 of 100
Soft-warning	3.0 mm (all five vessels)	Raises an LLM advisory; command continues under oversight	6 of 100
No-fly	1.0 mm (SMV, PV); 1.5 mm (HA, CA, SMA)	Blocks the command	6 of 100
Hard-stop	5.0 mm	Forces the ≤ 3 ms cross-arm E-stop at 0.0 N	5 of 100

Table 31: Table 6.10. The two safety-reporting clocks and the six Physical AI reporting triggers of 21 CFR §312.32(g), with the reporting clock and the evidentiary preservation for each. Every Physical AI adverse event preserves the hash-chained audit trail from minus 24 to plus 72 hours under 21 CFR part 11.

Event	Reporting clock	Evidentiary preservation
Fatal or life-threatening suspected adverse reaction	7 calendar days	Standard pharmacovigilance record
Other serious and unexpected suspected adverse reaction	15 calendar days	Standard pharmacovigilance record
1. Serious Physical AI adverse event	7 or 15 calendar days as applicable	Audit trail -24h to $+72$h
2. System-drug interaction causing incorrect drug administration	15 calendar days (regardless of seriousness)	Audit trail -24h to $+72$h
3. Cybersecurity incident	15 calendar days; 7 days if patient harm is possible	Audit trail -24h to $+72$h
4. Sustained model degradation	15 calendar days; trigger at ≥ 3 consecutive procedures or 24 cumulative hours	Audit trail -24h to $+72$h
5. Sim-to-real divergence beyond 2× tolerance	15 calendar days; trajectory over 4 mm or force over 1.0 N	Audit trail -24h to $+72$h
6. Digital-twin discrepancy beyond accuracy thresholds	15 calendar days	Audit trail -24h to $+72$h

Table 32: Table 6.11. Discontinuation and stopping rules, by the level at which each applies. Device and drug discontinuations are individual-participant or index-procedure rules; the two binding halt rules are study-level and rest with the DSMB. Conversion to open or manual technique is always available and is never itself a protocol violation.

Level	Condition	Action
Device (index procedure)	Surgeon-judged conversion to open or manual technique for safety or oncologic adequacy	Convert; not a protocol violation
Device (index procedure)	Repeated excursions against the ≤ 3 N per-arm or ≤ 18 N cumulative force caps	Discontinue device; convert
Device (index procedure)	Breach of a vascular no-fly zone or a recurring pattern of no-fly approaches	Discontinue device; convert
Device (index procedure)	Sim-to-real divergence beyond $2\times$ tolerance (over 4 mm or over 1.0 N)	Discontinue device; §312.32(g) report
Device (index procedure)	Any hard-stop cross-arm E-stop event	Halt; resume only after cause identified and safety matrix re-passed
Drug (participant)	Drug-attributable DLT in the 28-day window; other unacceptable toxicity; Grade C fistula; pregnancy; intercurrent illness; participant request	Discontinue daraxonrasib; continue safety and survival follow-up
Study (DSMB)	Device- or procedure-related SAE in at least 1 of 3 within a cohort	DSMB review and halt
Study (DSMB)	Two device-related deaths at any time	DSMB review and halt

Table 33: Table 6.12. The seven Physical AI record types retained under 21 CFR §312.57 and 21 CFR part 11, with the content and retention basis of each. Records are retained for at least two years after a marketing application is approved or, if none is filed, two years after the investigation is discontinued and the FDA is notified.

Record type	Content and retention basis
LLM advisory record	Each advisory verdict (ACCEPT, BLOCK, ESCALATE) with the hash-pinned seed and commit; retained for autonomy-bearing audit
Robot command record	Every issued command with the per-arm and cumulative force state at the heartbeat boundary
Sensor and telemetry record	The 640-channel stream (100 kHz force; 10 kHz other) including trajectory accuracy against the 0.05 mm specification
Safety-limit-event record	Every vascular-zone verdict, force-cap excursion, heartbeat or watchdog event, and E-stop activation
Verification-before-generation gate-surface record	The ten-gate suite decisions surrounding each advisory
Human-override and intervention record	Every manual override, conversion, and clinician confirmation
Binding hash-chained audit trail	The append-only chain that binds all of the above; preserved -24h to $+72$h around each Physical AI adverse event

Table 34: Table 6.13. Mapping of the informed-consent process to the basic and additional elements of 21 CFR §50.25, with the Physical-AI-specific content that distinguishes this consent from a drug-only consent.

Consent element (21 CFR §50.25)	Content in this trial
Purpose and investigational nature	First-in-human dose-finding and early-feasibility study; primary purpose is safety and feasibility, not individual benefit
Reasonably foreseeable risks	Daraxonrasib toxicities of the RAS(ON) class; major pancreaticobiliary surgical risks; system-specific device risks
Alternatives	Standard open or conventional robotic pancreaticoduodenectomy without the investigational advisory layer
Confidentiality	HIPAA Safe Harbor de-identification; records available to the FDA
Voluntary participation and withdrawal	Right to withdraw at any time without penalty or loss of entitled benefit
Additional: device disclosure	Distinct disclosure of the on-premises LLM-directed eight-arm system as a second-opinion advisory oracle outside the vendor kinematic stack
Additional: staged autonomy	Clinician-controlled and supervised levels only at this phase; full autonomy prohibited per 21 CFR §312.21(e)
Additional: Physical AI opt-out	Documented right to the operative intervention without the Physical AI advisory layer where clinically feasible, without penalty

Table 35: Table 6.14. The clinical and device facilities supporting the investigation, with the function and the governing regulatory basis of each. The facilities satisfy the adequacy requirement of 21 CFR §312.23(a)(6)(iii)(b) for both the drug and the device components.

Facility	Function	Regulatory basis
High-volume pancreaticobiliary surgical service	Performs the pancreaticoduodenectomy and the open fallback	21 CFR §312.23(a)(6)
Robotic surgical program	Hosts and operates the eight-arm platform	21 CFR §812 (IDE)
Investigational-product pharmacy	Controlled, temperature-monitored storage; drug-accountability log	21 CFR §312.57
Anatomic-pathology service	Standardized synoptic margin protocol with masked adjudication	R0 endpoint
Central imaging capability	RECIST v1.1 reads by a blinded reviewer	Exploratory PFS/OS
Clinical laboratory	Serum daraxonrasib trough assay; safety panels	Safety assessments
On-premises compute environment	Hosts the repository-pinned advisory LLM in network isolation under deny-by-default	21 CFR §312.21(e)
21 CFR part 11 records environment	Maintains the hash-chained, append-only audit trail	21 CFR part 11

Table 36: Table 6.15. The three-tier independent safety-oversight architecture, with the membership, review cadence, and authority of each tier. The binding stop and halt authorities rest with bodies independent of the conduct of the study.

Tier	Membership and cadence	Authority
Independent Safety Monitor (ISM)	Designated for each procedure; independent of the operating team; real-time	Stop authority at the case level
Data and Safety Monitoring Board (DSMB)	Surgical, oncologic, biostatistical, and device-safety members; at each cohort boundary and any triggered review	Binding halt authority; applies the two halt rules
Physical AI Safety Review Committee	Reviews autonomy-bearing behavior, gate-surface decisions, USL, and software changes; cadence not exceeding 90 days and ad hoc	Recommends USL reassessment and oversight changes

Table 37: Table 6.16. The investigator commitments made by executing FDA Form 1572, with the regulatory basis of each. The commitment to read and understand the Physical AI System Description is specific to this trial class.

Commitment	Regulatory basis
Conduct the study in accordance with the protocol and personally conduct or supervise the investigation	21 CFR §312.60
Inform participants that the drug and device are investigational; ensure consent and IRB requirements are met	21 CFR §§312.60, 50.25
Report adverse events to the sponsor	21 CFR §312.64
Read and understand the investigator's brochure and the Physical AI System Description	21 CFR §312.23(g)
Ensure all associates and staff are informed of their obligations	21 CFR §312.53(c)
Maintain adequate and accurate records available for inspection	21 CFR §312.68
Comply with all other requirements of 21 CFR part 312	21 CFR part 312

Table 38: Table 6.17. The principal financial conflict of interest and its structural mitigation, with the independent body to which each binding decision is assigned. No binding safety, escalation, or halt decision rests with the conflicted Sponsor-Investigator.

Binding decision	Independent body and basis
Study-level halt	DSMB, independent of the conduct of the study; applies the two prespecified halt rules
Case-level stop	Independent Safety Monitor, independent of the operating team; per-procedure real-time authority
Autonomy and USL changes	Physical AI Safety Review Committee; cadence not exceeding 90 days and ad hoc
Routine monitoring	CRO independent of the device manufacturer
Audit	Designee independent of the conduct of the study
Financial disclosure	21 CFR part 54; Forms 3454/3455; updated through one year after completion

7 Chemistry, Manufacturing and Control Information

7.1 Chemistry, Manufacturing and Control

This section presents the Chemistry, Manufacturing and Control (CMC) information for the investigational drug daraxonrasib (RMC-6236) supporting the Phase 1, first-in-human, open-label, single-arm investigation described elsewhere in this Investigational New Drug application (IND v1.0, repository v4.3.0). It is organized and submitted in accordance with 21 CFR §312.23(a)(7), which requires sufficient information about the composition, manufacture, and control of the drug substance and the drug product to assure the proper identification, quality, purity, and strength of the investigational drug, scaled to the early phase of investigation, the limited number of participants, and the short duration of exposure characteristic of a Phase 1 study [8]. Consistent with the FDA guidance on the content and format of CMC information for a Phase 1 study, the emphasis here is on the information needed to assure participant safety, including identity, strength, quality, purity, and stability, rather than on the detailed validation data appropriate to later phases [8]. The graded approach embodied in 21 CFR §312.23(a)(7)(i) recognizes that the amount of CMC information needed varies with the phase of investigation, the proposed duration, the dosage form, and the amount of information otherwise available; this submission supplies the identity, strength, quality, purity, and stability data appropriate to a short perioperative dosing schedule in up to eighteen treated participants at a single academic site, with approximately thirty-six participants screened to reach that treated population [8]. As the investigation proceeds and the manufacturing process and analytical methods mature, the CMC information will be updated through information amendments under 21 CFR §312.31 as required, and any change that materially affects the safety of the investigational product, alters the intended strengths, or modifies the container-closure system will be reported before the affected product is administered to participants.

The investigational regimen couples this oral drug with an on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) platform that is the subject of the parallel investigational device exemption content of this submission. The device is a durable instrument and is not a drug, a biologic, or a consumable manufactured lot; its design, specification, calibration, and quality controls are addressed in the device description and the Physical AI System Description fields of 21 CFR §312.23(g), not in this CMC section [1, 2]. The clean separation of the durable robotic instrument and its hash-pinned advisory model from the manufactured drug substance and drug product is deliberate: the verification-before-generation discipline that governs the device, including its ten-gate ACCEPT, BLOCK, and ESCALATE suite and its alignment with the Verification Before Generation in Physical AI Oncology Trials Act of 2026, operates on software and kinematics and has no bearing on the chemical identity or pharmaceutical quality of the tablet [6, 7]. Accordingly, the four CMC subsections below address only the orally administered drug substance and drug product, the non-applicable placebo, and labeling, followed by the environmental assessment. The CMC information architecture and the mapping of each subsection to its content is shown in Figure 17.

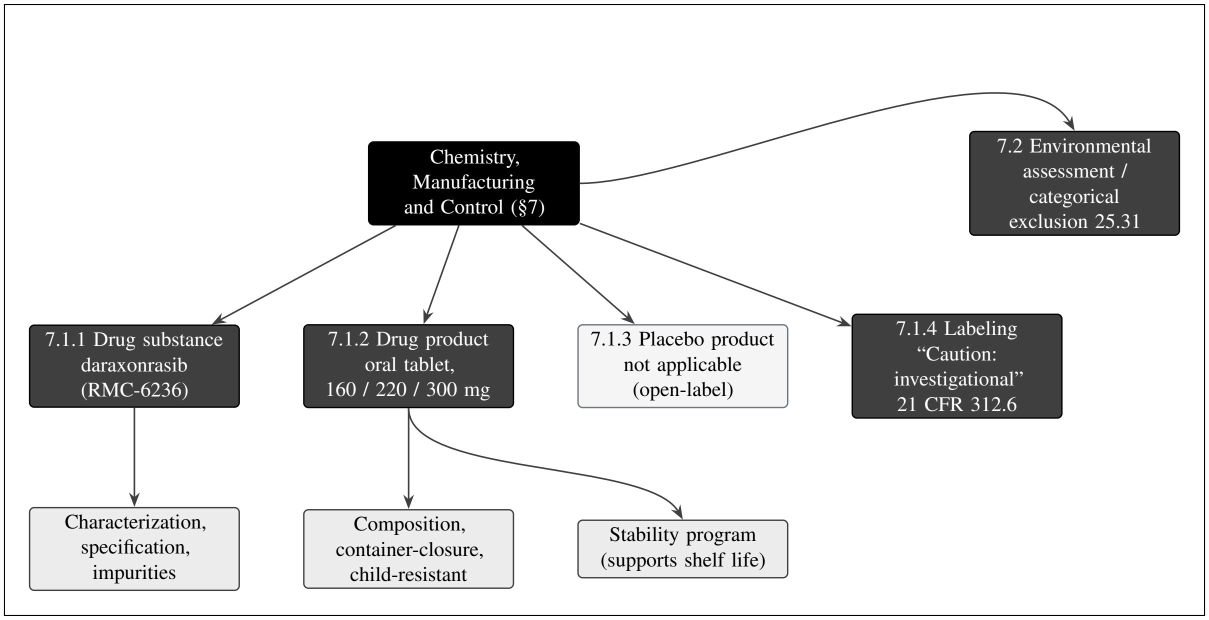

Figure 17. The Chemistry, Manufacturing and Control information architecture for §7. The drug substance (daraxonrasib, RMC-6236) is described by characterization, specification, and impurity controls; the drug product is the oral tablet at the 160, 220, and 300 mg strengths with its composition and child-resistant, light-protective container-closure under 21 CFR §312.6, supported by a stability program; the placebo product is not applicable for this open-label study; labeling carries the investigational caution statement; and an environmental assessment or categorical exclusion under 21 CFR §25.31 is provided.

The remainder of §7.1 follows the drug substance through the drug product to labeling, with each manufactured attribute tied to the test, analytical procedure, and acceptance criterion that controls it. Throughout, the governing principle is that every release and stability acceptance criterion exists to assure one of the five Phase 1 quality attributes named in 21 CFR §312.23(a)(7)(i), namely identity, strength (potency), quality, purity, and stability, and that any attribute not yet supported by a finalized validated method is carried as a provisional interim Phase 1 specification that will be tightened, not loosened, as data accumulate [8]. Table 7.1 maps each of these five attributes to the drug-substance and drug-product controls described below, so that a reviewer can trace any single quality concern from the regulatory requirement to the specific test that addresses it.

Table 39: Mapping of the five Phase 1 CMC quality attributes of 21 CFR §312.23(a)(7)(i) to the drug-substance and drug-product controls described in this section.

Quality attribute	Drug-substance control	Drug-product control
Identity	IR, UV to visible, HPLC retention time, NMR, HRMS, chiral HPLC against reference standard	HPLC retention time plus a spectroscopic identity test on the finished tablet
Strength (potency)	HPLC assay, 98.0 to 102.0 percent on the anhydrous, solvent-free basis	HPLC assay, 95.0 to 105.0 percent of label claim; content uniformity by USP
Quality	Solid-state form, particle size, water, residue on ignition, microbial limits	Appearance, hardness, friability, disintegration, dissolution, microbial limits
Purity	Organic impurities (ICH Q3A), genotoxic impurities (ICH M7), residual solvents (ICH Q3C), elemental impurities (ICH Q3D)	Degradation products (ICH Q3B) controlled to interim Phase 1 limits
Stability	Parallel ICH Q1A program establishing the retest period	ICH Q1A and Q1B program supporting the assigned shelf life

Table 7.1. Each regulatory quality attribute is addressed by at least one drug-substance control and one drug-product control. Interim Phase 1 acceptance criteria are scaled to the early phase of investigation and will be tightened as batch and process knowledge accumulate.

Two cross-cutting commitments frame the CMC information that follows. First, every acceptance criterion presented in this section is an interim Phase 1 criterion: it is set to assure participant safety at the current state of process and analytical knowledge and is expected to tighten, never to loosen, as additional batch data, finalized validated methods, and longer-term stability data accumulate. Second, any change to the drug substance or drug product that could affect the safety, identity, strength, quality, purity, or stability of the investigational product is managed by formal change control and reported to the IND by information amendment under 21 CFR §312.31 before the affected material is administered to participants. Table 7.11, at the close of §7.1, lists the principal CMC changes that trigger an information amendment together with the regulatory basis and the timing of the report; the remainder of this subsection should be read against that change-control framework, so that each specification and each control is understood as the current, amendable expression of an evolving but always safety-protective body of CMC knowledge [8].

7.1.1 Drug Substance

Nomenclature, description, and pharmacological class. The drug substance is daraxonrasib, also designated by the developmental code RMC-6236, an orally bioavailable small molecule of the RAS(ON) multi-selective (pan-RAS) inhibitor class. Daraxonrasib acts by binding the active, guanosine triphosphate (GTP) bound state of mutant RAS in a tri-complex with the endogenous chaperone cyclophilin A, thereby occluding the RAS to effector interface and suppressing downstream mitogen-activated protein kinase and phosphoinositide 3-kinase signaling across the common KRAS oncoprotein variants found in pancreatic ductal adenocarcinoma (PDAC), including the G12D, G12V, and

Table 40: Drug-substance nomenclature, classification, and regulatory descriptors for daraxonrasib (RMC-6236).

Descriptor	Value
Nonproprietary name	Daraxonrasib
Developmental code	RMC-6236
Pharmacological class	RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor
Mechanism of action	Binds the GTP-bound active state of mutant RAS in a tri-complex with cyclophilin A; occludes the RAS to effector interface
Targeted variants in this study	KRAS G12D, G12V, and G12R
Molecular nature	Non-biologic, chirally defined organic small molecule
Physical state	Solid (crystalline)
Route of administration	Oral, once daily
Controlled-substance status	Not a controlled substance
Radioactive-drug status	Not a radioactive drug
Regulatory milestone	FDA Breakthrough Therapy designation (June 2025), KRAS G12 metastatic PDAC
Anchoring recommended Phase 2 dose	300 mg once daily (RASolute 302)

Table 7.2. Nomenclature and regulatory descriptors of the drug substance. The substance is a small-molecule pan-RAS inhibitor and carries no controlled-substance or radioactive-drug obligations; abuse-potential and radioactive-drug considerations are addressed further in §10.

G12R substitutions targeted by this protocol [1, 4]. The agent received FDA Breakthrough Therapy designation in June 2025 for previously treated metastatic PDAC bearing a KRAS G12 mutation, and 300 mg administered orally once daily is the recommended Phase 2 dose established in the pan-KRAS RASolute 302 program to which the dose and exposure assumptions of this protocol are anchored [1, 4]. The drug substance is a non-biologic, chirally defined organic small molecule; it is not a radioactive drug, is not a controlled substance, and is not a peptide, protein, or nucleic acid. The drug substance is presented as a solid, and in the finished product it is formulated as an immediate-release oral tablet. Table 7.2 collects the nomenclature, classification, and regulatory descriptors of the drug substance in the form expected for the drug-substance description of 21 CFR §312.23(a)(7).

Manufacturer and method of preparation. The drug substance is manufactured by, or under contract to, the sponsor's designated active pharmaceutical ingredient manufacturer under current good manufacturing practice (cGMP) appropriate to the investigational stage, and is released against the drug-substance specification summarized in Table 7.3. The synthetic route is a defined, multi-step convergent organic synthesis with isolated, characterized intermediates and controlled reagents, catalysts, and solvents; in-process controls govern the critical steps, and the final isolation and drying steps establish the solid-state form, particle size distribution, and residual-solvent profile of the substance. The name, address, and establishment information of the drug-substance manufacturer, the detailed reaction scheme, the list of starting materials and reagents, the in-process controls, and the batch records are provided in the supporting CMC documentation on file with the sponsor and are cross-referenced here consistent with the Phase 1 level of CMC detail [8]. A statement that the drug substance is manufactured in compliance with cGMP, and the certificate of analysis for the clinical batch, accompany each shipment. Consistent with the Phase 1 graded approach, the submission identifies the regulatory starting materials, describes the controls applied at each critical processing step, and commits to reporting any change to the route of synthesis, the source of a starting material, or the established solid-state form by information amendment under 21 CFR §312.31 before such material is used to manufacture clinical drug product [8]. Because solid-state form and particle size distribution are determined at the final isolation and drying steps and in turn govern dissolution and oral exposure, these two attributes are treated as critical quality attributes of the drug substance and are controlled both as in-process parameters and as release tests. The regulatory starting materials are defined points in the synthesis at which the supplier, the specification, and the control strategy are fixed, and the steps downstream of those starting materials are conducted under cGMP appropriate to the investigational stage with isolated, characterized intermediates; this defined boundary allows the impurity-control strategy of ICH Q3A and the genotoxic-impurity assessment of ICH M7 to be applied consistently from a known, controlled origin to the final drug substance [18]. The certificate of analysis that accompanies each clinical batch reports the result of every test in the drug-substance specification against its acceptance criterion, and the batch is released only when every result conforms; non-conforming material is not released for use in clinical drug-product manufacture.

Characterization. The structure and identity of daraxonrasib are established by a panel of orthogonal physicochemical methods. Structural elucidation and confirmation use proton and carbon nuclear magnetic resonance spectroscopy, high-resolution mass spectrometry, infrared spectroscopy, and ultraviolet to visible spectroscopy; elemental analysis confirms empirical composition. Chiral purity is confirmed by chiral high-performance liquid chromatography (HPLC) to assure the correct stereochemical configuration of the defined stereocenters. The solid-state form (polymorph) is characterized by X-ray powder diffraction and differential scanning calorimetry, and the manufacturing process is controlled to deliver the intended crystalline form, because solid-state form governs dissolution and hence the oral exposure on which the pharmacokinetically gated restart advisory of the protocol depends. Hygroscopicity, aqueous and physiological-pH solubility, the partition coefficient, and the pKa are determined to support the biopharmaceutic understanding of the tablet. These characterization data constitute the reference standard against which clinical batches are compared. The biopharmaceutic relevance of these properties is concrete in this protocol: the perioperative pause-and-restart advisory compares the measured serum daraxonrasib trough against a 0.5 ng/mL threshold, and across the 32-iteration deterministic sweep the modeled baseline trough is 0.45 ng/mL, sub-threshold in every iteration [1]. Because that comparison turns on the oral exposure delivered by the tablet, any drift in solid-state form, particle size, or dissolution that lowered bioavailability would shift the modeled trough and the advisory it informs; controlling these attributes at release therefore protects not only pharmaceutical quality but the integrity of a safety-relevant clinical decision. The advisory itself remains a second-opinion, human-confirmed recommendation and never an autonomous dosing action, consistent with the prohibition on full autonomy under 21 CFR §312.21(e) [2].

Analytical procedures and reference standard. The analytical procedures used to release and to test the stability of the drug substance are the chromatographic, spectroscopic, and physical methods listed in the specification of Table 7.3, applied at the Phase 1 level of method qualification appropriate to a first-in-human study. The assay and related-substances methods are reversed-phase HPLC procedures with ultraviolet detection that have been demonstrated to be stability-indicating, that is, able to resolve the parent drug substance from its known and potential degradation products so that any increase in a degradant on storage is detected and quantified; the chiral HPLC method resolves the defined stereoisomer from its undesired enantiomer; and the headspace gas chromatography, inductively coupled plasma mass spectrometry, Karl Fischer, and X-ray powder diffraction procedures address residual solvents, elemental impurities, water content, and solid-state form respectively. At the Phase 1 stage these procedures are qualified rather than fully validated, with specificity, linearity over the working range, accuracy, precision, and the limits of detection and quantitation established to the extent needed to support the acceptance criteria; full method validation under ICH Q2 is completed as the program advances and is reported by information amendment. A qualified primary reference standard of daraxonrasib, fully characterized by the orthogonal panel described above and assigned a potency on the anhydrous, solvent-free basis, is established and is used to calibrate the assay and identity tests; working reference standards are qualified against the primary standard. Each clinical batch is compared against this reference standard, and the certificate of analysis for each batch reports the result of every test in the specification against its acceptance criterion. Where a new clinical batch is introduced during the investigation, comparability against earlier batches is assessed against the same specification and reference standard so that batch-to- batch consistency is demonstrated before the new batch is administered.

Specification and control of impurities. The drug-substance specification, with tests, analytical procedures, and acceptance criteria appropriate to a first-in-human Phase 1 study, is given in Table 7.3. Organic impurities (process-related and degradation-related), residual solvents, and elemental impurities are controlled to limits consistent with the applicable International Council for Harmonisation (ICH) quality guidelines (ICH Q3A for impurities in new drug substances, ICH Q3C for residual solvents, and ICH Q3D for elemental impurities), as adapted to the limited duration and participant exposure of a Phase 1 investigation [18, 8]. Any individual or total impurity exceeding its qualification threshold is qualified by reference to the nonclinical toxicology program summarized in the pharmacology and toxicology section of this IND. Genotoxic-impurity risk is assessed and controlled under the ICH M7 framework, with structural alerts evaluated and relevant potential mutagenic impurities controlled to the threshold of toxicological concern. The acceptance criteria below are interim Phase 1 specifications and will be tightened as process knowledge and additional batch data accumulate.

A worked example illustrates how the impurity acceptance criteria translate into an absolute daily exposure at the doses studied. At the highest dose level, dose level 3 (DL3), the once-daily dose is 300 mg; an individual organic impurity controlled to not more than 0.5 percent therefore corresponds to at most 1.5 mg of that impurity per day, and the not-more-than 2.0 percent total-impurity limit corresponds to at most 6.0 mg of total impurities per day. At the starting dose level 1 (DL1) dose of 160 mg the same percentage limits correspond to at most 0.8 mg per individual impurity and 3.2 mg total per day, and at dose level 2 (DL2) the 220 mg dose corresponds to at most 1.1 mg and 4.4 mg respectively. These absolute daily amounts, taken across the short perioperative dosing schedule, are the quantities that the nonclinical qualification is required to support, and they remain within the qualification framework of ICH Q3A as adapted for a Phase 1 exposure of limited duration in a population of up to eighteen treated participants [18, 8]. Table 7.4 records these worked daily impurity exposures at each dose level.

7.1.2 Drug Product

Description and composition. The drug product is an immediate-release, film-coated oral tablet supplied in the three strengths required to compose the once-daily doses of the 3+3 escalation: 160 mg for dose level 1 (DL1), 220 mg for dose level 2 (DL2), and 300 mg for dose level 3 (DL3), the last of which corresponds to the RASolute 302 recommended Phase 2 dose [1, 4]. The three strengths are dose-proportional, sharing a common qualitative composition and a common blend so that the only quantitative difference among strengths is the content of drug substance and the corresponding fill weight, with the tablets differentiated by size, debossing, or both to prevent dose confusion in this open-label study. The qualitative and representative quantitative composition is given in Table 7.5. The excipients are standard, compendial pharmaceutical excipients used at conventional levels; the formulation contains a filler or diluent, a binder, a disintegrant, a glidant, a lubricant, and a non-functional aqueous film coat. No excipient of human or animal origin presenting a transmissible spongiform encephalopathy risk is used, and no novel excipient is included. Because no novel excipient is present, no separate excipient safety qualification is required, and the excipient information is provided at the Phase 1 level of detail [8]. The functional roles of the excipients follow conventional immediate-release design: the microcrystalline cellulose and the second filler provide bulk and compactibility to the target fill weight; the binder confers granule and tablet cohesion; the croscarmellose sodium disintegrant promotes rapid break-up in the gastrointestinal tract to support the provisional not-less-than 80 percent dissolved in 30 minutes immediate-release criterion; the colloidal silicon dioxide glidant assures uniform powder flow into the press, which protects weight uniformity and hence content uniformity; the magnesium stearate lubricant, held to a conventional low level, prevents sticking and picking without retarding disintegration; and the hypromellose-based aqueous film coat is cosmetic and non-functional, applied to a target weight gain and not intended to modify release. Because the coat is non-functional, its only release-relevant requirement is that it not retard dissolution, which is confirmed by the dissolution test of the finished-product specification.

The single-strength, dose-proportional, common-blend design has a direct safety and accountability benefit in an open-label 3+3 escalation. Because the three strengths differ only in drug-substance content and fill weight, a single validated blend and a single set of in-process controls support all three presentations, and the physical differentiation by size and debossing provides a visible, dispensing-pharmacy-verifiable distinction among the assigned dose levels so that a participant at DL1 cannot be inadvertently dispensed a DL3 tablet. Table 7.6 summarizes the three strengths, their dose levels, and the identifying features that distinguish them at the investigational-product pharmacy.

Manufacture. The drug product is manufactured under cGMP appropriate to the investigational stage by the sponsor's designated drug-product manufacturer. The process is a conventional solid-oral-dosage process: dispensing and screening of the drug substance and excipients, blending (with wet or dry granulation as defined in the batch record), lubrication, compression of the final blend into core tablets on a rotary tablet press under in-process controls for tablet weight, hardness, thickness, and friability, and aqueous film coating of the cores in a perforated coating pan to a target weight gain, followed by drying. In-process controls at compression include tablet weight and weight uniformity, hardness, thickness, disintegration, and visual inspection; the coating step is controlled for weight gain and appearance. The batch formula, the master and executed batch records, the in-process control limits, and the name and address of the drug-product manufacturer are provided in the supporting CMC documentation and are cross-referenced here at the Phase 1 level of detail [8]. A certificate of analysis accompanies each clinical batch, and each batch is released against the drug-product specification before use. Table 7.7 lists the principal in-process controls applied during manufacture, the manufacturing step at which each is exercised, and its purpose, so that a reviewer can see how finished-product quality is built in rather than tested in.

Drug-product specification. Each clinical batch is tested and released against a drug-product specification that includes appearance, identity of the drug substance (by HPLC retention time and a spectroscopic method), assay (95.0 to 105.0 percent of label claim by HPLC), content uniformity by the USP uniformity-of-dosage-units test, degradation products controlled to interim Phase 1 limits consistent with ICH Q3B (not more than 0.5 percent per specified or unspecified degradant and not more than 2.0 percent total), dissolution by the USP apparatus and medium validated for the product (for example, not less than 80 percent dissolved in 30 minutes as a provisional immediate-release criterion to be confirmed by the developing dissolution method), water content, and microbial limits within USP acceptance criteria for an oral non-sterile product. The dissolution method and acceptance criterion are provisional and will be finalized as biopharmaceutic and stability data accrue. The full interim Phase 1 drug-product release and stability specification is collected in Table 7.8, with each attribute paired to its analytical procedure and its acceptance criterion; the same panel of stability-indicating attributes is monitored on the stability program described below, so that any change on storage is detected against the same criteria that govern release.

Container-closure system. The drug product is packaged in a child-resistant, light-protective, moisture-protective container-closure system, namely high-density polyethylene bottles with a child-resistant, induction-sealed closure and an appropriate desiccant, or equivalent child-resistant unit-dose blister packaging. The container-closure components contacting the product are of compendial pharmaceutical grade and are suitable for an oral solid dosage form; the child-resistant feature satisfies the special-packaging expectation for an investigational oral oncology agent, and the light-protective and moisture-protective features support the stability claim. Packaging and labeling operations are performed under cGMP, and the investigational caution statement and the label content described in Table 7.9 and in §7.1.4 appear on the immediate and any secondary container in accordance with 21 CFR §312.6 [8]. Each container-closure feature is tied to the quality attribute it protects: the high-density polyethylene bottle and induction seal together with the desiccant control moisture ingress and so protect assay and degradation-product limits on storage; the light-protective feature, qualified by the ICH Q1B photostability challenge, protects against photodegradation; and the child-resistant closure satisfies the special-packaging expectation for an oral oncology agent dispensed for at-home once-daily administration during the perioperative schedule.

Stability program. Stability of the drug product is supported by an ongoing stability program conducted under ICH Q1A conditions on representative batches in the proposed container-closure system, using stability-indicating analytical methods. The program comprises long-term, intermediate, and accelerated conditions, with attributes monitored at each station including appearance, assay, degradation products, dissolution, and water content; a photostability evaluation under ICH Q1B and an in-use stability evaluation for the multi-dose bottle presentation are included. The program and its current status are summarized in Table 7.10. The provisional shelf life (expiry or retest period) assigned to the clinical drug product is supported by the available long-term and accelerated data and is applied through expiry dating on the label; product is dispensed only within the labeled expiry or retest date. Stability is monitored throughout the investigation, and the shelf life will be extended through information amendments as additional long-term data become available [8]. The drug substance is also placed on a parallel ICH Q1A stability program to establish its retest period. The accelerated arm at 40 degrees Celsius and 75 percent relative humidity provides early detection of degradation pathways and supports provisional dating and the assessment of any temperature excursion incurred during shipment or site storage; should a significant change occur at the accelerated condition, the intermediate arm at 30 degrees Celsius and 65 percent relative humidity is read to characterize the behavior under the intermediate condition, consistent with the ICH Q1A decision framework. The in-use stability evaluation is specifically relevant to the multi-dose bottle presentation dispensed for at-home once-daily administration during the perioperative schedule: it establishes the period after first opening during which the desiccated, light-protective bottle continues to meet the assay, degradation-product, dissolution, and water-content criteria, and that labeled in-use period is reflected in the storage and handling directions of §7.1.4. Because the perioperative regimen pauses dosing before surgery and restarts it only on the advisory-determined day, a participant's bottle may be opened, set aside during the operative and immediate postoperative interval, and returned to use at restart; the in-use program is designed to cover precisely this open-then-resume pattern, and the unit-dose blister alternative is available where an even more conservative moisture control is preferred. Any temperature excursion incurred during shipment or site storage is assessed against the accelerated-arm data under a documented excursion protocol, and product is returned to use only when the excursion is shown to fall within the supported range; otherwise the affected units are quarantined and destroyed under the drug-accountability procedure.

7.1.3 Placebo Product

This is an open-label, single-arm, first-in-human study; there is no comparator arm, no masking of treatment assignment, and consequently no placebo product. The absence of a placebo is consistent with the dose-finding and early-feasibility objectives of a Phase 1 investigation and with ICH E9 principles for early-phase trials [18, 8]. In accordance with the ReGARDD IND content map, this subsection is retained for completeness and to preserve the table-of-contents numbering, and the placebo product is recorded here as not applicable. The bias-control measures appropriate to an open-label design, including the masking of the pathologist who determines R0 resection status and the central blinded radiographic reader, are described in the study protocol and in §6.1 of this IND rather than through a placebo. These outcome-assessor masking measures are methodological controls on endpoint adjudication, not pharmaceutical products, and accordingly impose no manufacturing, container-closure, or labeling obligation under this CMC section; in particular, no placebo-matching tablet is manufactured, and the drug-product labeling of §7.1.4 is constructed so that nothing appearing on labeling provided to a masked assessor reveals the assigned dose level or any outcome. Should a future amendment introduce a randomized or placebo-controlled component, a corresponding placebo product description, composition, and matching strategy, including the appearance, debossing, and container-closure matching needed to preserve any masking, would be submitted by information amendment under 21 CFR §312.31 at that time.

7.1.4 Labeling

The investigational drug product is labeled in accordance with 21 CFR §312.6, which provides that the immediate package of an investigational drug intended for human use bear a label with the statement "Caution: New Drug. Limited by Federal (or United States) law to investigational use," and that the label and labeling bear no false or misleading statement and make no claim that the drug is safe or effective for the purposes under investigation [8]. Because the study is open-label, the participant-facing label identifies the product and the assigned once-daily dose level; no information that would unblind a masked outcome assessor (for example, the masked pathologist or the central radiographic reader) appears on any label or in any labeling provided to those assessors. The required content of the immediate-container label and of the supplied labeling is summarized in Table 7.9. The label is affixed during cGMP packaging operations and is verified at the investigational-product pharmacy against the dispensing record before any unit is released to a participant. The investigational device and its single-use accessories carry the investigational-device caution statement under 21 CFR part 812 and are not relabeled or repackaged at the site; their labeling is addressed in the device content of this submission. The directions for use on the label reinforce the perioperative dosing schedule: the once-daily dose is paused before surgery and restarted only on the day determined by the human-confirmed, LLM-bound restart advisory described in the protocol, so the label instructs the participant to follow the study team's dosing direction and not to resume dosing on their own, and it carries the "swallow whole; do not crush, split, or chew" instruction appropriate to a film-coated immediate-release tablet.

To close §7.1, the change-control framework introduced above is set out explicitly. The CMC information in this section reflects the manufacturing process and analytical methods as currently established for the Phase 1 investigation, and it will evolve as process knowledge accrues, as methods are fully validated under ICH Q2, and as longer-term stability data become available. Every such evolution is governed by formal change control, and any change that could affect the safety of the investigational product, alter the strengths supplied, modify the container-closure system, or change the established route of synthesis or solid-state form is reported to the IND by information amendment under 21 CFR §312.31 before the affected material is administered to participants [8]. Table 7.11 lists the principal CMC changes that trigger an information amendment, the regulatory basis for each, and the timing of the report. This change-control discipline mirrors, in the pharmaceutical domain, the re-validation discipline that governs the device side of this combined submission, where any change to the robotic platform or its hash-pinned advisory model requires the Phase 0 simulation-validation gate to be re-cleared before further patient-facing activity; in both domains a controlled change is permitted only after the relevant safety-protective evidence has been re-established [2].

7.2 Environmental Assessment

The sponsor claims a categorical exclusion from the requirement to prepare an environmental assessment or an environmental impact statement under 21 CFR §25.31(e), which categorically excludes from environmental review the action of issuing an investigational exemption for a new drug intended to be administered to human subjects, provided that the conditions of 21 CFR §25.15(c) and §25.15(d) are met and provided that no extraordinary circumstances exist that would result in a significant environmental effect. The investigation described in this IND meets these conditions. Daraxonrasib will be administered orally to a small investigational population of up to 18 treated participants at a single academic site, at clinical doses of 160 to 300 mg once daily over a limited perioperative schedule; the quantities of drug substance and drug product manufactured, distributed, and used are correspondingly small and are confined to the controlled clinical setting and its regulated pharmaceutical-waste stream. The estimated concentration of the drug substance entering the environment at the expected use level is below the threshold at which an environmental assessment is ordinarily required, and no extraordinary circumstances are present: the action does not significantly affect the quality of the human environment, does not involve a substance that is persistent, bioaccumulative, or otherwise of recognized environmental concern at the quantities used, and does not affect any species or habitat protected under federal environmental law [8].

The scale of material involved is bounded and can be stated explicitly. With a maximum of eighteen treated participants distributed across the three dose levels of the 3+3 design and a once-daily perioperative schedule, the total quantity of daraxonrasib that could be administered over the entire investigation is on the order of grams, not kilograms, and is dispensed, consumed, returned, or destroyed entirely within the controlled clinical setting and its regulated pharmaceutical-waste stream. A bounding arithmetic illustration confirms the order of magnitude: even on the conservative assumption that every one of the eighteen treated participants received the highest dose level of 300 mg once daily for a continuous month of thirty days, the aggregate administered drug substance would be eighteen times 300 mg times thirty days, equal to 162,000 mg, or about 0.16 kg, distributed across the entire study and across the perioperative schedule in which dosing is paused around surgery rather than continuous. The realized total is smaller still, because the lower dose levels deliver 160 mg and 220 mg and because the schedule includes the pre-surgical pause and the restart determined by the human-confirmed advisory rather than uninterrupted daily dosing. This bounding quantity, entering the environment only as the small excreted fraction after metabolism in a population confined to a single academic site and managed through a regulated pharmaceutical-waste stream, is far below any threshold at which an environmental assessment is ordinarily required, and it confirms that the predicted environmental introduction concentration of the drug substance is negligible. Investigational product that is unused, returned, expired, or otherwise unsuitable is destroyed only after sponsor authorization and in accordance with applicable federal, state, and local regulations governing the disposal of pharmaceutical waste, and the destruction is documented in the drug-accountability record, so that no meaningful quantity of the drug substance is released to the general environment. The durable robotic device and its single-use accessories are likewise managed within the regulated clinical-waste stream and introduce no drug substance into the environment. To the sponsor's knowledge, no data or information exist that would indicate that the proposed investigation may significantly affect the quality of the human environment. Accordingly, neither an environmental assessment nor an environmental impact statement is required for this submission, and the categorical exclusion under 21 CFR §25.31(e) is claimed. Should the scope of the investigation expand in a manner that would exceed the conditions of the categorical exclusion, for example a substantial increase in the number of participants, the manufactured quantity, or the duration of exposure, an environmental assessment under 21 CFR §25.40 would be submitted by information amendment before implementing the change.

Table 41: Drug-substance specification for daraxonrasib (RMC-6236): interim Phase 1 release and stability tests, analytical procedures, and acceptance criteria.

Attribute / test	Analytical procedure	Acceptance criterion (interim Phase 1)
Appearance	Visual	White to off-white solid; free of foreign matter
Identity (spectroscopic)	IR; UV to visible	Conforms to reference standard
Identity (chromatographic)	HPLC retention time	Retention time matches reference standard
Identity (structural)	1H and 13C NMR; HRMS	Consistent with assigned structure
Stereochemical purity	Chiral HPLC	Single defined stereoisomer; undesired enantiomer not more than 0.5 percent
Assay (content)	HPLC with UV detection	98.0 to 102.0 percent on the anhydrous, solvent-free basis
Organic impurities (each)	HPLC, related substances	Not more than 0.5 percent per specified or unspecified impurity (ICH Q3A)
Organic impurities (total)	HPLC, related substances	Not more than 2.0 percent total
Genotoxic impurities	LC-MS or GC-MS, ICH M7	Below the threshold of toxicological concern for specified mutagenic impurities
Residual solvents	Headspace GC	Within ICH Q3C Class limits for solvents used in synthesis
Elemental impurities	ICP-MS	Within ICH Q3D oral permitted daily exposure limits
Water content	Karl Fischer titration	Not more than 1.0 percent w/w
Residue on ignition / sulfated ash	USP general chapter	Not more than 0.2 percent
Solid-state form	X-ray powder diffraction	Conforms to the designated reference polymorph
Particle size distribution	Laser diffraction	Within the validated process range supporting dissolution
Microbial limits	USP total aerobic / yeast and mold	Within USP limits for an oral non-sterile substance

Table 7.3. Interim Phase 1 drug-substance specification. Acceptance criteria are scaled to the early phase of investigation and will be tightened as batch and process data accumulate; impurity limits reference ICH Q3A, Q3C, Q3D, and M7 as adapted for Phase 1.

Table 42: Worked daily impurity exposure at each 3+3 dose level, derived from the interim percentage acceptance criteria of the drug-substance specification.

Dose level	Daily dose	Max single impurity (0.5 percent)	Max total impurities (2.0 percent)
DL1 (start)	160 mg	0.8 mg per day	3.2 mg per day
DL2	220 mg	1.1 mg per day	4.4 mg per day
DL3 (RP2D anchor)	300 mg	1.5 mg per day	6.0 mg per day

Table 7.4. Worked impurity exposures. The percentage acceptance criteria of Table 7.3 translate into the absolute daily amounts shown; these are the quantities the nonclinical qualification under ICH Q3A is required to support across the short perioperative dosing schedule.

Table 43: Representative qualitative and quantitative composition of the daraxonrasib immediate-release oral tablet (per the 160 mg strength; the 220 mg and 300 mg strengths are dose-proportional from a common blend).

Component	Function	Quantity (160 mg strength) and notes
Daraxonrasib (RMC-6236)	Active pharmaceutical ingredient	160 mg per tablet (220 mg / 300 mg in higher strengths)
Microcrystalline cellulose	Filler / diluent	Quantity sufficient to target fill weight
Lactose monohydrate or mannitol	Filler / diluent	Compendial grade, conventional level
Povidone or hypromellose	Binder	Conventional level
Croscarmellose sodium	Disintegrant	Conventional level for immediate release
Colloidal silicon dioxide	Glidant	Conventional level
Magnesium stearate	Lubricant	Conventional level (typically not more than 1.0 percent)
Film coat (hypromellose-based)	Aqueous non-functional coat	Applied to target weight gain; removed as water on drying
Purified water	Coating vehicle	Removed during processing (not present in finished tablet)

Table 7.5. Drug-product composition. All excipients are compendial and used at conventional levels; the film coat is non-functional and cosmetic. The 220 mg and 300 mg strengths are manufactured dose-proportionally from a common blend, differing only in drug-substance content and fill weight and in tablet size or debossing for identification.

Table 44: Drug-product strengths mapped to the 3+3 dose levels, with the identifying features that distinguish them at dispensing.

Strength	Dose level	Once-daily dose	Identification feature
160 mg	DL1 (start)	160 mg	Smallest tablet; distinct debossing; verified at dispensing
220 mg	DL2	220 mg	Intermediate tablet size; distinct debossing
300 mg	DL3	300 mg	Largest tablet; distinct debossing; equals RASolute 302 recommended Phase 2 dose

Table 7.6. Strength-to-dose-level mapping. The dose-proportional common-blend design and the physical differentiation among strengths support accurate dispensing and drug accountability in this open-label study; no strength carries any masking obligation because the study is open-label.

Table 45: Principal in-process controls applied during manufacture of the daraxonrasib oral tablet.

In-process control	Manufacturing step	Purpose
Blend uniformity	Blending / lubrication	Assures uniform drug-substance distribution before compression, supporting content uniformity
Tablet weight and weight uniformity	Compression	Controls dose delivered per tablet; supports assay and uniformity of dosage units
Hardness	Compression	Assures mechanical integrity and consistent disintegration and dissolution
Thickness	Compression	Confirms consistent tooling and compaction
Friability	Compression	Limits chipping and dust during coating and packaging
Disintegration	Compression	Early surrogate confirming immediate-release behavior
Coating weight gain	Film coating	Controls cosmetic coat to target; non-functional coat must not retard release
Appearance	Coating / inspection	Confirms intact, defect-free, correctly identified tablets

Table 7.7. In-process controls. Quality is built into the product at each step; the listed controls govern the critical compression and coating operations and feed forward to the finished-product release tests of Table 7.8.

Table 46: Drug-product specification for the daraxonrasib oral tablet: interim Phase 1 release and stability tests, analytical procedures, and acceptance criteria.

Attribute / test	Analytical procedure	Acceptance criterion (interim Phase 1)
Appearance	Visual	Film-coated tablet of defined color and shape; correct debossing; defect-free
Identity (chromatographic)	HPLC retention time	Matches reference standard
Identity (spectroscopic)	UV to visible or IR	Conforms to reference standard
Assay (label claim)	HPLC with UV detection	95.0 to 105.0 percent of label claim
Content uniformity	USP uniformity of dosage units	Meets USP acceptance value
Degradation products (each)	HPLC, stability-indicating	Not more than 0.5 percent per specified or unspecified degradant (ICH Q3B)
Degradation products (total)	HPLC, stability-indicating	Not more than 2.0 percent total
Dissolution	USP apparatus, validated medium	Not less than 80 percent dissolved in 30 minutes (provisional)
Water content	Karl Fischer titration	Within the limit supporting stability
Microbial limits	USP total aerobic / yeast and mold	Within USP limits for an oral non-sterile product

Table 7.8. Interim Phase 1 drug-product specification. The dissolution method and limit are provisional pending the developing method; degradation limits reference ICH Q3B; the same stability-indicating attributes are monitored on the stability program of Table 7.10.

Table 47: Drug-product stability program for the daraxonrasib oral tablet: storage conditions, time points, attributes monitored, and purpose.

Condition	Storage	Time points (months)	Attributes monitored / purpose
Long-term (ICH Q1A)	25 C / 60 percent RH	0, 3, 6, 9, 12, 18, 24	Appearance, assay, degradants, dissolution, water; supports assigned shelf life
Intermediate (ICH Q1A)	30 C / 65 percent RH	0, 6, 12	Same panel; triggered if a significant change occurs at accelerated
Accelerated (ICH Q1A)	40 C / 75 percent RH	0, 1, 2, 3, 6	Same panel; supports provisional dating and excursion assessment
Photostability (ICH Q1B)	ICH option 1 light exposure	Single challenge	Confirms need for light-protective packaging
In-use stability	25 C / 60 percent RH, bottle opened	Per protocol after first opening	Supports the labeled in-use period for the multi-dose bottle
Temperature excursion	Per documented excursion	As incurred	Supports continued-use decisions for monitored excursions

Table 7.10. Drug-product stability program. Stability-indicating methods are used at every station; the provisional shelf life is supported by the long-term and accelerated data and is extended by information amendment as additional long-term data accrue.

Table 48: Investigational-product label and labeling content for the daraxonrasib oral tablet under 21 CFR §312.6.

Label element	**Content and basis**
Investigational caution statement	"Caution: New Drug. Limited by Federal (or United States) law to investigational use" (21 CFR §312.6)
Product identity and strength	Daraxonrasib (RMC-6236) oral tablet; 160 mg, 220 mg, or 300 mg as assigned
Protocol and IND reference	Protocol number and IND number; sponsor protocol identifier
Lot and batch identification	Manufacturing lot number and a unique container identifier for accountability
Storage conditions	"Store at controlled room temperature; protect from light and moisture; keep in original container"
Expiry or retest date	Date supported by the stability program; not dispensed beyond this date
Quantity and directions	Net contents and the once-daily directions for use; "Swallow whole; do not crush, split, or chew"
Child-resistant statement	"Keep out of reach of children"; child-resistant closure per special packaging
Sponsor identification	Name and address of the sponsor-investigator (ChemicalQDevice)
Unblinding control	No information appearing on labeling provided to masked assessors that would reveal dose level or outcome

Table 7.9. Label and labeling content. The immediate package bears the 21 CFR §312.6 investigational caution statement and makes no claim of safety or effectiveness; the label supports drug accountability and protects the masked outcome assessors of this open-label study.

Table 49: Principal CMC changes that trigger an information amendment under 21 CFR §312.31, with regulatory basis and reporting timing.

CMC change	Regulatory basis	Reporting timing
Change to the route of synthesis or a starting-material source	21 CFR §312.31; ICH Q3A	Reported by information amendment before affected drug substance is used
Change to the established solid-state form or particle size process range	21 CFR §312.31	Reported before affected material is administered; supported by dissolution data
Tightening of an interim acceptance criterion	21 CFR §312.31	Reported at the next information amendment as data support tightening
Introduction of a new clinical batch	21 CFR §312.31	Comparability assessed against specification and reference standard before administration
Change to the container-closure system	21 CFR §312.31	Reported before the new system is used; supported by stability bridging
Extension of the assigned shelf life	21 CFR §312.31	Reported as additional long-term stability data accrue
Finalization of a validated method (for example, dissolution)	21 CFR §312.31; ICH Q2	Reported when the validated method and its criterion replace the provisional one
Introduction of a placebo or comparator (future amendment)	21 CFR §312.31	Full placebo description, composition, and matching strategy submitted before use

Table 7.11. CMC change control. Each listed change is managed by formal change control and reported by information amendment under 21 CFR §312.31 before the affected material is administered, so that the CMC information remains current and safety-protective throughout the investigation.

8 Pharmacology and Toxicology Information

8.1 Pharmacology and Drug Distribution

This section presents the pharmacology and toxicology information supporting the Phase 1, first-in-human, open-label, single-arm investigation of perioperative daraxonrasib (RMC-6236) in resectable and borderline-resectable pancreatic ductal adenocarcinoma (PDAC) described elsewhere in this Investigational New Drug application (IND v1.0, repository v4.3.0). It is organized and submitted in accordance with 21 CFR §312.23(a)(8), which requires adequate information about the pharmacological and toxicological studies of the drug, on the basis of which the sponsor has concluded that it is reasonably safe to conduct the proposed clinical investigation, including an integrated summary of the toxicological effects of the drug in animals and in vitro and, where applicable, a full data tabulation and the names and qualifications of the individuals who evaluated the studies [8]. Consistent with the FDA guidance on the content and format of an IND for a Phase 1 study, the depth of the pharmacology and toxicology information presented here is scaled to the early phase of investigation, to the small number of participants (up to eighteen treated, of approximately thirty-six screened, at a single academic site enrolling at roughly one to two participants per month), and to the short, bounded duration of drug exposure characteristic of a perioperative dose-finding study [8]. The governing regulatory architecture of this submission is a combined IND under 21 CFR Part 312 paired with an investigational device exemption under 21 CFR Part 812 for the on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy platform, and the pharmacology and toxicology content of §8 addresses only the drug article; the device carries no pharmacologic dose and is treated under the device sections of this application [1, 2].

Daraxonrasib is a clinically advanced investigational agent with an established nonclinical and clinical pharmacology package generated within its late-phase sponsor program, and it carries FDA Breakthrough Therapy designation granted in June 2025 for previously treated metastatic PDAC harboring KRAS G12 mutations [1]. The present IND does not seek to recapitulate that established package; rather, it cross-references the existing nonclinical pharmacology and toxicology of daraxonrasib and supplements it with the study-specific computational pharmacology, drug-distribution, and toxicology evidence that is unique to the perioperative, curative-intent surgical context and to the combined drug-device use evaluated here [1, 2]. That supplementary evidence is the principal subject of this section and consists of three classes of in silico work: an AI digital-twin PDAC simulation [14], a quantitative systems pharmacology (QSP) metastatic-PDAC trial simulation with verification, validation, and uncertainty quantification (VVUQ) [13], and a 100,000-patient in silico Phase 3 five-arm PDAC trial conducted in triplicate [15, 16]. The combined regimen also pairs the oral drug with an on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy (Whipple) platform that is the subject of the parallel investigational device exemption content of this submission; the device has no pharmacologic dose and is not addressed as a pharmacology or toxicology article here, but the operative and pathology outputs of the device procedure are inputs to the perioperative drug pause-and-restart advisory and therefore bear on the drug-distribution discussion that follows [1].

The remainder of this section is organized into the three subsections required by the integrated-summary structure of 21 CFR §312.23(a)(8): a pharmacology summary and conclusions, an integrated toxicology summary, and a full data tabulation. The pharmacology summary states the RAS(ON) mechanism of action, the pharmacodynamics, and the drug-distribution basis for the perioperative pause-and-restart schedule, including the 32-iteration deterministic pharmacokinetic sweep anchored to a 0.45 ng/mL operative-window baseline serum concentration. The integrated toxicology summary states the safety margins supporting the conservative DL1 160 mg once-daily first-in-human starting dose and the dose-limiting toxicity (DLT) framework that bounds the escalation. The full data tabulation summarizes

Table 50: Mapping of §8 content to the constituent requirements of 21 CFR §312.23(a)(8), with the source of each element for this clinically advanced agent.

Regulatory requirement	Location in this IND	Source of the evidence
Pharmacological effects and mechanism of action (21 CFR §312.23(a)(8)(i))	§8.1.1 Pharmacology Summary and Conclusions	Established RAS(ON) tri-complex mechanism cross-referenced to the sponsor program; restated for the perioperative indication [1].
Absorption, distribution, metabolism, excretion (21 CFR §312.23(a)(8)(i))	§8.1.1, drug-distribution paragraph and Table 8.3	Study-specific 32-iteration deterministic perioperative pharmacokinetic sweep; oral once-daily exposure anchored to the 300 mg reference dose [2].
Integrated summary of toxicological effects (21 CFR §312.23(a)(8)(ii))	§8.1.2 Toxicology: Integrated Summary	Established nonclinical toxicology cross-referenced; supplemented by QSP and digital-twin computational toxicology [13, 14].
Full data tabulation and evaluator qualifications (21 CFR §312.23(a)(8)(ii))	§8.1.3 Toxicology: Full Data Tabulation	Established package incorporated by reference; study-specific computational works tabulated as supporting documents [8, 1].

the tabulated computational studies that are provided as supporting documents to this IND. The overall structure of §8 and its relationship to the computational-evidence base is described in Figure 17, which is rendered with the Investigator Brochure content of §5 and is not reproduced here; the present section relies on tables and prose.

Regulatory mapping of §8 to 21 CFR §312.23(a)(8). Before turning to the substantive pharmacology, it is useful to state explicitly how the content of this section maps to the three constituent requirements of the governing regulation, so that a reviewer can confirm completeness at a glance. Table 8.1 sets out that mapping. The regulation at 21 CFR §312.23(a)(8)(i) calls for a pharmacology and drug-distribution section describing the pharmacological effects and mechanism of action of the drug in animals and the absorption, distribution, metabolism, and excretion of the drug, if known; the regulation at 21 CFR §312.23(a)(8)(ii) calls for an integrated summary of the toxicological effects of the drug in animals and in vitro; and where appropriate a full data tabulation and a statement of the names and qualifications of the individuals who evaluated the studies. For a clinically advanced agent the depth of original nonclinical tabulation is reduced by cross-reference to the established sponsor package, and the study-specific computational and perioperative drug-distribution evidence is supplied de novo, as the table records [8, 1].

Table 8.1. Mapping of the pharmacology and toxicology section to the three constituent requirements of 21 CFR §312.23(a)(8). Established nonclinical content is incorporated by cross-reference; the perioperative drug-distribution and computational evidence is supplied de novo.

8.1.1 Pharmacology Summary and Conclusions

Mechanism of action and pharmacological class. Daraxonrasib (RMC-6236) is an orally bioavailable, RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor. It acts by binding the active, guanosine triphosphate (GTP) bound state of mutant RAS in a tri-complex with the ubiquitous chaperone protein cyclophilin A, and the resulting daraxonrasib to cyclophilin A to RAS(ON) complex sterically occludes the effector-binding interface, suppressing downstream mitogen-activated protein kinase and phosphoinositide 3-kinase effector signaling [1]. Because the agent targets the shared active conformation rather than a single mutant side chain, it provides broad coverage across the common KRAS oncoprotein variants found in PDAC, including the G12D, G12V, and G12R alleles that define the molecular eligibility for this study [1, 2]. The pharmacological class is therefore a RAS(ON) multi-selective inhibitor, and the mechanistic rationale for evaluating the agent in resectable and borderline-resectable PDAC is the suppression of micrometastatic outgrowth during the perioperative window when systemic therapy must otherwise be interrupted for surgery. The dose and exposure assumptions used throughout this IND are anchored to the pan-KRAS RASolute 302 program, in which 300 mg once daily is the established recommended Phase 2 dose, and to the broader late-phase program comprising RASolute 302 and the first-line RASolve 301 study [1]. Because daraxonrasib engages the shared GTP-bound conformation common to the principal mutant alleles rather than discriminating among individual side chains, the molecular eligibility used in this study, namely documented KRAS G12 status (G12D, G12V, or G12R), is satisfied by the same broad pan-KRAS criteria that define eligibility in RASolute 302, and no allele-specific dose adjustment is anticipated within the studied range [1, 2]. This multi-selectivity is the pharmacologic property that distinguishes the agent from allele-locked inhibitors and that motivates its evaluation as the systemic backbone of the perioperative, curative-intent regimen described in this IND.

The biochemical logic of the tri-complex mechanism merits a more granular statement, because it is the basis on which the perioperative schedule re-uses an established exposure-response relationship rather than generating a new one. In the canonical RAS cycle, the oncoprotein toggles between an inactive GDP-bound state and an active GTP-bound state; oncogenic G12 substitutions impair the intrinsic and GTPase-activating-protein-stimulated hydrolysis of GTP and thereby bias the protein toward the active, signaling-competent conformation. Allele-specific covalent inhibitors of the prior generation captured a single mutant in a defined pocket and were therefore limited to one or two alleles. Daraxonrasib instead recruits the abundant intracellular immunophilin cyclophilin A and presents a composite surface that binds the GTP-bound state across the spectrum of G12 and other RAS variants, so that a single agent suppresses the dominant oncogenic driver in the large majority of PDAC tumors, in which KRAS mutation is nearly universal and the G12D, G12V, and G12R alleles predominate [1, 4]. The pharmacological consequence relevant to §8 is that target engagement, and hence the pharmacodynamic effect, is governed by free systemic drug concentration acting on a target whose conformational availability is shared across the eligible alleles; this is what allows the perioperative pause-and-restart schedule to be reasoned about in terms of a single serum-concentration threshold rather than an allele-by-allele exposure model [2, 13].

Table 8.2 places daraxonrasib in its pharmacological class relative to the prior generation of RAS-pathway inhibitors and records, for each defining attribute, the property of daraxonrasib and the implication of that property for the present perioperative study. The table is intended to make explicit why the agent is a suitable systemic backbone for a curative-intent perioperative regimen in a broadly KRAS-mutant disease.

Table 51: Pharmacological class attributes of daraxonrasib and their implications for the perioperative PDAC study.

Attribute	Property of daraxonrasib	Implication for this study
Molecular target	GTP-bound (active) state of mutant RAS, captured in a tri-complex with cyclophilin A	Engages the shared signaling-competent conformation rather than a single side chain; one agent covers the eligible alleles [1].
Allele coverage	Multi-selective across KRAS G12 variants including G12D, G12V, G12R	Molecular eligibility (documented KRAS G12) is satisfied by the broad pan-KRAS criteria of RASolute 302; no allele-specific dose change anticipated [2].
Route and schedule	Oral, once daily	Compatible with a perioperative pause-and-restart schedule keyed to surgical recovery; exposure is interruptible and re-establishable [2].
Downstream effect	Suppression of MAPK and PI3K effector signaling	Mechanistic surrogate for suppression of micrometastatic outgrowth during the operative interruption window.
Reference exposure	300 mg once daily established as the RASolute 302 recommended Phase 2 dose	The entire studied range (160 mg to 300 mg) sits at or below a characterized exposure [1].
Class toxicities	On-target gastrointestinal and dermatologic / mucocutaneous events	Predictable, dose-related, and bounded by the 3+3 DLT framework and protocol supportive care [2].

Table 8.2. Pharmacological class attributes of daraxonrasib as a RAS(ON) multi-selective inhibitor and their implications for the perioperative, curative-intent PDAC study. The shared-conformation mechanism is what permits a single serum-concentration threshold to govern the perioperative schedule.

Pharmacodynamics. The pharmacodynamic profile of daraxonrasib follows directly from its mechanism: target engagement is a function of the fraction of GTP-bound mutant RAS that is captured in the inhibitory tri-complex, which in turn tracks systemic drug exposure. The pharmacodynamic markers relevant to this perioperative study, the direction of the on-treatment change, and the way each marker is used in the study are summarized in Table 8.3. The dominant on-target, class-characteristic toxicities, principally gastrointestinal events (nausea, diarrhea, stomatitis) and dermatologic and mucocutaneous events, are themselves pharmacodynamic consequences of pan-RAS pathway suppression and are managed by the supportive-care and dose-modification machinery of the protocol rather than by withholding the mechanism [2].

A defining feature of the perioperative pharmacodynamics is the deliberate, schedule-driven oscillation of exposure: continuous pre-operative dosing drives target engagement and effector suppression to a working steady state; the protocol-mandated pre-operative pause then allows systemic exposure to wash out below the 0.5 ng/mL trough threshold so that the operative window is conducted at a sub-threshold concentration; and the human-confirmed postoperative restart re-establishes exposure at the earliest point that does not jeopardize anastomotic healing. The pharmacodynamic intent of this oscillation is to

Table 52: Pharmacodynamic markers of daraxonrasib relevant to the perioperative PDAC study and their use in the protocol.

Pharmacodynamic marker	On-treatment direction	Use in this study
RAS(ON) tri-complex target engagement	Increases with exposure	Mechanistic basis for exposure-anchored dose escalation; not measured directly in participants but modeled in the QSP and digital-twin evidence [13].
Downstream MAPK / PI3K effector signaling	Suppressed	Mechanistic surrogate for antitumor effect; informs the micrometastatic-outgrowth rationale for restarting drug early postoperatively.
Serum daraxonrasib trough concentration	Rises to steady state on continuous dosing; falls during the protocol-mandated operative pause	Objective adherence confirmation and the quantitative input (relative to the 0.5 ng/mL threshold) to the perioperative restart advisory (§5, §6).
Gastrointestinal and dermatologic class effects	Increase with exposure	On-target pharmacodynamic toxicities detected and bounded by the 3+3 DLT framework; managed by protocol supportive care.
CA 19-9 tumor marker (exploratory)	Expected to fall with antitumor response	Exploratory biochemical correlate of systemic disease control across the perioperative course.

minimize the duration over which micrometastatic disease is released from pan-RAS suppression while honoring the surgical-safety constraint. Because the present study does not directly assay tumor-tissue RAS occupancy in participants, the pharmacodynamic chain from exposure to effector suppression to antitumor effect is represented in the QSP and digital-twin computational evidence rather than measured in vivo, and the serum trough is used as the single objective, real-time quantitative input to the schedule [13, 14, 2]. The exploratory CA 19-9 tumor marker and standard imaging response provide biochemical and radiographic correlates of systemic disease control across the perioperative course but are not gating for the schedule.

Table 8.3. Pharmacodynamic markers of daraxonrasib and their use in the perioperative PDAC study. Target engagement and effector suppression are mechanistic; the serum trough is the operative quantitative input to the restart advisory.

Drug distribution and the perioperative pharmacokinetic basis. Daraxonrasib is administered orally, once daily, in a perioperative schedule consisting of a defined pre-operative course, a protocol-mandated pause spanning the operative window, and a postoperative restart whose timing is governed by an LLM-bound, human-confirmed advisory [2]. The drug-distribution question that this schedule poses is when, after surgery, systemic exposure should be re-established so as to suppress micrometastatic outgrowth at the earliest point that does not jeopardize anastomotic healing. The advisory takes two inputs, the pancreaticojejunostomy (PJ) International Study Group of Pancreatic Surgery (ISGPS) postoperative pancreatic-fistula grade (A, B, or C) and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold, and it recommends a restart on postoperative day T+7d, T+14d, or T+21d, or a continued hold [2]. The advisory is bound to a deterministic seed and a specific repository commit and is always human-confirmed; it is never an autonomous controller, consistent with the prohibition on full autonomy at 21 CFR §312.21(e) [3]. The drug-distribution logic of the schedule is therefore explicit and re-traceable: the pre-operative pause is required to clear systemic exposure ahead of the operation, the operative-window trough is measured against a single fixed threshold rather than estimated, and the postoperative restart is keyed jointly to that trough and to the dominant determinant of fistula risk, the pancreaticojejunostomy, so that systemic exposure is re-established neither before the highest-risk anastomosis has declared its healing trajectory nor so late that the therapeutic window for micrometastatic suppression is lost [2, 13].

The two-input, four-output structure of the advisory is set out in full in Table 8.4, which gives the decision rule for every combination of ISGPS fistula grade and trough status. The decision matrix is fully enumerated and deterministic; the LLM advisory functions only as a second-opinion oracle whose recommendation is checked against this hard-coded matrix, and any disagreement escalates to human adjudication rather than overriding the matrix [2, 7]. The deferral logic embodied in the matrix is conservative in the direction of surgical safety: a higher fistula grade always defers the restart by at least one week relative to the same trough status at a lower grade, and a Grade C fistula never restarts within the schedule without a separate human determination, mapping instead to the longest deferral or to the protocol drug stop rule.

Table 53: Perioperative restart-advisory decision matrix: the two inputs (ISGPS pancreatic-fistula grade and serum daraxonrasib trough relative to 0.5 ng/mL) and the recommended restart day.

ISGPS PJ grade	Trough below 0.5 ng/mL	Trough at or above 0.5 ng/mL	Rationale
Grade A (biochemical leak only)	Restart T+7d	Restart T+14d	Lowest-risk anastomotic course; earliest re-establishment of exposure when washout is confirmed, deferred one week if exposure has not yet cleared.
Grade B (clinically relevant fistula)	Restart T+14d	Restart T+21d	Clinically relevant fistula requires a longer healing interval; restart deferred to protect the pancreaticojejunostomy.
Grade C (severe fistula)	Restart T+21d or continued hold	Continued hold	Severe fistula; longest deferral or hold, with separate human determination; defers to the protocol drug stop rule.

Governance. The matrix is hard-coded and deterministic; the LLM advisory is a hash-pinned second-opinion oracle bound to a deterministic seed and repository commit, isolated outside the vendor kinematic stack, and always human-confirmed. Full autonomy is prohibited (21 CFR §312.21(e)); disagreement between advisory and matrix escalates to human adjudication.

Table 8.4. The perioperative restart-advisory decision matrix. A higher fistula grade always defers the restart relative to the same trough status at a lower grade; Grade C never restarts within the schedule without a separate human determination.

The pharmacokinetic basis for this schedule was characterized in a 32-iteration deterministic simulation sweep, the results of which are summarized in Table 8.5. Across all 32 iterations the baseline serum daraxonrasib concentration at the operative timepoint was 0.45 ng/mL, that is, below the 0.5 ng/mL trough threshold, reflecting the washout produced by the protocol-mandated pre-operative pause [2]. Under these sub-threshold operative-window conditions the advisory recommended a restart on postoperative day T+7d in 29 of 32 iterations (90.6 percent), corresponding to an ISGPS Grade A fistula with a sub-threshold trough; it recommended T+14d in 3 of 32 iterations (9.4 percent), corresponding to an ISGPS Grade B fistula or a delayed serum rise; and it recommended T+21d in 0 of 32 iterations (0.0 percent) in this sweep. An ISGPS Grade C fistula maps to a T+21d restart or a continued hold and to the drug stop rule of the protocol. The restart-day distribution and the uniform 0.45 ng/mL baseline are the quantitative drug-distribution conclusions that support the perioperative schedule and the advisory thresholds.

A short worked numeric example makes the drug-distribution arithmetic concrete. The operative-window baseline of 0.45 ng/mL sits 0.05 ng/mL below the 0.5 ng/mL threshold, a 10 percent margin (0.05 of 0.50), which is the quantitative expression of the pre-operative washout that the pause is designed to produce. In the sweep, the 29 of 32 T+7d recommendations and the 3 of 32 T+14d recommendations sum to 32 of 32, confirming that every iteration in this sweep fell into the two earliest restart categories, with no iteration reaching T+21d or a hold; this is the expected behavior when the operative-window inputs are dominated by Grade A fistulae at a confirmed sub-threshold trough, per the matrix of Table 8.4. The 90.6 percent T+7d proportion (29 of 32) and the 9.4 percent T+14d proportion (3 of 32)

Table 54: Perioperative pharmacokinetic sweep: 32-iteration deterministic restart-advisory distribution at a uniform 0.45 ng/mL operative-window baseline serum daraxonrasib concentration.

Advisory output	**Iterations (of 32)**	**Proportion**	**Determining condition**
Restart T+7d	29	90.6%	ISGPS Grade A fistula with sub-threshold trough (serum 0.45 ng/mL $<$ 0.5 ng/mL); earliest safe re-establishment of systemic exposure.
Restart T+14d	3	9.4%	ISGPS Grade B fistula or a delayed serum rise; restart deferred one week to protect anastomotic healing.
Restart T+21d	0	0.0%	ISGPS Grade C fistula maps here or to a continued hold; not triggered in this sweep given the uniform Grade A / B inputs.
Continued hold	0	0.0%	Grade C fistula with persistent risk; defers to the protocol drug stop rule.
Baseline (all 32)	32	100%	Operative-window serum daraxonrasib 0.45 ng/mL (sub-threshold) in every iteration, reflecting pre-operative washout.

are exact and reproducible because the sweep is deterministic and seed-bound; re-running the sweep at the pinned commit returns the identical distribution, which is the property that makes the advisory auditable under the deterministic-and-human-confirmed governance model [2, 7].

Table 8.5. The 32-iteration deterministic perioperative pharmacokinetic sweep. The operative-window baseline serum daraxonrasib concentration was 0.45 ng/mL (below the 0.5 ng/mL trough threshold) in all iterations; the advisory recommended T+7d in 29 of 32, T+14d in 3 of 32, and T+21d in 0 of 32.

Absorption, distribution, metabolism, and excretion in the perioperative context. The absorption, distribution, metabolism, and excretion (ADME) properties of daraxonrasib that are material to this study are those that govern how quickly the pre-operative pause clears systemic exposure and how quickly the postoperative restart re-establishes it. Daraxonrasib is orally bioavailable and is dosed once daily, which is consistent with steady-state accumulation on continuous dosing and with washout over a small number of days once dosing is held; the operative-window baseline of 0.45 ng/mL observed uniformly across the 32-iteration sweep is the quantitative signature of that washout under the protocol pause [2]. For the purposes of this Phase 1 study the detailed clinical ADME parameters, including the absolute and relative bioavailability, the volume of distribution, the metabolic pathways, and the routes of excretion, are those characterized in the established late-phase sponsor program and are incorporated by reference; the present submission does not generate new clinical ADME data but uses the established once-daily exposure behavior to justify the timing of the pause and the restart [1, 8]. The perioperative drug-distribution contribution that is unique to this IND is therefore not a new ADME dataset but the explicit, deterministic mapping from the two measurable perioperative inputs (fistula grade and serum trough) to a restart day, validated across the 32-iteration sweep and represented mechanistically in the QSP model [13, 2].

A second drug-distribution consideration specific to the combined regimen is the potential for a system-drug interaction between the operative course on the robotic platform and the systemic pharmacology of daraxonrasib. There is no pharmacokinetic interaction in the conventional sense, because the device administers no pharmacologic agent; the interaction of regulatory interest is operational, namely that an operative complication detected by the device and pathology workflow (most importantly a clinically relevant pancreatic fistula) defers the re-establishment of systemic drug exposure. This operational coupling is precisely what the restart advisory encodes, and it is also the reason that a system-drug interaction is one of the six enumerated Physical AI reporting triggers under 21 CFR §312.32(g), reportable within fifteen calendar days; the drug-distribution schedule and the device safety surveillance are thereby linked through the advisory and through the safety-reporting framework [2, 3].

Pharmacology conclusions. On the basis of the established RAS(ON) multi-selective mechanism, the broad pan-KRAS coverage encompassing the G12 alleles that predominate in PDAC, the exposure-anchored pharmacodynamics summarized in Table 8.3, the enumerated restart matrix in Table 8.4, and the deterministic perioperative pharmacokinetic sweep summarized in Table 8.5, the sponsor concludes that the pharmacology of daraxonrasib supports the proposed perioperative, once-daily oral schedule and the exposure thresholds used by the restart advisory. The 0.45 ng/mL sub-threshold operative-window baseline and the resulting predominant T+7d restart recommendation provide a quantitative basis for re-establishing systemic exposure early, while the deferral logic for higher fistula grades preserves anastomotic healing, so that the schedule is reasonably expected to balance micrometastatic suppression against surgical risk in the study population [2, 13].

8.1.2 Toxicology: Integrated Summary

Conservative first-in-human starting dose and its safety margin. The first-in-human starting dose in this study is DL1, 160 mg once daily, which is deliberately set below the RASolute 302 reference recommended Phase 2 dose of 300 mg once daily [1]. The escalation then steps up through DL2 at 220 mg once daily to DL3 at the 300 mg once-daily reference dose, so that the entire dosing range evaluated in this study (160 mg to 300 mg) lies at or below an exposure that has already been characterized in the late-phase sponsor program [1, 2]. The starting dose therefore carries a built-in safety margin in two senses: it is approximately 53 percent of the established reference dose by milligram amount (160 of 300 mg), and it sits within, rather than above, the exposure range for which the established nonclinical and clinical toxicology of daraxonrasib already supports administration. Because the agent is clinically advanced and its established toxicology package supports dosing at and above the doses studied here, the conservative DL1 starting dose, combined with the bounded perioperative exposure duration, provides the integrated-summary basis for the sponsor's conclusion that it is reasonably safe to begin the proposed clinical investigation [8, 1].

The arithmetic of the safety margin warrants an explicit statement, because it is the quantitative core of the integrated toxicology conclusion. The 160 mg starting dose is 160 of 300, or 53.3 percent, of the reference recommended Phase 2 dose; equivalently, it is a 46.7 percent reduction below that reference. The intermediate DL2 at 220 mg is 73.3 percent of the reference (220 of 300) and a 26.7 percent reduction; the top DL3 at 300 mg equals the reference exactly, at 100 percent. The step from DL1 to DL2 is a 60 mg increment (a 37.5 percent increase over 160 mg), and the step from DL2 to DL3 is an 80 mg increment (a 36.4 percent increase over 220 mg), so the relative increments are comparable and modest, and at no point does the escalation exceed the characterized reference exposure. This is a materially more conservative posture than a conventional first-in-human escalation that would begin at a small fraction of a no-observed-adverse-effect-level-derived dose and climb into uncharacterized territory; here every studied dose is at or below an exposure already administered to humans in the late-phase program [1, 8].

Table 55: Integrated toxicology summary: dose levels, safety margin relative to the reference dose, and the 3+3 dose-limiting toxicity framework.

Level	Dose (oral QD)	Fraction of 300 mg reference	Toxicology basis and escalation rule
DL1	160 mg	53%	Conservative first-in-human starting dose; below the RASolute 302 reference; 3 enrolled, 28-day DLT window; 0/3 escalate, 1/3 expand to 6.
DL2	220 mg	73%	Intermediate level; entered only after DL1 clears the DLT window; same 3+3 expansion and escalation logic.
DL3	300 mg	100%	Equal to the established RASolute 302 recommended Phase 2 dose; highest level studied; MTD if fewer than 2/6 DLTs.
DLT rule. MTD is the highest level with fewer than 2 of 6 DLT-evaluable DLTs; RP2D is selected at or below the MTD. Two device-or-procedure related deaths at any time, or one device-or-procedure serious adverse event within a cohort, trigger DSMB review and a binding halt.			

Dose-limiting toxicity framework. The toxicology of the escalation is bounded prospectively by a standard 3+3 dose-limiting toxicity (DLT) framework with sentinel staggered enrollment, rather than retrospectively. Three participants are enrolled at each dose level and observed across a 28-day DLT window. If no DLT occurs among the first three, the cohort escalates; if one of three experiences a DLT, the cohort expands to six; a single DLT among six permits escalation, whereas two or more DLTs at a level fix the maximum tolerated dose (MTD) at the preceding level. The MTD is the highest level at which fewer than two of six DLT-evaluable participants experience a DLT, and the recommended Phase 2 dose is selected at or below the MTD [2]. The dose levels and their relationship to the reference dose and the DLT framework are summarized in Table 8.6. The class-characteristic on-target toxicities of pan-RAS inhibition, principally gastrointestinal events (nausea, diarrhea, stomatitis) and dermatologic and mucocutaneous events, are the toxicities most likely to manifest as DLTs and are precisely the events the 28-day window is designed to detect and bound; perioperative exposure additionally raises a theoretical risk to wound and anastomotic healing if restart timing is suboptimal, which is the specific hazard the pause-and-restart advisory (Table 8.5) is designed to mitigate [2].

The maximum sample size of the dose-finding portion is eighteen treated participants, comprising three dose levels at up to six DLT-evaluable participants each (3 levels times 6 equals 18), with approximately thirty-six screened to yield that treated number, and with staggered sentinel enrollment so that the first participant at each level is observed through an initial safety interval before the remaining participants of that cohort are dosed. The precision that this sample size affords is modest and non-confirmatory, consistent with the descriptive, estimation-based statistical posture of the study under ICH E9: a level with zero DLTs in three participants carries an exact 95 percent Clopper-Pearson upper confidence bound of approximately 71 percent on the true DLT rate, and zero DLTs in six participants carries an upper bound of approximately 46 percent; across the full eighteen treated, a zero-event result corresponds to a 0 percent point estimate with an exact 95 percent interval of approximately 0 to 18 percent, and a single event corresponds to a 5.6 percent point estimate with an interval of approximately 0.1 to 27 percent [2]. These figures are stated so that the toxicological inference drawn from the 3+3 escalation is understood to be a safety-screening inference, not a precise estimate of the toxicity rate.

Table 56: Anticipated toxicity profile of perioperative daraxonrasib, mechanism, and protocol management.

Anticipated toxicity	Mechanism / origin	Protocol detection and management
Gastrointestinal (nausea, diarrhea, stomatitis)	On-target pan-RAS effector suppression in gastrointestinal epithelium	Dominant class effect; most likely DLT category; protocol supportive care, antiemetics and antidiarrheals; dose interruption or reduction if dose-modification criteria met [2].
Dermatologic / mucocutaneous (rash, mucositis)	On-target effector suppression in skin and mucosa	Common class effect; graded over the 28-day window; topical and supportive management; dose modification if severe.
Wound / anastomotic healing risk	Schedule hazard if exposure re-established before pancreaticojejunostomy healing declared	Not a pharmacologic toxicity per se; mitigated structurally by the deferral logic of the restart advisory keyed to ISGPS grade (Table 8.4).
Cytopenias and laboratory abnormalities	Class and disease-related	Routine on-study laboratory surveillance; reported through the safety framework; dose modification per protocol.
System-drug interaction events	Operative complication deferring drug restart	Reportable Physical AI trigger within 15 calendar days (21 CFR §312.32(g)); coupled to the advisory [3].

Table 8.6. Integrated toxicology summary. The 160 mg starting dose is 53 percent of the 300 mg reference dose, and the entire 160 mg to 300 mg range lies at or below an exposure characterized in the established daraxonrasib program; the 3+3 framework bounds the escalation prospectively.

Anticipated toxicity profile, severity grading, and management. The integrated toxicology summary is most informative when the anticipated on-target toxicities are stated alongside their expected severity distribution and their protocol management, because these are the events the 28-day DLT window must characterize. Table 8.7 sets out the principal anticipated toxicities of pan-RAS inhibition in this perioperative context, the mechanism by which each arises, and the protocol response. The gastrointestinal and dermatologic or mucocutaneous events are the dominant, dose-related, class-characteristic toxicities and are the events most likely to constitute a DLT at the higher dose levels; they are managed in the first instance by protocol supportive care and, where they meet dose-modification criteria, by dose interruption or reduction rather than by abandoning the mechanism. The perioperative-specific hazard, namely a theoretical impairment of wound or anastomotic healing if systemic exposure is re-established before the pancreaticojejunostomy has declared a safe healing trajectory, is not a pharmacologic toxicity of the drug per se but a schedule hazard, and it is mitigated structurally by the deferral logic of the restart advisory rather than by supportive care [2].

Table 8.7. Anticipated toxicity profile of perioperative daraxonrasib. The gastrointestinal and dermatologic events are the dominant, dose-related class effects and the most likely DLT category; the anastomotic-healing hazard is a schedule hazard mitigated structurally by the restart advisory.

Binding halt rules and the safety-reporting interface. The integrated toxicology framework is reinforced by binding halt rules that operate independently of the DLT escalation logic and that bridge the drug and device safety surveillance. One device-or-procedure related serious adverse event within a cohort, or two device-or-procedure related deaths at any time, triggers review by the Data Safety Monitoring Board (DSMB) and a binding halt, with the DSMB holding halt authority at cohort boundaries and on a triggered basis [2]. This halt machinery sits within a three-tier oversight structure comprising the DSMB, a per-procedure Independent Safety Monitor with real-time stop authority, and a Physical AI Safety Review Committee on a cadence of no more than ninety days, all reporting to a reviewing institutional review board. The toxicological events characterized by the 3+3 window and the halt rules described above feed the safety-reporting framework required by 21 CFR §312.32: a fatal or life-threatening suspected adverse reaction is reported within seven calendar days, and any other serious and unexpected suspected adverse reaction within fifteen calendar days; in addition, the combined drug-device regimen carries six enumerated Physical AI reporting triggers under 21 CFR §312.32(g), with audit trails preserved from twenty-four hours before to seventy-two hours after each Physical AI adverse event under 21 CFR Part 11 [3, 2]. The integrated toxicology conclusion thus rests not only on the conservative dose and the DLT framework but on a surveillance and reporting apparatus that bounds the realized risk during conduct.

Computational toxicology and QSP evidence. The integrated toxicology conclusion is reinforced by study-specific computational toxicology evidence. A quantitative systems pharmacology (QSP) metastatic-PDAC trial simulation, conducted with code, verification, validation, and uncertainty quantification (VVUQ), and an accompanying playbook, models the exposure to response and exposure to toxicity relationships of daraxonrasib in a mechanistic systems framework and supports the plausibility of the exposure thresholds and the dose range studied here [13]. An AI digital-twin pancreatic-cancer simulation, developed for in silico regulatory-compliance and cost-efficiency purposes, provides a complementary patient-level toxicodynamic representation [14]. These computational works do not substitute for the established nonclinical toxicology of daraxonrasib; rather, they supplement it with study-specific, in silico characterization of the perioperative context, and they are tabulated as supporting documents in the full data tabulation below [13, 14].

The evidentiary weight that may properly be assigned to each computational work is governed by its verification and validation status, and the integrated toxicology summary is careful not to overstate it. The QSP metastatic-PDAC simulation carries a documented VVUQ record and an accompanying playbook, so its exposure-response and exposure-toxicity relationships are accompanied by an explicit account of model verification, validation against reference behavior, and quantified uncertainty; this is the strongest of the computational evidence streams for toxicological inference and is the one on which the plausibility of the dose range and the exposure thresholds principally rests [13]. The digital-twin simulation provides a patient-level toxicodynamic and disease-progression representation oriented toward regulatory compliance and cost-efficiency, and it is used as a corroborating, individual-patient view rather than as a primary basis for the dose decision [14]. The 100,000-patient in silico Phase 3, run in triplicate, contributes population-scale plausibility for the regimen and the trial design rather than a direct toxicological margin [15, 16]. None of these works is offered in lieu of the established nonclinical toxicology; each is offered as study-specific in silico supplementation, and each is identified, with its form and evidentiary contribution, in the full data tabulation of §8.1.3.

Table 57: Full data tabulation: study-specific computational-evidence supporting documents for the pharmacology and toxicology of the perioperative daraxonrasib PDAC regimen.

Computational study	Form	Evidence contributed to §8
AI digital-twin PDAC simulation [14]	Patient-level digital twin	In silico, regulatory-compliance-oriented toxicodynamic and disease-progression representation of PDAC; supports the perioperative drug-distribution and toxicology reasoning at the individual-patient level.
QSP metastatic-PDAC trial simulation with VVUQ [13]	Mechanistic QSP model + VVUQ + playbook	Exposure-to-response and exposure-to-toxicity relationships in a verified, validated, uncertainty-quantified systems-pharmacology framework; underpins the dose range and exposure thresholds.
100,000-patient in silico Phase 3, 5-arm PDAC, triplicate [15, 16]	Large-scale virtual trial (triplicate)	Population-scale in silico characterization of a five-arm PDAC Phase 3 across 100,000 virtual patients, run in triplicate; supports the population-level plausibility of the regimen and the trial design.
End-to-end PDAC digital-twin trial proposals [16]	Trial-design proposals	End-to-end digital-twin clinical-trial scaffolding adapted for the perioperative drug-device design described in this IND.

Cross-reference. The established nonclinical toxicology full data tabulation for daraxonrasib (RMC-6236) is provided by reference to the sponsor's existing nonclinical package; the qualifications of the evaluators are stated there. The 32-iteration perioperative pharmacokinetic sweep (Table 8.5) is tabulated above.

Integrated toxicology conclusion. Taking together the conservative 160 mg first-in-human starting dose set at 53 percent of an already-characterized reference exposure, the prospective 3+3 DLT framework with a 28-day window and sentinel staggered enrollment, the binding halt rules, the perioperative pause-and-restart advisory that mitigates the theoretical anastomotic-healing hazard, and the supporting QSP and digital-twin computational toxicology, the sponsor concludes that the toxicological profile of daraxonrasib supports the conclusion that it is reasonably safe to conduct the proposed Phase 1 investigation at the doses and schedule specified, within the meaning of 21 CFR §312.23(a)(8) [8, 3].

8.1.3 Toxicology: Full Data Tabulation

Scope and form of the tabulation. For a clinically advanced agent such as daraxonrasib, the full data tabulation required by 21 CFR §312.23(a)(8)(ii) for the established nonclinical toxicology studies is provided by cross-reference to the sponsor's existing nonclinical package, and the qualifications of the individuals who evaluated those studies are stated in that package [8, 1]. What is tabulated here, and provided as supporting documents to this IND, is the study-specific computational-evidence base that characterizes daraxonrasib and the perioperative drug-device regimen in silico. These works were generated by the sponsor-investigator over 2025 and 2026 and comprise the digital-twin, QSP, and large-scale in silico Phase 3 simulations described above. Their identity, the evidence each contributes, and the location of each as a supporting document are tabulated in Table 8.8.

Table 58: Verification, validation, and uncertainty-quantification status of the tabulated computational evidence and the evidentiary role assigned to each work.

Computational study	VVUQ status	Evidentiary role in this IND
QSP metastatic-PDAC simulation [13]	Documented VVUQ record and playbook (verification, validation, uncertainty quantification)	Primary computational basis for the plausibility of the dose range (160 mg to 300 mg) and the exposure thresholds; mechanistic exposure-response and exposure-toxicity.
AI digital-twin PDAC simulation [14]	Regulatory-compliance-oriented; patient-level representation	Corroborating individual-patient toxicodynamic and disease-progression view; not a primary basis for the dose decision.
100,000-patient in silico Phase 3, triplicate [15, 16]	Triplicate replication across a 5-arm population	Population-scale plausibility for the regimen and the trial design; not a direct toxicological margin.
End-to-end digital-twin trial proposals [16]	Design-stage scaffolding	Trial-design adaptation; not a toxicological dataset.

Note. None of the computational works substitutes for the established nonclinical toxicology of daraxonrasib, which is incorporated by reference. Computational evidence is offered as study-specific in silico supplementation only, with evidentiary weight stratified by VVUQ status.

Table 8.8. Full data tabulation of the study-specific computational evidence (digital-twin, QSP with VVUQ, and the 100,000-patient in silico Phase 3 PDAC triplicate) provided as supporting documents, with the established nonclinical toxicology incorporated by cross-reference.

Verification, validation, and uncertainty-quantification status of the tabulated evidence. Because the study-specific evidence is computational, the full data tabulation records not only the identity of each work but the verification and validation posture that determines how much weight it bears, in keeping with the emerging expectation that in silico evidence submitted in support of a clinical investigation be accompanied by an explicit VVUQ account [13, 7]. Table 8.9 sets out, for each tabulated work, the verification and validation status, the form of uncertainty treatment, and the evidentiary role assigned to the work in this submission. The QSP metastatic-PDAC simulation is the only tabulated work with a complete documented VVUQ record and playbook, and it is correspondingly the primary computational basis for the dose-range and exposure-threshold plausibility; the remaining works are corroborating. This explicit stratification of evidentiary weight by VVUQ status is intended to forestall any inference that population-scale or single-patient simulation substitutes for the established nonclinical toxicology, which it does not.

Table 8.9. Verification, validation, and uncertainty-quantification status of the tabulated computational evidence. The QSP simulation with a documented VVUQ record is the primary computational basis; the remaining works are corroborating and do not substitute for the established nonclinical toxicology.

Evaluators and incorporation by reference. The study-specific computational studies tabulated in Table 8.8 were designed, executed, and evaluated by the sponsor-investigator and the study computational team, whose qualifications are stated in the Investigator and Facilities data of this submission; the verification, validation, and uncertainty-quantification methodology applied to the QSP work is documented in that study's accompanying VVUQ record [13]. The established nonclinical pharmacology and toxicology of daraxonrasib, including the full data tabulations and evaluator qualifications for those studies, is incorporated by reference to the sponsor's existing nonclinical package consistent with the Phase 1 IND content guidance [8, 1]. As the investigation proceeds, any new nonclinical or computational toxicology information bearing on participant safety will be submitted through information amendments under 21 CFR §312.31 and, where a safety signal meets the reporting thresholds, through IND safety reports under 21 CFR §312.32 [3].

Lifecycle maintenance of the pharmacology and toxicology record. The pharmacology and toxicology information in this section is a living record over the life of the IND, and the sponsor states explicitly how it will be maintained. New information from the ongoing late-phase sponsor program that bears on the nonclinical or clinical pharmacology and toxicology of daraxonrasib will be reflected through information amendments under 21 CFR §312.31, and updates to the Investigator Brochure will carry the most current integrated risk picture into the consent and protocol machinery of §5 and §6. Should any safety signal arise in this study that meets the reporting thresholds, it will be submitted as an IND safety report under 21 CFR §312.32 within the seven-day or fifteen-day clock as applicable, including the Physical AI triggers under 21 CFR §312.32(g) that are unique to the combined drug-device regimen; the audit-trail preservation window of twenty-four hours before to seventy-two hours after each Physical AI adverse event under 21 CFR Part 11 ensures that the underlying data for any such report are retained and re-traceable [3, 2]. Any change to the perioperative restart advisory, including any change to the 0.5 ng/mL threshold, the deterministic seed, or the pinned repository commit, would require re-validation of the sweep and would be carried into this section through the same amendment machinery, so that the drug-distribution conclusions of §8.1.1 always correspond to the validated configuration in force [2, 7].

Conclusion of the pharmacology and toxicology information. The pharmacology summary (§8.1.1), the integrated toxicology summary (§8.1.2), and the full data tabulation (§8.1.3) together provide adequate information about the pharmacological and toxicological properties of daraxonrasib, scaled to a Phase 1 study, on the basis of which the sponsor has concluded that it is reasonably safe to conduct the proposed perioperative, once-daily oral investigation in resectable and borderline-resectable PDAC at the 160 mg, 220 mg, and 300 mg dose levels under the 3+3 dose-limiting toxicity framework, in satisfaction of 21 CFR §312.23(a)(8) [8, 3, 1].

9 Previous Human Experience

This section assembles the previous human experience that supports the proposed Phase 1 investigation, organized under 21 CFR §312.23(a)(9). It states the marketed status of daraxonrasib (RMC-6236), summarizes the prior and ongoing clinical research experience with the drug, situates the proposed combination within the contemporary clinical care experience for robotic pancreaticoduodenectomy, and identifies the safety-reporting framework under which post-submission human experience will continue to be characterized. Because this is a first-in-human investigation of the drug in a perioperative, curative-intent, device-directed surgical context, the evidence base that substitutes for direct prior experience in that exact setting is drawn from three sources: the late-phase metastatic-disease program for daraxonrasib, the established human experience with robotic pancreaticoduodenectomy benchmarked by the 2025 Dutch nationwide cohort, and the author's 2025 to 2026 computational program that supplies the in silico scaffolding for the device-directed component [1, 2]. Figure 19 maps this evidence base, and the narrative below develops each contributing strand in turn, consistent with the phased-development expectations for an initial Phase 1 submission [8].

The regulatory logic of this section follows directly from the structure of 21 CFR §312.23(a)(9), which requires the sponsor to provide a summary of previous human experience known to the applicant, with emphasis on the relationship of the prior experience to the proposed investigation. For a drug that has accumulated a substantial late-phase exposure history in one disease setting (metastatic PDAC) and that is now being introduced into an adjacent but materially different setting (resectable or borderline-resectable PDAC undergoing curative-intent, device-directed surgery), that relationship is not one of direct precedent but of careful, bounded extrapolation. The present section therefore distinguishes throughout between what the prior human experience establishes (molecular target validity, the upper bound of tolerable systemic exposure, and a reference safety profile for drug-attributed adverse events) and what it does not establish (the safety, tolerability, and feasibility of administering the drug across a major abdominal resection performed by an LLM-directed robotic platform). Everything that the prior experience does not establish is precisely what this first-in-human investigation is designed to characterize, and the descriptive, non-confirmatory statistical posture of the study follows from that gap rather than from any deficiency of the underlying program [8, 1].

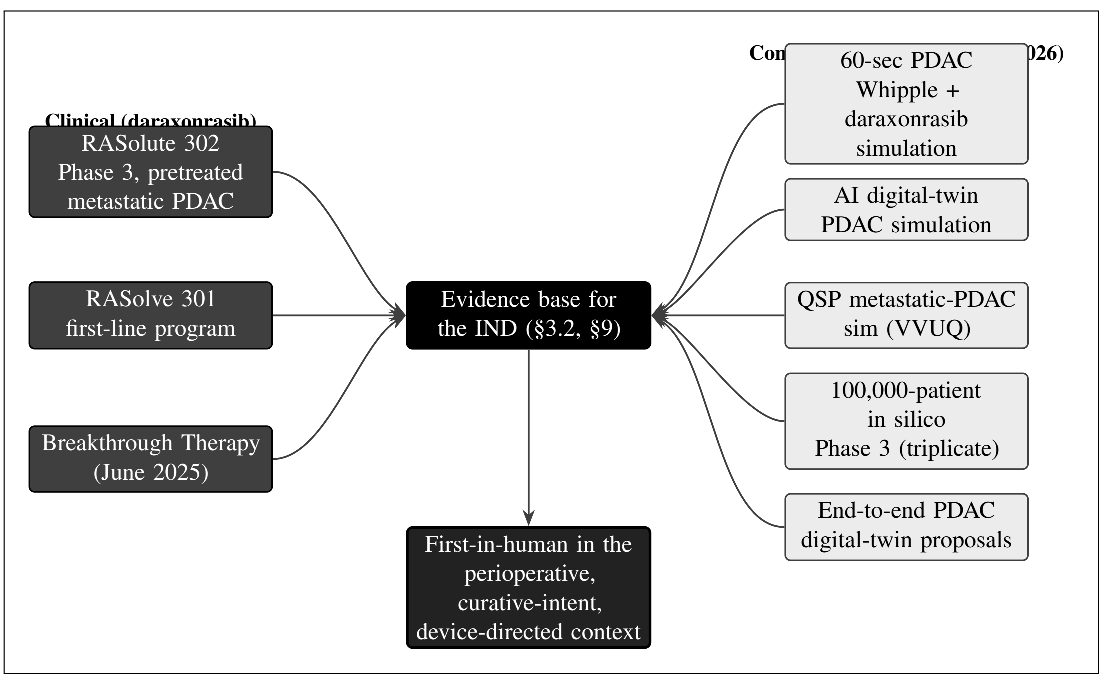

Figure 19. The previous human experience and evidence base. Clinically, daraxonrasib is supported by the RASolute 302 Phase 3 program in previously treated metastatic PDAC, the RASolve 301 first-line program, and its June 2025 Breakthrough Therapy designation; computationally, the author's 2025 to 2026 works supply the simulation scaffolding (the 60-second PDAC Whipple and daraxonrasib simulation, the AI digital-twin PDAC simulation, the quantitative-systems-pharmacology metastatic PDAC simulation with verification, validation, and uncertainty quantification, the 100,000-patient in silico Phase 3 triplicate, and the end-to-end PDAC digital-twin proposals). Daraxonrasib remains first-in-human in this perioperative, curative-intent, device-directed surgical context.

9.1 Marketed experience

Daraxonrasib (RMC-6236) is an investigational new drug and is not approved for marketing in the United States or in any other country for any indication. It is not available commercially, is not the subject of any New Drug Application approval, and therefore has no marketed experience, no postmarketing surveillance history, and no labeling derived from marketing authorization. All human exposure to date has occurred within controlled clinical investigations conducted under investigational status, and the safety and efficacy information available for the drug is accordingly derived entirely from that investigational program rather than from any marketed use [1]. Because there is no marketed experience, this subsection records that fact and points instead to the investigational program characterized in §9.2; no withdrawal, suspension, or restriction of any marketing authorization has occurred in any country, because none exists. The status reported here is current as of the data-cutoff supporting this IND (IND v1.0, repository v4.3.0); should marketing authorization be granted for any indication in any jurisdiction before this IND is in effect, that change would be communicated to the FDA through an information amendment under 21 CFR §312.31, and the marketed-experience characterization in this section would be revised accordingly.

Daraxonrasib is an orally administered, once-daily RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor. Mechanistically, it binds the active, GTP-bound (ON) state of multiple RAS variants in a tri-complex with the intracellular chaperone cyclophilin A and thereby suppresses downstream effector signaling, providing broad pan-KRAS coverage that includes the G12 alleles (G12D, G12V, and G12R) that predominate in pancreatic ductal adenocarcinoma (PDAC). This mechanism and pharmacological class are developed in full in the Introduction (§3.1) of this IND and are summarized here only insofar as they frame the human experience: the drug acts on a validated oncogenic driver in a disease whose five-year overall survival remains below 13 percent, the third leading cause of cancer death in the United States and the fourth in the European Union [4]. The pan-RAS mechanism is material to the previous-human-experience analysis for a specific reason. Because the drug suppresses signaling through the active state of multiple RAS isoforms rather than covalently occupying a single mutant allele, the human experience accumulated against the broad pan-KRAS eligibility of the metastatic program applies, in principle, to any tumor bearing a G12 driver, and the metastatic safety profile is therefore directly transferable as reference safety information for the resectable population enrolled here, even though the resectable population itself has not previously been exposed to the drug [1, 10].

The most significant regulatory milestone in the drug's investigational history to date is the grant of Breakthrough Therapy designation by the FDA in June 2025 for previously treated metastatic PDAC harboring KRAS G12 mutations. Breakthrough Therapy designation is reserved for an investigational drug intended to treat a serious or life-threatening condition for which preliminary clinical evidence indicates that the drug may demonstrate substantial improvement over available therapy on one or more clinically significant endpoints; the designation does not constitute marketing approval and does not alter the investigational status of the drug, but it does signal that the FDA has reviewed preliminary clinical data and found a credible signal of substantial improvement [1, 8]. For the purposes of this IND, the designation is relevant as a qualitative indicator of the maturity and promise of the human experience accumulated in the metastatic setting, not as a basis for any efficacy claim in the perioperative, resectable population studied here. No claim of efficacy, and no claim that the present study is expected to reproduce the metastatic benefit, is made or implied anywhere in this IND on the basis of the designation; the designation is recorded solely to characterize the depth and regulatory standing of the prior human experience. Table 9.1 records the marketed and designation status concisely.

Attribute	Status for daraxonrasib (RMC-6236)
Marketing authorization	None in any country; investigational new drug only.
Drug class	Oral RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor; binds GTP-bound active mutant RAS in a tri-complex with cyclophilin A.
Route and schedule	Oral, once daily.
Molecular target population	PDAC with documented KRAS G12 mutations (G12D, G12V, G12R); broad pan-KRAS coverage.
FDA Breakthrough Therapy	Granted June 2025 for previously treated metastatic PDAC with KRAS G12 mutations.
Marketed safety experience	None; all human safety data derive from the investigational program (§9.2).
Postmarketing surveillance	Not applicable; no marketed use exists.
Status-change reporting	Any future marketing authorization reported by information amendment, 21 CFR §312.31.

Table 9.1. Marketed and regulatory-designation status of daraxonrasib. The drug is investigational only; the June 2025 Breakthrough Therapy designation is a development milestone, not a marketing authorization.

The distinction between investigational status and marketing authorization carries operational consequences for this IND that are worth making explicit, because they shape how the prior human experience is permitted to be used. First, because the drug has no approved labeling, there is no labeled dose, labeled indication, or labeled safety warning that the present protocol can adopt by reference; instead, every dose, eligibility criterion, and safety expectation is justified from the investigational program and from the Investigator Brochure (§5), and is stated de novo in the protocol (§6.1). Second, because there is no marketed use, the adverse-event reference profile used to judge expectedness in this study is the Investigator Brochure reference safety information rather than approved labeling, consistent with the definition of an unexpected adverse reaction in 21 CFR §312.32(a). Third, because the drug is supplied only for investigational use, the chain-of-custody, accountability, and labeling provisions of 21 CFR §312.6 and §312.57 apply in full, and the perioperative pause-and-restart advisory described in §9.2 operates entirely within that investigational-supply framework rather than relying on any commercially dispensed product [3].

The status of the drug in other countries is summarized in §3.3 of this IND and is consistent with the marketed-experience statement above: daraxonrasib is investigational in every jurisdiction in which it is being studied, and no foreign marketing authorization, conditional approval, or accelerated-access scheme has been granted for any indication as of the data-cutoff supporting this submission. No foreign regulatory action, including refusal, withdrawal, suspension, or restriction of any application, has occurred that would bear on the human-experience characterization presented here. Should any such action occur after this IND is in effect, it would be reportable to the FDA in accordance with the information- amendment and safety-reporting obligations of 21 CFR §312.31 and §312.32, and the previous-human-experience analysis would be updated to reflect it. This cross-reference is recorded so that the marketed-experience subsection remains internally consistent with the foreign-status subsection and so that a reviewer reading §9 in isolation is not left with an incomplete picture of the drug's global regulatory standing [3, 1].

9.2 Prior Clinical Research Experience

The prior clinical research experience with daraxonrasib is concentrated in the late-phase metastatic-disease program, which comprises two pivotal studies and supplies the molecular eligibility, dosing, and safety reference framework adapted for this Phase 1 investigation. The two studies are RASolute 302, a Phase 3 study in previously treated metastatic PDAC, and RASolve 301, a first-line program; each is summarized below, followed by the way their experience is translated into the proposed protocol [1, 2]. The organizing principle of this subsection is that the metastatic program is treated as a source of bounding constraints rather than as a precedent: it bounds the molecular eligibility from above (the present study is no broader than the validated pan-KRAS criterion), it bounds the systemic exposure from above (the present study's highest dose does not exceed the metastatic recommended Phase 2 dose), and it supplies the reference safety information against which drug-attributed events are judged. Each of these bounding roles is developed below and then collected in Table 9.2.

RASolute 302 (Phase 3, previously treated metastatic PDAC)

RASolute 302 is the pivotal Phase 3 study of daraxonrasib in patients with previously treated metastatic PDAC. It establishes the human experience base for the drug at the dose level that the present protocol adopts as its highest escalation step: the RASolute 302 recommended Phase 2 dose corresponds to daraxonrasib 300 mg administered orally once daily, which this IND designates as dose level 3 (DL3) of its 3+3 escalation, with DL1 at 160 mg and DL2 at 220 mg. The molecular eligibility of RASolute 302 is broad pan-KRAS, encompassing the G12 alleles, and that breadth is carried forward directly into the present protocol: eligible participants must have histologically or cytologically confirmed PDAC with a documented KRAS G12 mutation (G12D, G12V, or G12R), anchoring the present study's molecular criterion to the validated RASolute 302 eligibility rather than introducing a narrower or novel mutational requirement [1]. The safety experience accumulated in RASolute 302, including the characterization of gastrointestinal effects (nausea, diarrhea, stomatitis), dermatologic and mucocutaneous events, and hepatic enzyme elevations, provides the reference safety information against which expectedness is judged for drug-attributed adverse events in this Phase 1 study, as detailed in the safety framework below.

The relationship between the RASolute 302 dose experience and the present dose ladder warrants a worked illustration, because it is the single most important quantitative link between the prior human experience and the proposed exposures. The present 3+3 ladder places its ceiling, DL3, at exactly the metastatic recommended Phase 2 dose of 300 mg once daily; the two lower steps are set at 220 mg (DL2, approximately 73 percent of the ceiling) and 160 mg (DL1, approximately 53 percent of the ceiling). Consequently, the entire planned exposure range of the present study lies at or below a dose that has already been administered to a Phase 3 population for an extended duration. A participant who escalates to DL3 in the present study is therefore not entering a previously unexplored systemic exposure; the novelty of the present study lies entirely in the surgical and device-directed context, not in the daraxonrasib dose. This is the deliberate design choice that allows a perioperative first-in-human-in-context study to begin without a separate single-agent dose-finding step: the systemic dose-response and tolerability have been established in the metastatic program, and the present study superimposes the surgical context on a known exposure envelope while still escalating cautiously from a conservative starting dose [1, 8].

RASolve 301 (first-line program)

RASolve 301 is the first-line program for daraxonrasib in PDAC and extends the human experience to the previously untreated metastatic population, including combination contexts. Its relevance to the present IND is twofold. First, it broadens the dose and exposure experience that informs the conservative, sub-RASolute-302 starting dose chosen for this first-in-human-in-context study (DL1 at 160 mg, well below the 300 mg recommended Phase 2 dose). Second, the perioperative pause-and-restart advisory specified in this protocol is designed to respect any first-line combination context that RASolve 301 may establish: the restart logic keys the resumption day to the postoperative pancreatic fistula grade and the serum daraxonrasib trough relative to the 0.5 ng/mL threshold, and the advisory framework is constructed so that it remains valid whether the drug is given as monotherapy or within a combination regimen [1, 2].

The first-line program is also material to the eligibility of the resectable and borderline-resectable population enrolled here, because that population is commonly treated with neoadjuvant systemic therapy before resection. The exemplar participant profile used throughout this IND (PAT-PDAC-0001: a 68-year-old with a 3.4 cm head-of-pancreas tumor abutting the superior mesenteric vein over approximately 75 degrees, KRAS G12D, microsatellite-stable, CA 19-9 of 412 U/mL, ECOG 1, after four cycles of neoadjuvant mFOLFIRINOX) illustrates exactly the kind of previously treated, biologically active disease that the first-line and pretreated programs together inform. The first-line experience does not change the present study's monotherapy posture for daraxonrasib in the perioperative window, but it ensures that the advisory and safety frameworks are forward-compatible with the combination contexts that the broader daraxonrasib program continues to develop [1, 10].

It is worth drawing out the precise sense in which the metastatic program does and does not constitute previous human experience for the present study, because the two are easily conflated. The metastatic program constitutes direct human experience with the drug at the molecular target (broad pan-KRAS, G12 alleles), at the systemic exposures spanned by the present ladder (160 mg to 300 mg once daily), and with the drug's characteristic adverse-event profile (gastrointestinal, dermatologic and mucocutaneous, and hepatic). It does not constitute human experience with the drug administered across the perioperative window of a major pancreatic resection, with the drug paused and restarted under a fistula-keyed advisory, or with the drug given to a patient whose abdomen has just been operated on by an LLM-directed robotic platform. The present study is first-in-human in each of these three context dimensions simultaneously, which is why the evidence base in Figure 19 terminates in a single dark node labeled first-in-human rather than in a claim of precedent. The metastatic human experience is the floor on which the present study stands; it is not a substitute for the human experience the present study is designed to generate [1, 2, 8].

Translation of prior experience into the proposed protocol

The prior clinical research experience is translated into the present protocol through three explicit linkages, summarized in Table 9.2. The molecular eligibility is anchored to the broad pan-KRAS RASolute 302 criteria; the dosing ladder is anchored to the RASolute 302 recommended Phase 2 dose as its ceiling, with two lower steps below it; and the reference safety information for drug causality and expectedness is taken from the metastatic program. The present study does not seek to reproduce or confirm the efficacy demonstrated in the metastatic setting; it is a descriptive, non-confirmatory, first-in-human investigation of the drug in a resectable or borderline-resectable, curative-intent population undergoing a device-directed pancreaticoduodenectomy, with co-primary endpoints directed at device and procedure safety, the maximum tolerated dose (MTD) and recommended Phase 2 dose (RP2D) via 3+3 escalation over a 28-day dose-limiting toxicity (DLT) window, and technical feasibility [8, 1].

Prior study	Population and status	Contribution to the present Phase 1 IND
RASolute 302	Phase 3, previously treated metastatic PDAC; broad pan-KRAS eligibility	Defines the molecular eligibility (KRAS G12) used here; its recommended Phase 2 dose (300 mg once daily) is adopted as DL3, the ceiling of the 3+3 ladder; supplies drug reference safety information.
RASolve 301	First-line metastatic PDAC program, including combination contexts	Broadens the dose and exposure experience supporting the conservative 160 mg starting dose; the perioperative pause-and-restart advisory is built to remain valid in any first-line combination context.
Breakthrough Therapy (June 2025)	Regulatory designation for previously treated metastatic PDAC with KRAS G12	Qualitative indicator of the maturity of the metastatic human experience; not a basis for any efficacy claim in the resectable population studied here.

Table 9.2. How the prior clinical research experience with daraxonrasib is translated into the eligibility, dosing, and safety reference framework of the present Phase 1 IND.

The dose ladder and its 3+3 escalation rule are stated in full in the General Investigational Plan (§4) and are reproduced here only to fix the relationship between prior experience and the proposed exposures: three participants are enrolled at each level; if zero of three experience a DLT the cohort escalates, if one of three does the cohort expands to six, and if one of six does the cohort escalates, whereas two or more DLTs in six define an exceeded MTD with the MTD declared at the prior level. The maximum treated sample is 18 participants (three levels times six), with approximately 36 screened, at a single academic site enrolling approximately one to two participants per month under staggered sentinel enrollment. Because the highest planned exposure (DL3, 300 mg once daily) does not exceed the already-characterized RASolute 302 recommended Phase 2 dose, the prior human experience directly bounds the upper end of the exposure range proposed here [1, 8]. Table 9.3 makes the dose mapping explicit and records, for each level, the relationship to the metastatic recommended Phase 2 dose and the role of the prior experience.

Dose level	Daraxonrasib (oral, once daily)	Fraction of metastatic RP2D	Relationship to prior human experience
DL1 (start)	160 mg	≈53%	Conservative starting dose, well below the established metastatic RP2D; supported by the breadth of the first-line and pretreated dose experience.
DL2	220 mg	≈73%	Intermediate step; remains below the metastatic RP2D ceiling.
DL3 (ceiling)	300 mg	100%	Equals the RASolute 302 recommended Phase 2 dose; the planned exposure does not exceed an already-characterized Phase 3 exposure.

Table 9.3. Mapping of the proposed 3+3 dose ladder onto the established metastatic recommended Phase 2 dose (RP2D) for daraxonrasib. The highest planned exposure (DL3) equals, and does not exceed, the metastatic RP2D, so the prior human experience bounds the upper end of the proposed exposure range.

A further consequence of this bounding relationship concerns the dose-limiting toxicity assessment itself. Because the systemic exposure at every dose level lies at or below the metastatic recommended Phase 2 dose, a dose-limiting toxicity observed in the present study is more likely to reflect the interaction between systemic daraxonrasib and the surgical insult, or the device-directed procedure, than a previously uncharacterized single-agent toxicity. This interpretation is not an assumption built into the dose-escalation rule (the 3+3 rule treats every DLT identically), but it is an interpretive lens that the principal investigator and the data safety monitoring board are expected to apply when adjudicating causality, and it is one reason the safety framework described in §9.3 carries a dedicated Physical AI stream alongside the conventional pharmacovigilance stream [8, 1]. The perioperative pause-and-restart advisory provides a concrete, quantitative example of how the prior exposure experience is operationalized at the bedside: across a 32-iteration deterministic sweep, the baseline serum daraxonrasib trough was 0.45 ng/mL (sub-threshold relative to the 0.5 ng/mL decision threshold) in every iteration, the advisory returned a T+7-day restart in 29 of 32 iterations (90.6 percent, corresponding to ISGPS Grade A fistula with a sub-threshold trough), a T+14-day restart in 3 of 32 iterations (9.4 percent, corresponding to Grade B or delayed recovery), and a T+21-day restart in none, with no iteration producing an indefinite hold. These figures are developed in full in the protocol and are cited here to show that the prior exposure experience is not merely referenced but is encoded into a deterministic, auditable restart rule [2, 1].

9.3 Clinical Care Experience

The clinical care experience relevant to this IND is the established human experience with robotic pancreaticoduodenectomy (the Whipple procedure), which provides the contemporary benchmark against which the device-directed component of this study will be evaluated, together with the safety-reporting framework that governs how new human experience is captured once the IND is in effect. The Whipple procedure is the most technically demanding curative-intent abdominal oncology operation: it requires resection and reconstruction of four luminal structures (the pancreas, the duodenum, the common bile duct, and the stomach or pylorus) and dissection around named vessels (notably the superior mesenteric vein and the portal vein), and its leading lethal complication is the fistula-sepsis-hemorrhage cascade graded under the International Study Group of Pancreatic Surgery (ISGPS) classification of postoperative pancreatic fistula [2]. The present study superimposes an investigational daraxonrasib exposure and an investigational LLM-directed robotic platform onto this established operation, and the clinical care experience with the operation itself is therefore the reference frame against which the incremental safety and feasibility of the device-directed component must be read.

The 2025 Dutch nationwide robotic-Whipple benchmark cohort

The contemporary human baseline for robotic pancreaticoduodenectomy is the 2025 Dutch nationwide cohort of 1000 robotic procedures, which anchors the secondary endpoint comparisons in this study. That cohort reported a conversion rate of 10.1 percent, Clavien-Dindo grade III or higher complications of 41.3 percent, ISGPS grade B or C (clinically relevant) postoperative pancreatic fistula of 24.4 percent, and 30-day mortality of 3.9 percent [2]. These four figures are used descriptively, not as a comparator arm: the present single-arm study is non-confirmatory, with no power calculation and no formal hypothesis test against the Dutch cohort, and all comparisons are framed as exact 95 percent Clopper-Pearson interval estimates around the small-sample proportions observed in the trial. The benchmark serves to demonstrate that even expert, high-volume robotic surgery leaves a substantial residual of conversions, high-grade complications, and clinically relevant fistulae, which are precisely the failure modes the device-directed component is designed to mitigate through its layered hard safety limits (the 10 kHz heartbeat bus, the cross-arm emergency stop of 3 ms or less, per-arm and cumulative tip-force caps, and the five-vessel vascular no-fly gate) [1]. Table 9.4 places the benchmark alongside the corresponding study secondary endpoints and the small-sample precision the study can achieve at its maximum size.

Outcome	**Dutch 2025 benchmark (n = 1000)**	**Role in the present study**
Conversion to open surgery	10.1%	Secondary endpoint; device-directed conversion (from malfunction or unsafe behavior) is also a co-primary feasibility input.
Clavien-Dindo grade III+ complications (90-day)	41.3%	Secondary endpoint; device- or procedure-related serious adverse events through 30 days, including Clavien-Dindo III+, are a co-primary safety endpoint.
ISGPS grade B or C pancreatic fistula	24.4%	Secondary endpoint and event of special interest; the pancreaticojejunostomy fistula grade is the input to the perioperative drug-restart advisory.
30-day mortality	3.9%	Secondary endpoint; 30-day and 90-day mortality are both followed.

Table 9.4. The 2025 Dutch nationwide robotic-Whipple benchmark cohort (n = 1000) and the corresponding secondary endpoints in the present study. The benchmark is used descriptively; all study estimates are reported with exact 95 percent Clopper-Pearson confidence intervals.

Because the study is small by design, its endpoint estimates carry wide exact confidence intervals, and the benchmark is therefore a contextual reference rather than a statistical comparator. At the maximum treated sample of 18, an observed rate of zero of 18 corresponds to 0 percent with a 95 percent Clopper-Pearson interval of 0 to 18 percent, and one of 18 corresponds to 5.6 percent with an interval of 0.1 to 27 percent; within a single dose cohort, zero of three corresponds to an interval of 0 to 71 percent and zero of six to an interval of 0 to 46 percent. These precision figures make explicit that the present study is designed to characterize the safety and feasibility of the combined intervention, not to establish superiority or non-inferiority against the Dutch benchmark [8]. Table 9.5 records these exact intervals so the reader can see directly how small-sample precision relates to the benchmark proportions: an observed study proportion can be numerically lower than a benchmark figure while its confidence interval still spans that benchmark, which is exactly why no inferential comparison is drawn.

Observed	Point estimate	Exact 95% CI (Clopper-Pearson)	Interpretive note
0 of 3	0%	0 to 71%	A single cohort cannot exclude a high underlying rate; informs cohort-level monitoring only.
0 of 6	0%	0 to 46%	An expanded cohort narrows the upper bound but remains wide.
0 of 18	0%	0 to 18%	The full study can bound an unobserved event rate below the benchmark conversion rate of 10.1%.
1 of 18	5.6%	0.1 to 27%	A single event yields an interval that still spans several benchmark figures; no inference is drawn.

Table 9.5. Exact 95 percent Clopper-Pearson confidence intervals for representative observed proportions at cohort and full-study sample sizes. The wide intervals confirm the descriptive, non-confirmatory posture of the study relative to the Dutch 2025 benchmark.

The surgical and oncologic care context, including the duct-to-mucosa pancreaticojejunostomy under controller ring-tension monitoring (0.30 to 0.60 N), the end-to-side hepaticojejunostomy (0.20 to 0.50 N), and the antecolic gastrojejunostomy (0.40 to 0.80 N), is specified in full in the protocol and is referenced here as the established clinical care experience the device is designed to reproduce within validated envelopes [2]. These reconstruction tension envelopes correspond to the operative phases of the procedure: phases P1 through P4 cover dissection and specimen extraction, phase P5 the pancreaticojejunostomy, phase P6 the hepaticojejunostomy, phase P7 the gastrojejunostomy, and phase P8 imaging, hemostasis, and drain placement. The device reproduces the human surgical technique within these envelopes rather than inventing a novel reconstruction, so the clinical care experience that validates the underlying operation also validates the maneuvers the device is asked to perform; the investigational element is the manner of execution (LLM-directed, robotic, force-bounded), not the operation. Importantly, the device-directed component operates under staged autonomy: in this Phase 1 study (Stage 1) the platform is restricted to clinician-controlled teleoperation and supervised operation only, with full autonomy prohibited under 21 CFR §312.21(e), and the on-premises open-weights LLM functions strictly as a second-opinion advisory oracle that is hash-pinned to a deterministic seed and commit and isolated outside the vendor kinematic stack [1, 6].

A further element of the clinical care experience is the vascular dissection itself, which is the operative step most directly responsible for the conversion-to-open and hemorrhage outcomes captured in the benchmark. The device-directed component encodes a five-vessel vascular no-fly gate covering the superior mesenteric vein, the portal vein, the hepatic artery, the celiac axis, and the superior mesenteric artery, sampled at 10 kHz, with a soft-warning advisory zone at 3.0 mm, a blocking no-fly zone at 1.0 mm for the superior mesenteric and portal veins and 1.5 mm for the hepatic artery, celiac axis, and superior mesenteric artery, and a hard-stop emergency-stop zone at 5.0 mm. Over a representative 100-tick verdict sample the gate returned 83 clear, 6 soft-warning, 6 no-fly, and 5 hard-stop verdicts. This quantitative gating is the device-directed analogue of the surgeon's situational judgment around named vessels, and it is the specific mechanism by which the device is intended to reduce the residual conversion and hemorrhage burden that the Dutch benchmark documents even in expert hands [1, 2].

The safety-reporting framework for ongoing human experience

The human experience generated by this study will be captured and reported under a two-stream safety framework that pairs a conventional pharmacovigilance stream for clinical events with a Physical AI safety stream for device- and AI-related events, each equally binding. On the clinical stream, any unexpected fatal or life-threatening suspected adverse reaction is reported to the FDA as soon as possible but no later than 7 calendar days after initial receipt of the information, and any other serious and unexpected suspected adverse reaction is reported no later than 15 calendar days after the determination that the event is reportable, consistent with 21 CFR §312.32 [3]. A Physical AI adverse event, per 21 CFR §312.32(f), is any untoward occurrence associated with the operation of the device during the investigation, whether or not caused by it, including robotic malfunction, AI prediction error, sensor failure, communication loss, unauthorized system access, digital-twin divergence, and any event requiring activation of the emergency stop. Six Physical AI reporting triggers are reported under §312.32(g): a serious Physical AI adverse event on the applicable 7-day or 15-day timeline; any system-drug interaction causing incorrect drug administration regardless of seriousness; any cybersecurity incident within 15 calendar days, or 7 days if it has the potential to cause patient harm; sustained AI model performance degradation below the pre-specified acceptance criteria for three or more consecutive procedures or a cumulative 24 hours; any sim-to-real divergence beyond twice the validated gap tolerance (trajectory deviation greater than 4 mm or force discrepancy greater than 1.0 N); and any digital-twin discrepancy beyond the pre-specified accuracy thresholds. For every Physical AI adverse event reported, the complete hash-chained audit trail covering the period from 24 hours before the event to 72 hours after the event is preserved under 21 CFR part 11 and made available to the FDA on request [3, 19]. Table 9.6 collects the two streams, the reporting clocks, and the governing regulatory citation so that the previous-human-experience section carries a complete, single reference for how new human experience will be reported.

Event class	Reporting clock	Definition and regulatory basis
Fatal or life-threatening suspected adverse reaction (clinical)	7 calendar days	Unexpected, related, fatal or life-threatening; 21 CFR §312.32(c)(1)(i)(A).
Other serious and unexpected suspected adverse reaction (clinical)	15 calendar days	Serious, unexpected, with reasonable possibility of drug causation; 21 CFR §312.32(c).
Serious Physical AI adverse event	7 or 15 calendar days	Device- or AI-related serious event; applicable clinical timeline applies; 21 CFR §312.32(f),(g).
System-drug interaction causing incorrect drug administration	15 calendar days	Reported regardless of seriousness; 21 CFR §312.32(g).
Cybersecurity incident	15 calendar days (7 if patient-harm potential)	Unauthorized access or compromise; 21 CFR §312.32(g).
Sustained AI model degradation	Per protocol trigger	Below acceptance criteria for ≥ 3 consecutive procedures or 24 cumulative hours; 21 CFR §312.32(g).
Sim-to-real divergence beyond 2x tolerance	Per protocol trigger	Trajectory >4 mm or force >1.0 N; 21 CFR §312.32(g).
Digital-twin discrepancy beyond thresholds	Per protocol trigger	Beyond pre-specified accuracy thresholds; 21 CFR §312.32(g).

Table 9.6. The two-stream safety-reporting framework governing how new human experience is captured once the IND is in effect, with reporting clocks and the governing 21 CFR §312.32 provisions. Every Physical AI adverse event preserves a hash-chained audit trail from 24 hours before to 72 hours after the event under 21 CFR part 11.

Once the IND is in effect and the 30-day clock has elapsed, post-submission authoring is driven by this safety data rather than by the initial submission, and the same generation method that produced the IND maintains the document suite in internal consistency. Figure 18 shows the post-submission maintenance flow: serious adverse events and suspected adverse reactions feed the 7-day and 15-day IND safety reports; emerging pharmacokinetic and pharmacodynamic data and cohort completion feed the annual Development Safety Update Report (DSUR) and amendments; data safety monitoring board (DSMB) cohort reviews feed the dose-escalation minutes and the RP2D memorandum; the principal investigator assesses causality and scientific justification across all of these; and the LLM then synchronizes the protocol, consent, and site documents so the whole suite remains internally consistent [2, 18].

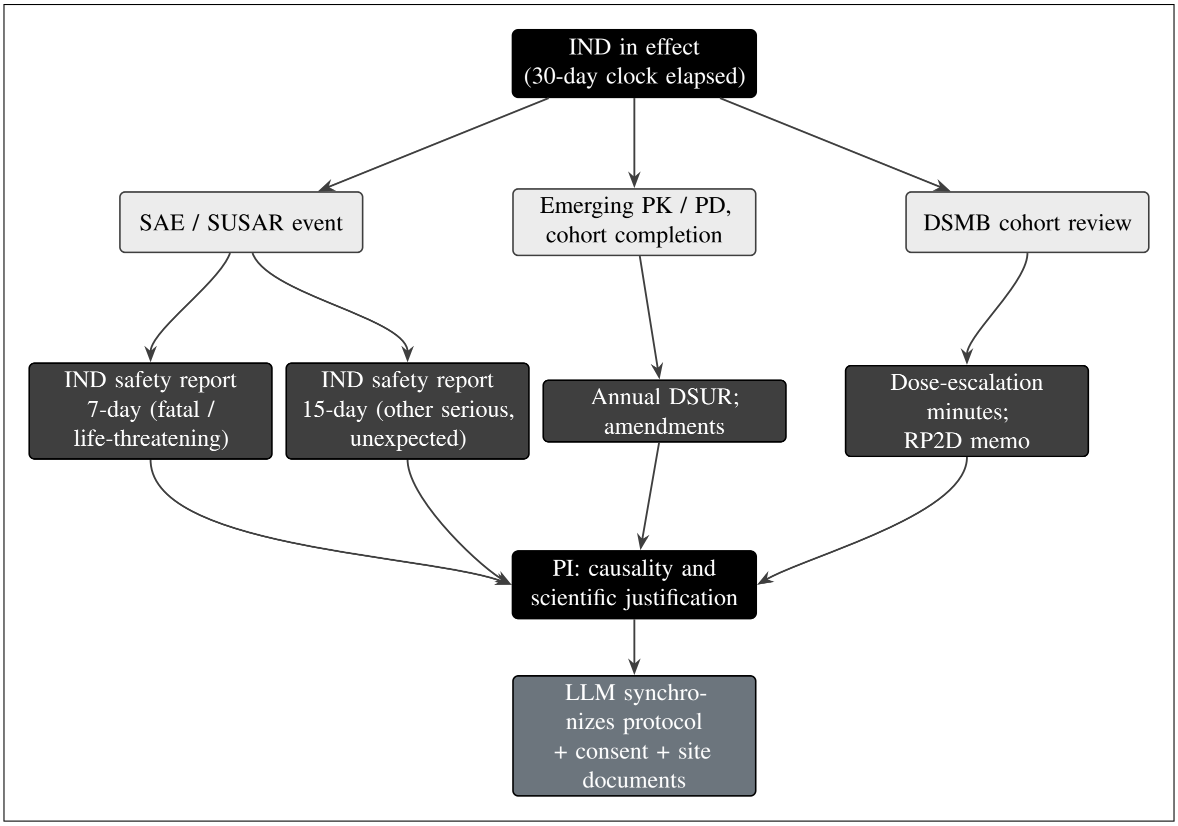

Figure 18. Post-submission IND maintenance authoring once the IND is in effect (the 30-day clock elapsed). Safety data drive the authoring: IND safety reports on the 7-day (fatal or life-threatening) and 15-day (other serious and unexpected) clocks, the annual DSUR, amendments, and dose-escalation minutes, with the principal investigator assessing causality and scientific justification and the LLM synchronizing the protocol, consent, and site documents so the whole suite stays internally consistent.

The three-tier oversight that reviews this accumulating human experience comprises the DSMB (acting at cohort boundaries and on triggered review, with halt authority), an Independent Safety Monitor (per-procedure, real-time, with stop authority), and a Physical AI Safety Review Committee (convening at intervals not exceeding 90 days during active enrollment). The binding halt rules are that one or more device- or procedure-related serious adverse events within a cohort, or two device-related deaths at any time, trigger DSMB review and possible halt [2, 6]. This governance ensures that the previous human experience characterized in this section is continuously updated and acted upon as the study generates the first direct human experience with the drug in this device-directed, curative-intent context, the gap that Figure 19 identifies as first-in-human [7]. Table 9.7 records the three oversight tiers, their cadence, and their authority so the reader can see how the abstract reporting clocks of Table 9.6 are connected to concrete halt and stop authority over the study.

Oversight tier	Cadence	Authority and binding triggers
Data Safety Monitoring Board (DSMB)	Cohort boundaries and on triggered review	Halt authority; $\geq$1 device- or procedure-related serious adverse event within a cohort, or 2 device-related deaths at any time, triggers review and possible halt.
Independent Safety Monitor	Per procedure, real-time	Stop authority during an individual procedure.
Physical AI Safety Review Committee	At intervals not exceeding 90 days during active enrollment	Reviews Physical AI events, model performance, and sim-to-real fidelity; escalates to the DSMB as warranted.

Table 9.7. The three-tier oversight reviewing the accumulating human experience, with cadence and binding authority. The DSMB and Independent Safety Monitor carry halt and stop authority respectively; the Physical AI Safety Review Committee provides AI-specific review on a cadence not exceeding 90 days.

The computational program summarized in Figure 19 does not generate human experience, but it is the verification-before-generation scaffolding that makes the first human exposures defensible, and it is therefore part of the previous experience that supports this submission in the broad sense contemplated by 21 CFR §312.23(a)(9). Before any patient-facing robotic activity, the device must clear a Phase 0 simulation-validation gate that requires a universal safety level of at least 7.0, at least 1000 simulations across at least two independent physics frameworks (drawn from Isaac, MuJoCo, Gazebo, and PyBullet), and demonstrated sim-to-real fidelity of less than 2 mm in trajectory and less than 0.5 N in tip-force, with re-validation after any change [14, 13]. The author's 2025 to 2026 works (the 60-second PDAC Whipple and daraxonrasib simulation, the AI digital-twin PDAC simulation, the quantitative-systems- pharmacology metastatic-PDAC simulation with verification, validation, and uncertainty quantification, the 100,000-patient in silico Phase 3 triplicate, and the end-to-end PDAC digital-twin proposals) supply the in silico evidence that this gate evaluates, and the 16-LLM code-generation evaluation against the FDA adverse event reporting system provides the model-reliability context for the advisory oracle [15, 16, 12]. Taken together, the clinical prior experience bounds the systemic exposure, the clinical care experience bounds the surgical context, and the computational program bounds the device behavior, so that the first direct human experience generated by this study is acquired within a fully characterized and continuously monitored envelope [1, 2, 7].

9.4 References

The references supporting this section are listed in full in the back matter and are cited inline above. The clinical and computational evidence base is documented in the daraxonrasib robotic-Whipple and on-premises Whipple sources [1, 2]; the phased-development and Phase 1 expectations in the Phase 1 guidance [8]; the PDAC epidemiology and metastatic-program context in the disease and program references [4]; the safety-reporting and Part 11 framework in the 21 CFR adaptations [3, 19, 18]; the verification-before-generation and clinical-trust framing in the legislative and trust references [6, 7]; the LLM-directed oncology trial authoring and site-documentation framing in the operational references [10, 9, 17, 5, 11]; and the author's 2025 to 2026 computational program in the digital-twin, quantitative-systems-pharmacology, in silico Phase 3, and digital-twin-proposal works [14, 13, 15, 16, 12].

10 Additional Information

This section addresses the categories of additional information enumerated for an original Investigational New Drug (IND) submission under 21 CFR §312.23(a)(10), namely drug dependence and abuse potential, radioactive drugs, pediatric studies, and any other information bearing on the safe and ethical conduct of the proposed clinical investigation. Because this is a combined IND and Investigational Device Exemption (IDE) first-in-human study of orally administered daraxonrasib (RMC-6236) delivered alongside an on-premises large language model (LLM) directed eight-arm robotic pancreaticoduodenectomy, the Other Information subsection additionally consolidates the Physical AI governance, cybersecurity, record retention, and three-tier oversight content that the autonomy-bearing investigational device requires for Phase 1 acceptance. Each statement below is grounded in the protocol (v1.0.0, DOI 10.5281/zenodo.20780121), is internally consistent with the General Investigational Plan, the Proposed Clinical Research, and the Previous Human Experience sections of this IND (repository v4.3.0, IND v1.0), and is aligned with FDA Phase 1 guidance [8] and with the regulatory adaptations developed for Physical AI investigations [3, 19, 18].

The four enumerated categories of 21 CFR §312.23(a)(10) are treated in turn, and each is given an explicit applicability determination so that the reviewing division can locate, on a single pass, the basis on which the category is either satisfied or found not applicable. Drug dependence and abuse potential is addressed and found to require no abuse-liability program. Radioactive drugs is found not applicable because the investigational agent contains no radionuclide. Pediatric studies is addressed by a documented deferral consistent with an adult-only, serious-and-life-threatening oncology indication that does not arise materially in children. The Other Information subsection then carries the Physical AI overlay, which is the principal substantive addition that distinguishes this submission from a conventional small-molecule Phase 1 IND and which the autonomy-bearing investigational device makes necessary for safe and ethical conduct. The structure mirrors the enumeration of the regulation so that completeness can be confirmed by inspection.

10.1 Drug Dependence and Abuse Potential

Daraxonrasib has no known dependence or abuse potential and is not a controlled substance under the Controlled Substances Act. The investigational drug is an oral, once-daily RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor that binds the guanosine triphosphate (GTP) bound active form of mutant RAS in a complex with cyclophilin A and thereby suppresses downstream effector signaling [1, 2]. Its pharmacological mechanism is targeted oncogenic-signaling inhibition; it has no activity at the central nervous system receptors, transporters, or ion channels that mediate reinforcement, euphoria, sedation, or physical dependence, and it does not belong to a pharmacological class associated with recreational use, tolerance with escalating self-administration, or a withdrawal syndrome. The drug carries an FDA Breakthrough Therapy designation (June 2025) for previously treated metastatic pancreatic ductal adenocarcinoma (PDAC) with KRAS G12 mutations, and the prior and ongoing human experience summarized in the Previous Human Experience section has not identified any signal of abuse, dependence, drug-seeking behavior, or diversion.

The applicability analysis follows the FDA assessment framework for abuse potential. That framework directs a sponsor to consider, first, the pharmacological class and mechanism of the molecule and its structural relationship to known drugs of abuse; second, the receptor-binding and central-nervous-system activity profile; third, the nonclinical behavioral pharmacology, including any signal in self-administration, drug-discrimination, and physical-dependence and withdrawal models; and fourth, the clinical experience to date, including any reports of euphoria, dissociation, mood elevation, or aberrant drug-seeking behavior. Applied to daraxonrasib, each of these considerations is negative. The molecule is a RAS(ON) inhibitor with no structural or pharmacological kinship to opioids, stimulants, cannabinoids, sedative-hypnotics, dissociatives, or hallucinogens; its molecular target is the active conformation of mutant RAS rather than any neurotransmitter system implicated in reward; and the accumulated human experience reflects a tolerability profile dominated by mechanism-related and gastrointestinal events rather than by any central effect that would support reinforcement. The conclusion that no abuse-liability program is required follows directly from the convergence of these four lines of evidence, not from the absence of any single study.

Accordingly, no abuse-liability program is proposed for this IND. No dedicated nonclinical abuse-potential studies (for example, self-administration, drug discrimination, or physical-dependence and withdrawal studies under the FDA abuse-potential assessment framework) are warranted, and no controlled-substance scheduling recommendation applies. The investigational product is dispensed under the accountability controls described in the Chemistry, Manufacturing, and Control Information and Proposed Clinical Research sections, with drug accountability logs maintained at the single academic site, so that the limited quantities used for the planned cohorts (up to eighteen treated participants across three dose levels) are reconciled from receipt through administration, return, and destruction. Should any unanticipated central-nervous-system finding suggestive of abuse potential emerge during the conduct of the study, it would be captured through the routine adverse event and serious adverse event reporting pathways under 21 CFR §312.32 and reviewed by the Data Safety Monitoring Board (DSMB).

A worked illustration of the accountability arithmetic makes the control concrete. With a maximum of eighteen treated participants and a once-daily oral regimen, the study exposure footprint is small and fully reconcilable. The drug accountability log records, per participant and per cohort, the quantity received, the quantity dispensed at each visit, the quantity returned unused, and the quantity destroyed, with each transaction time-stamped and counter-signed. Because the three dose levels (160, 220, and 300 mg once daily) are dispensed in defined quantities tied to the twenty-eight-day dose-limiting-toxicity (DLT) window and to the perioperative pause-and-restart advisory, the expected dispensed quantity for any participant is computable in advance, and any discrepancy between expected and reconciled quantity is an audit signal that is reviewed before the next dispensing. This closed-loop reconciliation, rather than scheduling or a dedicated abuse-deterrent formulation, is the proportionate control for a non-scheduled targeted oncology agent.

For completeness with respect to 21 CFR §312.23(a)(10)(i), the IND states affirmatively that the investigational agent is not listed in any schedule of the Controlled Substances Act, that no application for scheduling is pending or contemplated, and that no labeling, packaging, or distribution control beyond the ordinary investigational-product accountability of 21 CFR §312.57 and §312.62 is required on the basis of abuse potential. The perioperative pause-and-restart of the drug, in which dosing is held before surgery and resumed on a human-confirmed advisory, further limits continuous exposure and is itself recorded, so that the total quantity dispensed, held, resumed, and returned is auditable at every step. No take-home quantity is dispensed beyond the defined per-visit allotment, and any drug returned at a visit is reconciled before the next allotment is released. Because the study is conducted at a single academic site under direct institutional pharmacy control rather than through outpatient community dispensing, the diversion surface is small and the accountability chain is short, which is the appropriate level of control for a non-scheduled targeted oncology agent administered to a closely monitored surgical population.

Table 59: Abuse-potential assessment summary for daraxonrasib against the four considerations of the FDA abuse-potential framework, with the determination for each. The convergence of the four negative determinations supports the conclusion that no abuse-liability program is required for this IND.

Consideration	Finding for daraxonrasib	Determination
Pharmacological class and structure	RAS(ON) multi-selective (pan-RAS) small-molecule inhibitor; no structural or class kinship to opioids, stimulants, cannabinoids, sedative-hypnotics, dissociatives, or hallucinogens.	No abuse class
Central-nervous-system activity	No activity at receptors, transporters, or ion channels that mediate reinforcement, euphoria, sedation, or physical dependence; target is GTP-bound active mutant RAS in complex with cyclophilin A.	No reward signal
Nonclinical behavioral pharmacology	No signal warranting self-administration, drug-discrimination, or physical-dependence and withdrawal studies; no dedicated abuse-potential studies indicated.	Studies not warranted
Clinical experience to date	Prior and ongoing human experience (Breakthrough Therapy, June 2025) shows no abuse, dependence, drug-seeking behavior, or diversion.	No clinical signal

Table 10.1. Abuse-potential assessment summary. Each of the four considerations of the FDA abuse-potential framework returns a negative determination, supporting the conclusion that no abuse-liability program, no dedicated nonclinical abuse-potential studies, and no controlled-substance scheduling recommendation are required for this IND.

10.2 Radioactive Drugs

This subsection is not applicable. Daraxonrasib is not a radioactive drug and contains no radionuclide. The investigational product is a non-radioactive oral small-molecule tablet administered once daily; it is neither a positron-emitting nor a single-photon-emitting radiopharmaceutical, and it is not the subject of a Radioactive Drug Research Committee (RDRC) determination under 21 CFR §361.1. No radiation dosimetry, no radiation safety committee review, and no radioactive drug-specific manufacturing, handling, storage, or waste-disposal controls are required for the investigational agent. No information under 21 CFR §312.23(a)(10)(ii) is therefore required.

Imaging performed during the study is part of routine clinical care and is not an investigational radioactive drug administered under this IND. Diagnostic computed tomography, magnetic resonance imaging, and any standard-of-care nuclear medicine staging that may precede enrollment are performed and interpreted according to institutional standards and are outside the scope of this subsection; any radiopharmaceutical used in such standard-of-care imaging is administered under the institution's own authorizations and not under this IND. To avoid ambiguity for the reviewing division, the operative and device elements of the study are also addressed. The intraoperative imaging, hemostasis, and drain placement of operative phase P8 are optical and instrumental and involve no administered radioactivity; the eight-arm robotic system and its 640 sensor channels are electromechanical and optical instruments, not radiation-emitting devices within the meaning of a radioactive drug determination. The vascular no-fly gate, the force-envelope controls, and the cyber-physical telemetry stream described in the Other Information subsection are likewise non-radioactive. No element of the investigational product or the investigational device introduces a radionuclide into the participant, and the radioactive-drug category is therefore satisfied as not applicable in its entirety.

10.3 Pediatric Studies

The study population is restricted to adults aged eighteen years or older. Eligible participants have histologically or cytologically confirmed PDAC that is resectable or borderline-resectable, a documented KRAS G12 mutation (G12D, G12V, or G12R), an Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1, and are candidates for pancreaticoduodenectomy (Whipple resection), consistent with the broad pan-KRAS eligibility framework of the RASolute 302 program [1]. No participant under the age of eighteen will be enrolled, consented, screened, dosed, or operated upon under this protocol, and pediatric assent provisions are therefore inapplicable in practice even though the informed-consent architecture (21 CFR part 50 with ICH E6(R3)) provides for assent where it would otherwise apply [19].

A first-in-human Phase 1 evaluation in a pediatric population is neither warranted nor proposed at this stage, and a pediatric study plan is deferred. The deferral rationale rests on three considerations. First, the disease under study, PDAC, is overwhelmingly a disease of older adults; pancreatic ductal adenocarcinoma in children is exceedingly rare, the five-year overall survival of PDAC is below thirteen percent, and the condition is the third leading cause of cancer death in the United States and the fourth in the European Union [4], so the target indication does not arise materially in the pediatric population. Second, this is a first-in-human study of an autonomy-bearing investigational device combined with an investigational dosing regimen; the prudent and ethical sequence is to establish the device and procedure serious adverse event profile, the maximum tolerated dose (MTD) and recommended Phase 2 dose (RP2D), and the technical feasibility of the LLM-directed robotic Whipple in adults before any consideration of younger populations, and full autonomy is in any case prohibited at this stage under 21 CFR §312.21(e) [3]. Third, the staged-autonomy model of this trial class confines the present study to clinician-controlled teleoperation and supervised operation (Stage 1), reserving higher-autonomy operation for future study; extension to a pediatric population, if ever pursued, would follow successful completion of the adult program and would be the subject of a separate pediatric study plan with its own age-appropriate formulation, dosing, device sizing, and assent and parental-permission provisions.

The deferral is also coherent with the device-engineering constraints of the investigational system. The eight-arm robotic platform, with its 56 degrees of freedom and its fixed force-envelope caps of no more than 3 N per arm and no more than 18 N cumulative, is validated for the anatomy of an adult pancreaticoduodenal field and for the vessel-proximity tolerances of an adult superior mesenteric vein, portal vein, hepatic artery, celiac artery, and superior mesenteric artery. The vascular no-fly margins of 1.0 mm at the superior mesenteric vein and portal vein and 1.5 mm at the hepatic artery, celiac artery, and superior mesenteric artery, together with the Phase 0 simulation-validation gate, are calibrated to adult geometry. A pediatric extension would require revalidation of the entire cyber-physical envelope, the digital twin, and the simulation corpus against pediatric anatomy and would not be a simple eligibility amendment. This engineering reality reinforces the clinical and ethical case for deferring any pediatric evaluation until the adult program has established the safety and feasibility foundation on which any such revalidation would have to build.

Because this is an investigation of a serious and life-threatening oncology indication that does not occur meaningfully in children, the study qualifies for the conditions under which a pediatric assessment is not required for an adult-only first-in-human Phase 1 program; any subsequent pediatric commitments would be addressed at the appropriate later stage of development in consultation with the FDA. No pediatric data are submitted with this original IND.

10.4 Other Information

The autonomy-bearing investigational device requires governance, cybersecurity, record-retention, and oversight controls beyond those of a conventional small-molecule Phase 1 study. This subsection consolidates that Physical AI overlay. The on-premises LLM operates strictly as a second-opinion advisory oracle: it is hash-pinned to a deterministic seed and a repository commit, isolated outside the vendor kinematic stack, and never granted control authority over the robotic arms; full autonomy is prohibited under 21 CFR §312.21(e) [3], and every proposed robot-patient command passes a verification-before-generation ten-gate suite returning accept, block, or escalate, aligned with the Verification Before Generation in Physical AI Oncology Trials Act of 2026 (H.R. 9510) [6]. The governance design follows the trust and accountability principles developed for clinical Physical AI [7, 5] and the site-readiness and national-platform work for Physical AI oncology trials [11, 9].

10.4.1 Physical AI Governance and the Three-Tier Oversight Structure

Independent safety oversight operates on three coordinated tiers layered over the reviewing Institutional Review Board (IRB) and the Sponsor-Investigator. The DSMB, composed of members independent of the conduct of the study with surgical, oncologic, biostatistical, and device-safety expertise, reviews accumulating safety data at each cohort boundary and at any triggered review and holds binding halt authority. The Independent Safety Monitor (ISM) is designated for each procedure and provides real-time, case-level safety oversight independent of the operating team, with the authority to call an immediate stop. The Physical AI Safety Review Committee meets on a cadence of ninety days or less (and ad hoc on any triggering event) to review the autonomy-bearing behavior of the system, including the telemetry and audit record, the verification-before-generation gate-surface decisions, the Unified Safety Level (USL) readiness rating, and any software change, and recommends USL reassessment or oversight changes to the Sponsor-Investigator. The three tiers are coordinated so that no autonomy-bearing change and no escalation decision proceeds without appropriate independent review. The Sponsor-Investigator (ChemicalQDevice, Kevin Kawchak, Chief Executive Officer) holds the combined IND and IDE; the conflict of interest arising from the Sponsor-Investigator's relationship to the developer of the Physical AI system is disclosed in full under 21 CFR part 54 and is mitigated structurally by this independent oversight, so that safety and escalation decisions are not made by a financially interested party alone.

The coordination among the tiers is deliberate and is designed to remove any single point of discretion over a safety-critical decision. The reviewing IRB exercises human-subjects protection and reviews the protocol, the informed consent, and the significant-risk device determination, and it may suspend or terminate its approval. The DSMB applies the prespecified halt rules at each cohort boundary and at any triggered review and recommends continue, modify, pause, or halt. The ISM exercises real-time authority at the level of the individual procedure and may stop a case without waiting for a scheduled review. The Physical AI Safety Review Committee exercises the autonomy-specific oversight that a conventional DSMB is not equipped to perform, reading the telemetry and audit record and the gate-surface decisions and recommending USL reassessment when the evidence requires it. Because the Sponsor-Investigator and the developer relationship are the same financial interest, the structural mitigation is that the binding halt authority, the per-procedure stop authority, and the autonomy-readiness reassessment authority all reside in bodies independent of that interest, so that no safety-critical determination can be made by the interested party alone. Table 60 summarizes the oversight bodies, their cadence, and their authority, and Figure 20 renders the governance structure.

Table 60: Three-tier independent safety oversight and the human-subjects-protection structure for the combined IND and IDE study. Each body's review cadence and the authority it exercises are stated; the binding halt rules drive the continue, modify, pause, or halt decision.

Body	Review cadence	Role and authority
Sponsor-Investigator (ChemicalQDevice, K. Kawchak)	Continuous	Holds combined IND and IDE; protocol custody, safety reporting under 21 CFR parts 312 and 812, and compliance assurance; conflict of interest disclosed and managed under 21 CFR part 54.
Reviewing IRB	Per submission and continuing review	Human-subjects protection; reviews protocol, consent, and the significant-risk device determination; may suspend or terminate approval.
Data Safety Monitoring Board (DSMB)	Each cohort boundary and any triggered review	Independent surgical, oncologic, biostatistical, and device-safety members; applies the prespecified halt rules; recommends continue, modify, pause, or halt; binding halt authority.
Independent Safety Monitor (ISM)	Per procedure, real-time	Case-level safety oversight independent of the operating team; authority to call an immediate stop.
Physical AI Safety Review Committee	Cadence of 90 days or less, plus ad hoc	Reviews telemetry and audit records, verification-before-generation gate-surface decisions, USL readiness, and software changes; recommends USL reassessment or oversight changes.

Table 10.2. The oversight bodies, their cadence, and their authority for the combined IND and IDE first-in-human study.

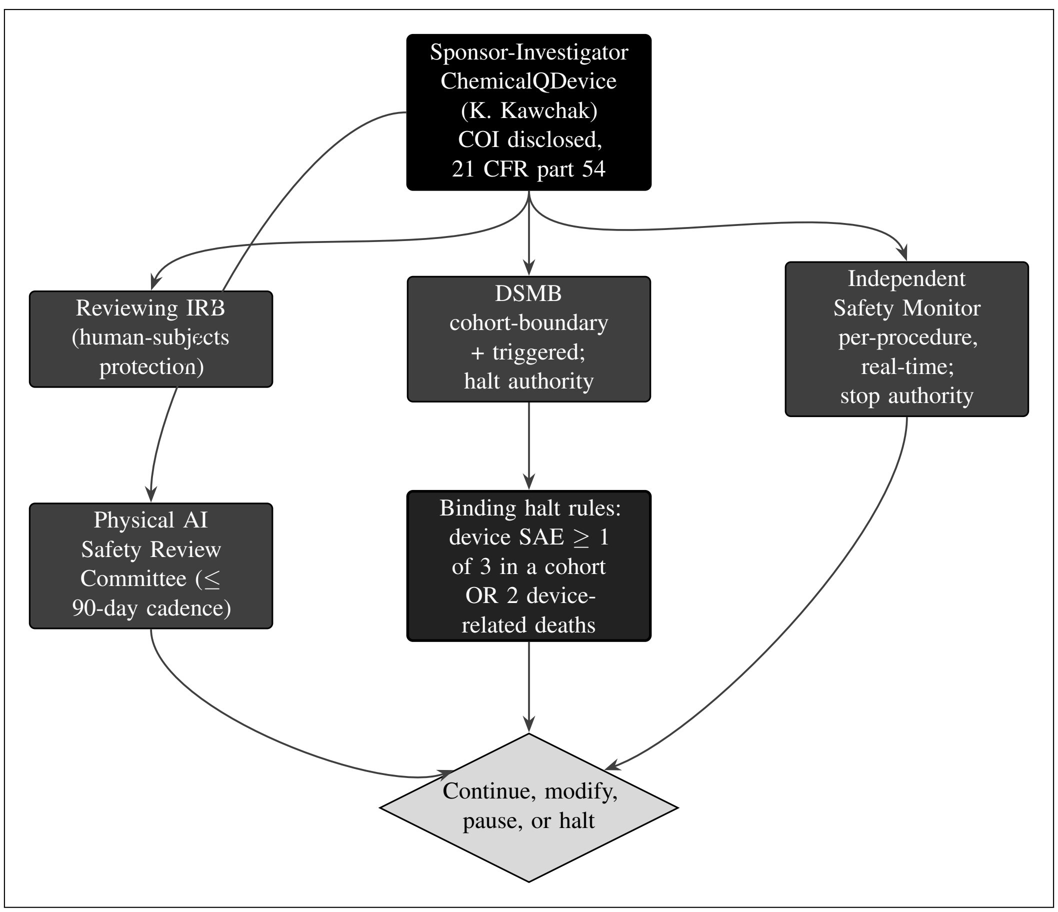

Figure 20. Study governance and three-tier safety oversight. The Sponsor-Investigator (ChemicalQDevice, Kevin Kawchak, with the conflict of interest disclosed under 21 CFR part 54) operates under a reviewing IRB and three independent bodies: the Data Safety Monitoring Board (cohort-boundary and triggered reviews, with halt authority), the Independent Safety Monitor (per-procedure, real-time, with stop authority), and the Physical AI Safety Review Committee (a cadence of 90 days or less). The binding halt rules, a device-related serious adverse event in at least one of three within a cohort or two device-related deaths at any point, drive the continue, modify, pause, or halt decision.

10.4.2 Binding Halt Rules and Stopping Authority

Two prespecified, binding halt rules govern the study and are applied by the DSMB. First, a device-related or procedure-related serious adverse event in at least one of three participants within a dose cohort triggers a mandatory DSMB review with authority to halt. Second, two device-related deaths at any point in the study trigger a mandatory DSMB review with authority to halt. On invocation of either rule, enrollment and any patient-facing robotic activity stop pending review. These rules operate independently of, and in addition to, the dose-finding rules of the 3+3 escalation (dose levels 160, 220, and 300 mg once daily, with a twenty-eight-day dose-limiting-toxicity window) and the per-procedure stop authority of the Independent Safety Monitor. The cyber-physical safety envelope provides further hard stops that are not discretionary: force caps of no more than 3 N per arm and no more than 18 N cumulative across the eight arms, a 10 kHz heartbeat with a sixty-four-byte frame and a one-hundred-microsecond watchdog, arm-park within fifty microseconds, a cross-arm emergency stop within three milliseconds to a commanded 0.0 N, and a hardware emergency stop within five hundred milliseconds consistent with 21 CFR §312.404. The vascular no-fly gate samples at 10 kHz and enforces a soft-warning advisory zone at 3.0 mm, blocking no-fly margins of 1.0 mm at the superior mesenteric vein and portal vein and 1.5 mm at the hepatic artery, celiac artery, and superior mesenteric artery, and a hard-stop emergency-stop zone at 5.0 mm. Any breach of a hard cyber-physical limit, a failure of the verification-before-generation gate surface, or a loss of audit integrity constitutes a clinical-hold ground for the autonomy-bearing system, on which all patient-facing robotic activity ceases and the decommissioning and data-preservation procedure is executed.

The interaction between the binding clinical halt rules and the deterministic cyber-physical limits is worth making explicit, because the two operate on different time scales and through different mechanisms. The cyber-physical limits are microsecond-to-millisecond hardware and firmware interlocks that act before any human or committee can deliberate; the cross-arm emergency stop reaches a commanded 0.0 N within three milliseconds and the hardware emergency stop completes within five hundred milliseconds. These are not discretionary and do not wait for a vote. The binding halt rules, by contrast, are deliberative thresholds that the DSMB applies after the fact to accumulating safety data, and they determine whether the study as a whole continues, is modified, is paused, or is halted. A single hard-stop activation in the operating field is handled by the envelope in milliseconds and is recorded in the safety-limit-event record; the pattern of such activations, together with the clinical outcomes, is what the DSMB reviews against the binding halt rules at the cohort boundary. The two mechanisms are complementary: the envelope protects the individual participant in real time, and the halt rules protect the cohort and the study by governing whether exposure continues. Table 61 sets out the deterministic envelope thresholds, their latency budgets, and the regulatory anchor for each.

Table 61: Deterministic cyber-physical safety envelope thresholds and latency budgets, distinguished from the deliberative binding halt rules. The envelope acts in real time at the level of the individual procedure; the binding halt rules are applied by the DSMB to accumulating data at the cohort and study level.

Control	Threshold or latency	Function and anchor
Per-arm force cap	$\leq$ 3 N per arm	Limits tissue-contact force; recorded in the safety-limit-event record.
Cumulative force cap	$\leq$ 18 N across eight arms	Bounds aggregate force across 56 degrees of freedom.
Control heartbeat	10 kHz, 64-byte frame, 100 us watchdog	Liveness of the cyber-physical loop; loss triggers arm-park.
Arm-park latency	$\leq$ 50 us	Safe halt of an arm on watchdog timeout.
Cross-arm emergency stop	$\leq$ 3 ms to commanded 0.0 N	Coordinated stop of all eight arms.
Hardware emergency stop	$\leq$ 500 ms	Independent hardware-level stop (21 CFR §312.404).
Vascular no-fly gate (soft)	3.0 mm advisory, 10 kHz sampling	Soft-warning advisory zone at the five named vessels.
Vascular no-fly gate (block)	1.0 mm SMV and PV; 1.5 mm HA, CA, SMA	No-fly block margin at the five named vessels.
Vascular no-fly gate (hard-stop)	5.0 mm emergency-stop zone	Triggers emergency stop on incursion.

Table 10.3. The deterministic cyber-physical safety envelope. These real-time interlocks act at the individual-procedure level on microsecond-to- millisecond time scales and are not discretionary; the DSMB binding halt rules act on accumulating data at the cohort and study level.

A representative 100-tick sample of the vascular no-fly gate during dissection, drawn from the protocol, illustrates how the gate verdicts distribute in practice: of one hundred sampled ticks, eighty-three return clear, six return a soft-warning advisory, six return a no-fly block, and five return a hard-stop emergency-stop verdict. The eleven blocking or stopping verdicts in this sample are each written to the safety-limit-event record with the vessel, the margin, and the latency of the response, so that the DSMB can review both the rate of gate engagement and the fidelity of the response when it convenes at the cohort boundary. This is the mechanism by which a real-time interlock event becomes part of the deliberative record that governs the binding halt determination.

10.4.3 Cybersecurity and System Integrity

Cybersecurity is treated as a patient-safety control rather than an administrative adjunct. The LLM-directed system is deployed on premises and isolated outside the vendor kinematic stack, with no requirement for external network connectivity during a procedure; the advisory model is hash-pinned to a deterministic seed and a repository commit so that the validated configuration in use is exactly the configuration that was tested and recorded. Access to identifiable records and to the cyber-physical control surface is governed by a deny-by-default authorization model, under which no role can read, write, or export a record except where an access right has been explicitly granted, and electronic signatures are bound to the records they authenticate. Every access, change, and export is written to a hash-chained audit trail in which each entry cryptographically incorporates the hash of the prior entry, so that the record is tamper-evident and any alteration is detectable. A cybersecurity incident is one of the six Physical AI reporting triggers under 21 CFR §312.32(g) and is reportable within fifteen calendar days, or within seven calendar days where there is potential for patient harm; the audit trail is preserved from twenty-four hours before to seventy-two hours after each Physical AI adverse event under 21 CFR part 11.

The six Physical AI reporting triggers under 21 CFR §312.32(g) together extend the conventional Subpart J reporting framework to the autonomy-bearing behaviors of the system, and each carries a defined clock so that the obligation is unambiguous. A serious Physical AI adverse event is reported on the fatal-or-life-threatening seven-calendar-day clock or the other-serious-and-unexpected fifteen-calendar-day clock, parallel to the conventional suspected-adverse-reaction reporting of 21 CFR §312.32. A system-drug interaction, in which the device behavior and the investigational drug interact in a manner bearing on safety, is reported within fifteen calendar days. A cybersecurity incident is reported within fifteen calendar days, or within seven where there is potential for patient harm. Sustained AI model degradation, defined as degradation persisting across at least three consecutive procedures or twenty-four cumulative hours, is reported because it indicates a drift of the advisory oracle away from its validated behavior. A sim-to-real divergence exceeding twice the tolerance, namely a trajectory error above 4 mm or a force error above 1.0 N, is reported because it indicates that the digital twin and the physical system have parted company beyond the validated envelope. A digital-twin discrepancy is reported on the same pathway. The common feature of the six triggers is that each denotes a departure of the autonomy-bearing system from its validated, deterministic behavior, whether that departure is adversarial (the cybersecurity incident), gradual (the sustained model degradation), or a loss of fidelity between the model and the physical system (the sim-to-real divergence and the digital-twin discrepancy). The seven-and-fifteen-day clocks parallel the conventional 21 CFR §312.32 reporting so that the autonomy overlay does not create a separate or slower pathway, and the seven-day cybersecurity clock where there is potential for patient harm reflects the elevated urgency of a control-surface compromise. Table 62 enumerates the six triggers, their definitions, and their clocks.

Table 62: The six Physical AI reporting triggers under 21 CFR §312.32(g), with the definition and the reporting clock for each. These triggers extend the conventional Subpart J reporting framework to the autonomy-bearing behaviors of the LLM-directed robotic system.

Trigger	Definition	Clock
Serious Physical AI adverse event	A serious adverse event in which the autonomy-bearing system is implicated, on the conventional 21 CFR §312.32 seriousness and expectedness logic.	7 days (fatal or life-threatening); 15 days (other serious, unexpected)
System-drug interaction	A safety-relevant interaction between the device behavior and the investigational drug.	15 days
Cybersecurity incident	A compromise or attempted compromise of the system, records, or control surface.	15 days; 7 days if potential for patient harm
Sustained AI model degradation	Degradation persisting across at least three consecutive procedures or 24 cumulative hours.	15 days
Sim-to-real divergence beyond 2x tolerance	Trajectory error above 4 mm or tip-force error above 1.0 N.	15 days
Digital-twin discrepancy	A discrepancy between the digital twin and the physical system bearing on safety.	15 days

Table 10.4. The six Physical AI reporting triggers under 21 CFR §312.32(g). Each trigger carries a defined reporting clock, and each generates an entry in the hash-chained audit trail preserved from 24 hours before to 72 hours after the event under 21 CFR part 11.

10.4.4 The Seven Physical AI Record Types and 21 CFR Part 11

Seven Physical AI record types are captured and retained for this trial class, in addition to the conventional clinical-trial records. Table 63 enumerates each record type, its content, and its primary regulatory anchor. All seven are bound to a deterministic seed and a repository commit by the hash-chained audit trail (record type seven), so that any reported behavior can be replayed deterministically and any requested record reconstructed. All electronic records, signatures, and the cyber-physical telemetry stream are maintained under 21 CFR part 11, with validated systems, secure time-stamped audit trails that do not obscure prior entries, role-based access controls, and electronic signatures bound to the records they authenticate. Records are retained for the period required by 21 CFR §312.57, namely at least two years after a marketing application is approved for the indication or, if no application is filed or approved, two years after the investigation is discontinued and the FDA is notified; the raw, losslessly reconstructable command and sensor tier is preserved so that a requested record can be reconstructed on FDA request under 21 CFR §312.57. Data released for analysis, deposition, or publication are de-identified by the HIPAA Safe Harbor method, with removal of all eighteen categories of protected health information identifiers.

The volume and tiering of the telemetry record warrant a concrete description, because the determinism guarantee depends on retaining enough of the stream to reconstruct any reported behavior. The system carries 640 sensor channels, eighty per arm across eight arms, sampling force at 100 kHz and other channels at 10 kHz. The sensor and telemetry record (record type three) is tiered from compact summaries, suitable for routine review, down to a raw, losslessly reconstructable level that preserves the full stream around any event of interest. The robot command record (record type two) captures each command issued to the eight arms and their 56 degrees of freedom with timestamps sufficient for deterministic replay, and the LLM advisory record (record type one) captures each prompt, the supplied context, and the advisory output of the second-opinion oracle, hash-pinned to seed and commit and always advisory rather than control-bearing under 21 CFR §312.21(e). The verification-before-generation gate-surface record (record type five) captures each accept, block, or escalate decision of the ten-gate suite; the human-override and intervention record (record type six) captures each manual override, each takeover by the backup operator, and each stop invoked by the Independent Safety Monitor; and the safety-limit-event record (record type four) captures the emergency-stop activations, the force-envelope excursions, and the vascular no-fly gating events. The hash-chained audit trail (record type seven) binds the preceding six together so that the entire record is tamper-evident and deterministically replayable.

Table 63: The seven Physical AI record types captured and retained for the LLM-directed robotic Whipple, in addition to the conventional clinical-trial records. All records are retained under 21 CFR part 11 for the period required by 21 CFR §312.57.

No.	Record type	Content and regulatory anchor
1	LLM advisory record	Each prompt, the supplied context, and the advisory output of the second-opinion oracle, hash-pinned to seed and commit; advisory only, never control authority (21 CFR §312.21(e)).
2	Robot command record	Each command issued to the eight arms (56 degrees of freedom), with timestamps for deterministic replay.
3	Sensor and telemetry record	The cyber-physical stream from 640 sensor channels (80 per arm; 100 kHz force, 10 kHz other), tiered from compact summaries to the raw losslessly reconstructable level.
4	Safety-limit-event record	Emergency-stop activations and latencies, positional and force-envelope excursions ($\leq$ 3 N per arm, $\leq$ 18 N cumulative), and vascular no-fly gating events (soft 3.0 mm, no-fly 1.0 to 1.5 mm, hard-stop 5.0 mm).
5	Verification-before-generation gate-surface record	Each accept, block, or escalate decision of the ten-gate suite, aligned with H.R. 9510 (Verification Before Generation in Physical AI Oncology Trials Act of 2026).
6	Human-override and intervention record	Each manual override, takeover by the backup operator, and stop invoked by the Independent Safety Monitor.
7	Hash-chained audit trail	Cryptographically chained binding of records 1 to 6 to a deterministic seed and repository commit; tamper-evident; preserved from -24 h to $+72$ h around each Physical AI adverse event (21 CFR part 11).

Table 10.5. The seven Physical AI record types, their content, and their regulatory anchors. Record type seven binds the preceding six to a deterministic seed and repository commit so that any reported behavior can be replayed deterministically.

10.4.5 Quality Assurance, Re-Validation, and Decommissioning

Quality assurance spans the clinical and the cyber-physical domains. Before every procedure, the pre-procedure safety matrix is verified, the USL readiness rating is confirmed at or above the deployment threshold, and the emergency-stop, positional, and force envelopes are checked. The Phase 0 simulation-validation gate, which must be satisfied before any patient-facing robotic activity, requires a USL of at least 7.0, at least one thousand simulations across at least two independent frameworks (for example Isaac, MuJoCo, Gazebo, or PyBullet), and a sim-to-real agreement of less than 2 mm in trajectory and less than 0.5 N in tip force. After any software update, model change, or configuration change to the LLM-directed system, a Phase 0 re-validation is performed and the USL is reassessed before any further patient-facing use; no update reaches a participant without re-validation.

The re-validation discipline is the operational counterpart of the hash-pinning guarantee. Because the advisory model is hash-pinned to a deterministic seed and a repository commit, any change to the model, the configuration, or the surrounding software produces a new commit and therefore a new validated configuration that has not yet passed the Phase 0 gate. The rule that no update reaches a participant without re-validation closes the loop: a change cannot be deployed to a patient-facing procedure until the new configuration has satisfied the USL threshold of at least 7.0, the simulation-count threshold of at least one thousand runs across at least two frameworks, and the sim-to-real agreement thresholds of less than 2 mm in trajectory and less than 0.5 N in tip force. The Physical AI Safety Review Committee reviews each such change and the associated USL reassessment on its cadence of ninety days or less, and ad hoc on any triggering event, so that the autonomy-readiness rating in force at any time corresponds to a configuration that has actually been validated.

On any clinical hold or closure, enrollment and patient-facing robotic activity stop immediately, affected participants are transitioned to appropriate clinical care, and a controlled decommissioning and data-preservation procedure is executed under which the robotic system is locked out of patient contact, the deterministic configuration and repository commit are recorded, and the complete hash-chained command and telemetry record is sealed and retained for the regulatory retention period. The computational evidence supporting this assurance approach includes prior in silico and digital-twin work by the author, namely a sixty-second PDAC Whipple with daraxonrasib simulation, an AI digital-twin PDAC simulation, a quantitative systems pharmacology metastatic-PDAC simulation with verification, validation, and uncertainty quantification, a one-hundred-thousand-patient in silico five-arm Phase 3 PDAC study in triplicate, and end-to-end PDAC digital-twin trial proposals [13, 14, 15, 16], together with a sixteen-LLM code-generation evaluation against FDA adverse-event reporting data [12] and the predictive Physical AI oncology modeling that informs the sim-to-real tolerances [17].

The perioperative daraxonrasib pause-and-restart advisory is an instructive worked example of how the advisory oracle, the human confirmation step, and the record-retention discipline operate together within these quality boundaries. The drug is paused before surgery, and the restart day is determined from an LLM-bound, human-confirmed advisory whose inputs are the pancreaticojejunostomy ISGPS fistula grade (A, B, or C) and the serum daraxonrasib trough relative to a 0.5 ng/mL threshold, with outputs of restart at day seven, day fourteen, or day twenty-one, or hold. A thirty-two-iteration deterministic sweep recorded in the protocol returns a restart at day seven in twenty-nine of thirty-two iterations (90.6 percent, corresponding to ISGPS Grade A with a sub-threshold trough), a restart at day fourteen in three of thirty-two iterations (9.4 percent, corresponding to ISGPS Grade B or a delayed profile), and no iteration returning day twenty-one, with a baseline serum trough of 0.45 ng/mL (sub-threshold) across all iterations. The advisory is recorded in the LLM advisory record (record type one), the human confirmation is recorded in the human-override and intervention record (record type six), and both are bound by the hash-chained audit trail (record type seven), so that the determinism of the sweep can be reproduced on FDA request and the human-in-the- loop character of the decision is preserved in the record. This example shows the advisory oracle functioning exactly as 21 CFR §312.21(e) requires: informing, never controlling, a clinical decision that a human makes and records.

10.5 Selected References

The following selected references support the additional information in this section and are cited in full in the back matter. The FDA Phase 1 guidance frames the content and graded rigor appropriate to an original first-in-human IND [8], and the daraxonrasib and on-premises robotic Whipple sources establish the drug mechanism and the device context [1, 2]. The Physical AI regulatory adaptations of 21 CFR parts 312 and 50 and of ICH guidance govern the autonomy-bearing overlay, the opt-out, and the Subpart J reporting and record-retention controls [3, 19, 18]. The Verification Before Generation in Physical AI Oncology Trials Act of 2026 (H.R. 9510) anchors the verification-before-generation gate surface [6], and the trust, accountability, and site-readiness literature for clinical Physical AI informs the governance and oversight design [7, 5, 11, 9]. The PDAC epidemiology and the prior computational evidence supporting the deferral and assurance rationale are drawn from the indication and digital-twin sources [4, 13, 14, 15, 16, 12, 17]. A complete reference list appears in the back matter.

11 Relevant Information

This section provides the additional relevant information that the U.S. Food and Drug Administration may request under 21 CFR §312.23(a)(11), together with the method-level grounding and verification disclosures that apply because this Investigational New Drug application (IND v1.0) was assembled by an artificial intelligence build operating over the repository at version v4.3.0. The disclosures are placed here, at the close of the package, so that a reviewer can read the entire submission first, on its clinical, statistical, and regulatory merits, and then evaluate exactly how the document was constructed: how each figure and number was grounded to a named source file, how each figure was cited where its content is discussed, how the build is monitored in real time on a public version-control system, how the repository and file naming mirror the document structure, and where the residual limitations of the method lie. Nothing in this section substitutes for the clinical, statistical, and regulatory substance of the preceding sections; rather, it states the controls that give a reviewer confidence that the substance is internally consistent across the package, traceable to named source files, and reproducible from those sources by a third party. The verification posture described here is aligned with the verification-before-generation discipline that the program has advanced for physical artificial intelligence in oncology trials, the same discipline codified in the H.R. 9510 framing on which the device-directed component of the trial rests [6, 7], and it is consistent with the phased-development expectations that govern an initial Phase 1 submission [8]. The disclosures should be read as a transparency artifact offered in the spirit of 21 CFR §312.23(a)(11), not as a request for any special regulatory accommodation: every generated document, this IND included, remains subject to the same human medical, statistical, and regulatory review as a conventionally authored submission.

The structure of the section is as follows. We first state the method-level grounding and verification discipline and present, in summary and then in detail, the five name-matching and grounding verifications that the build applies to every artifact it emits (Figure 21 and Table 11.1). We then describe the finer-grained figure-grounding discipline that keeps each rendered figure provably identical to its Mermaid source (Figure 22 and Table 11.2). We next document the real-time auto-commit and auto-pull-request monitorability of the build, the repository, directory, and file name matching that makes the package self-describing (Table 11.3), and the principal limitations of the method together with the mitigation applied to each (Table 11.4). A short worked example ties the abstractions to a single concrete trial number (Table 11.5), and a closing compilation statement records how a reviewer can reproduce the compiled PDF directly from the repository.

11.1 Method-Level Grounding and Verification

The IND was produced by a single-prompt, multi-file build that proceeds through four stages, mermaid to draft to full to final, with the present final stage expanding each full-stage section toward the package length target and applying the senior-author formatting pass that a PDF compiler is required to verify. To keep that build verifiable rather than merely fluent, the method applies five name-matching and grounding verifications to every artifact it emits. The five verifications are summarized in Figure 21 and detailed, with a concrete IND example for each, in Table 11.1. They are not abstract quality slogans; each one is a check that a reviewer can independently reproduce by inspecting the public repository tree, the rendered document, and the commit and pull-request history. The five together yield a grounded, monitorable IND in which every figure names its source, every figure is cited where its section discusses it, every file is visible on GitHub as it is written, the repository and directory names mirror the document structure, and the section file names map one to one onto the regulatory sections they contain.

This discipline matters for a regulatory submission in a way that distinguishes it from generic large-language-model authoring. A submission to FDA is read for internal consistency: a dose level stated in the General Investigational Plan must match the dose level in the protocol synopsis, in the dose-escalation figure, and in the safety narrative; a force cap stated in the device description must match the cap in the safety-reporting triggers. Ungrounded generation is fluent but does not guarantee that the same number appears identically wherever it is reused, and a single drifted number in a Phase 1 oncology submission can change the reviewer's understanding of the risk. The grounding discipline forecloses that drift by binding every quantitative claim a figure carries to a named source file from which the surrounding prose draws the same value, so that the figure and the text cannot diverge [10, 16].

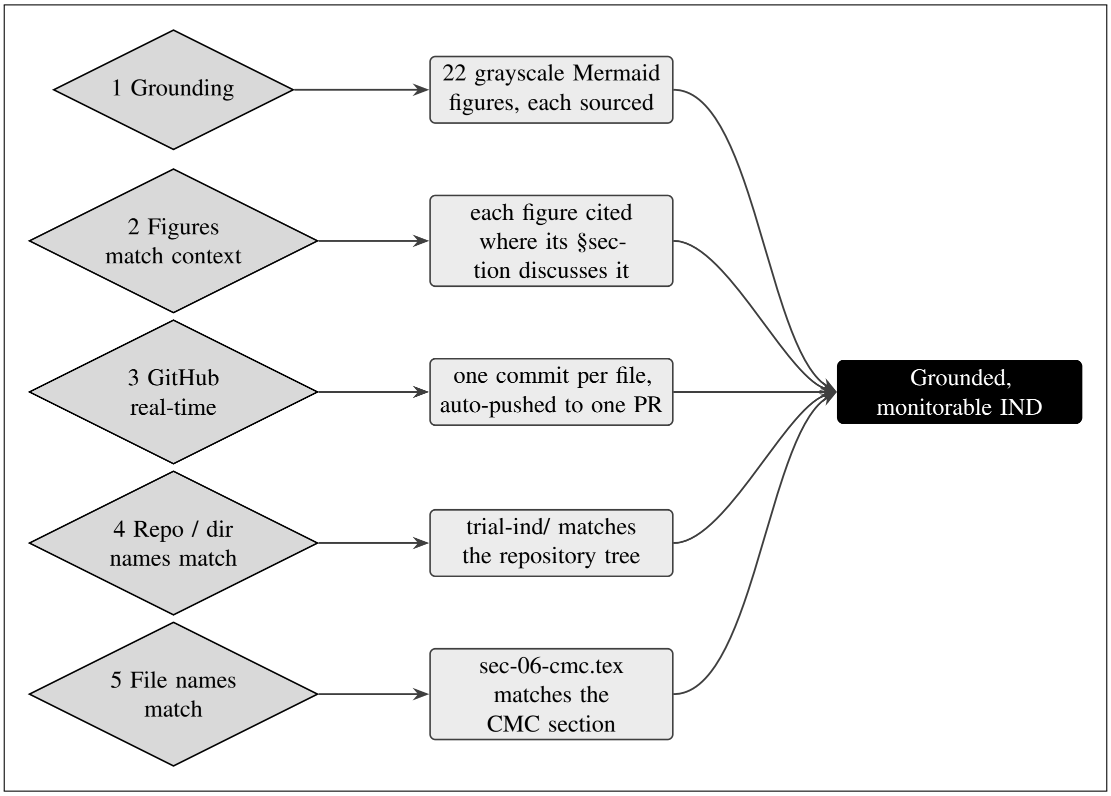

Figure 21. Five name-matching and grounding verifications, each with a concrete IND example: grounding (every one of the 22 grayscale figures names its source files); figures match context (each is cited where its section discusses it); GitHub real-time (one commit per file, auto-pushed to one continuously updated pull request); repository and directory names match (the trial-ind tree mirrors the document structure); and file names match (sec-06-cmc.tex maps to the Chemistry, Manufacturing and Control section). Together they yield a grounded, monitorable IND.

11.1.1 The Five Verifications in Detail

Verification 1, grounding. Every quantitative claim that a figure carries is traceable to a named source file rather than invented at generation time. The 22 grayscale figures of this IND each declare, in their Mermaid source under `trial-ind/mermaid/`, the source files from which their nodes, edge labels, and numbers were drawn, and those numbers are the same document counts, dose levels, force caps, sampling rates, and review clocks reused in the surrounding prose. For example, the dose levels of 160 mg, 220 mg, and 300 mg once daily, the 28 day dose-limiting-toxicity window, and the maximum enrollment of 18 treated subjects appear identically in the dose-escalation figure, in the General Investigational Plan, and in the protocol from which they originate. The same binding holds for the device parameters that recur across the safety sections: the eight robotic arms, the 56 total degrees of freedom

(8 arms times 7 each), the 640 sensor channels (80 per arm), the 100 kHz force sampling and 10 kHz sampling for other channels, the per-arm force cap of 3 N and the cumulative cap of 18 N, the 10 kHz safety heartbeat with its 100 microsecond watchdog, the cross-arm emergency stop at 3 ms to 0.0 N, and the hardware emergency stop at 500 ms under 21 CFR §312.404 all appear with identical values wherever they are reused. This binds the figures to the trial facts and prevents a figure from drifting away from the text it illustrates [10, 16].

Verification 2, figures match context. Each figure is cited in the sentence or paragraph that discusses it, in the section where that discussion belongs, so that no figure floats unreferenced and no claim in the prose lacks its illustrating figure. The daraxonrasib mechanism-of-action figure is cited in the introductory material that names the drug and its pharmacological class, the 3+3 dose-escalation automaton is cited in the General Investigational Plan, the chemistry information architecture is cited in the Chemistry, Manufacturing and Control section, the perioperative pause-and-restart advisory is cited where the restart-day decision is described, and the two figures of the present section are cited here, as Figure 21 and Figure 22. A reviewer can confirm this by following any figure number from its caption back to the citing sentence, and the converse: every figure number that appears in prose resolves to exactly one rendered figure.

Verification 3, GitHub real time. Each distinguishable file is committed and pushed at the moment it is written, and those commits accumulate on a single continuously updated pull request, so that the IND is observable on GitHub as it is constructed rather than only after completion. This gives the sponsor-investigator, and in principle an auditor, a real-time view of provenance: who or what wrote each file, in what order, and against which commit. The auto-commit and auto-pull-request behavior is the documentary analogue of the audit-trail and monitorability expectations that govern the device-directed component of the trial, under which the device preserves a window of records from 24 hours before to 72 hours after each physical artificial intelligence adverse event under 21 CFR part 11 [5, 9].

Verification 4, repository and directory names match. The directory tree named in the prose is the directory tree on disk. References to `trial-ind/mermaid/`, `trial-ind/full-ind/sections/`, `trial-ind/final-ind/sections/`, and `trial-ind/inputs/` denote the actual directories of the repository at version v4.3.0, so a reader can navigate from any path mentioned in the document straight to the corresponding location in the public tree without translation. The four stage directories, `mermaid/`, `draft-ind/`, `full-ind/`, and `final-ind/`, mirror the four build stages in the same left-to-right order in which the prose describes them.

Verification 5, file names match. The section file names map one to one onto the regulatory sections they contain. `sec-06-cmc.tex` is the Chemistry, Manufacturing and Control section, `sec-07-pharmacology-toxicology.tex` is the Pharmacology and Toxicology section, `sec-10-relevant-information.tex` is the present section, and `fig-09-five-grounding-verifications.md` is the Mermaid source for Figure 21. The naming convention makes the package self-describing and removes ambiguity about which file holds which content, so a reviewer can confirm that no section is missing and no file is misnamed simply by listing the directory.

#	Verification check	Concrete IND example
1	Grounding via Mermaid figures	The 22 grayscale figures in `trial-ind/mermaid/` each name their source files and carry the dose levels (160, 220, 300 mg QD), the 28 day DLT window, the force caps (3 N per arm, 18 N cumulative), and the 30 day FDA clock reused verbatim in the prose
2	Figures match IND context	Each figure is cited in the sentence that discusses it, in the section where that discussion belongs (mechanism in the Introduction, 3+3 automaton in the Plan, CMC architecture in the CMC section, these two figures here)
3	AI files viewable on GitHub in real time	One commit per file is pushed at generation, accumulating on a single continuously updated pull request, so each artifact appears on the branch as it is made
4	Repository and directory names match	The prose refers to `trial-ind/mermaid` and `full-ind/sections`, which are the directories on disk in repository v4.3.0
5	File names match repository file names	A cited `sec-06-cmc.tex` is the CMC section; `fig-09-five-grounding-verifications.md` is the Mermaid source for Figure 21

Table 11.1. The five name-matching and grounding verifications, each with a concrete example drawn from this IND. Every check is independently reproducible from the public repository tree, the rendered document, and the commit and pull-request history.

11.1.2 Why Grounding Is the Load-Bearing Control

Of the five verifications, grounding is the one on which the regulatory value of the package most depends, and it is worth stating precisely what it does and does not guarantee. Grounding guarantees that a number rendered in a figure or asserted in prose is the same number that appears in the named source file from which it was drawn; it does not, by itself, guarantee that the source number is clinically correct. Clinical correctness is established by the human medical, statistical, and regulatory review that every generated document must pass before use, and by the underlying protocol, investigator brochure, and chemistry record that the build ingests as curated inputs. What grounding contributes is the elimination of a specific and common failure mode of language-model authoring, namely the silent substitution of a plausible but unsourced value. Because the dose levels, the DLT window, the enrollment cap, the device force caps, and the safety-reporting clocks are each carried from a single source through every figure and every paragraph that reuses them, a reviewer who verifies one occurrence has effectively verified all of them, and any divergence would be visible as a mismatch against the named source. This is the property that lets a reviewer audit the package efficiently rather than re-checking every restatement of every number independently [10, 12].

The grounding discipline is reinforced by the fixed grayscale encoding of node roles in the figures. Because the eight-tone ramp is defined once in `indstyle.sty` and every figure draws its node fills from that ramp, the tonal meaning of a node, goal, process, accent, step, input, context, decision, or dark, is identical across all 22 figures and across all four build stages. A decision diamond in the 3+3 automaton carries the same tone as a decision diamond in the perioperative advisory or in the figure-grounding diagram of the present section, so a reviewer learns the visual vocabulary once and reads every figure with it. This consistency is itself a grounding property: it ties the visual encoding to a single source of truth in the style file rather than to per-figure choices that could drift [14].

11.2 Figure Grounding: Mermaid to TikZ

A second, finer-grained discipline governs the figures themselves: every figure in this IND is authored once as a grayscale Mermaid source in `trial-ind/mermaid/` and then reproduced as a TikZ figure of identical complexity in the LaTeX document, with no simplification of nodes, edges, edge labels, or quantitative content. The translation pathway and its verify-twice checkpoint are shown in Figure 22. The Mermaid source defines the nodes, edges, grayscale palette, and quantitative data; the conversion preserves all of these; the TikZ `mermaidfig` environment renders them using the `mm*` node styles defined in `indstyle.sty` (the same eight-tone ramp that maps one to one from the Mermaid classDef classes); and the result is verified twice, for text-box and arrow overlaps, for the specified arrow looseness, and for proper box spacing, before it appears identically in the draft, full, and final stages. A figure that fails the verify-twice checkpoint is returned to the TikZ step and corrected; only a figure that passes is allowed to render in the document.

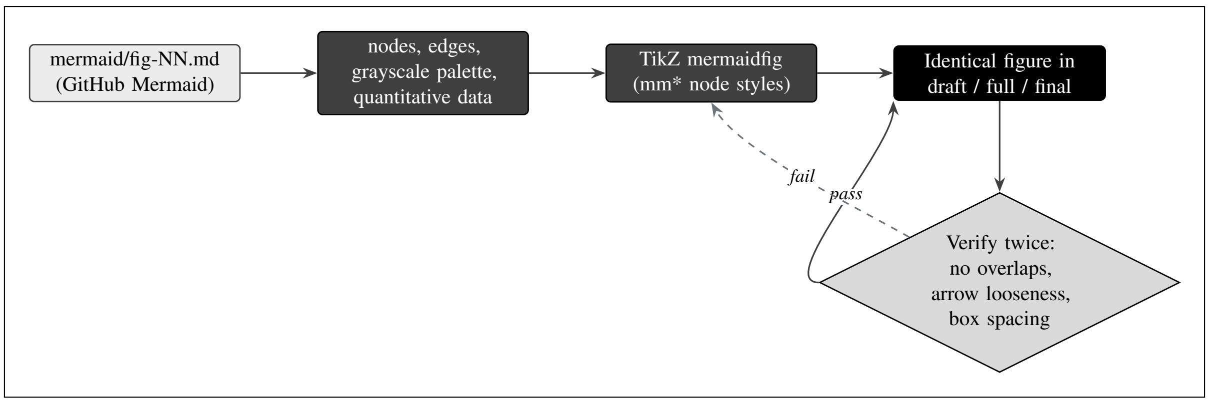

Figure 22. Figure grounding. Each Mermaid source in `mermaid/` is reproduced as a TikZ `mermaidfig` of the same complexity (the same nodes, edges, grayscale palette, and quantitative data), then verified twice for text-box and arrow overlaps, for the specified arrow looseness, and for proper box spacing, before it appears identically in the draft, full, and final stages. A figure that fails is returned to the TikZ step; only a figure that passes renders.

The practical effect of this discipline is that the figure a reviewer sees in the compiled PDF is provably the same diagram, with the same numbers, as the Mermaid source committed to the repository. There is no separate figure-rendering pipeline, no raster export, and no opportunity for a number to be transcribed differently between the source and the rendered figure. All figures are vector TikZ; the IND contains no PNG or JPG images and no `\includegraphics` calls. The grayscale ramp is fixed in the style file, so the tonal encoding of node roles (goal, process, accent, step, input, context, decision, dark) is identical across all 22 figures and across the four build stages [14].

11.2.1 The Verify-Twice Checkpoint

The verify-twice checkpoint is named for the fact that each figure is inspected on two independent criteria before it is admitted to the document, and that the inspection is repeated after any correction so that a fix to one criterion cannot silently break the other. The first criterion is geometric: no two text boxes overlap and no arrow crosses through a box. The build enforces explicit spacing rules that a reviewer can verify against the rendered coordinates, namely that side-by-side box centers are at least 4.0 cm apart in the horizontal direction, that stacked box centers are at least 1.7 cm apart in the vertical direction, that the node text width is 30 mm, and that decision diamonds, which are wider than rectangles for the same text, are given extra room. The second criterion is the routing of curved arrows: every curved edge uses a `to[out,in,looseness]` specification with a looseness held between 0.4 and 0.7, which keeps the curve gentle enough to avoid sweeping across a neighboring box while still separating parallel edges that converge on the same target. Table 11.2 records these geometric checks together with the threshold applied to each, so that a reviewer can confirm the layout discipline against the rendered figure rather than taking it on assertion [14, 11].

Geometric check	**Threshold**	**Purpose**
Side-by-side box centers	$\geq$ 4.0 cm in x	Prevents horizontal text-box overlap at the fixed 30 mm node text width
Stacked box centers	$\geq$ 1.7 cm in y	Prevents vertical overlap of stacked boxes and their captions
Node text width	30 mm fixed	Uniform box sizing so spacing thresholds apply identically to every figure
Decision diamonds (mmdec)	Extra room reserved	Diamonds are wider than rectangles for the same text and need clearance
Curved-arrow looseness	0.4 to 0.7	Keeps curves gentle so arrows do not sweep across neighboring boxes
Arrow-box intersection	None permitted	No edge may pass through a node it does not connect

Table 11.2. The verify-twice geometric checks applied to every figure before it is admitted to the document, with the threshold and purpose of each. The checks are repeated after any correction so that a fix to one cannot silently break another; only a figure that passes both inspections renders in the draft, full, and final stages.

Because the checkpoint is geometric and the thresholds are explicit, it is deterministic in a way that a subjective "does this look right" review is not. Two reviewers applying the 4.0 cm and 1.7 cm thresholds to the same rendered figure reach the same verdict, and a figure that passes in the draft stage continues to pass in the full and final stages because the node coordinates and the style file are unchanged across stages. This is what allows the program to assert that the figure is identical from Mermaid source to final PDF: not merely that it looks similar, but that it satisfies the same enumerated geometric and quantitative criteria at every stage [14].

11.3 Real-Time Auto-Commit and Auto-Pull-Request Monitorability

The build is monitorable in real time. As each section file, style file, figure source, and supporting file is written, it is committed individually and pushed to the remote, and the accumulating commits are surfaced on a single open pull request that updates continuously over the life of the build. This means that the provenance of the IND is not reconstructed after the fact from a final snapshot; it is visible while the document is being produced. A reader with access to the public repository can watch the `trial-ind/final-ind/` tree fill in section by section, can inspect the diff for any individual file, and can read the commit history as an ordered, timestamped record of how the package came together. This real-time monitorability is the documentary counterpart of the trial's own audit-trail commitments, under which the device preserves a window of records from 24 hours before to 72 hours after each physical artificial intelligence adverse event, and it is consistent with the broader transparency posture that the program has argued is necessary before higher-autonomy systems are used in oncology trials [6, 7, 11]. The same single-prompt, auto-commit, auto-pull-request method has been demonstrated across prior multi-file builds in the program, including the companion Phase 2 protocol and the legislative-text repositories, which establishes that the approach is reproducible rather than specific to this submission [10, 17].

It is important to be precise about what this monitorability does and does not provide in a regulatory sense. It provides a tamper-evident, timestamped, ordered record of document construction that an auditor can replay commit by commit; it does not provide, and is not offered as, the formal part 11 audit trail that governs the trial's clinical and device data, which is a separate system with its own controls. The two are analogues, not the same system: the build's version-control history documents how the application was written, while the trial's part 11 audit trail documents how clinical and device events are recorded during conduct of the study. Stating this distinction plainly avoids any implication that authoring a submission with a monitorable build relieves the sponsor-investigator of the independent obligation to maintain part 11 audit trails during the trial itself [5, 9].

11.4 Repository, Directory, and File Name Matching

The repository, directory, and file naming is itself a control. The IND lives under trial-ind/ in the public repository, and within it the four build stages occupy mermaid/, draft-ind/, full-ind/, and final-ind/, mirroring the mermaid to draft to full to final progression. Within each LaTeX stage the section files follow a fixed sec-NN-<name>.tex convention whose number and name match the ReGARDD table of contents, so that the file that holds the Chemistry, Manufacturing and Control section is named sec-06-cmc.tex, the file that holds the Pharmacology and Toxicology section is named sec-07-pharmacology-toxicology.tex, and the figure that this section reproduces first is sourced from fig-09-five-grounding-verifications.md. Table 11.3 records the directory and file-naming correspondences that make the package self-describing, so that a reviewer can move from any path named in the prose directly to the corresponding location on disk without translation, and can confirm that no section is missing and no file is misnamed.

Path in repository v4.3.0	**Holds**	**Name-match guarantee**
trial-ind/	The full IND across four stages	Top-level directory named for the IND it contains
trial-ind/mermaid/	The 22 grayscale figure sources	Stage 1 directory; fig-NN-<name>.md files
trial-ind/full-ind/sections/	The full-stage section files	Stage 3 sections; one sec-NN-<name>.tex per top-level section
trial-ind/final-ind/sections/	The final-stage section files	Stage 4 sections; the same one-to-one sec-NN mapping
sec-06-cmc.tex	Chemistry, Manufacturing and Control	File name matches the CMC section it contains
sec-10-relevant-information.tex	Relevant Information (this section)	File name matches the section it contains
fig-09-five-grounding-verifications.md	Source for Figure 21	Figure file name matches the figure it renders

Table 11.3. Repository, directory, and file name-matching for the IND (Verification 5). Every path named in the prose denotes the actual location on disk in repository v4.3.0, making the package self-describing and navigable.

The name-matching control has a second, quieter benefit for regulatory review: completeness checking. Because the ReGARDD table of contents enumerates the top-level sections, and because each top-level section corresponds to exactly one `sec-NN-<name>.tex` file whose number and name match the contents entry, a reviewer can verify that the package is complete by listing the `sections/` directory and confirming that the numbered files run consecutively with no gap. A missing section would appear as a missing file in the sequence, and a misfiled section would appear as a name that does not match its contents entry. The same logic applies to the figures: the 22 figure sources in `trial-ind/mermaid/` are numbered consecutively, so a gap in the figure numbering would be visible as a missing source file. This turns the naming convention into a lightweight but real completeness check that does not depend on reading the prose [9, 11].

11.5 Limitations of the AI-Generation Method

The grounding and verification controls above constrain the build but do not eliminate its limitations, and those limitations are stated plainly here so that a reviewer weighs the package accordingly. Table 11.4 enumerates the principal limitations together with the mitigation applied to each. The most important point is the one that no method-level control can change: the AI-generation method alters none of the clinical realities of the trial. It does not change tumor biology, it does not shorten the 28 day dose-limiting-toxicity observation window or the 30 and 90 day safety follow-up or the survival follow-up to 24 months, and it does not compress the fixed 30 day FDA review clock or the IRB meeting calendar. What the method compresses is the administrative and document-preparation bucket alone; the clinical and operational bucket and the fixed regulatory-review bucket are unchanged [8, 4]. Every generated document, this IND included, must still pass human medical, statistical, and regulatory review before use, and the build is an independent research artifact, not an active IND or investigational device exemption and not regulatory advice [3, 19].

Several narrower limitations follow from how the build operates. The method does not rely on live internet search for a project of this size, because specifying single external URLs is unreliable at scale; it depends instead on a curated repository of appropriately sized inputs, which bounds the source material a single build can absorb. Because the model does not always have access to a PDF compiler during generation, a portion of senior-author formatting, including table column widths, figure-caption spacing, and occasional page breaks, must still be applied by hand, and the final stage performs that pass explicitly. Total input is kept under roughly 5 MB with individual files held under per-file token limits, which caps how much source material a single build can ingest. High-dimensional or approximate inputs can make outputs less deterministic, and oversized or overly complex inputs can surface server errors that signal reduced quality; the build mitigates this by keeping inputs curated and appropriately sized. These constraints are the same ones characterized for the broader single-prompt method and are disclosed here for the specific case of an IND [18, 15, 12].

A further limitation deserves explicit emphasis for a Phase 1 oncology submission. The method's controls are controls on *document construction*, not on *clinical judgment*. They ensure that the dose-escalation rule rendered in the figure is the rule stated in the protocol, but they do not, and cannot, decide whether 160 mg once daily is the correct starting dose, whether the 28 day DLT window is the correct observation period, or whether the 3 N per-arm force cap is the correct safety limit. Those decisions are made by the human investigators, informed by the daraxonrasib clinical experience that supports the 300 mg once daily RASolute 302 recommended Phase 2 dose, by the preclinical and computational evidence base, and by the device engineering record, and they are subject to FDA and IRB review. The AI-generation method is a faithful and auditable scribe of those decisions; it is not a substitute for them. The boundary is the same staged-autonomy boundary that governs the device: just as the on-premises language model in the operating room is a second-opinion advisory oracle only, with full autonomy prohibited under 21 CFR §312.21(e), the document-generation build is an advisory authoring tool whose output is gated by human review [6, 7].

Limitation	Mitigation applied in this IND
The method changes no clinical reality (tumor biology, DLT window, survival follow-up)	The 28 day DLT window, 30 and 90 day safety follow-up, and 24 month survival follow-up are preserved exactly as specified in the protocol; the build compresses only document preparation
The method cannot shorten fixed regulatory clocks	The 30 day FDA review and the IRB meeting calendar are stated as binding and unchanged; faster authoring only starts the clock sooner
The method controls construction, not clinical judgment	Dose levels, DLT window, and force caps are authored faithfully from the protocol but decided by human investigators under FDA and IRB review
Generated content can be fluent but wrong without grounding	The five name-matching and grounding verifications and the figure verify-twice checkpoint bind every figure and number to a named source file
No live internet search at scale; URLs unreliable in bulk	Inputs are a curated repository of appropriately sized files rather than ad hoc external URLs
No PDF compiler during generation; some formatting needs a manual pass	Senior-author formatting (column widths, caption spacing, page breaks) is applied explicitly in the final stage
Bounded input budget (under 5 MB total; per-file token limits)	Source material is curated and apportioned so each build stays within limits without truncation
High-dimensional or oversized inputs reduce determinism and can surface server errors	Inputs are kept appropriately sized; complex content is split so the build remains deterministic and stable
The artifact is research, not an active regulatory submission	The package is labeled an independent research artifact, not an active IND or IDE and not regulatory advice; human review is required before any use

Table 11.4. Principal limitations of the AI-generation method and the mitigation applied to each in this IND. The clinical and fixed-regulatory limitations are intrinsic and cannot be removed; the method-level limitations are mitigated by grounding, verification, and a human review requirement.

11.5.1 A Worked Example of Grounding

To make the grounding discipline concrete rather than abstract, consider a single trial number followed from its source to its rendered occurrences. The recommended Phase 2 dose of daraxonrasib, 300 mg once daily, is established in the daraxonrasib clinical record (the RASolute 302 program) and is the value of the third and highest dose level, DL3, in this trial's 3+3 escalation. From that one source the number propagates, unchanged, to several places: it is DL3 in the dose-escalation figure; it is the top of the dose-level ladder in the General Investigational Plan; it is the dose at which a maximum tolerated dose determination would conclude if no more than one of six subjects experienced a dose-limiting toxicity at that level; and it is the dose carried into the precision discussion of the statistical plan, where the maximum enrollment of 18 subjects (three levels of six) bounds the confidence intervals. Table 11.5

traces this single value through its occurrences and pairs it with the companion enrollment and precision figures, so a reviewer can verify the entire chain from one source.

Grounded value	**Source of record**	**Where it recurs identically in the IND**
300 mg once daily (DL3)	Daraxonrasib RASolute 302 RP2D	Top dose level in the dose-escalation figure, the General Investigational Plan ladder, and the MTD/RP2D narrative
28 day DLT window	Protocol v1.0.0	Dose-escalation automaton, safety narrative, and the co-primary endpoint defining MTD/RP2D
18 treated subjects (3 levels $\times$ 6)	Protocol enrollment cap	Number-of-subjects figure, statistical precision table, and halt-rule denominators
Precision at 0 of 18	Exact Clopper-Pearson	0% observed, 95% CI 0% to 18%, stated identically in the statistical plan and any figure that cites it
Precision at 1 of 18	Exact Clopper-Pearson	5.6% observed, 95% CI 0.1% to 27%, stated identically wherever the single-event precision is discussed

Table 11.5. A worked grounding example. One source value, the 300 mg once daily recommended Phase 2 dose, propagates unchanged through the dose-escalation figure, the General Investigational Plan, and the statistical precision discussion, alongside the 28 day DLT window, the 18 subject enrollment cap, and the exact Clopper-Pearson precision figures, so a reviewer can verify the whole chain from a single source.

The worth of the example is that it shows grounding operating at the level of review, not just of authoring. A reviewer who confirms that DL3 is 300 mg once daily in the dose-escalation figure, and that the source of record for that value is the daraxonrasib RASolute 302 recommended Phase 2 dose, has simultaneously confirmed every other place the value appears, because the grounding discipline guarantees they are the same value drawn from the same source. The same holds for the 28 day DLT window, the 18 subject cap, and the exact Clopper-Pearson precision figures of 0 percent (95 percent confidence interval 0 percent to 18 percent) at 0 of 18 and 5.6 percent (95 percent confidence interval 0.1 percent to 27 percent) at 1 of 18. The grounding does not make these clinical or statistical choices correct; it makes them consistent and auditable, which is exactly the property a reviewer needs to evaluate them efficiently [10, 4].

11.6 Compilation Statement

All LaTeX sources for this IND compile in Overleaf with pdfLaTeX. The final-stage document is built from `main.tex`, the shared style file `indstyle.sty`, the bibliography `references.bib`, and the `sections/` directory of section files, with every figure rendered as vector TikZ inside the `mermaidfig` environment and every table set full width with the ragged-right column types defined in the style file. There are no raster images, no `\includegraphics` calls, and no external image dependencies, so the package compiles from its own sources without additional assets. The bibliography is processed with the `ieeetr` style, and the codified references throughout the package use the section symbol, for example 21 CFR §312.23, exactly as rendered by the style file. A reviewer can therefore reproduce the compiled PDF directly from the repository at version v4.3.0 by uploading the stage directory to Overleaf and compiling with pdfLaTeX, which closes the loop from grounded source to verifiable, reproducible document. The reproducibility is exact in the sense that the same sources produce the same document:

the figures are deterministic TikZ with fixed coordinates, the tables are fixed-width tabularx with explicitly stated column widths, and the style file pins the grayscale ramp, the page geometry, and the typographic penalties, so a third-party compile yields a PDF that matches the submitted one figure for figure and table for table [8, 1, 2].

References and Back Matter

This back matter closes the Initial Investigational New Drug Application v1.0 for the LLM-Directed PDAC Robotic Daraxonrasib Phase 1 study, assembled within repository v4.3.0. The complete reference list renders below through the `ieeetr` bibliography style from `references.bib`. Every entry carries a clickable URL and, where a digital object identifier is assigned, a clickable DOI URL with the DOI text shown in full; long links break on any character so that no link runs off the right margin of the text block. The reference base supporting this application is intentionally comprehensive and spans seven evidentiary strands: the FDA Phase 1 and combination drug-device regulatory guidances [8]; the daraxonrasib (RMC-6236) clinical and mechanistic literature underpinning the RASolute 302 and RASolve 301 programs and the June 2025 Breakthrough Therapy designation [1]; the 21 CFR Part 312 (IND) and Part 50 (informed consent) and ICH Physical Artificial Intelligence regulatory adaptations [3, 19, 18]; the five author computational works dated 2025 to 2026 (the 60-second PDAC Whipple plus daraxonrasib simulation, the artificial intelligence digital-twin PDAC simulation, the quantitative systems pharmacology metastatic-PDAC simulation with verification, validation, and uncertainty quantification, the 100,000-patient in silico Phase 3 five-arm PDAC triplicate, and the end-to-end PDAC digital-twin trial proposals) [15, 13, 14, 16]; the H.R. 9510 Verification Before Generation in Physical AI Oncology Trials Act of 2026 and the national platform and clinical-trust frameworks [6, 7, 11]; the sixteen-large-language-model code-generation evaluation against the FDA Adverse Event Reporting System [12]; and the proposed clinical research protocol v1.0.0 (DOI 10.5281/zenodo.20780121) that this IND operationalizes.

The back matter elements that follow document authorship and contributor attribution, the open-weights build model used to assemble this application, the ethical and independence disclosures that bound the entire repository, the data availability and licensing terms, and the recommended citation. Table -.1 records the provenance of each major content strand so that a reviewer can trace every quantitative claim in IND Sections 3 through 11 back to its grounding source. Table -.2 then maps the back matter disclosure elements to the controlling federal authorities under 21 CFR, and Table -.3 enumerates the four build-stage source trees that constitute the archived, reproducible record of this application.

Acknowledgments

This Investigational New Drug Application was assembled with Claude Code (Opus 4.8 Ultracode build, 1M context) as the build model, operating under the single-prompt mermaid to draft to full to final pipeline described in IND Section 11 and committing each stage in real time. At each stage the build model expanded a grayscale Mermaid specification into a draft section scaffold, then into a full regulatory draft, and finally into this senior-author polished form, with automated commits and pull requests opened for human review after every stage so that the provenance of every sentence and every quantitative value remained auditable. The author thanks ChatGPT 5.5 (Thinking Extended) and Gemini 3.1 Pro for research-source assistance and for manual editing support; these systems were used only for source retrieval and editorial review and exercised no control over the clinical design, the safety architecture, or the regulatory determinations recorded herein. This v1.0 application was edited and approved by Kevin Kawchak and added to repository v4.3.0 under kevinkawchak/physical-ai-oncology-trials/tree/main/trial-ind. The repository contributors are @kevinkawchak, @claude, @google-gemini, and @openai. No funding body, sponsor, or regulatory authority directed, reviewed, or endorsed the preparation of this document.

Evidentiary strand	**Grounding source and key facts**	**Supports IND sections**
Daraxonrasib mechanism and class	Oral RAS(ON) multi-selective (pan-RAS) inhibitor binding GTP-bound mutant RAS in complex with cyclophilin A; once-daily dosing [1]	3.1, 8.1
Dose-escalation design	3+3 escalation across DL1 160 mg, DL2 220 mg, DL3 300 mg once daily; 28-day DLT window; RP2D at or below MTD [8]	4, 6.1
Robotic Whipple device	On-premises LLM-directed eight-arm system, 56 degrees of freedom, 640 sensor channels, force caps 3 N per arm and 18 N cumulative [2]	5, 7
Verification before generation	Ten-gate ACCEPT/BLOCK/ESCALATE suite; LLM as second-opinion advisory oracle only; full autonomy prohibited under 21 CFR §312.21(e) [6, 7]	5, 10, 11
Perioperative pause-and-restart	ISGPS fistula grade plus serum trough versus 0.5 ng/mL; 32-iteration sweep yielding T+7d 29/32, T+14d 3/32, T+21d 0/32 [9]	6.1, 9
Computational evidence base	100,000-patient in silico Phase 3, QSP metastatic-PDAC simulation, and digital-twin trials [15, 13, 14, 16]	8, 9
Regulatory framework	21 CFR Part 312 and Part 50 Physical AI adaptations; ICH harmonization [3, 19, 18]	1, 6, 10, 11

Table -.1. Provenance map linking each evidentiary strand to its grounding reference and to the IND sections it supports.

Author and ORCID

Kevin Kawchak, Chief Executive Officer, ChemicalQDevice, San Diego, California. ORCID iD: https://orcid.org/0009-0007-5457-8667. Correspondence: kevink@chemicalqdevice.com. In the proposed study the author would serve as Sponsor-Investigator on behalf of ChemicalQDevice under the combined obligations of a sponsor and an investigator set out in 21 CFR §312.3(b), assuming the sponsor duties of 21 CFR §§312.50 through 312.59 and the investigator duties of 21 CFR §§312.60 through 312.69 simultaneously. The resulting financial interest is disclosed in full under 21 CFR Part 54 and on FDA Form 3455, and is mitigated by the independent three-tier oversight structure detailed in the proposed clinical research and additional information sections: a Data Safety Monitoring Board with binding halt authority at each cohort boundary and on any triggered review, an Independent Safety Monitor with real-time per-procedure stop authority, and a Physical AI Safety Review Committee convening on a cadence not exceeding 90 days. Because the Sponsor-Investigator does not sit on any of these bodies and cannot override a stop or halt determination, the conflict of interest is structurally separated from the decisions that protect enrolled subjects, consistent with the financial-interest mitigation expectations under 21 CFR Part 54.

Ethical Disclosures

This is an independent research and regulatory-authoring demonstration. No real patients were enrolled, no protected health information was used, and no Institutional Review Board approval was sought or obtained. The exemplar case PAT-PDAC-0001 (a synthetic 68-year-old with a 3.4 cm head-of-pancreas lesion abutting the superior mesenteric vein, KRAS G12D, microsatellite stable, CA 19-9 of 412 U/mL, ECOG 1, after four cycles of neoadjuvant mFOLFIRINOX) and all quantitative simulation outputs, including the 32-iteration pause-and-restart sweep and the 100-tick vascular no-fly verdict distribution, are synthetic and illustrative and were generated in silico for the sole purpose of exercising the regulatory and safety logic of this application. This document is not an active Investigational New Drug Application or Investigational Device Exemption, is not a submitted or approved protocol, and does not constitute medical, surgical, or regulatory advice. It is not endorsed by the United States Food and Drug Administration, the National Institutes of Health, the Department of Health and Human Services, any Institutional Review Board, the International Council for Harmonisation, or any sponsor. Any future first-in-human conduct of the design described here would require a separately submitted IND under 21 CFR Part 312 and a separately submitted IDE under 21 CFR Part 812, a 30-day FDA review period that elapses without the imposition of a clinical hold under 21 CFR §312.42, and prospective IRB approval under 21 CFR Part 56 together with informed consent obtained under 21 CFR Part 50 and ICH E6(R3). The staged-autonomy framework that constrains this Stage 1 study to clinician-controlled teleoperation and supervised operation, with the LLM acting only as a second-opinion advisory oracle and full autonomy prohibited under 21 CFR §312.21(e), reflects a deliberate ethical choice to keep a qualified human surgeon in unambiguous control of every committed action.

Back matter element	Disclosure content	Controlling authority
Sponsor-Investigator role	Combined sponsor and investigator obligations assumed by a single party	21 CFR §312.3(b); §§312.50 to 312.69
Financial interest	Sponsor-Investigator equity interest disclosed and mitigated by independent oversight	21 CFR Part 54; FDA Form 3455
Human subjects status	No enrollment, no PHI, no IRB approval; synthetic exemplar only	21 CFR Part 50; Part 56
Future conduct gate	Separate IND plus IDE, 30-day review with no clinical hold, prospective IRB approval	21 CFR §§312.40 to 312.42; Part 812
Autonomy limit	Stage 1 teleoperation and supervised operation; advisory oracle only	21 CFR §312.21(e)
Data and records	CC BY 4.0 archive; reproducible build sources; no proprietary data	21 CFR Part 11 (electronic records)

Table -.2. Back matter disclosure elements mapped to the controlling federal authorities that govern each element.

Data Availability

All application sources, spanning the `mermaid`, `draft-ind`, `full-ind`, and `final-ind` build stages, are available in the public GitHub repository and archived on Zenodo under the Creative Commons Attribution 4.0 International (CC BY 4.0) license at DOI 10.5281/zenodo.xxxxxxxx. The four build stages constitute a complete and reproducible authoring record: the mermaid stage holds the grayscale figure specifications, the draft-ind stage holds the section scaffolds, the full-ind stage holds the regulatory drafts, and the final-ind stage holds this senior-author polished application. The underlying proposed clinical research protocol v1.0.0 is archived separately at DOI 10.5281/zenodo.20780121, and the five supporting computational works are archived at their own DOIs as cited in the bibliography below. No proprietary, embargoed, or patient-identifiable data are contained in or required to reproduce this application; every numeric value can be regenerated from the deterministic, hash-pinned simulation seeds recorded in the source tree, consistent with the electronic-records integrity expectations of 21 CFR Part 11.

Rights and Permissions

Cite This Article

Kawchak K. Phase 1 PDAC IND: AI Generation. Zenodo. 2026; 10.5281/zenodo.xxxxxxxx. IND v1.0, repository v4.3.0.

Build stage	**Contents and role in the reproducible record**	**Archive**
`mermaid`	Grayscale Mermaid figure specifications (8-tone classDef ramp) that map 1:1 to the TikZ figures	Zenodo CC BY 4.0; DOI 10.5281/zenodo.xxxxxxxx
`draft-ind`	Section scaffolds with drafting instructions and the ReGARDD section file map	Same repository and DOI
`full-ind`	Full regulatory drafts with complete tables, figures, and citations	Same repository and DOI
`final-ind`	Senior-author polished application; full-width ragged-right tables; verified figures	Same repository and DOI

Table -.3. The four build-stage source trees that together form the archived, reproducible record of IND v1.0 under repository v4.3.0.

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